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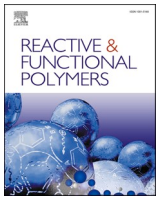

Review

Poly(glycidyl methacrylate): Mechanisms of ring-opening and radical polymerization, green synthetic strategies, and advanced functional applications

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ABSTRACT

Glycidyl methacrylate (GMA) is a bifunctional monomer combining a polymerizable methacrylate group with a reactive epoxide, enabling both radical polymerization and epoxide ring-opening (RO) reactions for the synthesis of functional polymers with tunable architectures. This review provides a comparative, mechanism-oriented analysis of GMA polymerization. Ionic polymerization proceeds via cationic and anionic pathways, where monomer-activated anionic systems offer superior control over molecular weight and effective suppression of side reactions, whereas cationic activated monomer (AM) processes provide faster kinetics and simpler operation but reduced structural precision.

Radical polymerization is discussed in terms of conventional free radical polymerization (FRP), externally stimulated radical polymerization (ESRP), and reversible deactivation radical polymerization (RDRP). ESRP approaches enable milder conditions, improved temporal control, and more sustainable radical generation compared to FRP, with plasma-initiated systems being particularly effective for GMA due to efficient surface activation and preservation of epoxy functionality. Within RDRP strategies, RAFT is the most suitable for GMA owing to its metal-free nature and broad functional group tolerance, whereas ATRP and NMP may be limited by catalyst requirements or narrower monomer scope. Externally stimulated controlled radical polymerization (ESCRP) further extends RDRP strategies by integrating external energy inputs with reversible deactivation mechanisms, thereby enhancing control and mitigating inherent limitations. Overall, these mechanistic differences govern functional group retention, polymer architecture, and material properties, thereby guiding the selection of the most appropriate polymerization strategy for targeted applications.

1. Introduction

Macromonomers with dual functionality have attracted considerable interest in polymer science because they provide access to multiple polymerization pathways and thus offer a high level of structural tunability. In general, a macromonomer "macromolecular monomer" can be defined as a low-molecular-weight polymer that retains the reactivity of a monomer while already exhibiting some of the features of a polymer [1–4]. This dual character makes it an effective link between simple monomers and macromolecules, allowing the design of more complex materials. Glycidyl methacrylate (GMA) is a representative example, as it contains both an epoxide ring and a methacrylate group within the same structure. The coexistence of these two functional moieties endows GMA with remarkable versatility, since it can undergo polymerization via both ring-opening (RO) and radical mechanisms, leading to a broad spectrum of architectures and properties [5,6].

The epoxide group of GMA is characterized by its high degree of reactivity, which can result in RO through a variety of mechanisms. The predominant mechanisms include cationic, anionic, and monomer-activated pathways. As with many chemical processes, the outcome of

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these reactions is strongly dependent on external parameters such as solvent polarity, medium acidity or basicity, and the nucleophilicity of the environment [7–11]. Studies on the ring-opening polymerization (ROP) of GMA remain relatively limited. This gap may be attributed to the intrinsic complexity of ROP mechanisms, which can involve multiple competing pathways occurring simultaneously. Such mechanistic competition, together with the occurrence of various side reactions, often leads to reduced control over the reaction and may result in the formation of different structural isomers, thereby complicating both the polymerization process and the resulting structure–property relationships. The methacrylate double bond can also, in principle, undergo anionic polymerization; however, its reactivity is significantly lower than that of the epoxide group. As a result, this pathway is less favorable and more difficult to control. The reduced reactivity of the vinyl functionality often leads to increased susceptibility to side reactions, thereby limiting the overall efficiency and selectivity of the process.

At the same time, the methacrylate functionality can be engaged in radical polymerization. Besides conventional free-radical polymerization (FRP) [12–16], a wide set of externally stimulated radical polymerization techniques (ESRP); photo-initiation, ultrasound assistance, microwave heating, plasma activation, electrochemical initiation and radiation initiation have expanded the synthetic toolbox [17–19]. These energy-assisted approaches offer more sustainable alternatives, operating under milder conditions and aligning more closely with green chemistry principles.

In parallel, controlled radical polymerization methods, particularly atom transfer radical polymerization (ATRP), reversible addition fragmentation chain transfer (RAFT) and nitroxide-mediated polymerization (NMP) have emerged as powerful strategies to synthesize poly (GMA) with precise molecular weights, low dispersities, and well-defined topologies [20,21]. However, the application of NMP to GMA remains less explored, partly due to the relatively high temperatures required, which may promote undesired epoxide RO [22]. In addition, ATRP systems rely on transition metal catalysts (typically Cu-based), which may not only raise concerns regarding residual metal contamination and purification requirements, but may also promote undesired epoxide RO under certain conditions [23].

To overcome these limitations, combining RDRP techniques with ESRP has emerged as an effective strategy to minimize side reactions while enhancing both control and sustainability. Controlled techniques ensure molecular-level accuracy, whereas ESRP techniques emphasize greener, energy-efficient processing. Their combination provides a dual advantage: structural precision and environmental responsibility. This integration is increasingly viewed as a promising direction for developing next-generation materials based on this multifunctional monomer [24–26].

Characterization of poly(GMA) has played a central role in connecting synthesis with functionality. Structural and functional analysis by Fourier transform infrared (FTIR) and nuclear magnetic resonance (NMR) spectroscopy, thermal behavior studies through differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA), and morphological investigations using scanning electron microscopy (SEM), transmission electron microscopy (TEM), and atomic force microscopy (AFM) have all contributed to a deeper understanding of structure–property relationships [27,28]. Such insights are essential for guiding the rational design of advanced materials. This role becomes even more critical for GMA-based systems, where the coexistence of RO and radical polymerization pathways, combined with possible side reactions and structural rearrangements, can lead to complex and sometimes ambiguous architectures. In this context, advanced and complementary characterization techniques are necessary not only to confirm polymer structure, but also to distinguish between competing mechanistic pathways and to accurately correlate synthetic conditions with the final material properties. In addition, the morphology of GMA-based systems has been widely reported to vary significantly depending on the polymerization route, reaction conditions, ranging from homogeneous structures to highly heterogeneous or phase-separated morphologies, which further highlights the strong structure–processing–morphology relationship characteristic of this monomer [29,30].

Recent developments in polymer science highlight sustainability as a central theme. The use of recyclable or bio-based catalysts, environmentally friendly solvents, and energy-efficient processes demonstrates the growing integration of green chemistry into polymerization strategies [31,32]. GMA serves as an excellent model to explore this transition because its dual reactivity allows conventional approaches to be reinterpreted in more sustainable contexts. In this context, green and sustainable polymerization mechanisms are gaining increasing attention; however, the valorization of sustainable routes toward GMA monomer production remains largely underexplored. Most existing studies focus on polymerization processes, while the upstream synthesis and sustainable functionalization of GMA are rarely addressed, leaving an important gap in achieving a fully sustainable material lifecycle.

The diversity of polymerization pathways available for GMA significantly broadens the range of accessible material structures, which in turn expands its application potential. This structural versatility has enabled GMA-derived polymers to find use across a wide variety of fields beyond fundamental polymer chemistry. In particular, poly(GMA)-based materials have seen increasing application in biomedical engineering, environmental remediation, industrial coatings and adhesives, and electrochemical devices [33–36]. These emerging applications highlight the importance of combining mechanistic understanding with advanced controlled polymerization techniques and environmentally friendly methodologies to fully exploit the potential of GMA in the development of sustainable materials.

Despite the large number of studies on GMA-based systems, several limitations remain. Most works focus separately on either RO or radical polymerization, resulting in a fragmented understanding of structure–property relationships. In addition, comparative analyses between conventional, controlled, and energy-assisted polymerization strategies are still scarce, particularly with regard to efficiency, sustainability, and the occurrence of side reactions. Consequently, a unified framework linking mechanistic control, processing conditions, and environmental considerations is still lacking.

In this context, and unlike most recent reviews in polymer science that primarily address radical polymerization architectures [37], polymer networks [38], or sustainability concepts [39] without specifically considering the dual reactivity of GMA-as well as the review by Muzamil et al. [40], which mainly focuses on post-polymerization modification of poly(GMA), the present work aims to provide a comprehensive and integrative perspective. Specifically, it proposes a unified and comparative framework that simultaneously considers ROP, radical polymerization, and controlled radical strategies within a single discussion. Furthermore, this review extends beyond existing literature by addressing key aspects often treated separately, including mechanistic competition, side reactions, ESRP techniques, and sustainability considerations such as green solvents, green catalysts, and the largely underexplored upstream production of the GMA monomer.

Building on this perspective, the present review is organized as follows. First, the fundamental mechanisms of epoxide ROP are examined, including cationic, anionic, and monomer-activated pathways. Next, the radical polymerization of GMA is discussed, ranging from conventional FRP to ESRP and RDRP techniques such as ATRP, RAFT, and NMP. This is followed by a section dedicated to the characterization of poly(GMA), covering spectroscopic, thermal, and morphological analyses. Subsequently, recent advances in green and sustainable polymerization are reviewed, with particular emphasis on catalysts, solvents, and bio-based systems. Finally, the broad range of applications of poly(GMA) in biomedical, environmental, industrial, and electrochemical fields is presented, highlighting the relationship between polymerization mechanisms, material structure, and functional performance.

2. The fundamentals of epoxide ring-opening mechanisms

2.1. Approaching nucleophiles; the mechanism of epoxide ring opening

Heterocyclic monomers are capable of undergoing RO reactions through nucleophilic attack. Among them, epoxides are three-membered cyclic ethers containing an oxygen atom. Despite the absence of a classical leaving group, epoxides are highly reactive toward nucleophiles due to their significant ring strain and associated ring instability [41,42]. Since nucleophilic substitution reactions are central to the ionic polymerization of GMA, which will be discussed in the following sections, it is essential to understand the dominant RO pathways (SN1- or SN2-type mechanisms), as well as the factors governing their selectivity.

Epoxide RO reactions can proceed under either basic or acidic conditions, as summarized in Scheme 1. Basic media involve strong nucleophiles, whereas acidic conditions activate the epoxide toward attack by weaker nucleophiles through protonation of the oxygen atom [10]. These two catalytic environments significantly influence both the reaction pathway and the stereochemical outcome of the products. In both cases, SN2-type mechanisms are generally predominant for asymmetrical epoxides such as GMA due to steric and electronic effects, although SN1-type pathways may also occur under strongly acidic conditions [43,44].

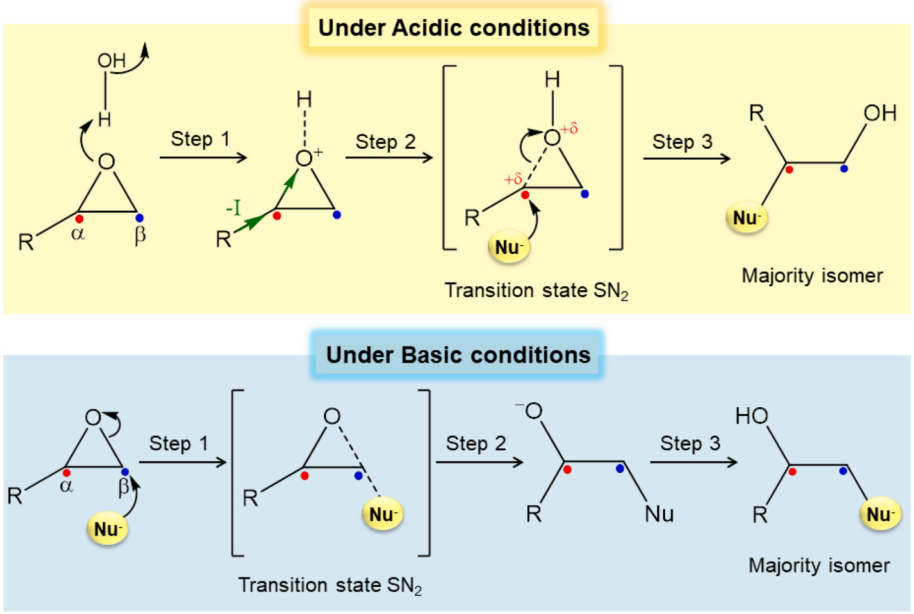
Under acid-catalyzed conditions, the SN2-type RO proceeds through a two-step process as shown in Scheme 1. First, protonation of the epoxide oxygen generates a better leaving group and increases the electrophilicity of the adjacent carbon atoms. The positive charge is preferentially stabilized at the more substituted carbon (α-position) through inductive effects, enhancing its susceptibility to nucleophilic attack (step 1). Subsequently, nucleophilic attack occurs simultaneously with C—O bond cleavage, forming a transition state characterized by partial bond formation and bond breaking (step 2). Completion of this step leads to RO and formation of the major regioisomer as illustrated in Scheme 1-step 3. A minor product may also form when nucleophilic attack occurs at the β-position [43].

An alternative SN1-type pathway can also occur under acidic conditions following epoxide protonation as depicted in Scheme 2. In this case, RO proceeds via the formation of a carbocation intermediate, either at the β-position (primary carbocation) or at the α-position (secondary carbocation), followed by nucleophilic attack. The reaction preferentially proceeds through the more stable, more substituted carbocation, leading predominantly to the major regioisomer via the secondary carbocation pathway (Scheme 2) [42].

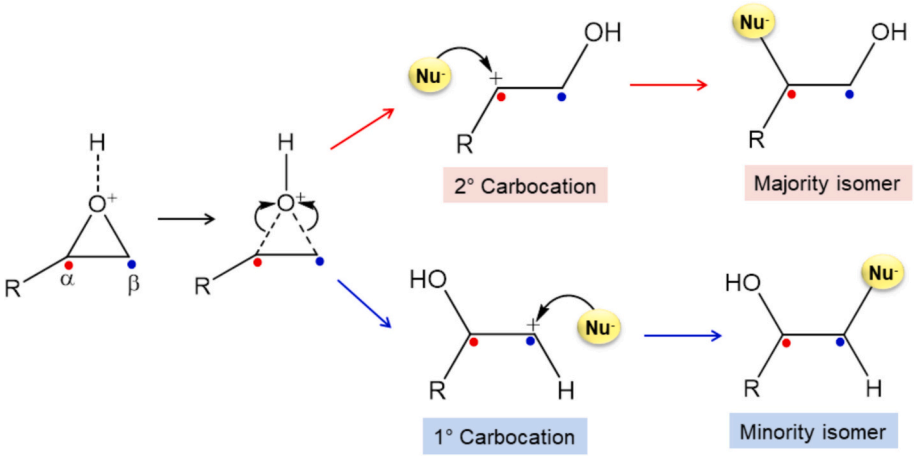
Under basic conditions, in the absence of sufficient acid to protonate the epoxide oxygen, nucleophilic attack proceeds preferentially at the least sterically hindered carbon atom (β-position), as illustrated in Scheme 1. The reaction follows a classical SN2 pathway (step 1), in which the nucleophile attacks the epoxide carbon simultaneously with C—O bond cleavage. This concerted process leads to oxirane ring opening and formation of an alkoxide anion intermediate (step 2). Subsequently, an intramolecular or medium-assisted proton transfer occurs, leading to protonation of the alkoxide species and formation of the final alcohol product as the major regioisomer (step 3). These alkoxide ions are strong bases (pKa ≈ 17); therefore, proton transfer occurs rapidly in the presence of weakly acidic protons in the system. Consequently, these intermediates do not promote further epoxide activation, and no polymerization is expected under these basic conditions [40]. Two regioisomers can be obtained under these conditions: nucleophilic attack at the β-position leads to the major product, whereas attack at the α-position results in the minor regioisomer, as shown in Scheme 1 [43].

In overview, the regioselectivity of asymmetric epoxide RO, including that of GMA, is governed by the nature of the nucleophile and the reaction conditions. In SN2-type mechanisms, weak nucleophiles under acidic conditions preferentially attack the more substituted carbon (α-position), whereas strong nucleophiles under basic conditions favor attack at the less sterically hindered carbon (β-position).

From a comparative perspective, SN2-type mechanisms proceed via a concerted pathway without the formation of discrete intermediates, resulting in higher regioselectivity governed primarily by steric and electronic effects. This makes them more suitable for controlled transformations of asymmetric epoxides and therefore preferred for the synthesis of well-defined polymer structures derived from GMA. In contrast, SN1-type mechanisms involve the formation of a carbocation intermediate, which enables reactions under relatively milder nucleophilic conditions but often leads to lower regioselectivity due to carbocation stabilization, rearrangements, and competing reaction pathways.



Scheme 1. Nucleophilic substitution mechanism type 2 under acidic and basic conditions

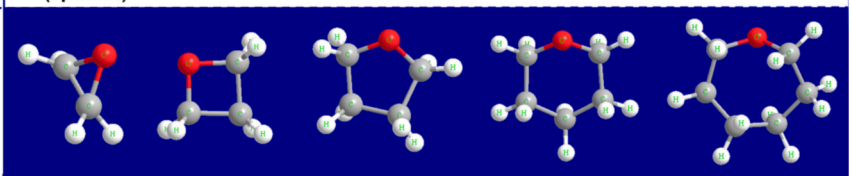


Scheme 2. Nucleophilic substitution mechanism type 1 under acidic conditions.

2.2. *Ring strain and reactivity of the epoxide group*

Ring-opening polymerization (ROP) is widely recognized as one of the most efficient methods for producing linear polymers from cyclic ethers. Among the different mechanisms involved, the cationic pathway is generally considered the predominant route for most cyclic ethers. However, oxiranes (epoxides) constitute a notable exception, as they can undergo both cationic and anionic polymerization depending on the reaction conditions and the nature of the initiating system. The RO ability of these monomers is governed by a combination of thermodynamic and kinetic factors [45], which are particularly favorable in the case of epoxides compared to other cyclic ethers. In this context, GMA, bearing an oxirane ring, represents a versatile monomer that can be polymerized via both anionic and cationic mechanisms. Therefore, understanding the relationship between these governing factors and the intrinsic reactivity of the epoxide ring is essential before discussing the detailed ROP mechanisms.

Among these factors, ring strain plays a central role in determining the reactivity of cyclic ethers [46]. This intrinsic property depends primarily on ring size, i.e., the number of atoms constituting the cycle as

Cyclic ether	Oxirane (epoxide)	Oxetane	Tetrahydropyran	Tetrahydrofuran	Oxepane
3D molecular structure					
2D chemical structure					
Ring size	3	4	5	6	7
Ideal sp^3 angle	109.5°	109.5°	109.5°	109.5°	109.5°
Actuel C-C-O C-O-C angle	~ 60°	88°-91°	100°-107°	109°-110°	112°-116°
Deviation from ideal angle	~ 49.5°	~ 21.5°-18.5°	~ 9.5°-2.5°	~ 0.5°	~ 2.5°-6.5°
Ring strain energy kJ mol^{-1}	≈112	≈106	≈23	≈5	≈42

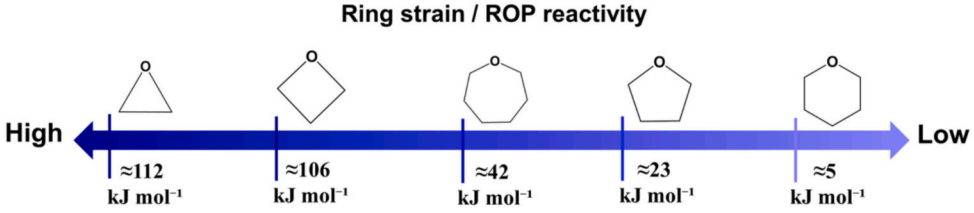


Fig. 1. Structures of cyclic ethers showing bond angles and associated ring strain energy.

illustrated in Fig.1. Smaller rings exhibit significantly higher strain due to geometric constraints. For instance, oxirane (112 kJ·mol^{-1}) and oxetane (106 kJ·mol^{-1}) display much higher ring strain than larger cyclic ethers such as oxepane (42 kJ·mol^{-1}), tetrahydrofuran (23 kJ·mol^{-1}), and tetrahydropyran (5 kJ·mol^{-1}) [47]. This increased strain originates from pronounced angle distortion (60° for oxirane and 90° for oxetane) and bond deformation in three- and four-membered rings. In contrast, five- and six-membered rings, such as tetrahydrofuran and tetrahydropyran, can adopt more stable conformations, leading to significantly reduced ring strain. This stabilization is attributed to favorable interactions, including hyperconjugative interactions between C—H bonds and the lone pairs of the endocyclic oxygen atom, as well as interactions between adjacent C—H bonds and CH_2 groups [45].

As a direct consequence, cyclic ethers with higher ring strain exhibit enhanced RO reactivity. This behavior can be rationalized in terms of thermodynamic parameters governing the conversion of cyclic monomers into linear polymer chains. As ring size increases, the reduction in ring strain leads to a decrease in the thermodynamic driving force for polymerization. This is reflected in both entropy and enthalpy contributions. For example, oxirane exhibits a more negative entropy of polymerization ($\Delta S^\circ_{p_o|\gamma} = -147$ J·mol^{-1}·K^{-1}) compared to tetrahydrofuran ($\Delta S^\circ_{p_o|\gamma} = -74$ J·mol^{-1}·K^{-1}) which reflects a greater loss of configurational freedom upon conversion from a small cyclic monomer to a more ordered polymer chain. At the same time, oxirane displays a significantly more exothermic enthalpy of polymerization ($\Delta H^\circ_{p_o|\gamma} = -140$ kJ·mol^{-1} vs -19.1 kJ·mol^{-1}), indicating a larger release of energy due to the efficient relief of ring strain during polymerization. Although the entropy change is unfavorable due to the loss of configurational freedom, the highly exothermic enthalpy associated with ring strain release dominates, resulting in an overall thermodynamically favorable polymerization process. Consequently, the equilibrium monomer concentration is much lower for oxirane (Meq $= 7.9 \times 10^{-15}$ mol·L^{-1}) compared to tetrahydrofuran (Meq $= 3.3 \times 10^{-1}$ mol·L^{-1}). indicating that the thermodynamic equilibrium is strongly shifted toward polymer formation in the case of oxirane, with nearly complete monomer conversion and a negligible tendency for depolymerization under equilibrium conditions [45,48,49].

In addition to thermodynamic considerations, the nucleophilicity of the cyclic ether also represents a key parameter influencing its reactivity. This property can be estimated from the basicity (pKb), since more basic species are generally better electron donors and thus more nucleophilic; consequently, nucleophilicity decreases as pK_b increases [50]. Cyclic ethers with lower nucleophilicity (and thus lower basicity) are less prone to electrophilic activation and are therefore more likely to undergo nucleophilic attack, favoring anionic polymerization pathways. This is particularly evident for oxirane ($pK_b = 7.55$), which is less basic than oxetane ($pK_b = 3.60$). Consequently, oxiranes preferentially undergo anionic polymerization, whereas oxetanes are more readily polymerized via cationic mechanisms [47].

Overall, the combined analysis of thermodynamic and kinetic parameters clearly indicates that oxiranes exhibit the highest propensity for ROP among cyclic ethers, due to their significant ring strain, favorable enthalpic driving force, and suitable electronic activation, which together make their reactivity more pronounced compared to larger cyclic ethers such as oxetane and tetrahydrofuran.

However, this apparent reactivity advantage must be interpreted with caution, as it does not necessarily ensure better control over polymerization processes. These considerations highlight the complex interplay between ring strain, thermodynamic parameters, and electronic effects in governing the reactivity of cyclic ethers. Cyclic ethers generally exhibit higher reactivity than their linear counterparts due to significant ring strain, whereas linear ethers are characterized by the absence of such strain, with bond angles close to the ideal tetrahedral value (~109.5°). In contrast, highly strained systems such as epoxides (~60° bond angles) benefit from a strong thermodynamic driving force for RO, which significantly enhances their reactivity. However, this same driving force can also increase susceptibility to side reactions, particularly under strongly nucleophilic conditions. In linear ethers, where ring strain is absent, reduced reactivity often limits undesired side reactions such as chain transfer; however, in some cases, increased basicity and coordination ability of oxygen atoms can still promote competitive transfer processes, especially in polar media. This indicates that reactivity and selectivity are not directly correlated, but rather depend on the balance between activation and stability of the propagating species.

Furthermore, although epoxides often favor anionic polymerization pathways due to their electrophilic activation, cationic polymerization typically exhibits faster kinetics (10^4–10^6 L·mol^{-1}·s^{-1}) compared to anionic systems (1–10^4 L·mol^{-1}·s^{-1}). However, this kinetic advantage in cationic systems does not necessarily lead to better structural control, as rapid propagation is frequently accompanied by increased side reactions such as rearrangements and termination events. In contrast, anionic systems, despite their slower kinetics, can provide higher structural precision when the reactivity of the carbanionic species is properly regulated.

Based on these apparent limitations, the following sections provide a more detailed and systematic discussion of the underlying mechanistic differences between cationic and anionic ROP pathways, with particular emphasis on their governing factors, the role of reaction conditions, and their implications for polymerization control, chain growth behavior, side reactions, and the resulting polymer architecture.

2.3. Ionic polymerization mechanism

Ionic polymerization of epoxide-containing monomers, such as GMA, represents a versatile and efficient strategy for the synthesis of functional polyethers. Depending on the nature of the active species, the ROP of the epoxide group can proceed via either cationic or anionic pathways. In general, the reaction proceeds through electrophilic activation in the case of cationic mechanisms, whereas nucleophilic attack governs the anionic process.

Under acidic conditions, cationic polymerization is known to proceed via two competing mechanisms that may occur simultaneously, the predominance of which strongly depends on the reaction conditions. Careful control of the reaction medium, including specific reaction conditions such as solvent and monomer concentration, can selectively favor one pathway over the other, thereby enabling control over polymer structure and properties. The first mechanism is identified as the “end of the activated chain” (ACE) pathway, while the second corresponds to the “activated monomer” (AM) mechanism. These two pathways differ significantly in terms of reaction control and susceptibility to side reactions.

Under basic conditions, two distinct anionic RO mechanisms can be distinguished. The first corresponds to the conventional anionic polymerization pathway, initiated by strong nucleophiles such as alkoxides or hydroxides, where propagation proceeds via successive nucleophilic attack on the epoxide ring. The second mechanism involves the activated monomer pathway, in which the monomer is first activated through interaction with a protic or Lewis acidic species, thereby increasing its susceptibility toward nucleophilic attack.

The activated monomer mechanism can, under appropriate conditions, provide improved control over the polymerization process due to a better regulation of the reactivity of the propagating species and a reduction in undesired side reactions. In contrast, conventional anionic polymerization generally exhibits a relatively slow polymerization rate, and the attainable molecular weight is often limited due to the occurrence of chain transfer reactions, particularly in the case of substituted epoxides such as GMA.

Thus, three types of ionic polymerization for epoxide RO can be considered, involving four distinct mechanisms, as presented in Fig. 2.

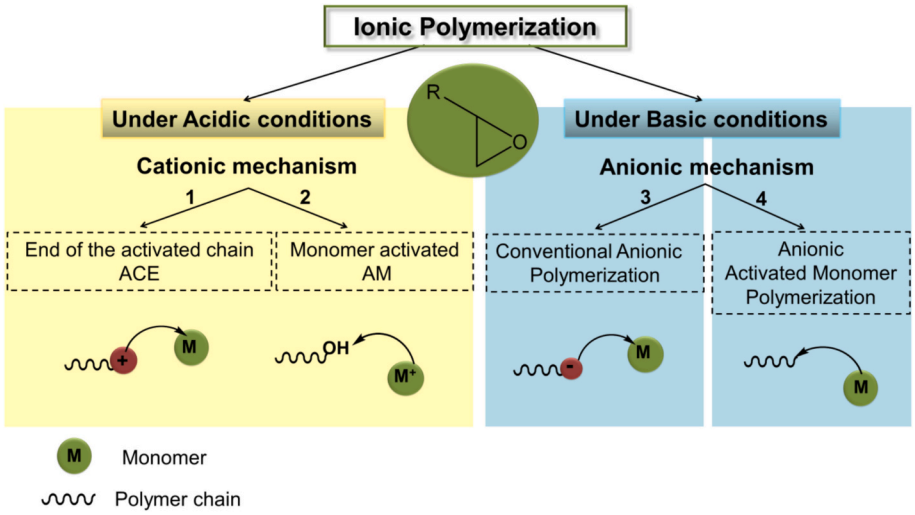


Fig. 2. Overview of the cationic and anionic ROP pathways of the epoxide group in GMA.

According to the above discussion, cationic polymerization can proceed through competing mechanisms that may occur simultaneously under the same catalytic system, where selective control of the reaction medium allows the preferential formation of one pathway over another. In contrast, anionic polymerization generally occurs under different catalytic systems and in distinct reaction media, leading to a clearer separation between the conventional anionic and activated monomer mechanisms.

In addition to epoxide ROP, vinyl polymerization of methacrylate groups in GMA can also proceed via anionic chain-growth mechanisms, leading to poly(methacrylate)-based architectures. In this pathway, propagation occurs through a carbanion-mediated Michael-type addition to the activated C═C double bond, resulting in a stabilized enolate-type propagating species. However, in multifunctional monomers such as GMA, this vinyl anionic polymerization may compete with epoxide RO reactions, leading to complex reaction pathways and reduced control over polymer structure.

Another important factor governing side reactions is the pH of the reaction medium, which can significantly influence the occurrence of transesterification processes. In this context, the hydrolysis of GMA can occur under basic conditions, indicating that reversible mechanisms may take place in solution depending on the reaction environment. In addition, such processes can lead to the formation of new by-products in the reaction medium, and can also be affected by the nature of the solvent, where certain solvents may promote or stabilize these reversible transformations.

In light of these aspects, the following section presents a detailed description of the ionic polymerization mechanisms of epoxide-containing monomers, with emphasis on GMA. The discussion covers the fundamental differences between cationic and anionic pathways, including their mechanistic features, reaction medium effects, and governing parameters, highlighting their impact on the final polymer structure and properties. Special attention is given to the competition between cationic mechanisms under acidic conditions, as well as the two distinct anionic pathways operating under basic conditions. In addition, the influence of reaction conditions on polymerization control, chain transfer processes, and side reactions such as transesterification and hydrolysis is addressed. Furthermore, the anionic vinyl polymerization of the methacrylate functionality in GMA, which proceeds via a carbanion-mediated chain-growth mechanism, will also be described to provide a comprehensive understanding of its dual reactivity. The detailed mechanisms of these processes, including associated transfer reactions, will be described with a focus on GMA as the model monomer.

2.3.1. Cationic polymerization mechanism (principles of cationic polymerization)

Cationic polymerization of epoxide monomers is a highly sensitive ionic process strongly influenced by the nature of the catalyst, reaction medium, and structural features of the monomer. In the case of multifunctional epoxides such as GMA, the overall mechanistic landscape becomes particularly complex due to the coexistence of multiple competing activation and propagation pathways.

The initiation step can be divided into two successive stages. The first stage occurs when the monomer interacts with an electrophilic species such as a proton, leading to the formation of a positively charged intermediate known as a secondary oxonium ion (Fig. 3). In the literature, three main activation pathways have been identified for epoxide monomers: protonation [51], alkylation [52], and acylation [53,54], depending on the nature of the catalyst and reaction conditions. The general representations of these activation modes are illustrated in Fig. 3.

Protonation corresponds to the coordination of the epoxide oxygen atom with a proton [51,55]. This process is typically promoted by Brønsted acids or Lewis acids in the presence of a co-catalyst such as water, which serves as a proton source. The two other activation

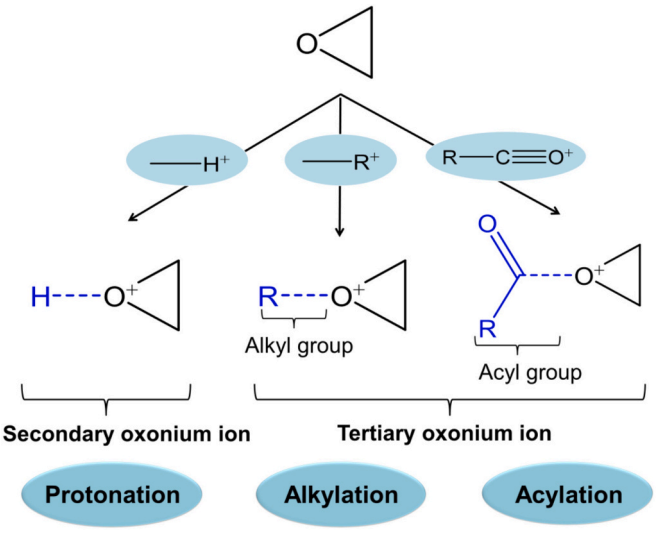


Fig. 3. General activation modes of epoxide monomers leading to cationic species formation.

pathways, alkylation and acylation [56,57], involve the covalent attachment of an alkyl or acyl group to the oxygen atom of the epoxide, respectively. Alkylation is commonly performed using alkyl halides [58], although triethyloxonium tetrafluoroborate ($(C_2H_5)_3O^+$ BF_4^-), a Meerwein salt [59], is one of the most efficient and widely used alkylating agents due to its high solubility in polar media and strong electrophilic character, enabling mild and efficient reaction conditions [60]. Acylation is generally achieved in the presence of strong Lewis acids (MX) such as BF_3, SbF_5, $AlCl_3$, or $ZnCl_2$, combined with an acyl halide as the acyl source [61]. The corresponding activation mechanism is illustrated in Fig. 4.

Acylation is generally considered more favorable than alkylation due to the introduction of a carbonyl group, which enhances the electrophilicity of the activated monomer and increases its susceptibility toward nucleophilic attack [62,63]. This effect may lead to a resonance-stabilized limiting structure involving partial positive charge delocalization over the oxygen atom (Fig. 5-a). However, the existence of a doubly charged oxonium species is not supported experimentally in the literature [64].

An approximate rationale for the absence of such highly charged species can be illustrated by analogy with the water molecule, for which the hydronium ion (H_3O^+) is thermodynamically favored over hypothetical higher charged species such as H_4O^{2+}. The formation of such a doubly protonated structure would lead to significant instability due to strong electrostatic repulsion between positively charged hydrogen atoms, as well as an unfavorable increase in coordination around the oxygen center (four hydrogen atoms instead of the stable three-coordinate configuration). On this basis, it can be inferred that the generation of a partial positive charge on the oxygen atom is sufficient to significantly enhance the electrophilicity of the acylation-activated epoxide without requiring the formation of highly charged, unstable limiting structures.

In addition, protonation leads to the formation of a secondary oxonium ion, whereas alkylation results in a tertiary oxonium ion as presented in Fig. 5-b. The introduction of alkyl substituents increases both steric hindrance and electron-donating effects, which generally reduces the electrophilic character of the activated monomer compared to protonated epoxides. Consequently, alkylated species tend to exhibit lower reactivity toward nucleophilic attack. Overall, based on these mechanistic considerations, protonation and acylation are regarded as the most efficient activation pathways for epoxide monomers, as they generate sufficiently electrophilic species that strongly favor nucleophilic RO reactions.

The second stage of the initiation process corresponds to the first propagation event and plays a decisive role in defining the overall polymerization mechanism. At this stage, two competitive mechanistic pathways may develop, leading to the coexistence of two main propagation modes, namely the activated chain end (ACE) mechanism and the

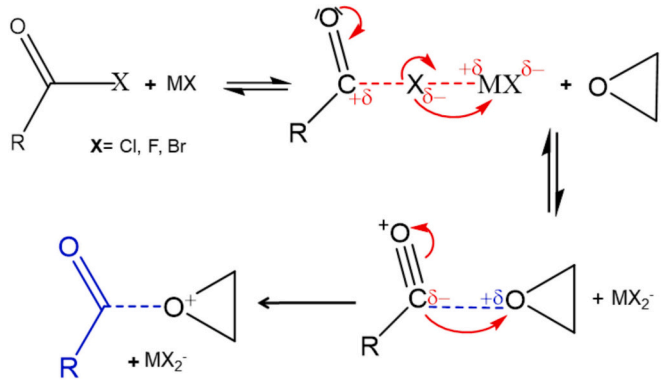


Fig. 4. Epoxide activation via acylation in the presence of Lewis acids and acyl halides.

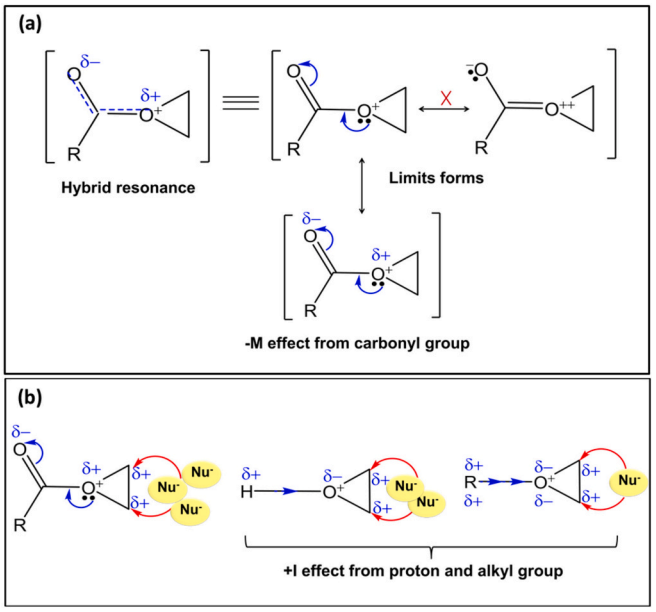


Fig. 5. (a) Attractive mesomeric effect of the carbonyl group in the acylation process. (b) Comparison of the suitability of the three epoxide activation processes for nucleophilic attack and the inductive donor effect exerted by the proton and alkyl group in the protonation and alkylation process.

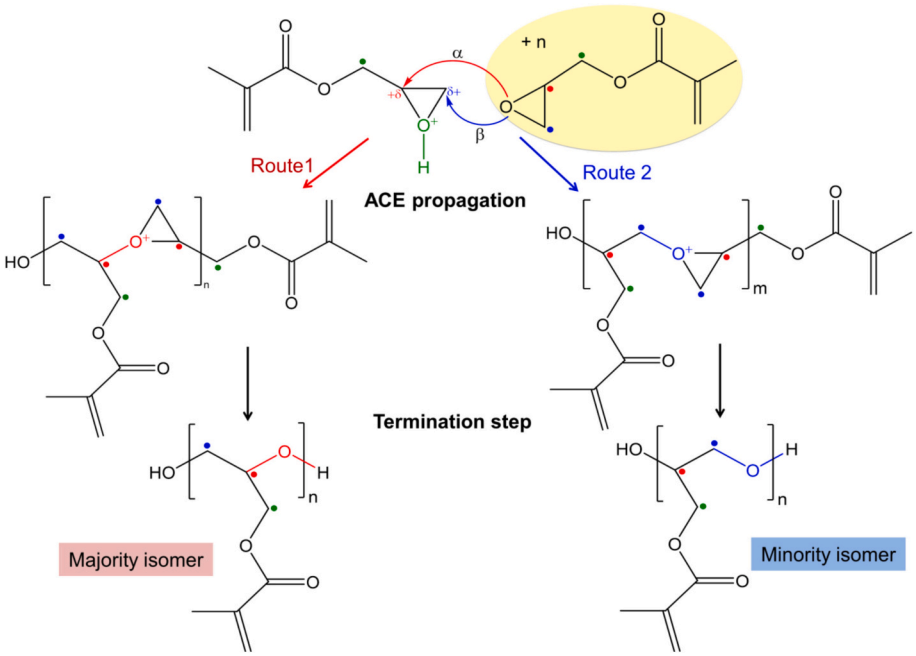
activated monomer (AM) mechanism [45,65–67]. In this context, the second stage of initiation can also be regarded as the first true propagation step, which ultimately governs the dominant mechanistic pathway.

In the case of the ACE mechanism (Scheme 3), the oxonium ion formed during the initiation step attacks another monomer at the α-carbon, leading to the formation of a new tertiary oxonium ion. This step can be considered as the continuation of chain activation, where the cationic active center remains located at the growing polymer chain end. These active species are sufficiently reactive to enable successive monomer insertions, thereby sustaining chain propagation.

Due to the asymmetric structure of the GMA epoxide ring, propagation proceeds via an SN2-type RO mechanism, which gives rise to two possible regioisomeric pathways. The nucleophilic attack may occur either at the tertiary carbon, leading to the major isomer (Route1), or at the secondary carbon, leading to the minor isomer (Route2). As a consequence of these competing pathways, the ACE mechanism is generally non-regioselective, resulting in atactic polymer structures. Even in cases where one pathway is kinetically favored, complete stereochemical control is not achieved, and a degree of structural randomness persists (Fig.6). The formation of these two poly(GMA) isomers via the ACE mechanism has been confirmed in several studies [68–70].

The AM mechanism involves the presence of a Brønsted acid in combination with an alcohol or polyol acting as a nucleophilic initiator (Scheme 4). In this pathway, the epoxide ring is first activated during the initiation step through protonation of the oxygen atom by the acid, leading to the formation of an oxonium ion, as previously described. In the subsequent step, the protonated (activated) monomer is attacked by the alcohol or polyol initiator, resulting in the formation of a covalent bond and the generation of a protonated chain end bearing a primary or secondary hydroxyl group, depending on the nature of the initiating species (Scheme 4). In this context, primary hydroxyl groups are typically formed when nucleophilic attack occurs at the α-position, whereas secondary hydroxyl groups are obtained when the attack proceeds via the β-position.

The propagation process then proceeds through successive addition of activated monomer units to the hydroxyl termini of the growing



Scheme 3. Simplified representation of cationic polymerization by activated chain end mechanism.

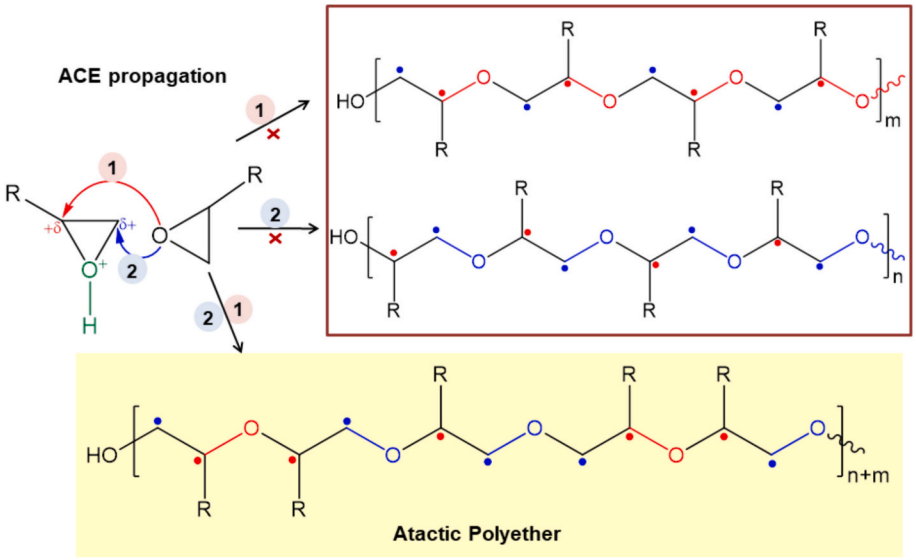


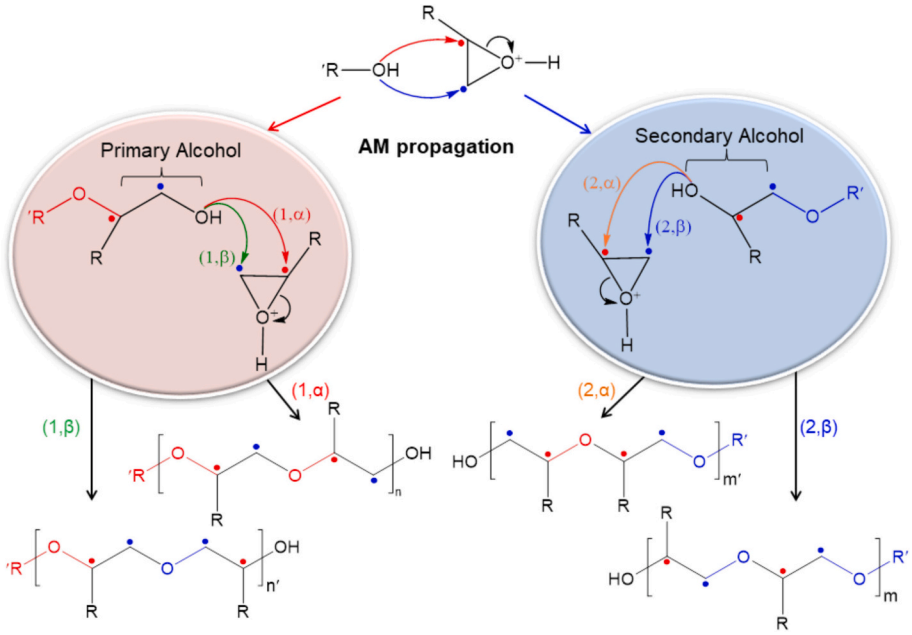
Fig. 6. Representation of the atactic arrangement of the polyether formed by the ACE mechanism.

polymer chain, with continuous regeneration of the active oxonium species. This stepwise growth mechanism ensures chain extension via repeated nucleophilic attack–proton transfer cycles, as reported in the literature [71,72].

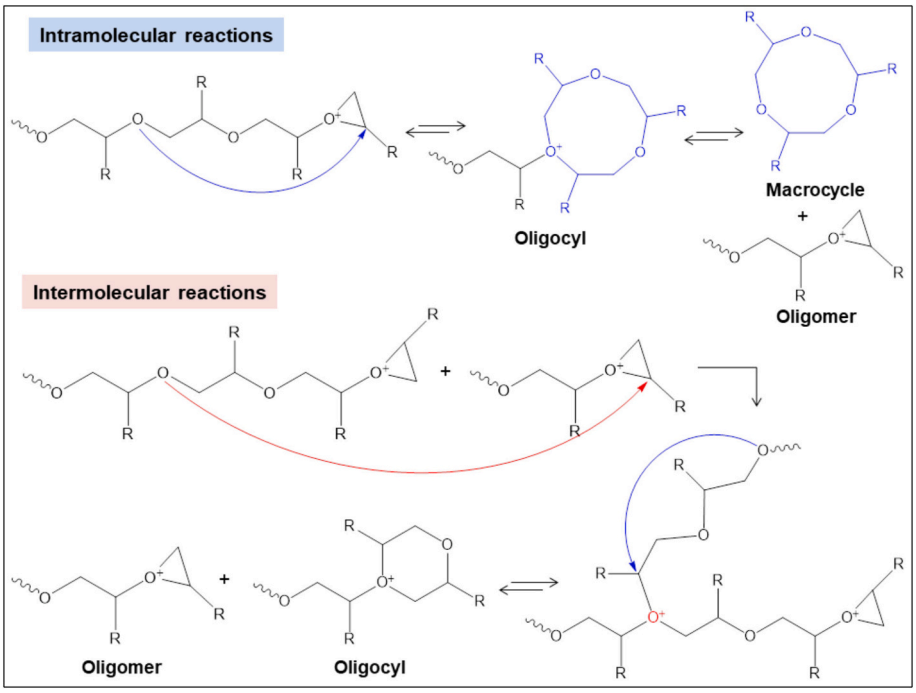
The mechanism becomes more complex in this case because GMA is a monomer bearing an unsymmetrically substituted oxirane ring, similar to propylene oxide [73] and epichlorohydrin [74], in contrast to ethylene oxide, which is considered an unsubstituted epoxide [75]. In such systems, four possible modes of addition can be identified when a substituent is present on the epoxide function, as illustrated in Scheme 4. Specifically, the hydroxyl group at the chain end may originate from either a primary (1,y) or secondary (2,y) alcohol formed during the initiation step, while the electrophilic attack on the activated epoxide ring may occur at either the methine (x,α) or methylene (x,β) carbon positions (α- and β-positions, respectively) [67].

These four possible addition pathways may generally occur in competition, with no single pathway being overwhelmingly dominant. In other words, their associated rate constants are expected to be of similar magnitude, resulting in a non-selective propagation process. The slight variations in reactivity can be attributed to a combination of electronic and steric effects arising from both the activated monomer and the terminal hydroxyl group of the growing chain [1]. Nevertheless, certain pathways may be favored depending on the nature of the substituent on the epoxide ring. In particular, additions involving primary alcohol-terminated chains are generally favored due to lower steric hindrance compared to secondary alcohols. As a consequence of this multiplicity of competing addition routes, polymers formed via the AM mechanism typically exhibit a linear structure with an atactic microstructure.

Transfer reactions are among the most common side reactions occurring during cationic polymerization. In the case of the ACE mechanism (Scheme 5), the oxonium ion located at the propagating



Scheme 4. Simplified representation of cationic polymerization by activated monomer mechanism.



Scheme 5. Schematic representation of the back-biting process with its two types of reaction: intramolecular and intermolecular.

chain end exhibits sufficient electrophilicity to react not only with weakly nucleophilic cyclic ethers but also with the oxygen atoms present along polyether chains. This reactivity is related to the higher basicity of ether oxygen atoms in linear polyethers compared to those in cyclic epoxide structures. As a consequence, these chain-end oxonium species can induce molecular weight redistribution through a process commonly referred to as “back-biting”. This intramolecular transfer leads primarily to cyclization reactions, resulting in the transformation of linear polyether chains into cyclic species as illustrate in Scheme 5. Therefore, the final products may include not only low-molecular-weight oligomers but also macrocyclic and oligocyclic structures with variable ring sizes, depending on the number of oxygen atoms involved in cyclization.

Importantly, the macrocycles formed through ether linkages may still retain reactivity, allowing them to further participate in transfer reactions and be reincorporated into growing polymer chains. Thus, back-biting contributes to a dynamic equilibrium between chain growth, degradation, and molecular weight redistribution processes. Mechanistically, back-biting reactions can proceed through two distinct pathways: intramolecular and intermolecular transfer. Intramolecular transfer occurs when the oxonium ion reacts with an ether oxygen located within the same polymer chain, leading to cyclization. In contrast, intermolecular transfer involves reactions between two different polymer chains, resulting in chain exchange and molecular

weight redistribution (Scheme 5) [9,11].

On the other hand, in contrast to the ACE-type mechanism, the cyclization phenomenon is significantly reduced in the AM mechanism due to the absence of charged species along the growing polymer chain [76]. Nevertheless, an undesirable side reaction may still occur, involving the nucleophilic attack of the oxygen atom of the growing chain onto the activated monomer, as illustrated in Scheme 6 (transfer reaction). However, for substituted epoxides, this secondary reaction is less frequently observed. This behavior can be attributed to the higher basicity of the terminal hydroxyl group as well as steric hindrance, which reduces the probability of intrachain reactions.

The comparison between these two mechanisms leads to the observation that the ACE mechanism is generally less favorable than the AM mechanism. This difference can be attributed to the location of the active centers: in the AM mechanism, the active species are located on the monomer units, whereas in the ACE mechanism, they are situated at the growing polymer chain end. As a consequence, these charged chain-end species are more prone to side reactions such as the "back-biting" process compared to neutral or less reactive species involved in the AM pathway.

One of the major challenges in cationic polymerization systems is that AM and ACE mechanisms often compete simultaneously. The predominant pathway is governed by reaction kinetics, particularly the relative rates of reaction between hydroxyl (HO^-) groups and protonated monomer species in the AM mechanism, and between monomer units and protonated monomer species in the ACE mechanism, as described in Eqs. (1) and (2), respectively [66,77].

$$v_{AM} = k_{AM}\,[OH]\,[M^+] \tag{1}$$

$$v_{ACE} = k_{ACE}\,[M]\,[M^+] \tag{2}$$

$$\frac{vAM}{vACE} = \frac{kAM\,[OH]}{kACE\,[M]} \tag{3}$$

Where v_{AM} and v_{ACE} are the rates of activated monomer and activated chain-end mechanisms, respectively; k_{AM} and k_{ACE} are the corresponding rate constants; [OH], [M], and [M^+] denote the concentrations of hydroxyl groups, monomer, and cationic active species.

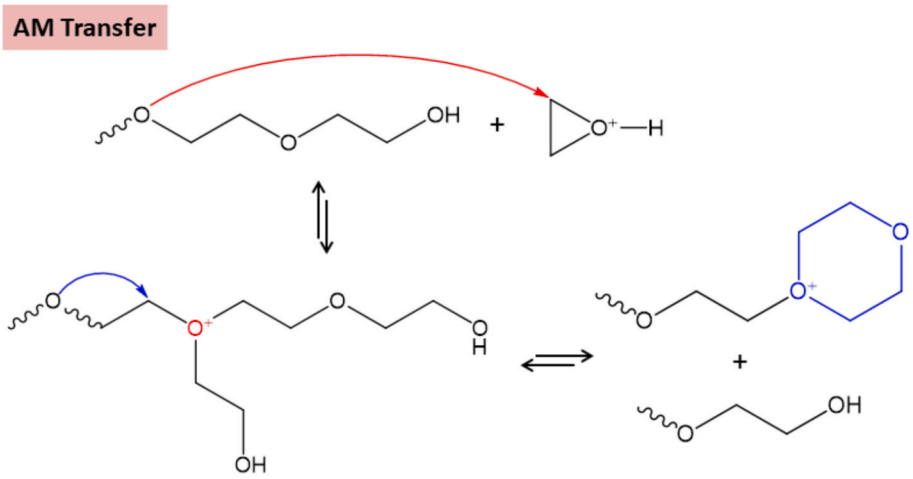
In practice, AM propagation can be favored when the reaction between hydroxyl (HO) groups and cationic species proceeds significantly faster than the reaction between monomer units and the same cationic centers as predicted by Eq. (3). To promote the desired mechanism, the polymerization conditions can therefore be adjusted by ensuring that the concentration of hydroxyl groups is much higher than that of monomer units [OH] >> > [M]. This requirement can be achieved either by increasing the amount of alcohol in the reaction medium, which typically leads to lower molar masses due to excessive chain initiation, or by gradually adding the monomer during the polymerization process. The latter approach allows the formation of polyethers with moderate molar masses; however, it often results in significantly longer reaction times [66]. Overall, by controlling these parameters, cyclization reactions can be reduced or even suppressed in cationic polymerization of cyclic ethers.

The termination step is required to stop chain propagation, particularly in the case of the ACE mechanism, where the positive charge is located at the end of the growing polyether chain. In contrast to the AM mechanism, in which the propagating species are essentially neutral after nucleophilic attack, termination in cationic systems mainly involves deactivation of the active oxonium centers to prevent further monomer incorporation and the formation of new growing chains. In practice, the termination process is generally not well-controlled and is often considered a secondary reaction pathway that may occur concurrently with transfer reactions, rather than a strictly independent step.

In conclusion, the cationic polymerization of epoxide monomers, particularly GMA, is governed by a highly complex mechanistic network involving multiple initiation pathways and two competing propagation modes, namely ACE and AM mechanisms. The coexistence of these pathways, combined with the intrinsic reactivity of unsymmetrically substituted epoxides, leads to a non-trivial balance between propagation, transfer, and side reactions. Among these processes, chain transfer reactions (notably back-biting and cyclization) play a central role in determining molecular weight distribution and structural complexity, often resulting in atactic and structurally heterogeneous polymers. Although the AM mechanism generally offers better control by reducing chain-end reactivity, both AM and ACE pathways may coexist depending on reaction conditions, limiting full mechanistic selectivity. Therefore, precise control over acidity, nucleophile concentration, and monomer addition strategy remains essential to steer the polymerization toward more defined architectures. A deeper mechanistic understanding is still required to achieve predictable and controllable cationic polymerization of multifunctional epoxide systems such as GMA.

2.3.2. *Conventional anionic polymerization mechanism*

2.3.2.1. Anionic polymerization of epoxide group. Anionic ROP of epoxides is a well-established and highly controlled route for the synthesis of polyethers, generally offering improved reaction control and reduced side reactions compared to cationic processes. As discussed previously, epoxides are more prone to nucleophilic attack than electrophilic activation, thereby favoring anionic polymerization pathways. This



Scheme 6. Representation of a possible transfer reaction for an unsubstituted epoxide.

behavior is particularly relevant for epoxides, which preferentially undergo anionic polymerization under suitable conditions, whereas larger cyclic ethers such as oxetanes are more often associated with cationic mechanisms due to their distinct structural and electronic features.

The anionic ROP of epoxides was first reported by Staudinger in 1933 [78] and later further developed by Flory in 1940 [79]. Since then, it has been established as a more controlled technique for the polymerization of reactive epoxide monomers toward nucleophilic species [80]. The most commonly used initiators for this type of polymerization are alkali metal derivatives, including hydrides, alkyls, aryls, amides, and, in particular, alkoxides of sodium, potassium, and cesium. In this context, the initiation rate is strongly dependent on the basicity of the corresponding alkoxide species, with higher basicity generally leading to faster initiation kinetics [79,81,82].

Studies on substituted epoxide monomers have demonstrated that epoxide functionalities embedded in more complex macromolecular structures, such as GMA-based polymers, can be efficiently activated under anionic conditions. In particular, the pendant epoxide groups of poly (GMA) can be converted into reactive alkoxide species upon treatment with strong bases such as sec-BuLi, enabling subsequent anionic ROP reactions. For example, Satti et al. [83] reported a successful grafting-from strategy in which the epoxide groups of poly(GMA) were first activated to generate poly(GMA–Li^+), followed by the anionic ROP of hexamethylcyclotrisiloxane, leading to well-defined poly(GMA-g-dimethylsiloxane) copolymers with a high degree of structural control and multiple siloxane units incorporated per epoxide site.

Despite the feasibility of such pathways, it is important to note that relatively few studies have focused on the direct anionic polymerization of GMA as a monomer. Instead, most reported works exploit the epoxide functionality of GMA in post-polymerization modification strategies rather than true chain-growth polymerization. In this context, Tsarevsky et al. [84] demonstrated that epoxide groups in GMA-containing copolymers undergo efficient nucleophilic RO reactions with sodium azide, yielding functional azido-alcohol derivatives in high yields. These functionalities were further utilized in click chemistry reactions to produce well-defined graft copolymers bearing poly (ethylene oxide) side chains, highlighting the high reactivity and versatility of the epoxide group toward nucleophilic activation outside classical polymerization processes.

Recent studies also provide additional insight into the effect of epoxide substitution on reactivity. Park et al. [80] reported the anionic ROP of functional epoxides under solid-state mechanochemical conditions, showing that while certain bulky epoxides may exhibit enhanced conversion under ball milling, their behavior in conventional solution polymerization remains fundamentally different. In particular, steric hindrance and electronic effects associated with substitution generally reduce propagation efficiency in solution, further supporting the lower chain-growth efficiency observed for substituted epoxides such as GMA under classical anionic conditions.

A clear comparison with unsubstituted epoxides further emphasizes this effect. Under anionic polymerization conditions, ethylene oxide remains the benchmark monomer, enabling highly efficient and well-controlled propagation to high molar masses, typically reaching Mn values in the range of 10^5–10^6 g·mol^{-1} with narrow dispersities (Mw/Mn ≈ 1.1–1.5) [85,86]. In contrast, substituted epoxides such as GMA generally exhibit reduced propagation efficiency due to steric and electronic constraints, as well as competing interactions introduced by functional substituents, leading to significantly less efficient chain growth under comparable conditions.

To better understand the origin of these pronounced differences in polymerization efficiency between unsubstituted and substituted epoxides, it is necessary to revisit the fundamental mechanistic features governing conventional anionic ROP. In particular, the balance between nucleophilic attack, ion-pair stability, and steric/electronic effects plays a decisive role in controlling propagation kinetics and suppressing or promoting side reactions. These factors become increasingly critical in functionalized systems such as GMA, where additional substituents can significantly alter the reactivity of the epoxide ring and the overall chain-growth process. Accordingly, the following section discusses in detail the key mechanistic aspects underlying anionic ROP.

In the initiation step, a nucleophilic SN2-type attack by oxyanion species occurs on the epoxide ring, leading to its opening and the formation of a new alkoxide species capable of attacking additional monomer molecules during the propagation step (Scheme 7). In the case of a mono-substituted epoxide, the nucleophilic attack preferentially occurs at the less sterically hindered carbon atom. During propagation, the successive addition of monomer units results in a random distribution of substituent groups, leading to the formation of polyethers with an atactic microstructure. The termination step is typically achieved by the addition of protic compounds containing labile hydrogen atoms, such as alcohols or water, yielding polyethers with hydroxyl end groups [82].

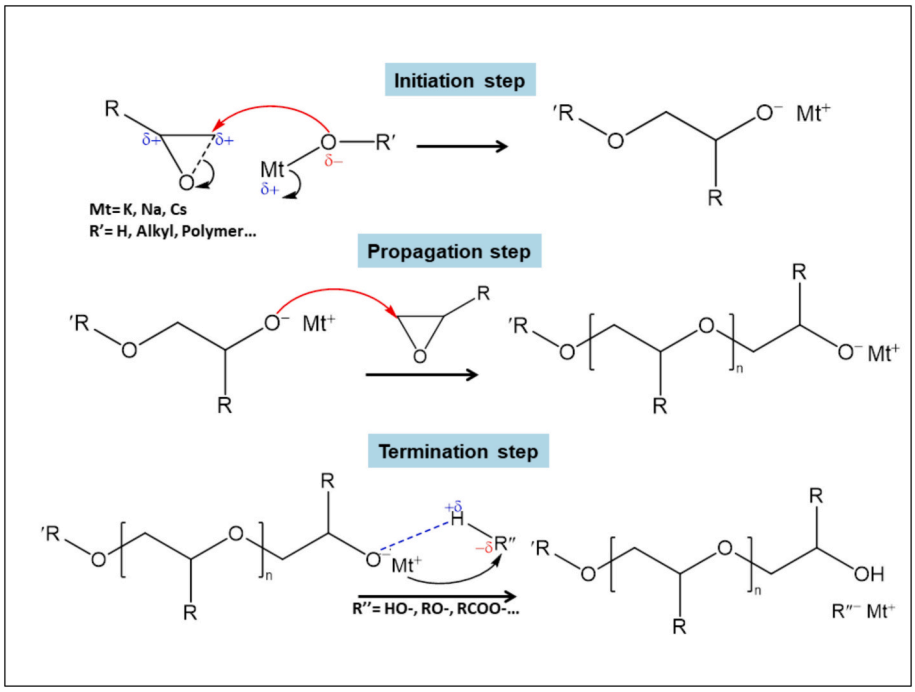
Despite the fact that oxirane derivatives generally favor anionic polymerization over the cationic route, as discussed in the previous section, this approach suffers from certain limitations. In particular, the polymerization rate is relatively slow and the attainable molecular weight is often limited due to the occurrence of chain transfer reactions, especially in the case of substituted epoxides. Several efforts have been devoted to improving this polymerization route in order to minimize chain transfer reactions and achieve better control over molar mass, although these limitations persist for certain catalytic systems [87].

Chain transfer reactions have been widely reported in the literature for monomers such as propylene oxide [88] and glycidyl ethers [89,90], These systems can therefore be used as representative models of mono-substituted epoxides to better understand similar side reactions occurring during the polymerization of GMA.

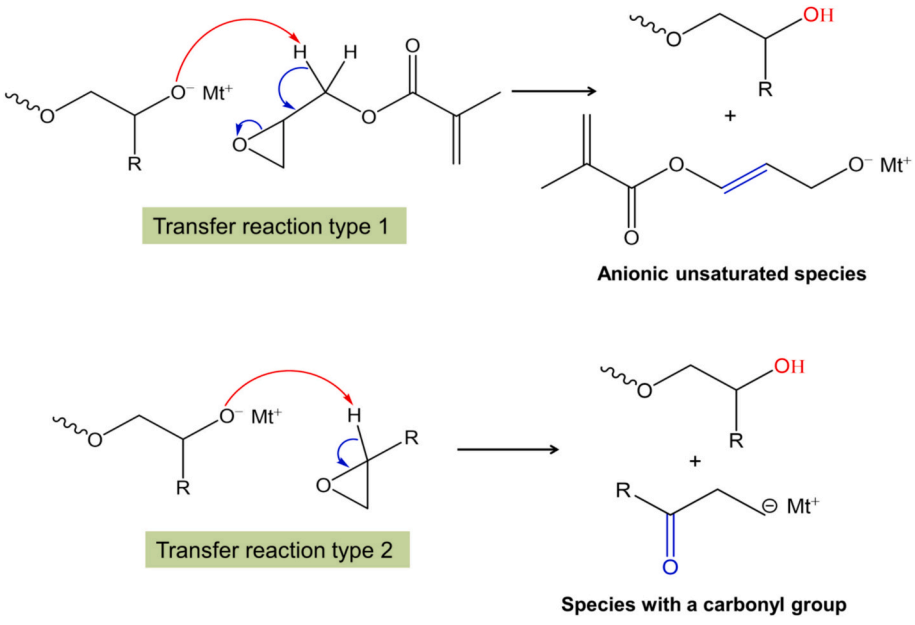
During the propagation of the polyether chain, the strongly basic conditions of the reaction medium, resulting from the high nucleophilicity of propagating alkoxide species, can promote chain transfer reactions (Scheme 8). One common pathway involves proton abstraction from the substituent group of the monomer unit by the oxyanion, leading to chain termination and the formation of a new initiating species, along with an unsaturated anionic byproduct capable of reacting with another monomer molecule (Scheme 8, type 1). Another type of transfer reaction involves proton abstraction from the epoxide ring itself rather than from the substituent group, resulting in the formation of a new species containing a carbonyl functionality (Scheme 8, type 2) [91].

Ultimately, it can be concluded that, in the case of substituted oxiranes, anionic polymerization using alkali metal alkoxides is inherently limited in its ability to achieve high molar masses under conventional conditions. This limitation arises from the pronounced tendency of substituted epoxides to undergo competing side reactions, particularly chain transfer processes such as proton abstraction, which continuously disrupt chain growth and regenerate new active species. As a result, the system predominantly yields low-molar-mass polyethers with relatively broad molar mass distributions. In contrast, ethylene oxide, owing to its unsubstituted structure and reduced susceptibility to such side reactions, can be polymerized in a significantly more controlled manner under similar conditions, enabling the formation of high-molar-mass polymers with narrower dispersity and more well-defined architectures.

2.3.2.2. Anionic polymerization of the methacrylate group. Methacrylate monomers constitute an important class of activated vinyl compounds widely used for the synthesis of functional polymers. Their reactivity is governed by the presence of an electron-withdrawing ester group directly conjugated to the carbon–carbon double bond. This structural feature induces a strong polarization of the C═C bond through combined inductive (—I) and conjugative effects, increasing the electrophilicity of the β-carbon and facilitating nucleophilic addition. At the same time, the resulting propagating carbanionic species formed after addition is stabilized by mesomeric delocalization toward the carbonyl group, which plays a key role in sustaining chain growth. As a result of



Scheme 7. Mechanism of epoxide RO using conventional anionic polymerization.



Scheme 8. Transfer reactions during anionic ROP mono-substituted epoxides monomers.

this electronic activation, methacrylates are well-suited for anionic chain-growth polymerization, where propagation proceeds via successive nucleophilic additions to the vinyl group. In favorable cases, this process can exhibit a high degree of control, enabling the formation of polymers with well-defined structures [92]. However, the efficiency and predictability of this polymerization pathway remain strongly dependent on the chemical structure of the monomer, particularly the nature of substituents that may influence both the reactivity of the double bond and the stability of the propagating species.

In this context, GMA represents a particularly intriguing monomer due to its bifunctional nature, combining a methacrylate double bond with a highly reactive epoxide group. While the methacrylate functionality is electronically activated for anionic polymerization as described above, the presence of the epoxide introduces additional reactivity that can compete with or interfere with the propagation of the vinyl chain. This dual reactivity makes GMA a complex system, in which the balance between vinyl addition and side reactions involving the epoxide group plays a decisive role in determining the overall polymerization behavior.

Typical initiating systems capable of promoting this pathway include strong organometallic bases such as organolithium compounds (e.g., n-butyllithium, sec-butyllithium) [93], often used in combination with organocuprate systems such as n-BuLi/(Cy_2N)Cu, as well as additives like lithium chloride to modulate ion pairing and improve control [94].

These systems generate highly reactive carbanionic species that can add to the activated methacrylate double bond and propagate the polymer chain under carefully controlled low-temperature conditions. More recently, alternative initiating systems based on thiourea anion/Lewis acid complexes have been reported to enable controlled anionic (co) polymerization of (meth)acrylates under mild conditions, while still operating through a carbanion-mediated propagation mechanism [95].

Several studies have explored the anionic polymerization of GMA via its double bond using such initiating systems. For instance, Feng and Han [96] reported the anionic polymerization and copolymerization of GMA using n-BuLi/$(Cy_2N)Cu$ in THF at temperatures below −15 °C. Under optimized conditions, high monomer conversion (up to 94.6%) was achieved; however, the resulting polymers exhibited relatively low number-average molar masses (Mn ≈ 5600 g•mol^{-1}) with moderate dispersity (Đ ≈ 1.46). In copolymerization with methyl methacrylate, incorporation of GMA reached approximately 52%, confirming its participation in the anionic process, but also highlighting the limited control over chain growth and molecular weight development. These relatively low Mn values further emphasize the difficulty of achieving efficient and well-controlled propagation when GMA is involved in anionic vinyl polymerization.

Similarly, Hild and Lamps [97] reported the synthesis of di- and triblock copolymers through sequential anionic polymerization of styrene and GMA. In their approach, well-defined polystyrene macroanions were first generated and subsequently used to initiate the polymerization of GMA, leading to poly(styrene)-b-poly(GMA) architectures. The authors demonstrated that propagation proceeds preferentially from the polystyryllithium species toward GMA, whereas the reverse initiation is not feasible, highlighting a marked asymmetry in reactivity. The use of modifiers such as 1,1-diphenylethylene was required to reduce chain-end reactivity and improve block formation control.

Similarly, Zhang et al. [98] reported the sequential anionic copolymerization of styrene and GMA using n-BuLi in the presence of LiCl in toluene/THF mixtures, yielding poly(styrene)-b-poly(GMA) block copolymers. Although relatively well-defined structures were obtained, a slight broadening of molar mass distribution was observed upon incorporation of GMA units, reflecting its more complex reactivity compared to classical vinyl monomers.

A representative comparison was reported by Antoun et al. [99] who investigated the anionic polymerization of various methacrylates, including methyl methacrylate, tert-butyl methacrylate, and GMA, using LiCl-complexed sec-butyllithium in THF at low temperature. Their results clearly demonstrated that, under identical conditions, only tert-butyl methacrylate exhibits a truly living behavior, whereas GMA displays significantly reduced control, with deviations between theoretical and experimental molar masses and limited reinitiation efficiency. This difference highlights the strong influence of the epoxide group in GMA, which can interact with the propagating carbanionic species or undergo side reactions, thereby disturbing the propagation of the methacrylate chain. As a result, even when polymerization of the double bond is achieved, the system often departs from ideal living characteristics. More generally, these findings emphasize that, in contrast to conventional methacrylates, the anionic polymerization of GMA is intrinsically less predictable and more sensitive to reaction conditions.

To rationalize these deviations from ideal living behavior, a detailed mechanistic analysis of the anionic polymerization of the methacrylate double bond in GMA is required, including both the electronic factors governing propagation and the competing side reactions that disrupt carbanion stability and reaction control.

The anionic polymerization of GMA via its methacrylate double bond proceeds through a nucleophilic chain-growth mechanism that is strongly influenced by the electronic structure of the ester group. The vinyl functionality in GMA is activated toward nucleophilic attack by the electron-withdrawing character of the adjacent ester moiety. This activation arises primarily from the inductive (—I) effect of the carbonyl group, which withdraws electron density through σ-bonds and increases the electrophilicity of the β-carbon (terminal CH_2) of the double bond, as illustrated in Scheme 9. In addition, conjugation within the C=C–C=O system further contributes to the polarization of the double bond, reinforcing the susceptibility of the β-carbon to nucleophilic attack.

The initiation step involves the generation of a carbanionic species, typically from strong organolithium initiators such as sec-butyllithium, often in the presence of lithium chloride to modulate ion pairing [100]. The nucleophilic carbanion adds preferentially to the β-carbon of the methacrylate double bond in a Michael-type addition, leading to the formation of a new carbon–carbon bond and a propagating carbanion located at the α-position relative to the carbonyl group [101], as presented in Scheme 9. The stability of this propagating species is ensured by a mesomeric (−M) effect, whereby the negative charge is delocalized toward the carbonyl group, forming an enolate-like structure. This resonance stabilization is essential for sustaining the propagation step and enabling chain growth.

Propagation proceeds through successive additions of monomer units via the same mechanism, ideally leading to a living polymerization system. However, in the case of GMA, the presence of a pendant epoxide group significantly affects the reactivity and control of the process. The epoxide ring is highly strained and more electrophilic than the methacrylate double bond, making it particularly susceptible to nucleophilic attack. As a result, several competing side reactions can occur.

First, direct nucleophilic attack on the epoxide ring leads to RO and formation of alkoxide species, which alters the nature of the active center and disrupts the vinyl propagation pathway. Second, these newly formed alkoxides can participate in secondary reactions such as chain transfer or intermolecular reactions, further reducing control over molar mass and dispersity. Third, coordination effects between lithium cations and the epoxide oxygen or carbonyl group may modify the ion-pair structure and influence the reactivity of the propagating species. Finally, the coexistence of multiple reactive sites within the same monomer introduces competition between vinyl addition and RO processes, leading to decreased selectivity and reduced efficiency of chain growth.

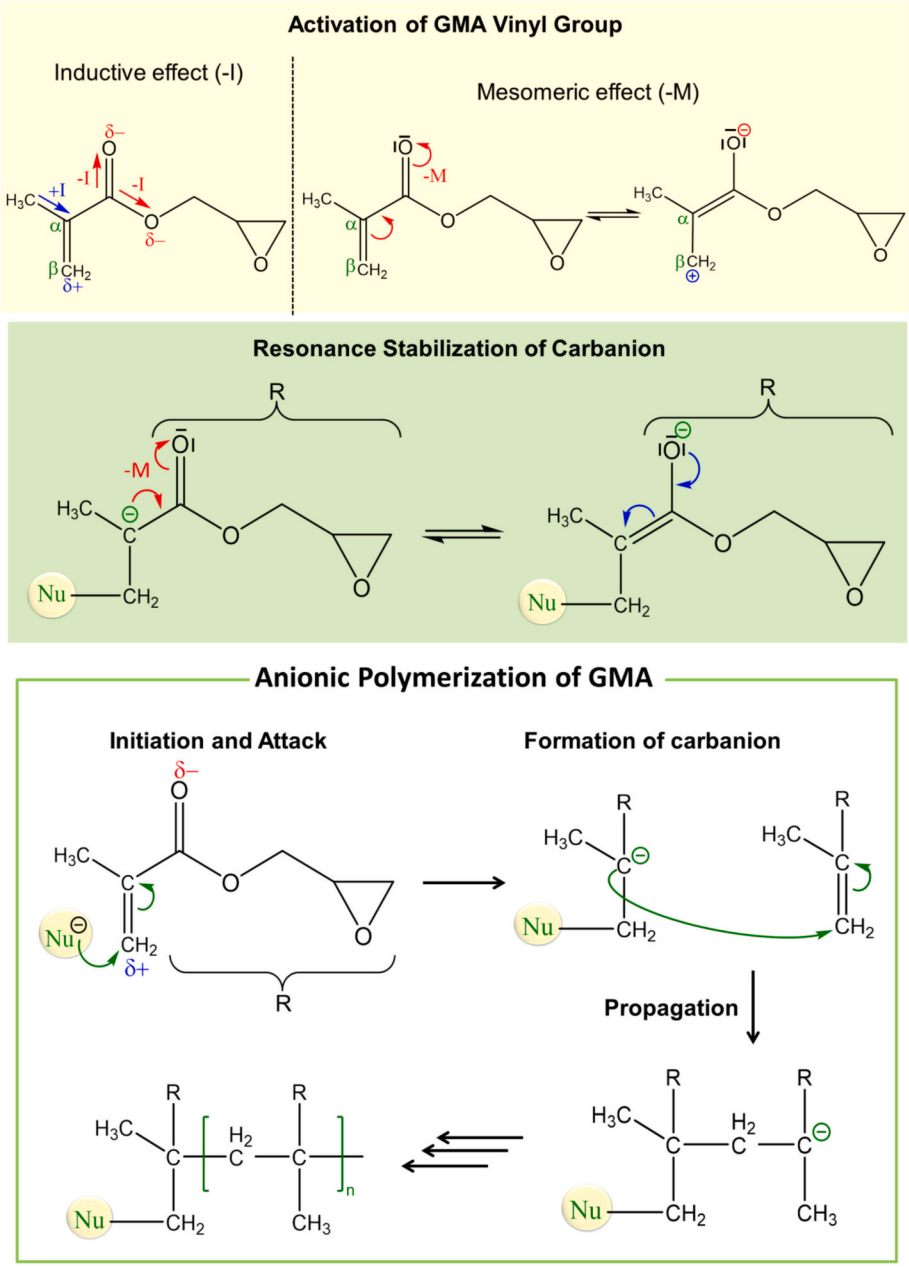
Overall, although the methacrylate double bond in GMA is electronically activated for anionic polymerization through inductive and conjugative effects, the concurrent high reactivity of the epoxide functionality introduces significant limitations. These competing processes result in reduced control, lower achievable molar masses, and deviation from ideal living behavior, thereby restricting the practical applicability of this polymerization pathway. Recent advances in alternative initiating systems and mild anionic polymerization strategies may however provide new opportunities to improve control over such bifunctional monomers.

2.3.3. *Monomer activated anionic polymerization mechanism*

Monomer-activated anionic polymerization is an advanced strategy that combines features of anionic and coordination polymerization. In this mechanism, the monomer is first activated through coordination to a Lewis acidic center prior to nucleophilic attack [102,103]. This approach has been developed to overcome the limitations of conventional anionic polymerization, particularly in terms of reactivity and achievable molar mass.

Historically, since the pioneering work of Pruitt and Baggett in 1955, iron chloride-based catalytic systems have been widely investigated for the synthesis of high-molecular-weight polyethers [104,105]. Shortly thereafter, Price and Osgan [106], demonstrated that such catalysts could produce crystalline and stereoregular poly(propylene oxide), highlighting their ability to control the configuration of asymmetric centers and generate isotactic structures. These findings have attracted considerable attention and stimulated extensive research into coordination-based catalytic systems for epoxide polymerization.

Epoxide polymerization using coordination catalysts involves a wide range of metal derivatives, including main-group metals such as zinc, cadmium, and aluminum, as well as transition metals such as iron [107].



Scheme 9. Anionic vinyl polymerization pathway of GMA: electronic activation and carbanion stabilization.

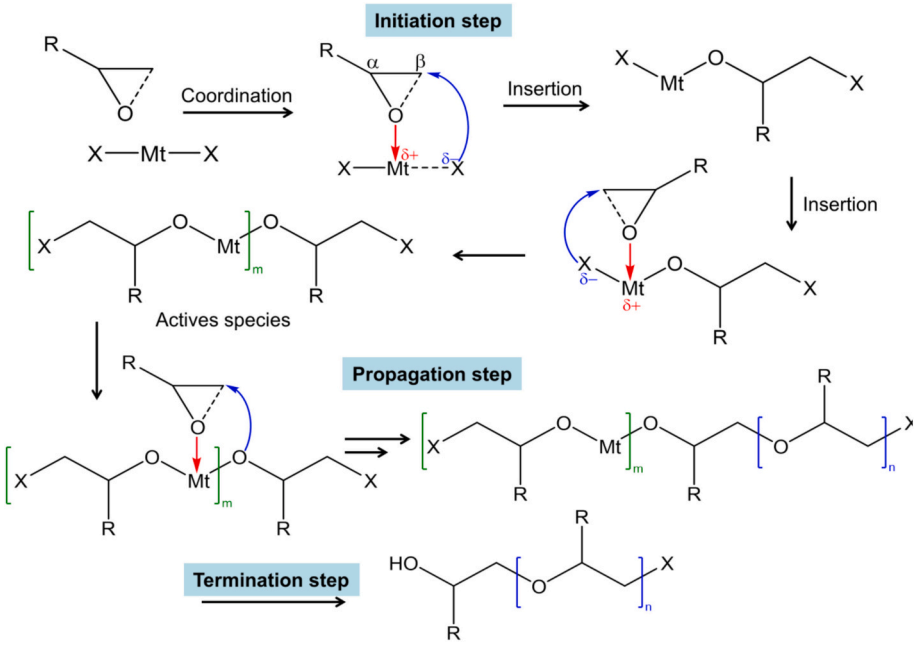
The most widely studied catalysts are those consisting of diethylzinc [108,109] or triethylaluminium [110–112] combined with a co-catalysts as anionic initiator such as; alkali metal salts, alkali metal alkoxides, onium salts, ammonium salts etc.

These catalytic systems have been successfully applied to a broad range of epoxides including propylene oxide [113], glycidyl methyl ether [114], epichlorohydrin [115], allyl glycidyl ether [116], glycidol [117]. In the case of GMA, this approach has been particularly attractive due to its bifunctional nature, allowing selective RO of the epoxide group while preserving the methacrylate functionality. Under monomer-activated anionic conditions, controlled homopolymerization and copolymerization have been achieved, typically leading to polymers with Mn values on the order of $\sim 2 \times 10^4$ g·mol^{-1} within short reaction times (2 h), while maintaining relatively narrow dispersities [7].

To better understand the advantages offered by these coordination-based catalytic systems compared to conventional anionic mechanisms, it is essential to examine their underlying mechanism in greater detail. In particular, these systems exhibit enhanced propagation rates and improved stereocontrol, including the ability to generate more isotactic structures. Such performance arises from the cooperative activation of the monomer and the growing chain end through coordination to a Lewis acidic metal center [47]. A detailed mechanistic description is therefore necessary to rationalize the origin of this increased reactivity and selectivity.

This polymerization process is initiated by prior coordination of the epoxide monomer to the metal center (M–X) through the lone pair of the oxygen atom, forming a metal–monomer complex (Scheme 10). This coordination activates the oxirane ring by increasing the electrophilicity of the carbon atoms, thereby lowering the energy barrier for RO. Once activated, the nucleophilic species associated with the metal center attacks the epoxide ring, preferentially at the least substituted (β) carbon atom, leading to RO and formation of a metal–alkoxide chain end, which acts as the active propagating species [108] (Scheme 10).

A key feature of this mechanism is that the incorporation of the



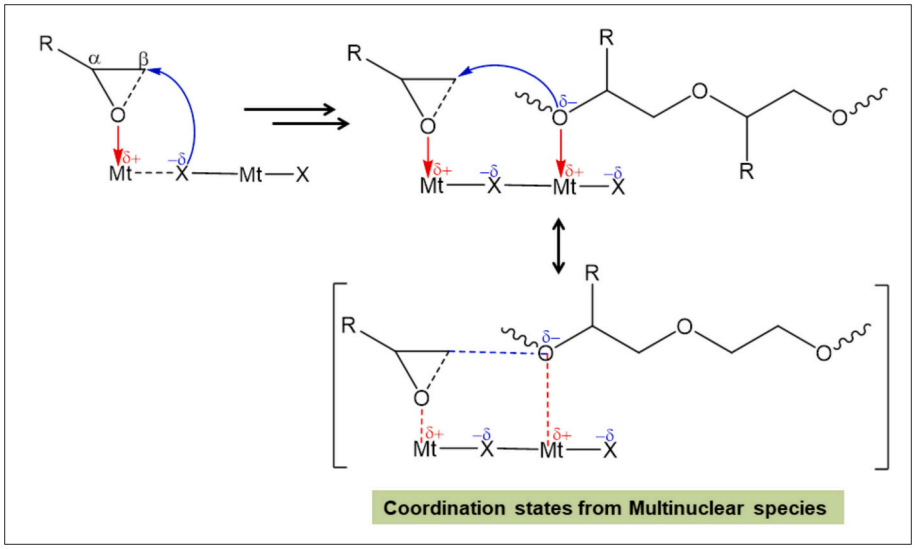
Scheme 10. Mechanism of ROP of GMA via monomer activated anionic polymerization using mononuclear catalysts.

coordinating oxirane into the active Mt–X bond proceeds via a multi-centred transition state rather than a four-centred one, involving simultaneous bond formation and bond cleavage [107,118]. Within this mechanistic framework, two types of catalytic species can be distinguished depending on the number of metal centers involved: mononuclear (X–Mt–X) as presented in Scheme 10 and multinuclear (Mt–X–Mt–X) complexes as shown in Scheme 11. These structural differences play a crucial role in determining the polymerization behavior.

Multinuclear species provide multiple coordination environments and distinct active sites, which enhance monomer activation and promote faster propagation. As a result, they generally favor the formation of higher-molecular-weight polyethers (Scheme 11). In contrast, mononuclear catalysts (Scheme 10) offer a more uniform active site environment, often leading to improved control over chain growth and more "living-like" polymerization behavior [119].

Using these catalytic systems, polymerization can yield two distinct fractions exhibiting different crystallization behaviors (Fig. 7): a crystalline fraction with significantly higher molar mass and an amorphous fraction with lower molar mass. The crystalline fraction has been identified as an isotactic polyether with regular head-to-tail linkages, formed via C_β-attack during oxirane RO. In contrast, the amorphous fraction corresponds to an atactic polymer, arising from C_α-attack on the monomer, which leads to head-to-head and tail-to-tail linkages and therefore a loss of stereochemical regularity [120].

This difference can be rationalized by the regioselectivity of the nucleophilic attack on the epoxide ring. When the attack occurs at the Cα position, two distinct RO pathways are possible, leading to either retention or inversion of configuration at the stereogenic center, which results in a mixture of S and R configurations along the polymer chain and consequently an atactic structure. In contrast, nucleophilic attack at the C_β position proceeds through a single dominant RO pathway, preserving the initial configuration of the monomer (either S or R). This selective pathway leads to a regular stereochemical arrangement along the chain and results in a single crystalline isotactic fraction. The two



Scheme 11. Coordination states from multinuclear catalysts via monomer activated anionic polymerization of epoxide.

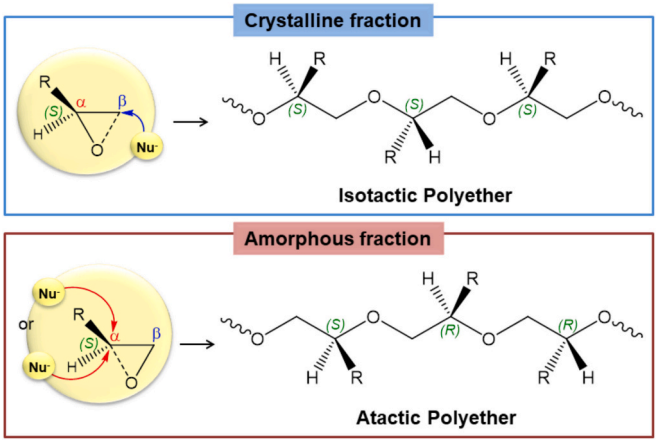


Fig. 7. Representation of possible isotactic and atactic polyether structures generated using monomer activated anionic polymerization of epoxide.

possible isotactic and atactic structures are schematically illustrated in Fig. 7.

In conclusion, monomer-activated anionic and coordination-assisted catalytic systems represent an efficient strategy for the controlled ROP of epoxides. These approaches rely on prior monomer activation through coordination to a Lewis acidic center, leading to the formation of highly reactive complexes. Such activation enhances monomer reactivity, improves ion-pair separation, and accelerates propagation under relatively mild conditions. Compared to conventional anionic polymerization systems, these catalytic approaches can achieve higher molecular weights and faster kinetics by effectively suppressing side reactions, particularly chain-transfer processes. In addition, they can provide improved control over chain microstructure, in some cases enabling the formation of more ordered crystalline fractions that are not typically accessible via conventional methods. Despite these advantages and the growing number of studies on epoxide monomers, investigations specifically focused on GMA within these catalytic systems remain relatively limited. Nevertheless, one drawback of these systems is that the amount of catalyst is typically relatively high, making removal difficult afterwards.

Overall, these features highlight both the versatility and the limitations of such catalytic systems for the synthesis of well-defined polyethers and functional polymer architectures. Additional investigations focused on these systems would be valuable to further advance their development.

At the end of this section devoted to the ionic polymerization of GMA, a comparative evaluation of the four possible mechanistic pathways is required in order to clearly highlight the respective advantages and limitations of each polymerization route in terms of complexity, selectivity, and applicability. In this context, ROP can proceed through either cationic or anionic pathways, as the selection of the mechanism strongly depends on the nature of the monomer and the targeted polymer properties. A systematic comparison will first be carried out within each class of mechanisms (cationic and anionic separately), before proceeding to a global comparison between both ionic approaches.

In cationic polymerization, two mechanisms ACE and AM are generally considered to operate simultaneously. However, the AM pathway is typically more favorable than the ACE mechanism. This difference arises from the nature of the active species: in the AM mechanism, the reactive centers are associated with activated monomer units, whereas in the ACE mechanism, they are located at the growing polymer chain end. As a consequence, the charged chain-end species involved in the ACE pathway are more prone to side reactions, such as back-biting and chain transfer, leading to reduced control over molar mass and polymer structure.

From a structural standpoint, the ACE mechanism typically leads to a more limited number of isomeric products (generally two), whereas the AM mechanism can generate a larger variety of isomers (up to four), thereby increasing the complexity of the reaction pathway and complicating the identification and characterization of the resulting products. Despite this increased complexity, the AM mechanism remains advantageous due to its reduced susceptibility to side reactions and improved overall selectivity. However, this complexity may limit predictive control and complicate process optimization, particularly for multifunctional monomers.

A comparison between the two main anionic pathways—conventional anionic polymerization and monomer-activated (coordination-assisted) polymerization—highlights significant differences in terms of reactivity, control, and achievable molar mass. Conventional anionic polymerization relies on strong nucleophiles and typically provides good control for simple, unsubstituted epoxides but becomes less efficient for functional monomers due to steric and electronic effects. In contrast, monomer-activated anionic systems exhibit enhanced performance, enabling access to higher molecular weights and faster kinetics through effective suppression of side reactions, particularly chain-transfer processes. However, this improved efficiency is accompanied by practical limitations, as these systems generally require relatively high catalyst loadings, which can complicate catalyst removal and downstream purification.

A direct comparison between anionic and cationic pathways further highlights their complementary nature in terms of application. While anionic systems generally provide superior control over polymer architecture, cationic routes particularly via the AM mechanism are often easier to implement and proceed under faster and milder conditions, albeit with reduced control. In addition, although anionic mechanisms ensure better selectivity toward epoxide RO, functional monomers such as GMA may still present challenges due to the sensitivity of the methacrylate double bond under nucleophilic conditions, requiring careful optimization of reaction parameters. Importantly, all cationic and conventional anionic routes typically yield linear atactic polyethers, whereas isotactic structures are mainly accessible through specific anionic coordination-controlled systems.

Overall, these considerations demonstrate that no single mechanistic pathway can be regarded as universally optimal. Instead, the choice between cationic and anionic routes must be carefully balanced according to the required level of structural control, functional group tolerance, and synthetic practicality, particularly when multifunctional monomers such as GMA are involved.

2.4. *pH- and solvent-dependent reaction pathways*

The polymerization of GMA is not limited to the RO of the epoxide group but is also influenced by several competing side reactions that can significantly affect both the reaction pathway and the properties of the resulting polymers. Among these, hydrolysis and transesterification play a key role, as they involve the methacryloyl functionality and may compete with the desired propagation process. The extent of these reactions is strongly governed by the reaction conditions, particularly the pH of the medium, solvent nature, and water content, and the presence of nucleophilic species. Therefore, understanding these competing processes is essential for achieving controlled polymerization and tailoring the structure of GMA-based materials.

GMA is a bifunctional monomer composed of a glycidyl (epoxide) group and a methacryloyl (ester) functionality, as illustrated in Fig. 8. During polymerization, the desired epoxide RO process is generally considered relatively slow and irreversible; however, it may compete with faster side reactions that can significantly influence the overall reaction pathway. This behavior was first highlighted by Van Dijk-Wolthuis et al. (1995) [121], who demonstrated that reversible processes may occur simultaneously with epoxide RO polymerization under specific reaction conditions. In particular, these reversible pathways are mainly associated with transesterification reactions, which are distinct

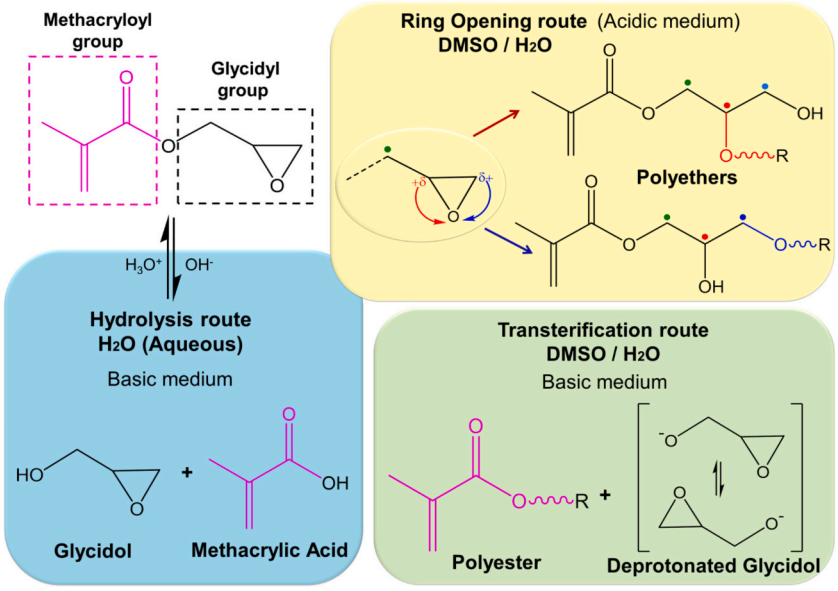


Fig. 8. The RO and transterification pathways of GMA under basic and acidic conditions.

from, yet closely related to, the hydrolytic and RO behavior of GMA. Such reactions are favored in polar aprotic solvents (e.g., DMSO) and under mildly basic conditions (pH $\geq$ 7), where solvent polarity enhances nucleophilic reactivity. In these media, strong solvation of cations reduces ion pairing, while anions such as alkoxides or alcoholates remain relatively unsolvated and therefore highly nucleophilic, facilitating efficient attack on the ester carbonyl group and promoting ester exchange reactions.

In polar aprotic solvents such as DMSO, transesterification can proceed efficiently under basic conditions through highly nucleophilic alkoxide species without necessarily involving hydrolysis; however, due to the hygroscopic nature of the medium, trace amounts of water may introduce competing hydrolysis and epoxy RO reactions, particularly under strongly basic conditions [8,122]. In contrast, in protic solvents [123] (e.g., water or alcohol-containing media), extensive hydrogen bonding and direct solvation of anions significantly reduce nucleophilicity and shift the reaction landscape toward competitive hydrolysis of the methacryloyl ester and nucleophilic RO of the epoxy group. The outcome of these reactions is strongly dependent on the pH of the medium: under basic conditions, hydroxide ions promote both ester hydrolysis and epoxy RO, whereas under acidic conditions, protonation of the epoxy group enhances its susceptibility toward nucleophilic attack, favoring the RO (Fig.8). Overall, water acts as the essential reactant for hydrolysis, while the solvent environment and pH collectively govern the competition between transesterification, hydrolysis, and epoxy RO pathways.

Interestingly, even in mixed solvent systems containing comparable amounts of polar aprotic (DMSO) and protic (water) components, the reaction pathway is not solely governed by solvent composition. Under acidic conditions (pH $\approx$ 4), protonation of the epoxide group significantly enhances its electrophilicity, thereby favoring epoxy RO over transesterification, despite the presence of an aprotic solvent [124,125]. This observation demonstrates that pH can outweigh solvent effects and act as the dominant parameter controlling reaction selectivity.

Several studies have further investigated the influence of these parameters during the copolymerization of GMA with monomers bearing hydroxyl or carboxylic acid functionalities. These investigations revealed that the methacryloyl group can react with alcohol-containing species through transesterification, leading to the formation of polyester segments in addition to the polyether backbone generated via epoxide RO [126] (Fig.8). As a result, complex polymer architectures may be obtained due to the coexistence of multiple reaction pathways. Notably, glycidol formed during hydrolysis does not significantly participate in subsequent polymerization reactions, despite the presence of both epoxide and hydroxyl functionalities. This limited reactivity has been attributed to the relative stability of its anionic form, in which the negative charge is delocalized over the oxygen atoms [127], as illustrated in Fig. 8. As a result, its reactivity is considerably lower compared to that of GMA, particularly the methacryloyl functionality, which exhibits higher reactivity under typical polymerization conditions.

The majority of studies indicate that the occurrence of these side reactions is primarily governed by the pH of the reaction medium. In particular, hydrolysis of GMA can take place even under neutral conditions, whereas strongly acidic environments (e.g., pH $\approx$ 3) tend to suppress this process. Li et al. [128] investigated the evolution of the pH during the reaction in an aqueous phosphate-buffered saline medium and provided valuable insight into the dynamic nature of the system. Their results showed that the initial pH of the solvent was approximately 7.4; however, upon addition of GMA, the pH rapidly increased to about 8.5 within one day. This shift toward basic conditions promotes the hydrolysis of GMA, leading to the formation of methacrylic acid. As the reaction proceeds, the accumulation of acidic species gradually lowers the pH of the medium, eventually reaching acidic conditions after approximately three days.

As a consequence, the hydrolysis process progressively slows down and becomes less favorable. After about five days, the system stabilizes at a pH of around 6.4, at which point the competition between hydrolysis and polymerization is significantly reduced, allowing the epoxide RO polymerization to dominate. These observations highlight the self-regulating nature of the system and confirm the dynamic interplay between hydrolysis and transesterification, as well as the reversible character of the latter.

In contrast, when the reaction medium is maintained under strongly basic conditions (e.g., pH $\approx$ 8), additional transformations may occur. In such cases, the primary products formed via epoxide RO may undergo further hydrolysis, leading to cleavage of the methacryloyl group along the polymer chain and the formation of new hydroxyl functionalities. These species may subsequently participate in reversible transesterification reactions, potentially regenerating epoxide-containing structures or inducing further structural rearrangements within the system [128].

In summary, the occurrence of transesterification and hydrolysis

during GMA polymerization is strongly dependent on the pH of the reaction medium. These side reactions can take place under neutral and basic conditions (pH $\geq$ 7), whereas they are effectively suppressed in highly acidic environments. Therefore, strict control of the pH during the reaction is essential to minimize undesired pathways and achieve selective polymerization.

Importantly, the final product distribution is governed by the competition between these reaction pathways rather than a single dominant mechanism. Under basic conditions with uncontrolled pH, multiple products may be generated due to the coexistence of competing reaction pathways. In addition to the two primary isomers resulting from epoxide RO, which lead to the formation of linear polyether structures, transesterification involving the methacryloyl functionality can give rise to polyester segments. Furthermore, hydrolysis-related processes contribute to the formation of glycidol as a secondary by-product, increasing the complexity of the reaction system. In this context, epoxy RO is generally the predominant pathway, while transesterification and hydrolysis contribute to secondary product formation. These simultaneous transformations highlight the strong influence of the reaction environment on product distribution and polymer architecture. In contrast, in highly acidic media, these side reactions are effectively suppressed, allowing the polymerization to proceed predominantly through oxirane RO and resulting mainly in the formation of the two expected isomers.

Building on the understanding of the reaction mechanism and the influence of solvent and pH on the competing pathways, from an industrial perspective, the control of these parameters is essential for ensuring process scalability, reaction selectivity, and reproducibility. Although polar aprotic solvents such as DMSO are highly effective in promoting selective transesterification at laboratory scale, their industrial implementation is often restricted by economic, environmental, and safety considerations. In contrast, aqueous or protic systems are more suitable for large-scale applications; however, they require precise pH regulation to control competing reaction pathways, including hydrolysis, transesterification, and epoxy RO. Therefore, careful optimization of both solvent choice and pH conditions is necessary to achieve stable and well-controlled reaction performance under industrial conditions.

2.5. Analytical perspectives and critical outlook for further synthesis

Based on previously reported studies and their critical analysis, several insights and potential strategies addressing the challenges associated with the ionic polymerization of epoxides have emerged. However, the ROP of GMA remains insufficiently explored, and significant mechanistic ambiguities persist. More importantly, the current literature lacks a unified understanding of the interplay between competing pathways, which severely limits the predictability and rational design of GMA-based systems.

To enhance the utilization of this versatile monomer-particularly in light of the intrinsic mechanistic complexity that may restrict its broader application we propose herein a set of hypotheses and strategic directions that require systematic experimental validation. It must be emphasized that most existing studies are highly system-dependent, and therefore any generalization of mechanistic behavior remains premature at this stage.

As outlined earlier, three principal synthetic pathways are available for epoxide polymerization: (i) cationic polymerization in acidic media, and (ii) anionic polymerization in basic media, proceeding via either conventional or coordinative mechanisms. While the anionic coordinative pathway is often regarded as superior for many epoxides due to its ability to yield high-molecular-weight polymers with improved control, this paradigm cannot be directly transposed to GMA without critical reassessment. Indeed, the multifunctional nature of GMA introduces additional reactivity that fundamentally alters the balance between propagation and side reactions. In particular, GMA is highly sensitive to hydrolysis and transesterification under basic conditions, which may introduce competing reaction pathways and reduce selectivity in these systems.

In this context, cationic polymerization may appear more suitable, as it enables better control of the reaction environment through maintenance of acidic conditions. However, this apparent advantage is counterbalanced by intrinsic limitations of cationic systems, including poor control over molar mass distribution, high sensitivity to impurities, and the prevalence of transfer reactions. Therefore, neither pathway can currently be considered universally optimal, highlighting the need for application-driven selection rather than mechanism-driven preference.

Nevertheless, hydrolysis of GMA leads to the formation of glycidol, a water-miscible compound compatible with green chemistry principles. While often treated as a parasitic reaction, this transformation should not be systematically regarded as detrimental. On the contrary, it may represent an opportunity to introduce additional functionality into the polymer backbone. However, this potential benefit remains insufficiently demonstrated and is highly dependent on reaction conditions and product architecture. This dual role of hydrolysis as both a side reaction and a potential functionalization pathway is largely overlooked in current literature and deserves deeper investigation.

Ultimately, the selection of the optimal ROP pathway for GMA whether under acidic or basic conditions should be dictated by the targeted application and desired material properties. Critically, a comprehensive framework correlating reaction conditions, dominant mechanisms, and resulting polymer structures is still missing, representing a major bottleneck in the field.

One of the most significant challenges in GMA polymerization lies in the accurate identification and characterization of the resulting products. This difficulty arises from the coexistence of multiple competing mechanisms. In cationic systems, both AM and ACE mechanisms may operate simultaneously. When combined with hydrolysis and transfer reactions, the system evolves into a highly complex reaction network rather than a single, well-defined pathway. This complexity is rarely fully resolved in experimental studies, leading to potential misinterpretation of polymer structures and mechanisms. To address this issue, promoting the predominance of a single mechanism is highly desirable. The AM mechanism is generally considered more advantageous, as it limits transfer reactions more effectively than the ACE pathway. However, achieving AM predominance often requires the addition of monomer or solvent during the reaction. This introduces a critical trade-off: while favoring AM, such additions may reduce the effective acidity of the medium and indirectly promote competing reactions, including transesterification. Importantly, transesterification should not be viewed solely as a detrimental side reaction. Depending on the system, it may contribute to improved material properties by introducing structural heterogeneity, flexibility, or additional functional groups.

In systems where hydrolysis-derived by-products are undesirable, strict control of medium acidity is essential. Establishing strongly acidic initial conditions (e.g., nominal pH $\approx$ 3) may help maintain sufficient proton activity throughout the reaction. However, the use of pH as a descriptor in non-aqueous or heterogeneous systems is fundamentally limited and should be interpreted with caution. A more rigorous approach would involve quantifying acidity in terms of proton activity or catalyst strength rather than nominal pH values. An alternative strategy to suppress hydrolysis involves performing the reaction under solvent-free (bulk) conditions. In such cases, the ACE mechanism is expected to dominate due to the absence of protic species required for AM initiation. While this approach simplifies the reaction environment, it may also reduce the ability to finely tune polymerization kinetics and structure. Additionally, maintaining a highly acidic environment is more straightforward in homogeneous systems, whereas heterogeneous catalysts may introduce local acidity variations. Despite this limitation, heterogeneous systems remain attractive from a sustainability and recyclability perspective, highlighting a trade-off between control and practicality.

From another perspective, the possible participation of glycidol formed via hydrolysis in the polymerization process introduces an additional level of complexity. Glycidol can react through both its epoxide and hydroxyl functionalities, potentially leading to branched architectures [66]. In cationic polymerization, the AM mechanism is known to favor branched polyethers, whereas ACE leads to linear structures as illustrate in Fig.9. However, in the case of GMA, such branching has not been conclusively demonstrated. The hypothesis that glycidol incorporation could enhance structural complexity and material performance remains intriguing but currently lacks direct experimental validation. This raises a fundamental question: why is glycidol polymerization not clearly observed under typical GMA polymerization conditions? It is unlikely that simple electronic effects, such as charge delocalization, are sufficient to explain this behavior. More plausible explanations include kinetic competition with GMA, steric hindrance, or catalyst selectivity, all of which remain insufficiently investigated. This represents a critical knowledge gap in the field.

Finally, the strategies and mechanistic considerations proposed in this section aim to rationalize and simplify the inherent complexity of GMA polymerization. However, it must be acknowledged that these proposals remain largely hypothetical and require rigorous experimental validation. Future research should prioritize the establishment of clear structure–mechanism-property relationships, supported by advanced characterization techniques and systematic variation of reaction parameters. Without such efforts, the full potential of GMA as a multifunctional monomer will remain underexploited.

3. Radical polymerization of GMA

3.1. Classic free radical polymerization

The high reactivity of the methacrylate double bond in GMA makes it well-suited for conventional free radical polymerization (FRP). This technique commonly relies on thermal initiation using organic peroxides such as benzoyl peroxide (BPO) or azo compounds like azobisisobutyronitrile (AIBN), 4,4′-azobis(4-cyanovaleric acid) (ACVA) and dicumyl peroxide (DCP). The polymerization process can be carried out in various media, including bulk, solution, suspension, or emulsion. Table 1 provides a summary of key studies on GMA polymerization, along with the respective experimental conditions employed.

While FRP is valued for its operational simplicity and suitability for large-scale industrial processes, it offers limited control over polymer architecture. Polymers synthesized via this method typically exhibit broad molecular weight distributions and random chain structures [129,130]. Additionally, the reactive epoxide group in GMA can undergo undesirable side reactions such as crosslinking especially under acidic or basic conditions [40] which can further compromise control over polymer structure and hinder reproducibility. Despite these limitations, conventional FRP remains a practical choice for the synthesis of GMA-based polymers, particularly when precise control over molecular characteristics is not a critical requirement. This method continues to be widely used in applications such as coatings, adhesives, and thermosetting resins.

More specifically, during radical polymerization, the epoxide functionality may react with trace nucleophiles or impurities present in the reaction medium, leading to chain defects, branching, or intermolecular coupling. These processes can effectively act as additional termination-like pathways by disrupting chain propagation and increasing structural heterogeneity. At higher conversions or elevated temperatures, such side reactions may become more pronounced, sometimes leading to partial crosslinking or loss of chain-end fidelity. Overall, the dual reactivity of GMA combining a polymerizable double bond with a chemically labile epoxide group makes it more prone to side reactions than conventional methacrylates. This behavior limits the level of control achievable via FRP and explains the growing interest in controlled radical polymerization techniques when well-defined GMA-based architectures are required.

From a kinetic standpoint, pulsed laser polymerization (PLP) has emerged as a powerful and widely recognized technique for the precise investigation of GMA polymerization kinetics. Unlike conventional FRP, which relies on continuous radical generation from thermal initiators, PLP employs short, high-energy UV laser pulses to generate radicals in a temporally controlled manner. Each pulse produces a well-defined burst of radicals followed by rapid termination, effectively “freezing” the growth of propagating chains and enabling analysis at very low conversion [131]. This approach has been successfully applied to the study of styrene/GMA copolymerization [132], notably by Wang and Hutchinson, who combined PLP with size exclusion chromatography (SEC)

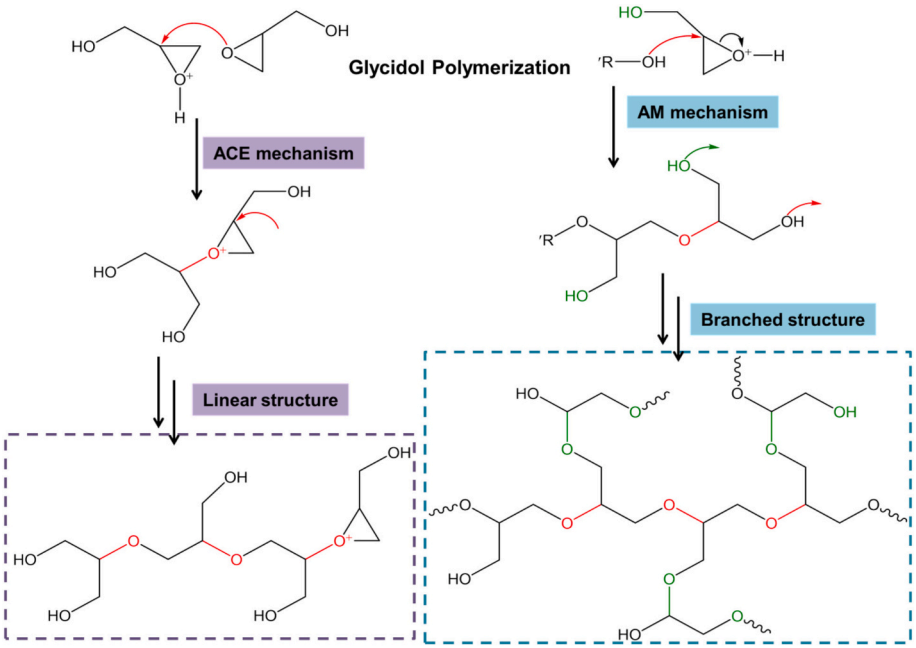


Fig. 9. Representation of branched and linear polyethers structure generated from AM and ACE mechanism using glycidol monomer.

Table 1
Free radical homopolymerization and copolymerization of GMA.

Copolymer	Comonomer (full name)	Initiator	Polymerization Conditions	Polymerization medium	Reference
DNR-g-GMA-co-MMA	Depolymerized natural rubber, methyl methacrylate	AIBN, BPO (1-3 wt % relative to DNR)	80 °C, toluene 2 h, N_2	Solution	[135]
GMA-co-BA	N-Butyl acrylate	BPO (0.8 mol%)	60 °C, Ar, 24 h	Bulk	[136]
GMA-co-ODA	Octadecyl acrylate	AIBN (2 wt%)	70 °C, toluene, N_2	Solution	[137]
GMA-co-DVB	Divinylbenzene	AIBN (4 wt%)	70 °C, acetonitrile, N_2, 24 h	Solution	[138]
GMA-co-BCPM	4-benzyloxycarbonylphenyl methacrylate	BPO	70 °C, ethyl methyl ketone, N_2	Solution	[139,140]
GMA-co-MMA	methyl methacrylate	BPO	90 °C, 1,4-dioxane,2 h	Solution	[16]
GMA-co-FPMA	o−/*p*-Formylphenyl methacrylate	BPO (0.09 mol%)	80 °C, dioxane, N_2	Solution	[141]
GMA-co-AN	Acrylonitrile	BPO (1 wt%)	70 °C, 1,4-dioxane, Ar, 8 h	Solution	[28]
GMA-co-AN	Acrylonitrile	AIBN (0,05 wt%)	60 °C	Bulk	[142]
GMA	–	AIBN	60 °C, THF	solution	[142]
GMA	–	BPO (2 wt%)	85 °C, acetonitrile	solution	[69]
GMA-g-DNR	Depolymerized natural rubber	AIBN	80 °C, toluene, N_2, 2 h	Grafting in solution	[143]
GMA-co-Styrene	Styrene	PLP	50–140 °C, xylene, 3-180 s.	Bulk, solution	[132]
GMA-co-MAn	Maleic anhydride	BPO (0,74 mol %)	80 °C, THF, Ar, 24 h	Solution	[144]
GMA-co-MAn	Maleic anhydride	BPO (1 mol% and 1 Wt%)	80 °C, THF, Ar, 8 h	Solution	[145]
Poly(GMA) microbeads	GMA and Ethylene glycol dimethacrylate (crosslinker)	BPO (1 wt%)	80 °C, aqueous medium, 2 h,	Suspension	[146]
HDPE-g-GMA	High Density Polyethylene resins	Pre-irradiated HDPE (10–40 kGy)	40 °C, aqueous emulsion, Tween-20 (0.5–12 wt %), N_2, 1 h	Emulsion grafting	[147]
Poly(GMA)-maghemite microspheres	–	ACVA (2 wt%)	70 °C, aqueous phase with CM-dextran stabilized maghemite nanoparticles, N_2, 20 h	Emulsion	[148]
PVC-g-poly(GMA)	Poly(vinyl chloride)	DCP	Isothermal: 100–120 °C (5 h); Non-isothermal: 25–180 °C (10 °C/min), plastisol medium (PVC + DINP), N_2	Bulk / in-situ plastisol	[149]
GMA-co-MTM	3-methylthienyl methacrylate	AIBN (1 wt%)	60 °C, DMF, Ar	solution	[150]
GMA-co-AMS	α-methylstyrene	AIBN	65 °C, THF, Ar, 16 h	solution	[151]
TMA-co-GMA	2-methyl-N-1,3-thiazole-2-ylacrylamide	BPO (1 wt%)	65 °C, 1,4-dioxane,N_2	solution	[152]
NFM-co-GMA	2-(4-nitrophenyl)-2-oxoethyl-2-methacrylate	AIBN (1 wt%)	65 °C, 1,4-dioxane, N_2	solution	[13]
MMOAB-co GMA	methyl-4-[[(methacryloyloxy)acetyl]oxy] benzoate	AIBN (1 wt%)	65 °C, 1,4-dioxane, 10 h	solution	[12]
MBAOM-co-GMA	2-[(methoxy-1,3-benzothiazole-2-yl) amino]-2-oxoethyl methacrylate	AIBN	65 °C, 1,4-dioxane, N_2	solution	[14]
BMAOME-co-GMA	2-[[(2H-1,3-benzodioxol-5-yl)methyl] amino)-2-oxoethyl 2-methylprop-2-enoate	AIBN (1 wt%)	65 °C, 1,4-dioxane, N_2	solution	[153]
HAOEME-GMA	2-(4-hydroxyanilino)-2-oxoethyl-2-methylprop-2-enoate	AIBN	65 °C, 1,4-dioxane	solution	[15]

and NMR spectroscopy to obtain detailed kinetic information. Such a methodology minimizes common sources of error encountered in conventional FRP, including uncontrolled radical flux, initiator decomposition, and monomer depletion, thereby allowing accurate determination of propagation rate coefficients and providing insight into depropagation phenomena. The robustness and reliability of PLP have been further validated in subsequent studies [131,133,134], establishing it as a reference method for the kinetic analysis of methacrylate monomers. Although PLP is not intended for preparative polymer synthesis, it provides essential mechanistic understanding that complements conventional FRP and supports the development of more controlled polymerization strategies.

In conclusion, although conventional FRP remains the most widely used and industrially relevant approach for the synthesis of GMA-based polymers, its inherent lack of control and susceptibility to side reactions significantly limit its ability to produce well-defined macromolecular architectures. These limitations are primarily related to uncontrolled radical activity and the additional reactivity of the epoxide functionality, which can further complicate reaction pathways.

In this context, the following sections will focus on more controlled radical polymerization methodologies that have been developed to overcome these drawbacks. Particular attention will be given to RDRP techniques, including ATRP, NMP and RAFT polymerization, which provide improved control over molecular weight, dispersity, and polymer architecture.

In addition, emerging approaches based on external energy input will be discussed, including light-, electro-, and mechanically mediated radical polymerization systems, which are increasingly regarded as green and sustainable methods due to their reduced energy consumption, lower catalyst loadings, and improved environmental compatibility. Finally, hybrid strategies combining controlled radical mechanisms with external energy activation will be highlighted, as they offer new opportunities to further enhance spatiotemporal control, efficiency, and functional group tolerance in the polymerization of GMA and related monomers, while aligning with the principles of green chemistry.

3.2. *Externally stimulated radical polymerization techniques (ESRP)*

In recent years, a series of non-conventional radical polymerization methodologies have emerged as sustainable alternatives to conventional FRP of GMA. These processes are commonly referred to as “externally stimulated radical polymerization techniques” (ESRP) because they rely on external physical or chemical stimuli rather than conventional thermal initiation to generate reactive radical species in a controlled manner. Such stimuli include light, ultrasound, microwave irradiation, electrochemical input, plasma discharge, pulsed laser activation, and ionizing radiation, enabling spatially and temporally regulated radical formation. Representative examples include photo-polymerization, ultrasound-assisted polymerization, microwave-assisted polymerization, electrochemically initiated polymerization, plasma-initiated polymerization, and radiation-induced polymerization, as summarized in Fig. 10.

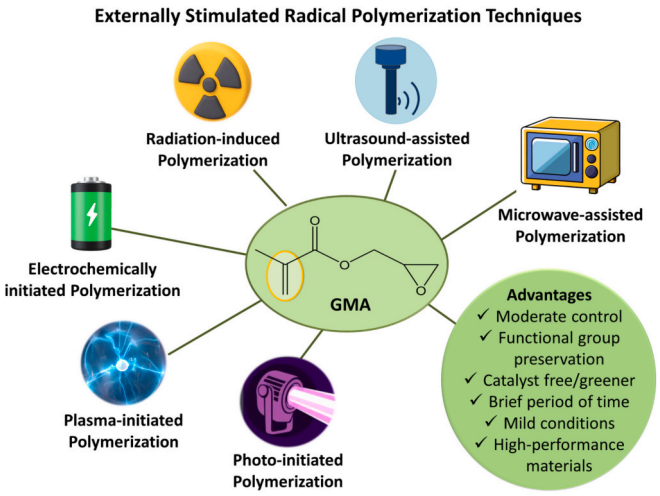


Fig. 10. Energy-assisted advanced radical polymerization methods for GMA.

Compared with classical FRP, which generally requires elevated temperatures, longer reaction times, and often provides limited control over polymer architecture and molecular weight distribution, ESRP methodologies offer several important advantages. Many of these techniques can be conducted under mild experimental conditions with improved reaction selectivity, reduced side reactions, and enhanced preservation of sensitive functionalities such as the epoxide group in GMA. Furthermore, several ESRP approaches can proceed without metal catalysts or complex mediating systems, thereby reducing toxicity, purification requirements, and environmental impact [154,155].

From a green chemistry perspective, these methodologies are particularly attractive because they promote improved energy efficiency, reduced use of hazardous reagents, and minimization of waste generation. In some cases, they can also be performed under solvent-free conditions or using environmentally benign solvents. Moreover, the precise control of radical generation under external stimulation provides enhanced temporal regulation of the polymerization process and improved process efficiency compared with conventional FRP [156–158].

Compared to conventional FRP, which typically lacks temporal control over radical formation, these advanced methodologies offer enhanced reaction regulation and improved process efficiency. Consequently, they represent environmentally benign and operationally flexible routes for the synthesis of functional GMA-based polymers with controlled architecture and tailored properties. In this context, the following section provides a detailed discussion of these techniques, emphasizing their underlying mechanisms, green chemistry compliance, and relevance to advanced material design.

Consequently, ESRP techniques represent environmentally friendly and operationally versatile routes for the synthesis of advanced GMA-based functional polymers with tailored architectures and properties. Their ability to combine polymerization efficiency, structural control, and reduced environmental burden makes them highly promising for the development of next-generation materials for biomedical, optical, coating, adhesive, and industrial applications. In this context, the following sections provide a detailed discussion of these techniques, emphasizing their mechanisms, advantages, limitations, and relevance to sustainable advanced material design.

3.2.1. Photo-initiated polymerization

Photo-initiated polymerization has emerged as a highly efficient and versatile strategy in comparison with conventional thermally initiated processes. The most significant benefits of the technology can be found in the accelerated polymerization, which is accomplished within a remarkably brief period of time, the exact spatial and temporal management of the curing process with the capacity to function under conditions that are mild, solvent-free, and energy-efficient [159,160]. It is evident that these attributes render photo-polymerization not only industrially attractive but also more sustainable from the perspective of green chemistry. The principle of photo-FRP is predicated on the initiation of a photo-initiator by ultraviolet or visible light, which in turn leads to the generation of radicals. Two primary mechanisms are distinguished: Type I photo-initiators undergo direct homolytic cleavage to release primary radicals, while Type II initiators transfer energy to a co-initiator that subsequently produces radicals. Subsequent to their formation, these radicals proceed to attack the carbon–carbon double bonds of vinyl or methacrylate groups, thereby initiating chain propagation and enabling rapid polymerization.

The unique benefits offered by this material have led to its wide ranging application in the field of GMA-based systems, facilitating advancements in this area of research. For instance, the study by Alemdar et al. [161] demonstrated that the incorporation of GMA into the chitosan structure, followed by crosslinking with polyethylene glycol diacrylate, resulted in the formation of thermally stable hydrogels. Furthermore, swelling experiments revealed that the water uptake capacity (swelling ratio) of the hydrogels strongly depended on the UV exposure time, highlighting the critical role of irradiation conditions in tailoring hydrogel performance. In addition, gelatin that has been functionalized with GMA and photopolymerized with Irgacure 2959 has been shown to produce 3D scaffolds that are biocompatible, with improved porosity. This, in turn, has been shown to enhance hepatocyte viability and proliferation, thus highlighting the strong potential of photo-crosslinked Gelatin-GMA in tissue engineering [162]. Furthermore, poly(vinyl alcohol)-GMA hydrogels initiated with Irgacure 2959 have been observed to exhibit smooth microstructures and high thermal stability [159]. Poly(methyl methacrylate-*co*-GMA) prepared under UV irradiation has been exploited for stable optical waveguides. In this study, photosensitive poly(MMA-*co*-GMA) films were obtained and mixed with bisphenol-A epoxy resin, which allowed precise tuning of the refractive index within the range of 1.481–1.588 at 1550 nm. The spin-coated films demonstrated high UV lithographic sensitivity, a glass transition temperature of 153 °C following crosslinking, and exhibited excellent thermal stability with a decomposition temperature of up to 324 °C. Furthermore, smooth-surfaced optical channel waveguides were fabricated directly by UV exposure and chemical development. When coated, these waveguides exhibited low propagation losses of less than 3 dB/cm at 1550 nm, thereby emphasizing their suitability for advanced photonic applications [163].

A study of dual-cure resins composed of GMA and poly(ethylene glycol) dimethacrylate with diphenyl(2,4,6-trimethylbenzoyl) phosphine oxide initiator has demonstrated their efficacy in the domains of digital light processing and 3D printing. The dual-stage curing process utilizes a combination of ultraviolet-triggered radical polymerization, enabling precise fabrication of complex structures, and a subsequent thermal etherification that reinforces the polymer network. This strategy has been demonstrated to yield materials with enhanced glass transition temperature (up to ~105 °C), elevated modulus (up to 1607 MPa), and the capacity to reprogram printed structures into permanent three-dimensional forms, signifying considerable promise for advanced additive manufacturing [164].

In more recent publication, GMA-modified bis(2-hydroxyethyl) terephthalate acrylates have been shown to undergo ultrafast photopolymerization, achieving a carbon–carbon double bond conversion rate of up to 98.62% within only a few seconds under 365 nm irradiation and 96.92% under 395 nm. The rapid curing process results in the formation of hydrophobic and mechanically robust films, with a contact angle of 83.9° compared to 25.1° for the aliphatic reference resin. In addition, the films exhibit exceptional thermal stability and improved tensile strength when combined with sensitizers such as N-vinylcarbazole [160].

In summary, photopolymerization of GMA can be considered a

versatile and sustainable strategy that combines rapid reaction rates, mild conditions, and precise spatial control. This technique facilitates the fabrication of well-defined hydrogels, scaffolds, coatings, and optical materials with tunable mechanical, thermal, and biological properties. The epoxy's functionality is preserved whilst exhibiting compatibility with advanced manufacturing methods, such as lithography and three-dimensional printing. This suggests that it has significant potential for use in biomedical, industrial, and photonic applications.

3.2.2. Ultrasound-assisted polymerization

Ultrasound-assisted radical polymerization has emerged as a novel, advanced alternative to conventional FRP. In this approach, high-frequency ultrasonic waves are utilized to induce acoustic cavitation. This process involves the formation, growth, and collapse of micro-bubbles, which results in the creation of localized hot spots characterized by elevated temperatures and pressures [165]. These transient conditions result in the homogeneous cleavage of solvent or monomer molecules, producing radical species that can efficiently initiate polymerization [18,166]. This technique enables the clean and controllable generation of free radicals without the need for conventional chemical initiators. Furthermore, ultrasound-assisted polymerization offers several advantages, including mild reaction conditions, enhanced reaction control, reduced chemical consumption, and compatibility with green chemistry principles [167].

In the specific case of GMA, ultrasound has been shown to overcome the limitations of conventional FRP in aqueous systems. This phenomenon was demonstrated by Sankar and Rajendran [17], who conducted a study on the polymerization of GMA (a hydrophobic monomer) utilizing a disite phase-transfer catalyst in conjunction with ultrasonic irradiation. The primary challenge is attributable to the low solubility of GMA in water, which impedes the efficacy of conventional biphasic FRP. In order to address this issue, the authors of the study introduced a bis-quaternary ammonium catalyst. This catalyst facilitated the efficient transfer of radical species, which had been generated by ultrasound in the aqueous phase where potassium peroxydisulfate was active, into the organic phase containing the monomer [18]. Concurrently, ultrasonic energy enhanced the decomposition of the initiator and improved interfacial mixing via cavitation. The synergistic strategy exhibited a substantial enhancement in polymerization rate and monomer conversion, even under mild reaction conditions. This approach enhances reaction efficiency and offers a more sustainable alternative to traditional FRP, especially in the context of hydrophobic monomers like GMA in heterogeneous systems. This ultrasound-assisted FRP technique, in combination with phase-transfer catalysis, has also been effectively applied to other methacrylic monomers, including MMA [168] and ethyl methacrylate [17]. This approach has shown comparable enhancements in polymerization rates, conversion efficiency, and control over polymer behavior. Furthermore, ultrasound initiation has been successfully employed in the miniemulsion polymerization of various methacrylate monomers, including MMA, n-butyl methacrylate, and 2-ethylhexyl methacrylate. This latter has resulted in well-defined latex particles with high molecular weights [169]. In consideration of the structural similarity and reactivity of GMA, this approach demonstrates considerable promise for future implementation in GMA-based systems, particularly with regard to the synthesis of functionalized polymer nanoparticles.

3.2.3. Microwave-assisted radical polymerization

Microwave-assisted radical polymerization has emerged as an efficient alternative to conventional thermal polymerization, offering several advantages that can be directly related to established green chemistry principles. In particular, this approach aligns with energy efficiency (principle 6) by enabling rapid and uniform heating through direct interaction of microwave irradiation with polar species in the reaction medium, thereby significantly reducing reaction times and overall energy consumption compared to conventional heating methods [170–172]. In addition, the technique can contribute to safer solvent use (principle 5), as many microwave-assisted processes are performed under solvent-free conditions or with reduced solvent volumes. Furthermore, shorter reaction times and improved reaction control help minimize side reactions and by-product formation, supporting waste prevention (principle 1). In certain systems, microwave activation may also reduce the need for additional chemical initiators or catalysts, partially addressing principle 9 (catalysis). The underlying mechanism of microwave-assisted radical polymerization is based on the direct coupling of microwave energy with polar molecules or ions, resulting in rapid, homogeneous heating and, in some cases, enhanced radical generation and propagation rates [173].

This approach has been widely applied to the synthesis of acrylates and methacrylate-based polymers [174,175]. In the case of GMA, microwave-assisted radical polymerization has been shown to facilitate efficient copolymerization and surface functionalization, thereby creating new opportunities for the development of advanced functional materials. Chyzy et al. [176] reported that the microwave-assisted synthesis of GMA-ethyl methacrylate oligomers produced well-defined structures with enhanced thermal and physicochemical properties in markedly shorter reaction times compared to conventional methods. These copolymers were further applied in biomedical applications. In a similar manner, the grafting of poly(GMA) chains onto carbon-based nanomaterials has been achieved through radical polymerization initiated under microwave irradiation. This method has been demonstrated to generate functionalized nanocomposites with enhanced surface reactivity [177]. As stated in a subsequent study carried out by Abdollahi Baghban et al. [178] microwave-assisted synthesis of methacrylated gelatin using GMA has been shown to reduce reaction time, energy consumption and reagent usage. Additionally, the effectiveness of the method was demonstrated by the successful production of hydrogels with a high degree of methacrylation. Furthermore, it was demonstrated that an increase in both the duration and the intensity of the microwave irradiation led to a substantial enhancement in crosslinking, consequently resulting in improved mechanical strength and thermal stability. This finding serves to substantiate the efficacy of this eco-friendly methodology for biomedical applications.

Recently, Gonzalez et al. [179] demonstrated that microwaves have the capacity to be utilized not for the initiation of radical polymerization but rather for the purpose of accelerating the post-functionalization of polymer networks containing GMA. In their work, a methacrylate-based network was first crosslinked through conventional photopolymerization, and microwaves were subsequently applied to activate the RO of residual epoxide groups from GMA. This strategy facilitated the efficient grafting of amine groups, resulting in the production of functionalized materials with properties that have been tailored for specific applications. Consequently, in this instance, the function of microwaves is not the initiation of radical reactions but rather the rapid and effective post-polymerization modification of GMA-based materials.

It can be concluded from the findings that microwave-assisted radical polymerization of GMA can be applied effectively in the customization of the properties of GMA-based materials. This, in turn, enables rapid development of specific functionalities suited to the desired applications.

3.2.4. Radiation-induced radical polymerization

Radiation-induced polymerization is a versatile and environmentally friendly polymerization technique aligned with key principles of green chemistry, particularly atom economy, reduced use of hazardous reagents, and energy efficiency. It does not require additional chemical initiators, proceeds under mild conditions, and enables precise control of grafting onto various substrates. The fundamental mechanism of this process is based on the interaction of high-energy radiation, such as γ-rays or X-rays, with matter, leading to ionization and the generation of reactive free radicals. These species can directly initiate polymerization either within the monomer phase or at the surface of a polymeric

substrate [180,181].

A number of studies have employed this strategy in GMA. For instance, γ-ray irradiation has been employed for the simultaneous grafting of GMA onto water hyacinth fibres, yielding grafting yields of almost 58% [182]. In another work, γ-ray-induced graft polymerization of GMA onto polyethylene hollow-fiber membranes has been shown to be an effective strategy for introducing sulfonic acid functionalities, thereby producing ion-exchange membranes with high efficiency for the adsorption of heavy metals [183]. In a similar vein, Li and Wang [184] reported the radiation-induced grafting of GMA onto high-density polyethylene, demonstrating that grafting efficiency is strongly dependent on irradiation dose, monomer concentration, and atmospheric conditions. These results highlight that high-energy irradiation not only generates reactive radical sites on both the monomer and the polymer substrate, but also enables spatially controlled surface activation without the need for chemical initiators. Importantly, one of the key advantages of radiation-induced grafting is the preservation of the epoxy group of GMA during the grafting process, which remains available for subsequent post-functionalization reactions such as amination, hydrolysis, or sulfonation, thereby significantly expanding the chemical versatility of the resulting materials. The findings of Li and Wang further demonstrated that radiation lamination improves interfacial adhesion, confirming the potential of this method to enhance both surface and mechanical properties of polyethylene-based systems.

In a separate research study, Nasef and Sugiarmanwan [147] undertook an investigation into the emulsion graft polymerization of GMA onto high-density polyethylene utilizing an electron beam accelerator. The study highlighted the influence of parameters such as monomer concentration, radiation dose, temperature, and surfactant content on the degree of grafting. It was demonstrated that surfactant concentration plays a key role in micelle size and grafting efficiency. This emulsion-based radiation method was found to be advantageous over conventional solvent systems, requiring lower doses and monomer concentrations while offering improved cost-effectiveness. Recently, the grafting of GMA onto sepiolite via radiation-induced methods has been shown to yield grafting levels of over 600%, with the resultant materials demonstrating notable Cu(II) adsorption capabilities [185]. Kanjilal et al. [186] also achieved γ-radiation-induced grafting of GMA onto polypropylene non-woven fabrics. The grafting yield was strongly influenced by solvent, monomer concentration, and irradiation parameters, highlighting the potential of this method for developing adsorbent materials. More recently, Shaheen et al. [187] developed radiation-grafted GMA bamboo fiber composites, which upon sulfonation, exhibited high copper adsorption capacity with 96% removal efficiency.

A comprehensive evaluation of the extant literature reveals a consensus among studies that radiation-induced radical polymerization of GMA facilitates efficient grafting onto a diverse array of natural and synthetic substrates. This process results in materials that exhibit enhanced surface reactivity, improved mechanical properties, and noteworthy adsorption capacities. The efficacy of this approach is contingent upon the irradiation conditions and reaction medium, thereby enabling the precise modulation of grafting yields and functional group incorporation. Consequently, this method serves to expand the scope of GMA-based polymers, rendering them applicable in diverse fields, including ion-exchange membranes, high-performance adsorbents, and functional composites.

3.2.5. *Electrochemically initiated radical polymerization*

Electrochemically initiated radical polymerization is a novel, eco-friendly technique for the modification of substrate surfaces with functional polymers. This approach is consistent with several principles of green chemistry, particularly the reduction of hazardous reagents, the elimination of chemical initiators, improved energy efficiency, and enhanced process controllability. This technique offers several advantages over conventional methods [188]. Firstly, it eliminates the use of chemical initiators. Secondly, it allows precise control over grafting density and polymer chain growth. Thirdly, and perhaps most importantly, it enhances interfacial adhesion between the substrate and polymer matrix, which is particularly valuable in composite applications. In addition, the electrochemical control of the reaction provides an energy-efficient and on-demand activation process, further reinforcing its sustainability and alignment with green chemistry strategies.

A representative early study on electroinitiated polymerization of GMA was reported by Samal and Nayak [189] in dimethylformamide (DMF) using ferric sulfate as supporting electrolyte. Their work demonstrated that GMA can be efficiently polymerized under electrochemical activation without the use of conventional chemical initiators, confirming the radical nature of the process. The authors systematically investigated the influence of key reaction parameters such as current intensity, monomer concentration, electrolyte concentration, and electrolysis time on polymer yield and molecular weight. It was shown that the polymerization yield increases with electrolyte concentration and electrolysis time, while a linear dependence was observed at moderate current values, indicating controlled radical generation at the electrode surface. However, at higher current densities, a decrease in molecular weight was observed, attributed to enhanced termination reactions due to excessive radical concentration. Among all parameters studied, the use of ferric sulfate as electrolyte and moderate current intensity (around 10 mA in their system) were identified as the most favorable conditions, providing higher conversion and better control over polymer growth. This study clearly highlights the critical role of electrochemical parameters in tuning both the efficiency and the structural characteristics of the resulting electropolymer, and supports the potential of ECRP as a controllable and sustainable route for functional polymer synthesis.

More recently, Molle et al. [190] reported the electrochemically initiated radical polymerization of GMA using a 4-fluorobenzenediazonium-based initiating system, enabling a more controlled radical generation pathway. Under these conditions, poly(GMA) was obtained with number-average Mn ranging from approximately 6000 to 15,000 $g{\cdot}mol^{-1}$ and Đ between ~1.5 and 3.1, depending on the monomer-to-initiator ratio and applied current. These results confirm the feasibility of ECRP for methacrylate systems but also highlight the inherent limitation in achieving narrow dispersity under purely radical electrochemical conditions.

In comparison, 2,6-difluorophenyl acrylate and pentafluorophenyl acrylate-based systems generally provide higher Mn values (up to ~11,000–19,000 $g{\cdot}mol^{-1}$ under standard conditions and significantly higher at increased monomer loading for pentafluorophenyl acrylate), together with comparatively lower dispersities (Đ ≈ 1.5–2.3 under optimized conditions). At first glance, this may suggest a better control over polymerization; however, such an interpretation can be misleading. The observed differences are primarily governed by the intrinsic electronic nature of the monomers rather than improved mechanistic control. Indeed, the strongly electron-deficient active ester character of 2,6-difluorophenyl acrylate and pentafluorophenyl acrylate enhances propagation efficiency, leading to higher apparent Mn, but simultaneously increases their susceptibility to perturbations in radical flux and electrochemical conditions. As a result, even minor variations in current density or radical concentration can significantly affect side reactions, including termination and decomposition pathways, reflecting a narrow operational window despite seemingly improved Đ values. In contrast, GMA exhibits a more chemically robust behavior under identical electrochemical conditions due to the stability of its epoxide functionality. This robustness translates into a wider tolerance toward electrochemical parameters, but also results in lower Mn values and less efficient chain-growth control. Therefore, the system demonstrates a clear trade-off: fluorinated active ester monomers favor higher propagation efficiency but require stricter control, whereas GMA offers operational stability at the expense of molecular weight control and uniformity.

In this context, electrochemically initiated radical polymerization of GMA is not only limited to solution polymerization but has also been extensively explored for surface modification purposes. Electrochemical

grafting of GMA onto conductive substrates, such as carbon fibres, has emerged as a targeted method to enhance interfacial adhesion in composite materials. In their Advances study, Hu et al. [19] demonstrated that GMA can be covalently anchored onto carbon fibres surfaces via electrochemical initiation, using aluminum chloride as the supporting electrolyte. The grafting efficiency exhibited a marked dependence on electrolyte concentration, reaction time, and current intensity. It was demonstrated that optimal parameters resulted in a substantial enhancement of interfacial properties. The proposed mechanism is a two-step process. Firstly, anodic oxidation of the carbon fibres surface is carried out, with the aim of generating oxygen-containing functional groups (e.g. –COOH, –OH, =C=O). Secondly, radical-mediated polymerization of GMA takes place under the applied potential. In a similar vein, Zhou et al. [191] reported that the combination of electrochemical activation with tailored surface pre-treatments facilitated the covalent grafting of GMA onto high-modulus carbon fibres. In this process, the applied potential generated surface-bound radicals that initiated polymerization of GMA, while the pre-treatments introduced oxygen-containing functional groups to further promote interfacial reactivity. The grafted fibres that resulted from the process demonstrated a significantly enhanced interfacial adhesion and an improvement in the composite mechanical properties.

In summary, electrochemically initiated radical polymerization represents a promising and environmentally friendly strategy for the synthesis and surface functionalization of GMA-based polymers. As demonstrated by Hu et al. and Zhou et al., it can also be extended to surface grafting applications, enabling covalent attachment through electrochemically generated reactive species and leading to improved interfacial and mechanical properties.

However, the overall process remains complex, as both bulk polymerization and surface modification are highly dependent on a delicate balance between radical generation, monomer reactivity, and electrochemical parameters, making precise control still challenging. Despite these advances, studies on electrochemical initiation of GMA are still limited, highlighting the need for further work to better understand the underlying mechanisms and to improve the control and versatility of this approach.

3.2.6. *Plasma-initiated radical polymerization*

Plasma-initiated radical polymerization has been widely employed as an effective technique for the polymerization of GMA. In particular, cold plasma (non-thermal plasma) is commonly used, where energetic electrons at near-ambient temperature generate a variety of reactive species, including radicals, ions, and excited molecules, without significantly increasing the bulk temperature of the system. These plasma-generated species, such as reactive oxygen and nitrogen species and surface-bound radicals, can initiate radical polymerization by abstracting hydrogen atoms or directly activating monomer units [192,193]. This mechanism is consistent with early fundamental studies on plasma–liquid interactions. For instance, Osada et al. (1983) [194] demonstrated that PRP of vinyl monomers, including acrylate systems, proceeds extremely rapidly in aqueous media, highlighting the strong influence of solvent effects on radical formation and stabilization. Their work also quantified radical generation using 2,2-diphenyl-1-picrylhydrazyl trapping experiments, estimating a radical liberation rate of approximately 1×10^{-5} mol.L^{-1} cm^{-2}, and proposed that the initiation process involves the interaction between gaseous plasma species and monomers in the liquid phase, with possible contributions from solvated ionic species.

This approach offers several advantages over conventional thermal initiation, including initiator-free radical generation, operation under mild conditions, and reduced energy input. In addition, the controlled formation of radical species at the plasma–monomer interface allows efficient initiation while minimizing side reactions and waste production, in agreement with green chemistry principles.

This behavior is consistent with recent developments in atmospheric-pressure plasma-initiated chemical vapor deposition, where a comparative investigation of MMA, butyl methacrylate (BMA), and GMA has been reported under identical pulsed plasma conditions. In this framework, Loyer et al. demonstrated that ultrashort pulsed plasma discharges (≈100 ns) enable a transition between a surface-dominated FRP regime at low duty cycles and a plasma-dominated regime at high duty cycles for all three methacrylates.

A clear difference in polymer growth is observed between the three monomers. At low pulse frequency (≈100 Hz), the apparent weight-average molar masses follow the order MMA (≈2600 g.mol^{-1}) < BMA (≈4800 g.mol^{-1}) ≪ GMA (≈94,000 g.mol^{-1}), highlighting the much stronger contribution of surface-controlled polymerization pathways in GMA compared to MMA and BMA. Similarly, the deposition rates are significantly lower for MMA (≈0.18 nm.s^{-1}) and BMA (≈0.24 nm.s^{-1}) than for GMA (≈2.5 nm.s^{-1}), confirming its higher reactivity under identical plasma conditions.

Among the studied monomers, GMA exhibits a distinct behavior due to the presence of the reactive epoxy functionality, which promotes additional radical formation pathways and enhances chain growth efficiency under plasma activation. In contrast, MMA and BMA mainly undergo conventional radical propagation with limited contribution from secondary plasma-induced reactions, resulting in lower molecular weights and less pronounced transition between kinetic and plasma regimes. Overall, these results demonstrate that GMA is the most reactive monomer under atmospheric-pressure plasma-initiated chemical vapor deposition conditions, while MMA and BMA show more moderate and predictable growth behavior. This comparative analysis clearly highlights that monomer structure, adsorption ability, and intrinsic propagation kinetics collectively govern the balance between surface polymerization and plasma-driven reactions, ultimately determining the structure and functionality of the resulting polymer films.

Beyond thin-film growth behavior under plasma-initiated deposition conditions, the reactivity of GMA has also been widely exploited in surface modification strategies, particularly through plasma-assisted grafting onto polymer substrates. In this context, dielectric barrier discharge (DBD)-based plasma deposition at atmospheric pressure has been extensively investigated using (GMA, methacrylic acid-[2,3-epoxypropyl ester]) as a model precursor in order to evaluate the ability of plasma processes to generate functional coatings while preserving the original chemical structure of the monomer. Dielectric barrier discharge plasma polymerization.

Under pulsed excitation conditions, the plasma dynamics are modulated in a way that limits excessive fragmentation, thereby favoring surface-controlled growth mechanisms. In optimized regimes, it has been reported that up to ~90% of the epoxy groups of GMA can be retained in the resulting plasma polymer films, while maintaining deposition rates on the order of 3–5 nm·s^{-1}. These conditions reflect a favorable balance between polymerization efficiency and functional group preservation, which is essential for post-functionalization applications.

From a mechanistic standpoint, film formation under pulsed plasma conditions is governed by the interplay between plasma-generated reactive species and surface processes occurring during and after the discharge pulses. It is generally accepted that intact or partially activated monomer molecules can interact with surface radicals formed during plasma pulses, contributing significantly to chain growth at the substrate interface. The afterglow phase further allows the participation of longer-lived reactive species, promoting propagation reactions while reducing over-fragmentation and uncontrolled crosslinking. This mechanism explains the relatively high retention of epoxy functionality compared to continuous plasma operation, where energetic species lead to more extensive bond scission and structural rearrangement.

In addition to plasma deposition studies, surface functionalization routes have also been extensively explored. GMA was successfully plasma-grafted onto polypropylene films [195], rendering them hydrophilic. In this approach, a short argon plasma exposure of only 2 min was

sufficient to activate the film surface, followed by an additional 2 min of GMA vapor exposure that accomplished the graft polymerization. The treatment time was found to be a crucial factor in the efficacy of the process, with the covalent anchoring of GMA being achieved in a remarkably short time. This remarkably short treatment time enabled the covalent anchoring of GMA, resulting in a hydrophilic surface with a surface energy of 55.8 mJ/m^2. Notably, the introduced hydrophilicity was stable and remained unchanged for at least three weeks, confirming the durability of this modification. Such findings highlight that plasma-initiated grafting of hydrophilic monomers like GMA is a rapid and effective means of surface modification.

In addition, plasma graft polymerization of GMA has been extended to aqueous-phase systems for the preparation of functional interfaces. In this context, GMA was *graft*-polymerized onto plasma-treated fibres after brief exposure to air, using dilute aqueous solutions containing sodium dodecyl sulfate to enhance monomer solubility. Subsequently, β-cyclodextrin was chemically immobilized onto the grafted surface, enabling efficient adsorption of phenolic compounds in aqueous media. This strategy highlights the potential of GMA-based plasma grafting for the construction of supramolecular and host–guest functional materials. In a similar manner, plasma-initiated grafting of GMA on polypropylene fibres followed by amination with diethylenetriamine yielded efficient adsorbents with a maximum cadmium uptake of 46.47 $mg \cdot g^{-1}$, as reported Chen haonan et al. [196]. On silicon (Si (100)) substrates, plasma polymerization of GMA demonstrated that the discharge parameters have a significant impact on the preservation of epoxide groups. The utilization of low radio frequency power (approximately 5 W) in conjunction with relatively high pressure (100–400 Pa) resulted in the formation of polymer films that exhibited epoxide functionalities analogous to those observed in the GMA homopolymer. Pretreatment with argon plasma has been demonstrated to promote covalent bonding of the deposited layer to the silicon surface. This, in turn, has been shown to enhance adhesion with subsequently applied polyimide coatings through reactive curing of the epoxy groups [197]. In contrast, on poly (tetrafluoroethylene) films, strong adhesion of GMA-based layers to polyimide was only achieved when oxygen plasma pre-activation was employed; the absence of pre-treatment or the use of argon plasma resulted in coatings with a lower binding strength that were more easily removed by solvent extraction [198]. This difference underscores the significance of substrate surface chemistry: while silicon readily forms covalent interactions under mild plasma conditions, poly (tetrafluoroethylene) requires the introduction of oxygen-containing functional groups to enable stable grafting and effective interfacial bonding. In a complementary direction, more recent studies have extended this approach to natural substrates such as cotton nonwoven fabrics. In these cases, plasma activation followed by grafting of GMA and subsequent amination with ethylenediamine has been shown to produce efficient and reusable adsorbents for arsenic removal. The grafting efficiency was found to be contingent upon a number of factors, including plasma power, reaction temperature, grafting time, and monomer concentration. It was determined that optimal conditions were necessary to yield flexible materials with high functional group density [199].

In consideration of the aforementioned points, it may be concluded that plasma-initiated radical polymerization provides a powerful and versatile route for the synthesis and functionalization of GMA-based materials under mild, initiator-free, and energy-efficient conditions. The use of non-thermal plasma enables the generation of reactive species at the plasma–monomer interface, allowing controlled initiation while limiting thermal degradation. In pulsed atmospheric-pressure systems, the balance between surface-driven polymerization and plasma-induced reactions is strongly dependent on monomer structure and discharge parameters, leading to markedly different behaviors between MMA, BMA, and GMA. Among them, GMA exhibits the highest reactivity and functional retention due to its epoxy group, which enhances radical formation and chain growth efficiency.

In addition, the use of dielectric barrier discharge configurations and pulsed excitation significantly improves functional group preservation, particularly for epoxy moieties, while maintaining high deposition rates. This controlled activation also enables efficient grafting onto various substrates, demonstrating the strong adaptability of plasma-initiated processes for designing functional surfaces and advanced polymer interfaces.

At the end of this section, a comparison between these techniques is necessary to better understand, based on the studies reported herein, the differences in performance and efficiency for GMA polymerization under external stimulation conditions.

Among the different ESRP approaches, a clear hierarchy can be drawn when considering the specific case of GMA. Photopolymerization and radiation-induced polymerization generally provide the most controlled systems, ensuring efficient radical generation while maintaining relatively high preservation of the epoxy functionality. In contrast, ultrasound- and microwave-assisted systems mainly act as acceleration tools, enhancing reaction kinetics but offering more limited control over radical specificity and polymer structure. Electrochemically initiated system provides good spatial and temporal control of radical generation; however, its efficiency strongly depends on electrode interfaces and electrolyte conditions, which may limit its general applicability for GMA systems.

Within this framework, plasma-initiated polymerization appears as the most versatile and reactive approach for GMA, particularly under atmospheric-pressure and pulsed conditions. It uniquely combines high radical density, surface activation, and solvent-free operation, enabling both rapid polymer growth and high functional group retention when optimized. Compared to the other techniques, plasma-based methods offer a broader tunability between surface-dominated and plasma-dominated regimes, making them especially effective for tailoring GMA reactivity and film architecture. However, this high reactivity must be carefully balanced, as excessive plasma exposure can also lead to fragmentation and loss of functionality. Overall, while each technique presents specific advantages, plasma-initiated systems stand out as the most powerful and adaptable approach for GMA-based polymerization when properly controlled.

When compared with classical FRP, all these externally stimulated techniques can be considered more aligned with green chemistry principles, mainly due to their reduced reliance on thermal initiators, lower energy consumption, and the possibility of milder operating conditions. However, this sustainability improvement does not necessarily correspond to better control over molecular architecture or polymer uniformity, which remains a key limitation depending on the technique used. Overall, while each method offers specific advantages, plasma-initiated systems stand out as the most powerful and adaptable approach for GMA polymerization, provided that process parameters are carefully optimized.

3.3. Controlled/reversible deactivation radical polymerization

In view of the inherent limitations of conventional FRPs, a range of RDRP techniques have been developed and successfully implemented in the field of GMAs. These techniques are distinguished by their mechanism, a process that differs from that of conventional FRP, which necessitates the thermal decomposition of initiators to generate free radicals. The mechanism of this type of controlled polymerization is more complicated, but it offers the advantage of precise molecular weight control, reduced dispersity, and the synthesis of well-defined polymer architectures [200,201].

Three methods have been developed in the field of controlled GMA polymerization: RAFT, ATRP and NMP. The ensuing discussion will center on these methods, elucidating the mechanism that will be translated by these three types of polymerization. It will also encompass a discussion of the types of initiators and polymer architectures that have been explored in the field of GMA, along with other more developed methods that combine controlled and advanced polymerization

using GMA as a monomer unit, with some prospects for the future. The following essay will provide a comprehensive overview of the relevant literature on the subject.

3.3.1. *Reversible addition–fragmentation chain transfer (RAFT)*

The RAFT polymerization process has been identified as one of the most robust and controllable methods for GMA polymerization, offering a superior degree of control over polymer chain length and dispersion. This process is an innovative method in which thiocarbonylthio compounds (e.g., dithiobenzoates, trithiocarbonates, xanthates, dithiocarbamates) are utilized as chain transfer agents. These compounds facilitate a reversible addition-fragmentation mechanism, enabling precise control over the process [202]. In addition, the RAFT as advanced process serves to illustrate the efficacy of the control mechanisms employed in the synthesis of copolymers, which are characterized by more defined architectures. A multitude of works cited in the extant literature have demonstrated the efficacy of this technique for developing GMA-based polymers with well-defined architectures including; Y-shaped diblock copolymer[203], linear block copolymer [204], spherical, worm-like and vesicle micelles [205] or even brush-block copolymer [206].

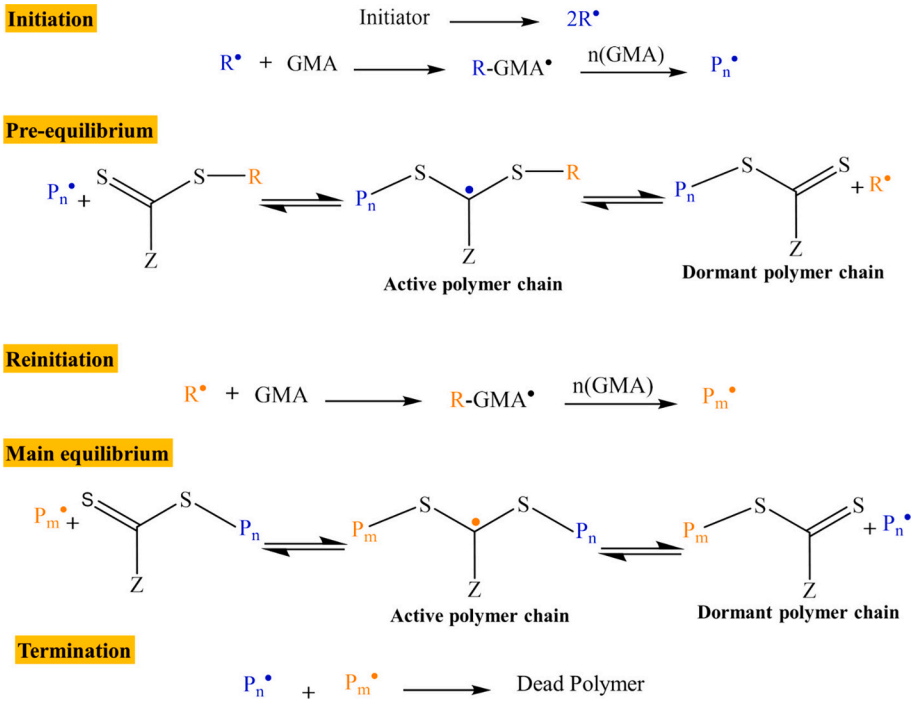
The mechanism of RAFT polymerization as described in the literature [202,207] is initiated by the generation of radical species (R• in blue Scheme 12) through the thermal decomposition of a conventional initiator (AIBN serves as an example of such an initiator). The aforementioned radicals undergo a reaction with the GMA monomer unit, thereby forming the initial activated monomer (R-GMA•). The activation process is characterized by its radical nature, which is evidenced by the propagation of the activation on the same unit. This propagation results in the formation of a long activated chain, which is denoted by P_n. In the ensuing stage, designated a pre-equilibrium, the propagating radical species (P_n) rapidly adds to a RAFT agent, which is typically a dithiobenzoates, forming a transient radical intermediate. Subsequently, this intermediate undergoes fragmentation, releasing a new radical species (R• in orange Scheme 12) capable of initiating the growth of another polymer chain (P_m). This reversible addition-fragmentation process establishes a main dynamic equilibrium between active and dormant polymer chains, thereby ensuring controlled chain propagation as illustrate in Scheme 12. Termination may be achieved through two distinct processes: recombination of two active radicals or disproportionation (hydrogen transfer). The outcome of these processes is the formation of dead chains that are incapable of reactivation. However, this stage is less frequent for the following reason: the active radical is constantly in equilibrium with a dormant form (via the RAFT intermediate). As a result, the concentration of free radicals is reduced, leading to a decrease in the number of collisions and, consequently, fewer terminations.

The *Z*-group in RAFT agents plays a crucial role in controlling radical polymerization by modulating both the stability and reactivity of the intermediate radical species. This influence is primarily exerted through a combination of mesomeric (resonance) and inductive effects.

In the context of RAFT polymerization, thiocarbonylthio compounds are indispensable for this process. The distinguishing characteristic of these compounds is the presence of a central thiocarbonylthio group, known as S—C (=S). This group is pivotal in facilitating a reversible equilibrium between active and dormant polymer chains, thereby enabling controlled radical polymerization [208]. In conjunction with a stabilizing or modulating Z group, this equilibrium is facilitated. This thiocarbonylthio functionality is a common feature of all RAFT agents, including: The following compounds are of particular interest: dithiobenzoates, trithiocarbonates, xanthates and dithiocarbamates [209,210]. In addition, given that the Z groupe has the capacity to act either as an electron donor or acceptor, it is imperative to acknowledge the substantial impact of this parameter on the stability of the intermediate radical, and the overall reactivity of the RAFT agent [210].

In light of the preceding discussion, this section aims to provide a comparative analysis of these four RAFT agents in order to identify the most compatible compound for GMA polymerization. It is hypothesized that this will facilitate a more precise assessment of the stability of these RAFT agents on this monomer unit. Subsequently, the effect of the Z group will also be studied on the basis of various experimental studies to determine the most effective Z group in GMA polymerization, taking into account several parameters in order to achieve enhanced control.

Among these thiocarbonylthio compounds, dithiobenzoates are particularly effective for controlling the polymerization of activated monomers such as GMA. In these agents, the *Z*-group is typically an



Schema 12. RAFT polymerization mechanism using GMA as the unitary monomer.

aromatic moiety such as a phenyl or naphthyl ring which acts as a resonance-withdrawing group (−M) (Fig. 11). This electron-withdrawing nature enhances the electrophilicity of the thiocarbonyl carbon, thereby facilitating the addition of the propagating radical. Furthermore, the resulting intermediate radical is stabilized through delocalization of the unpaired electron into the conjugated π-system of the aromatic *Z*-group, as illustrated in Fig. 11. This dual effect activation of the thiocarbonyl carbon and stabilization of the radical intermediate not only promotes efficient fragmentation but also contributes to precise control over molecular weight and narrow dispersity [209].

In contrast to dithiobenzoates, trithiocarbonates, which contain alkylthio (–SR′) groups as Z-substituents, exert an inductive electron-withdrawing effect (—I) due to the electronegativity of the sulfur atom (Fig. 11). This inductive effect enhances the electrophilicity of the thiocarbonyl carbon, thereby facilitating radical addition and enabling good control over polymerization. Although trithiocarbonates perform well with less activated monomers, their effectiveness with GMA is generally lower than that of dithiobenzoates [211]. Dithiocarbamates, on the other hand, incorporate dialkylamino (–NR_2) Z-groups that exhibit a strong electron-donating mesomeric effect (+M). This resonance donation increases electron density on the thiocarbonyl carbon, decreasing its electrophilicity and destabilizing the intermediate radical [209]. As a consequence, the fragmentation step becomes less favorable, leading to poor RAFT equilibrium and uncontrolled polymer growth.

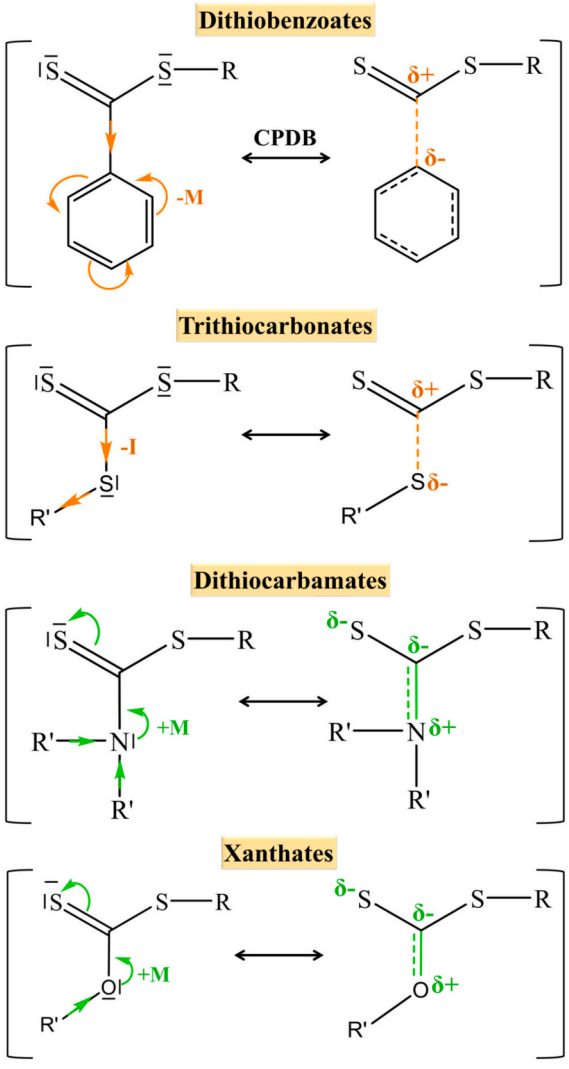


Fig. 11. Inductive and mesomeric effect resulting from thiocarbonylthio compounds as RAFT agents.

Similarly, xanthates feature alkoxy (–OR) Z-groups, which also donate electrons through mesomeric (+M) effect, though weaker than that of NR_2 (see Fig. 11). This excessive electron density destabilizes the intermediate radical and slows the reversible fragmentation, making xanthates suitable only for low-activity monomers like vinyl acetate, but ineffective for GMA.

In summary, the RAFT control efficiency in GMA polymerization follows a trend that closely correlates with the electron-withdrawing capacity of the *Z*-group. Groups with stronger electron-withdrawing effects better stabilize the intermediate radical and promote efficient fragmentation, leading to improved control over polymer growth:

Dithiobenzoates > Trithiocarbonates > Xanthates > Dithiocarbamates.

Focusing on dithiobenzoate-type RAFT agents widely acknowledged for their efficiency in mediating the controlled radical polymerization of activated monomers this section reviews key studies exploring their application in the polymerization of GMA. The goal is to compare their performance in terms of molecular weight control and dispersity, particularly at high monomer conversions, in order to identify the most suitable agent for synthesizing well-defined GMA-based polymers. The RAFT agents compared in this analysis are summarized in Table 2.

In a study by Benaglia et al. [212], the impact of the Z-group structure on RAFT polymerization was systematically evaluated using three dithiobenzoate-based agents: 2-cyanoprop-2-yl dithiobenzoate (CPDB), 2-cyano-4-methylpent-2-yl 4-cyanodithiobenzoate (CMCD) and 2-cyano-4-methylpent-2-yl diethoxyphosphoryldithioformate (CMDP). All three share the same branched R-group (2-cyano-4-methylpent-2-yl), selected for its excellent leaving group ability.

Comparing agents CPDB and CMCD reveals the significant role of Z-group electronic properties in controlling the polymerization. Agent CPDB, with a substituted phenyl Z-group (Fig. 12), provided only moderate control: at conversions between 31% and 36%, the resulting polymers showed high number-average molecular weights (Mn ≈ 18,600–22,000 g/mol) and relatively broad dispersities (Đ ≈ 1.23–1.24). In contrast, agent CMCD, bearing a para-cyano substituent on the phenyl ring, delivered markedly better control even at lower conversions (4.2–11%). The polymers obtained had significantly lower Mn values (≈1700–4600 g/mol) and narrower Đ (1.20 to 1.11). This enhanced performance is attributed to the strong electron-withdrawing character of the cyano group, which combines inductive (—I) and mesomeric (−M) effects (see Fig. 12). These effects stabilize the intermediate radical and accelerate the fragmentation step, thereby reinforcing the dynamic equilibrium between active and dormant chains and improving control over the polymerization. However, the high stability of the intermediate radical in agent CMCD can also lead to a noticeable reduction in the overall polymerization rate. This is because the enhanced stabilization tends to lower the radical's reactivity toward monomer addition an effect that becomes particularly relevant when working with moderately reactive monomers such as GMA. As a result, lower monomer conversions were observed with agent CMCD compared to agent CPDB. While CMCD offers superior control over molecular weight and dispersity, its slower polymerization kinetics make it more suitable for highly reactive monomers, where sufficient chain propagation can still occur efficiently. In contrast, agent CPDB, despite its less effective RAFT equilibrium control, allows faster polymerization and may therefore be more appropriate for GMA. These findings highlight the importance of tailoring the electronic nature of the *Z*-group to the specific reactivity of the monomer, to strike a suitable balance between polymerization rate and control.

The third agent CMDP exhibited intermediate performance between agents CPDB and CMCD. At higher monomer conversion (~45%), it produced polymers with a relatively narrow dispersity (Đ ≈ 1.14), suggesting better control than CPDB but not as refined as CMCD. Interestingly, the phosphoryl group in CMDP also introduces an additional advantage: it permits post-polymerization aminolysis, offering a reactive functional handle for subsequent chemical modifications. This

Table 2
Summary of RAFT polymerization of GMA using dithiobenzoate-type agents.

ATRP Agent	Z-Group Structure	Medium	Temperature (°C)	Conversion (%)	Mn (g/mol)	Dispersity (Đ)	Refernce
CPDB	Phenyl	Toluene (solution)	70	31–36	18,600-22,000	1.23–1.24	[212]
		Bulk	60	53–68	up to 10,000	1.1	[214]
CMCD	Phosphoryl dithioformate	Toluene	70	4.2–11	1700-4600	1.11–1.20	[212]
CMDP	p-CN Phenyl	Toluene	70	~45		1.14	
		Bulk	60	~97	24,700-26,000	1.22–1.41	
CPDN	Naphthalene	Bulk	80	~99	10,000-26,000	1.27–1.22	[213]
		Benzene (solution)	60	~64	lower Mn due	1.20–1.24	

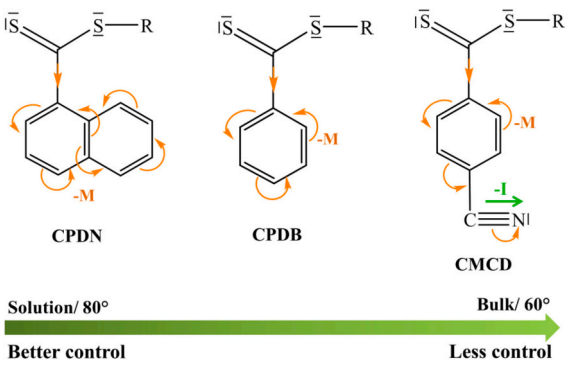


Fig. 12. Influence of Z-group on the mesomeric stabilization in dithiobenzoates.

versatility makes agent CMDP a promising compromise when moderate control and further functionalization of the polymer are desired.

In the study conducted by Zhu et al. [213] systematically examined how reaction temperature and medium influence the RAFT polymerization of GMA, using 2-cyanoprop-2-yl 1-dithionaphthalate (CPDN) as a chain transfer agent. At 60 °C under bulk conditions, polymerization control gradually diminished at high monomer conversion, with the Đ increasing from 1.22 to 1.41 as conversion approached 97%. Interestingly, when the same reaction was performed at a higher temperature of 80 °C, also in bulk, polymerization control improved significantly-Đ values decreased from 1.27 to 1.22, even at 99% conversion. In comparison, conducting the reaction in benzene solution at 60 °C yielded only moderate control, with Đ ranging from 1.20 to 1.24 at 64% conversion.

These findings suggest that conducting the polymerization in bulk at elevated temperatures (80 °C) provides the most favorable conditions for achieving well-controlled GMA polymerization with CPDN. This enhancement in control can be attributed to two main factors: First, elevated temperature accelerates the fragmentation step in the RAFT process, promoting a more dynamic and efficient exchange between active and dormant chains. Second, bulk conditions increase the local concentration of reactive species, thereby enhancing the frequency of radical transfer events. In contrast, polymerization in solution tends to dilute the radical concentration, potentially slowing the RAFT equilibrium and slightly compromising molecular weight control, as evidenced by broader Đ.

To better understand the influence of *Z*-group structure on RAFT polymerization control, it is informative to compare the studies by Zhu et al. [213] and Benaglia et al. [212] which employed CPDN and CPDB, respectively. Both RAFT agents share the same R-group (2-cyanoprop-2-yl), known for its excellent leaving ability, but differ in their Z-group structure: CPDN features a naphthalene ring, while CPDB contains a simpler phenyl group. A meaningful comparison can be drawn between the results obtained with CPDN at 60 °C in benzene and those with CPDB at 70 °C in toluene, as both represent polymerizations carried out in aromatic solvents under comparable solution-phase conditions. Despite the slight difference in temperature, the comparison helps isolate the effect of *Z*-group electronics.

Under these conditions, CPDB yielded relatively broad molecular weight distributions (Đ ≈ 1.23–1.24) at low conversions (31–36%), whereas CPDN afforded slightly narrower Đ (1.20–1.24) even at a much higher conversion (64%). This suggests that CPDN provides better control over the polymerization process, likely due to the extended π-conjugation of the naphthalene *Z*-group. The increased delocalization stabilizes the intermediate radical via a stronger mesomeric (+M) effect, thereby facilitating reversible chain transfer and enhancing the RAFT equilibrium (see Fig. 12).

Another relevant study that permits for a direct comparison with that of Zhu et al. [213] was reported by Gudipati et al. [214]. The comparison was conducted between CPDB and CPDB as an agent, but in a different medium from the first, in bulk and at 60 °C. Both studies provide valuable insight into the influence of *Z*-group structure on polymerization control under identical temperature and medium conditions. When using CPDB, the polymerization proceeded with good control, maintaining low and consistent Đ values around 1.1 throughout the reaction and reaching number-average molecular weights (Mn) up to 10,000 Da. A near-linear relationship between Mn and monomer conversion was observed, especially at higher [GMA]/[CPDB] ratios, indicating effective regulation of the polymer growth. In comparison, polymerizations carried out with CPDN showed broader Đ evolution increasing from 1.22 to 1.41 as the conversion approached 97% suggesting a gradual loss of control at high conversion, despite the Mn still increasing linearly. This divergence in performance points to a key trade-off: while the naphthalene *Z*-group in CPDN enhances radical stabilization through an extended +M mesomeric effect, this same stabilization may slow down the fragmentation step or increase termination events at high conversions, thereby compromising control. Overall, CPDB appears to provide more stable and reliable control under bulk polymerization conditions at 60 °C, whereas CPDN offers more consistent Mn evolution but may be less effective at maintaining narrow dispersity at high conversion levels.

As demonstrated in the preceding explanations, the divergent outcomes observed in the comparisons conducted by these studies have resulted in contradictory conclusions regarding the superiority of either CPDN or CPDB in terms of enhanced polymerization control. A more in-depth comparison of these two agents is necessary to identify the best agent. This comparison should focus on the nature of the reaction medium (solution or bulk). Several key observations have been made as a result of this research. The objective of these analyses is to enhance our comprehension of the impact of the Z group in CPDN and CPDB RAFT agents under varying reaction conditions.

In solution (in either toluene or benzene), both agents demonstrate good solubility, ensuring a homogeneous system that is conducive to well-controlled polymerization. CPDN, which has a naphthalene Z-group, has slightly lower Đ values than CPDB at significantly higher conversions. This highlights the beneficial role of its extended conjugation and + M mesomeric effect in stabilizing the intermediate radical as described previously. In contrast, in bulk polymerization, where the absence of solvent increases viscosity and amplifies compatibility differences, the agents perform differently than expected. CPDB maintains good control (Đ ≈ 1.1), indicating strong solubility and compatibility

with neat GMA. Conversely, CPDN shows a decline in control at high conversion (Đ reaching 1.41), despite favorable performance at lower conversion levels. This may be due to its reduced solubility or decreased miscibility in the highly viscous monomer environment, which hinders effective chain transfer.

These observations suggest that, while the naphthalene Z-group in CPDN offers superior electronic stabilization in solution, it may be at a disadvantage in bulk due to the low solubility and system homogeneity, which reduce its effectiveness in maintaining control at high monomer conversion.

Overall, the collected literature highlights that no single Z-group in RAFT agents is universally optimal for the polymerization of GMA, reflecting the inherent trade-off between radical stabilization and reactivity. While strongly electron-withdrawing Z-groups can enhance control through improved intermediate stabilization, they may simultaneously reduce propagation kinetics, whereas electron-donating groups often lead to loss of control despite higher reactivity. Consequently, the efficiency of RAFT polymerization is highly context-dependent, and the selection of the Z-group must be critically balanced with the reaction medium, temperature, and monomer activity to avoid over-stabilization or loss of equilibrium efficiency. This emphasizes that achieving well-defined GMA-based architectures requires not only an appropriate RAFT agent, but also a finely tuned optimization of all reaction parameters.

3.3.1.1. Externally stimulated RAFT techniques. The combination of advanced initiation methods with RAFT polymerization represents a powerful strategy that addresses the shortcomings of using either approach on its own. RAFT polymerization, in principle, provides precise control over molecular weight and architecture. However, it is most often dependent on thermal initiation, which restricts both the reaction conditions and the range of monomers that can be efficiently polymerized. In contrast, techniques such as γ-radiation, photo-initiation, electrochemical activation, and plasma initiation have been shown to generate radicals under milder and more sustainable conditions (as described in previous sections). However, when these approaches are applied independently, they generally offer limited control over chain propagation. This can result in broad molecular weight distributions and premature termination events.

The integration of these initiation strategies with RAFT has the potential to combine the advantages of both. RAFT ensures well-regulated chain growth, while the alternative initiation modes expand the versatility of activation, allowing polymerizations to proceed under milder, adjustable, and even solvent-free environments[215–218]. This synergy is of particular value in the context of GMA, where the preservation of the reactivity of the epoxy group during polymerization is of paramount importance.

For instance, Yin et al. [219] investigated the ambient-temperature RAFT polymerization of GMA under UV–visible light irradiation using a CPDB-based chain transfer agent and a (diphenyl(2,4,6-trimethylbenzoyl)phosphine oxide) photoinitiator. A detailed comparison of different irradiation ranges (visible light: 405–577 nm, long-wave UV–vis: 365–577 nm, and full-wave UV–vis: 254–577 nm) revealed a strong dependence of the polymerization control on the light source. Under visible light irradiation, the system exhibited highly controlled behavior with a linear semi-logarithmic kinetic profile up to ~87% monomer conversion, indicating a constant radical concentration and efficient RAFT equilibrium. The resulting polymers showed narrow dispersities (Mw/Mn ≈ 1.18–1.20) and monomodal GPC traces shifting regularly toward higher molecular weights. Importantly, the epoxide groups of GMA were fully preserved throughout the polymerization process, confirming the chemoselectivity of the system. In addition, successful chain extension experiments further demonstrated the living character of the RAFT process under these conditions.

Beyond solution polymerization, light-mediated RAFT strategies have also been extended to surface engineering with high spatial precision. Farquet et al. [26] developed a method for the fabrication of well-defined poly(GMA) nanostructures on fluoropolymer substrates using extreme ultraviolet radiation combined with RAFT-mediated surface polymerization. In this approach, extreme ultraviolet interference lithography was first used to create periodic surface activation patterns. Subsequent exposure to air generated peroxide functionalities on the irradiated regions, which acted as initiating sites for thermally driven surface-initiated RAFT polymerization. The use of RAFT control played a key role in regulating the growth of polymer chains from the surface. Compared to conventional FRP grafting, RAFT significantly reduced the thickness of the polymer layer by approximately one order of magnitude and enabled much finer spatial control, achieving feature sizes down to the nanometer scale (~50 nm). The study also showed that the grafting density increased with the absorbed radiation dose, reaching values on the order of 0.1–0.2 chains·nm^{-2}. At low irradiation doses, a transition from isolated "mushroom"-like chains to a more densely packed "brush" regime was observed. Overall, this work demonstrates that combining extreme ultraviolet patterning with RAFT polymerization enables precise control over both the architecture and density of surface-grown polymer structures, providing access to functional epoxy-bearing nanostructured interfaces with potential applications in micro- and nanofabrication.

In similar manner, the study by Zhou et al. [220] demonstrates the strong potential of photoinduced electron transfer-RAFT polymerization as a versatile and mild surface modification strategy for functional hydrogels. In the case of poly(vinyl alcohol), this approach enables the controlled grafting of GMA chains under visible light irradiation, followed by post-functionalization through epoxide RO with chitosan derivatives. This sequential strategy highlights several key advantages of photoinduced electron transfer-RAFT for GMA-based systems. First, the use of light-mediated RAFT allows polymerization to proceed under mild conditions, avoiding harsh thermal activation that could compromise the integrity of sensitive polymeric substrates such as poly(vinyl alcohol) hydrogels, second, the controlled radical mechanism ensures successful formation of well-defined polymer brushes on the surface, which can subsequently be further modified through the highly reactive epoxide groups of GMA. This post-functionalization capability is particularly valuable for biomedical applications, where additional functionalities such as antibacterial or antifouling properties can be introduced in a modular manner.

Building on previous work on light-mediated grafting systems, γ-Radiation has also been widely employed as an external energy source to initiate RAFT-mediated polymerization systems, enabling simultaneous radical generation in both solution and on solid substrates, and thus allowing efficient grafting-from approaches such as cellulose functionalization with GMA under controlled radical conditions [25]. This study focuses on the comparison between RAFT-mediated γ-irradiation polymerization of GMA in solution and grafting-from cellulose, with particular attention to the influence of irradiation dose on polymerization control and side reactions.

In solution, poly(GMA) exhibits a relatively controlled RAFT behavior as a function of absorbed dose. Using CPDB, the molecular weight increases from Mn ≈ 5300 g·mol^{-1} (Đ ≈ 2.1 at 2.1 kGy) to Mn ≈ 5500 g·mol^{-1} (Đ = 1.77 at 4.4 kGy), then Mn ≈ 13,100 g·mol^{-1} (Đ = 1.56 at 6.4 kGy) and up to Mn ≈ 43,200 g·mol^{-1} (Đ = 1.69 at 8.3 kGy). However, when using cumyl dithiobenzoate (CDB), broader dispersities (Đ up to ~3.5) are observed, reflecting less efficient control over radical growth under irradiation conditions.

For the grafting-from cellulose system, direct determination of Mn and Đ for grafted chains is not feasible due to their insolubility; thus, the evaluation is based on degree of grafting and surface characterization techniques. Despite this limitation, a clear dose-dependent behavior is observed. An intermediate irradiation range, approximately 4–6 kGy, provides the most favorable balance between radical generation and controlled graft propagation, within this window, grafting proceeds

efficiently while maintaining a partial RAFT control over chain growth. However, at higher absorbed doses (≥8 kGy), the system becomes increasingly dominated by radiation-induced secondary processes, including radical recombination, chain branching, and crosslinking of poly(GMA), leading to a loss of control, increased dispersity in solution, and the formation of insoluble network structures. This behavior is further attributed to the sensitivity of the epoxide group in GMA toward γ-radiation, which can promote additional side reactions such as RO, branching, and crosslinking, particularly at higher absorbed doses.

Accordingly, the precise control achieved in GMA-based RAFT polymerizations not only governs macromolecular growth but also directly determines the subsequent self-organization behavior of the resulting polymeric systems. The utilization of self-assembly techniques with this monomer has been the subject of extensive research, primarily due to its hydrophobic properties. A notable development is the integration of self-assembly with RAFT polymerization, which has emerged as an effective strategy for the controlled synthesis of well-defined nanostructures from GMA. RAFT polymerization provides precise control over molecular weight and polymer architecture, which in turn directly regulates the self-assembly behavior of amphiphilic block copolymers containing hydrophobic poly(GMA) segments. The inherent hydrophobicity of these blocks leads to spontaneous organization into diverse morphologies, such as spheres, worms, and vesicles, when dispersed in selective solvents [205,221]. Notably, the morphology of the resulting nanostructures can be meticulously adjusted by modifying polymerization parameters, encompassing the degree of polymerization of the core-forming block, the molecular weight of the macro-RAFT agent, and the reaction temperature. As Dai et al. [222] demonstrated, systematic modulation of these parameters allows the fabrication of nanostructures with controlled dimensions and internal organization. Furthermore, this combined strategy enables the in situ generation of functional nanostructures under mild polymerization conditions [223], while the epoxy functionality of GMA offers versatile routes for post-polymerization modification without affecting self-assembly [204]. A substantial body of research, including the comprehensive review by Li et al. [224], has underscored the promise of RAFT-mediated self-assembly of GMA-based copolymers in domains such as drug delivery, coatings, and catalysis. The combination of RAFT polymerization with self-assembly offers a scalable, one-pot route to uniform and functional nanomaterials with tunable size, morphology, and reactivity, opening new pathways for advanced material design [225].

This strong interplay between macromolecular design and hierarchical organization highlights how the control of polymer architecture can be further extended through external stimuli, enabling even more precise regulation of GMA-based systems at multiple length scales.

In view of these findings, an innovative approach has been established, combining externally stimulated RAFT techniques with the self-assembly process. A notable enhancement in versatility and reactivity is thus achieved in the synthesis of GMA-based nanostructures. Among the advanced techniques examined in this approach was photo-induced RAFT polymerization [226,227]. This process permits exact modulation of polymerization kinetics and self-assembly pathways in situ by merely regulating light exposure. This dynamic control facilitates on-demand initiation and cessation of polymer growth, which is highly advantageous for delicate monomers like GMA that possess reactive epoxy groups sensitive to harsh conditions. Furthermore, visible light-induced RAFT-mediated emulsion polymerization expands the process window to ambient temperature and aqueous or biphasic environments, increasing sustainability and compatibility with biological or environmentally sensitive applications. This approach is consistent with the principles of green chemistry, as it reduces energy consumption and the use of hazardous reagents. Tan et al. [228] demonstrated this approach by synthesizing well-defined epoxy-functionalized block copolymer nano-objects via visible light-initiated RAFT emulsion polymerization of GMA, achieving fine control over particle morphology and size. Consequently, the integration of photo-self-assembly process with RAFT polymerization of GMA provides a method for the fabrication of multifunctional nanostructures that is energy-efficient, scalable, and tunable with precision. These nanostructures are well-suited for advanced applications in fields such as coatings, catalysis, and biomedicine [229].

In conclusion, externally stimulated RAFT polymerization strategies represent a powerful and increasingly important green alternative to conventional thermally initiated processes. The use of external energy sources such as light, γ-radiation, and other activation modes enables polymerization under milder conditions, with reduced energy consumption and improved process sustainability. In addition, these approaches provide temporal and spatial control over radical generation, which is particularly advantageous for sensitive monomers such as GMA, where preservation of the epoxy functionality is essential.

Despite these significant advances, the combination of external energy sources with RAFT polymerization of GMA remains relatively limited compared to classical thermal systems. In particular, only a few external initiation strategies, mainly photo- and γ-radiation-based approaches, have been extensively explored in the context of GMA chemistry. This highlights an important gap in the current literature regarding the integration of other emerging external stimuli, such as electrochemical, plasma, or mechanochemical activation, with RAFT-controlled GMA polymerizations.

It is therefore expected that future research will further expand this field by developing new externally driven RAFT systems, enabling more sustainable, versatile, and precisely controlled polymerization platforms. Such developments will not only enhance the design of advanced GMA-based architectures but also strengthen the role of controlled radical polymerization as a key tool in green polymer chemistry and functional material synthesis.

3.3.2. *Atom transfer radical polymerization (ATRP)*

ATRP is a controlled/living radical polymerization technique that enables precise control over molecular weight, architecture, and dispersity. First developed in the mid-1990s, notably by Krzysztof Matyjaszewski [230–232], ATRP is based on a dynamic equilibrium between active radical species and dormant alkyl halide species, mediated by a redox catalytic system. Typically, this system involves a transition metal complex—most commonly Cu(I)/Cu(II)—coordinated with nitrogen-based ligands [233–235]. The mechanism relies on the reversible transfer of a halogen atom between the growing polymer chain and the metal complex, allowing for a controlled activation–deactivation process and minimizing irreversible termination reactions [236,237]. Common ATRP initiators include alkyl halides such as α-haloesters (e.g., ethyl α-bromoisobutyrate, ethyl 2-bromoisobutyrate; EBiB), benzyl halides, and tertiary alkyl halides, selected according to the monomer reactivity and reaction conditions.

In the case of GMA, ATRP is particularly well suited due to the methacrylate structure of the monomer, enabling the synthesis of well-defined poly(GMA) with narrow dispersities while preserving the reactive epoxide functionality. This epoxy group provides broad opportunities for post-polymerization modifications, including RO reactions, grafting, and crosslinking. However, the use of transition metal catalysts in ATRP introduces an intrinsic challenge, as the epoxide moiety may interact with the catalytic species, potentially leading to undesired side reactions such as RO under certain conditions. Despite these limitations, extensive studies have demonstrated that careful optimization of the polymerization conditions enables efficient control over molecular architecture while maintaining epoxide integrity.

In view of the specific reactivity of GMA, particular attention has been devoted to achieving precise control over its polymerization, in order to identify strategies to mitigate possible side reactions associated with this process. To address these aspects, the following section provides a detailed discussion of the mechanistic features of GMA polymerization via ATRP, with a particular focus on its reaction pathway and control elements. This is followed by an overview of recent advances integrating green and sustainable approaches aimed at further

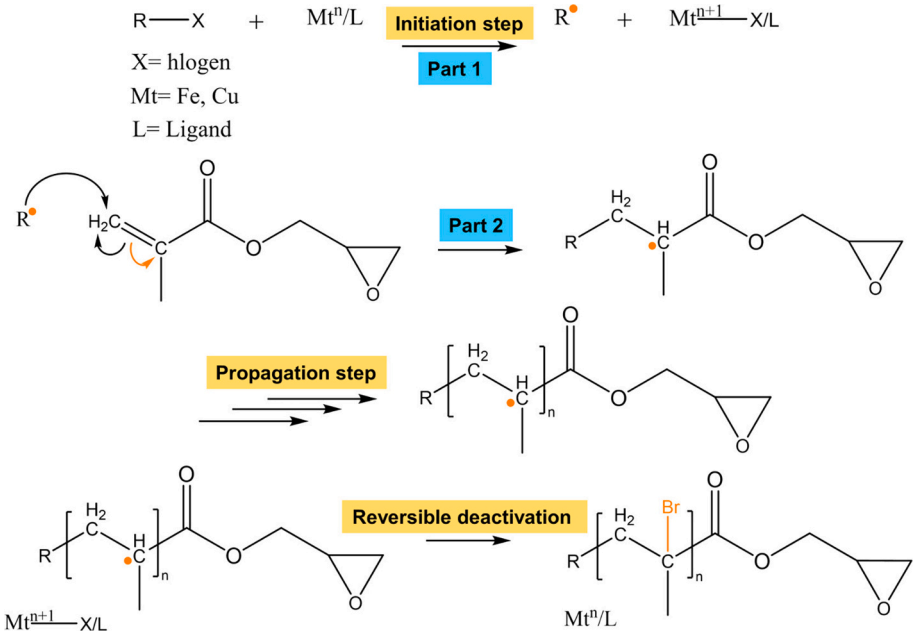
improving the efficiency and versatility of the process. Overall, these developments contribute to expanding the functional scope and application potential of GMA-based materials.

As elucidated by numerous studies in the relevant literature [238,239], the ATRP mechanism is defined as follows: The initiation step of this polymerization is divided into two parts, the first part is the formation of the active radical (R•) (Scheme 13-part1); which begins with the activation of a halogenated initiator of type R-X (where R is an organic chain and X is a halogen, often Br or Cl) in the presence of a metal complex in reduced form (Mt^n/L, usually a transition metal with a stabilizing ligand, such as a complex of Cu(I), Fe(II), Ru(II), etc. This step is based on the ability of the reduced metal Mt^n to strip the halogen from the initiator R-X, creating a free radical (R•) and an oxidized metal in the form of Mt^{n+1}-X (in an inactive state). In the second part (Scheme 13-part2), the active free radical produced will initiate polymerization, where the R• has the ability to react with a GMA monomer, opening its C═C double bond to form an end-of-chain radical. The chain propagates with the subsequent addition of GMA units (Scheme 13-propagation step). Notably, the epoxide ring of GMA remains intact throughout this process. The growing radical can be deactivated in a reversible manner by reacting with the oxidized metal complex (Mt^{n+1}–X/L), which results in the transfer of a halogen atom back to the polymer chain end (shema13-reversible deactivation step). This process regenerates the inactive species, and restores the reduced metal catalyst. The concentration of free radicals in solution is limited by this reversible activation deactivation equilibrium, thereby suppressing termination and chain transfer events. As a result, polymer chains grow in a controlled manner with narrow molecular weight distribution (low dispersity).

A variety of GMA-based copolymers with different architectures have been successfully synthesized via ATRP and its advanced variants. These include well-defined linear poly(GMA) homopolymers [240], as well as *block* copolymers such as poly(GMA)-*block*-poly(styrene), poly (GMA)-*block*-poly(ethylene glycol), poly(GMA)-*block*-poly(butyl acrylate), and poly(dimethylsiloxane)-*block*-PGMA [241–243]. In addition, random copolymers of GMA with other methacrylates have been reported, together with graft copolymers incorporating GMA units into functional side-chain structures [84,244]. More complex topologies, such as *star*-shaped and multi-arm polymers, have been achieved via "core-first" and "arm-first" ATRP approaches, yielding three-dimensional macromolecules with potential for crosslinking and nanomaterial design [245,246]. Furthermore, surface-initiated ATRP has facilitated the synthesis of dense polymer brushes of poly(GMA) grafted from substrates such as boron nitride nanotubes [247], or carbon nanotubes [24], providing versatile platforms for biointerfaces, hybrid coatings, and adsorption materials . Finally, the post-polymerization modification of epoxy groups (via RO reactions with amines, thiols, or alcohols) has allowed the preparation of highly functional, hierarchical, and multifunctional architectures, broadening the scope of poly(GMA)-based materials for biomedical, environmental, and smart material applications [248].

Star-shaped architectures of GMA-based polymers have been successfully synthesized by ATRP using two distinct strategies: the "core-first" and the "arm-first" approaches. In the "core-first" method, multifunctional initiators bearing three to six active halogen sites act as the central core from which polymer chains grow simultaneously as illustrate in Fig. 13. For instance, epoxy-functional star copolymers of butyl acrylate and GMA were obtained by ICAR-ATRP using tri-, tetra-, and hexafunctional initiators, resulting in well-defined *star*-shaped polymers with reactive epoxy functionalities preserved in the arms [249]. Conversely, the "arm-first" method entails the preliminary synthesis of linear macroinitiators (arms) by ATRP using monofunctional initiator, such as ethyl 2-bromoisobutyrate (EBiB), subsequently followed by their coupling or crosslinking with a multifunctional agent such as: divinylbenzene to yield the star core in situ (see Fig. 14). For instance, ATRP of GMA-containing macroinitiators, subsequently crosslinked with divinylbenzene, yielded well-defined star polymers with controlled arm length and narrow dispersity without requiring post-polymerization modification [250].

In order to develop a comprehensive understanding of the distinctions between these two approaches, it is essential to conduct an in-depth comparison. This comparison will facilitate the prediction of the effect of each approach on the synthesized of star-shaped polymer. The "core-first" approach is synthetically more straightforward, necessitating a single polymerization step. However, it provides less flexibility in the tuning of the length and composition of the arms and is strongly dependent on the purity and functionality of the initiator. The "arm-first" approach entails a more arduous process, encompassing a minimum of two stages, yet it ensures superior control over molecular weight distribution, arm uniformity, and star topology. In particular, when applied to GMA, the "arm-first" method ensures that the epoxy groups



Schema 13. Schematic representation of the ATRP radical polymerization mechanism using the "arm first" approach.

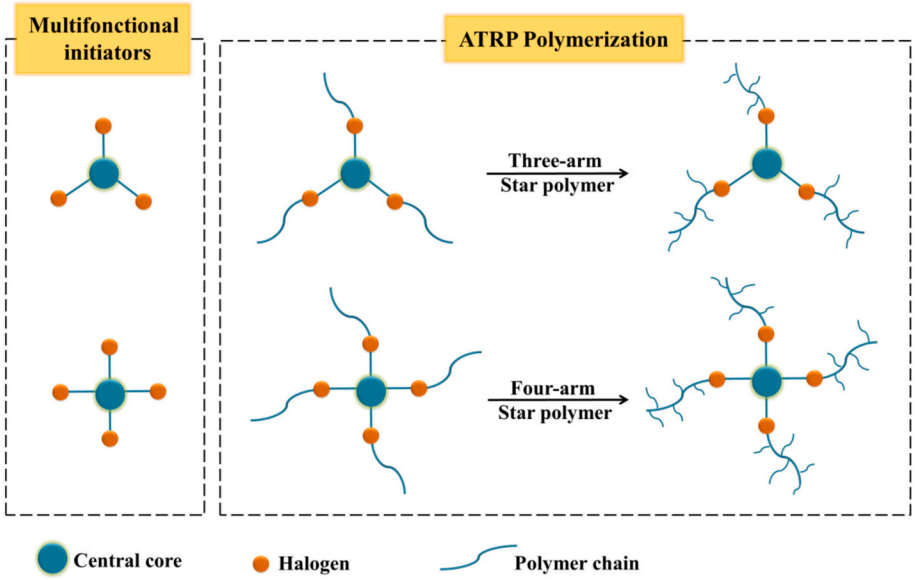


Fig. 13. Schematic representation of star polymer architectures formed by "core-first" ATRP polymerization.

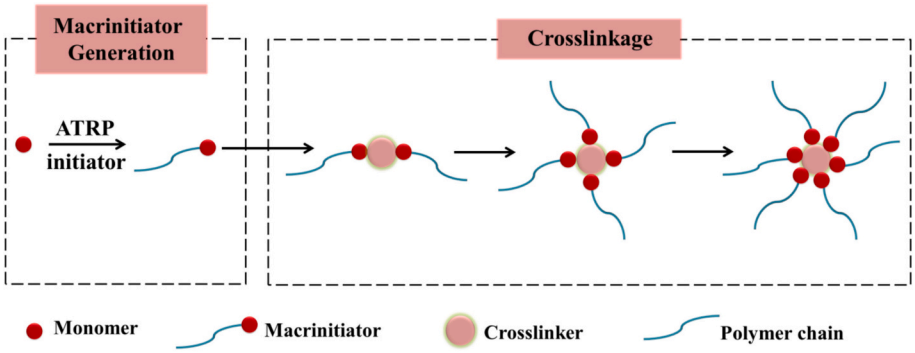


Fig. 14. Schematic representation of star polymer architectures formed by "arm-first" ATRP polymerization.

remain intact during star formation, while the "core-first" approach integrates the epoxide functionality directly from the growing arms. This makes the selection of either strategy is indicative of a trade-off between synthetic simplicity "core-first" and architectural precision "arm-first".

Despite the high efficiency of ATRP for the synthesis of well-defined poly(GMA), several challenges arise from the intrinsic reactivity of the epoxide functionality. In particular, undesired side reactions such as epoxide RO or partial crosslinking may occur under inappropriate catalytic or reaction conditions, leading to loss of solubility and poorly controlled architectures. These limitations are strongly influenced by the nature of the catalytic system, including both the ligand environment and solvent effects.

Among these parameters, the ligand plays a central role in determining the level of control in ATRP. Beyond simple coordination to the metal center, the ligand governs the dynamic activation–deactivation equilibrium that defines the controlled character of the process. In this mechanism, the Cu(I)/ligand complex activates the dormant alkyl halide species (R–X) to generate propagating radicals, while the Cu(II)/ligand complex ensures rapid and reversible deactivation of the growing chains. This equilibrium, described by Eq. 4, is essential to maintain a low steady-state radical concentration, thereby minimizing termination reactions and preserving control over the polymerization:

$$R\text{–}X + Cu(I)/L \rightleftharpoons R\bullet + Cu(II)/X\text{–}L \quad (4)$$

Among the ligands most commonly employed in ATRP of GMA are tris[2-(dimethylamino)ethyl]amine (Me_6TREN) [240], tris(2-pyridylmethyl)amine (TPMA) [240], N,N,N′,N″,N″-pentamethyldiethylenetriamine (PMDETA) [84,251–253], 1,1,4,7,10,10-hexamethyltriethylenetetramine (HMTETA) [254], and 2,2′-bipyridine (bpy) [255] (see Fig. 15). However, it is important to emphasize that, to date, no systematic study has directly compared the performance of different ligands specifically in the ATRP of GMA. Most available studies focus on individual ligand systems under specific conditions, making direct comparison difficult.

To overcome this limitation, a comparative mechanistic framework can be established based on the fundamental work of Tang, Wei, and Krzysztof Matyjaszewski [256] on the effect of ligand structure on activation rate constants in ATRP. Their study demonstrated that ligand structure strongly modulates the Cu(I)/Cu(II) equilibrium and the resulting radical concentration, According to the activation rate constants (k_{act}) (see Fig. 15), the ligand activity follows the general order Me_6TREN > TPMA > HMTETA > PMDETA > bpy. Although this relationship was established for model methacrylate systems, it provides a valuable basis for rationalizing ligand effects in GMA polymerization.

Highly active ligands such as Me_6TREN generate very fast activation–deactivation cycles, leading to an increased steady-state radical concentration. While this enhances polymerization rates, it also promotes secondary processes such as backbiting and intermolecular coupling, resulting in branching and partial crosslinking; the backbiting

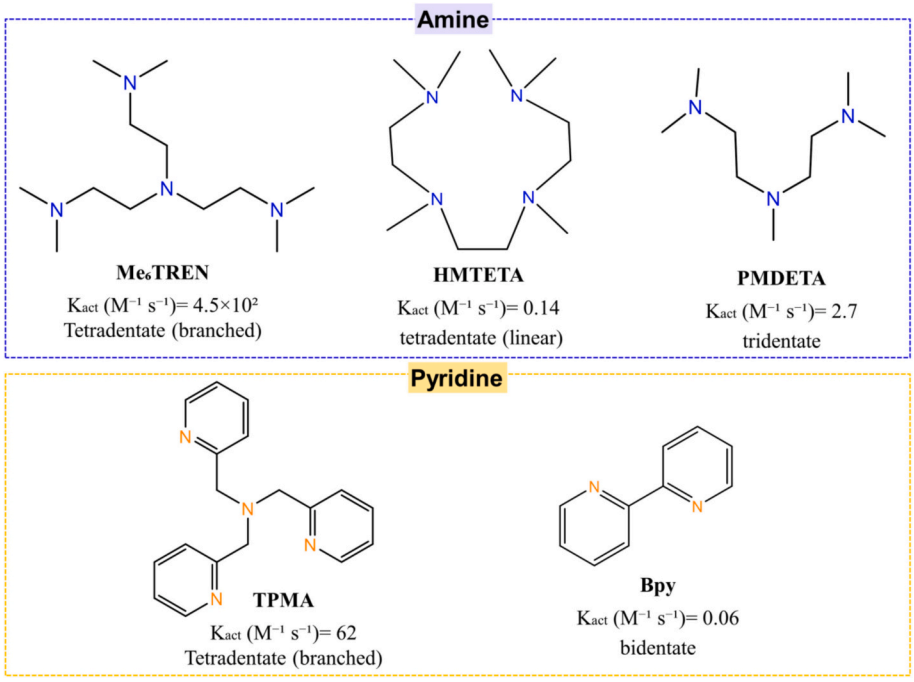


Fig. 15. Chemical structures of ligands used in Cu-mediated ATRP of GMA.

and intermolecular mechanism will be discussed in detail in the following section. In addition, the high catalytic activity increases the frequency of interactions between the epoxide groups and the copper complex. Both Cu(I) and Cu(II) species can act as Lewis acids and coordinate to the epoxide oxygen, thereby polarizing the ring and facilitating its opening in the presence of trace nucleophiles such as water, alcohols, or amines. Consequently, the susceptibility of the epoxide functionality to side reactions is indirectly enhanced under highly active catalytic conditions.

By contrast, TPMA provides a more balanced catalytic environment. Its tripodal amine coordination stabilizes both Cu(I) and Cu(II) oxidation states, ensuring efficient yet more regulated activation–deactivation equilibrium. This leads to a lower radical flux compared to Me₆TREN, thereby decreasing termination events and limiting epoxide RO side reactions. As a result, TPMA is often considered a suitable compromise for the ATRP of functional monomers such as GMA, where preservation of the epoxide group and good control over molecular weight are both essential.

PMDETA also enables efficient ATRP of GMA, although its more flexible coordination environment can lead to partial loss of control at higher conversions. This behavior is associated with a progressive shift in the ATRP equilibrium, where accumulation of Cu(II) species and increased medium viscosity reduce deactivation efficiency. Consequently, the concentration of active radicals becomes less uniformly regulated, favoring termination reactions, chain–chain coupling, and uneven chain growth. In addition, diffusion limitations at high conversion further reduce the relative rate of deactivation compared to propagation, leading to broader molecular weight distributions and the formation of branched or slightly crosslinked structures.

More rigid multidentate ligands such as HMTETA provide stronger and more defined coordination to the copper center, leading to a more stable Cu(I)/Cu(II) redox equilibrium. This enhanced stabilization improves the efficiency of the reversible deactivation step, maintaining a low and well-regulated radical concentration, which translates into narrower molecular weight distributions and better preservation of the epoxide functionality. In contrast, bipyridine-based ligands create a less active catalytic system due to their lower electron-donating ability and more rigid bidentate coordination. This results in slower activation kinetics and reduced radical flux, thereby minimizing side reactions and improving functional group tolerance. However, this increased control is achieved at the expense of slower polymerization rates.

Beyond ligand design, the reaction medium represents another key parameter governing the ATRP equilibrium and overall polymerization control. The choice of solvent plays a crucial role in the ATRP of GMA, as it directly affects monomer solubility, catalyst stability, and the position of the activation–deactivation equilibrium. Solvent polarity strongly influences the ATRP equilibrium constant (k_{ATRP}): increasing polarity stabilizes ionic species and raises k_{ATRP}, leading to higher steady-state radical concentrations and faster polymerization rates, but also increasing the likelihood of termination reactions and side processes [201,257,258]. In addition, the solvent must ensure sufficient solubility of both the GMA monomer and the Cu(I)/Cu(II)–ligand complex to maintain a homogeneous reaction medium and efficient deactivation. This aspect is particularly critical at high conversion, where viscosity increases and diffusion limitations can further compromise control.

In the case of GMA, solvent effects are even more significant due to the presence of the epoxide functionality, which is sensitive to nucleophilic attack. Polar protic solvents or trace impurities can promote undesired epoxide RO reactions, especially in the presence of copper complexes that may act as Lewis acids. Therefore, solvents must be carefully selected to balance catalytic activity and functional group preservation.

Solvent polarity and ligand structure act synergistically to regulate ATRP kinetics. This interplay has been experimentally demonstrated by Porello et al. [259] who investigated the ATRP of GMA/styrene under various reaction conditions. Using a CuBr/PMDETA catalytic system, they showed that controlled polymerization can be achieved both in bulk and in toluene at 60 °C, with low dispersities ($Đ < 1.3$) and a linear evolution of molecular weight with conversion. The use of bulk conditions in this copolymerization system is particularly relevant due to the presence of styrene, a relatively stable and non-functional monomer that tolerates high radical concentrations and enables mechanistic studies with minimal solvent interference. In contrast, GMA homopolymerization typically requires solution conditions to better preserve the epoxide functionality and improve control over the catalytic equilibrium. Their results further highlight that optimal ATRP control requires

simultaneous tuning of both ligand structure and reaction medium, as the use of toluene improves reaction homogeneity and catalyst performance compared to bulk conditions in more sensitive systems.

Highly active ligands such as Me$_6$TREN, which exhibit large activation rate constants (k_{act}), typically require moderately polar solvents (e.g., acetonitrile or DMSO) to ensure catalyst solubility; however, excessively polar environments may lead to overly high radical concentrations and reduced control [260]. In contrast, less active ligands such as PMDETA benefit from the use of mixed solvent systems, where a polar solvent (e.g., DMF) ensures solubility of the catalyst, while a less polar solvent (e.g., toluene) decreases overall polarity, lowers k_{ATRP}, and improves control over radical concentration [261]. More balanced ligands such as TPMA perform efficiently in polar or aqueous media, providing a good compromise between activity and control, whereas weaker ligands such as bpy require highly polar solvents to maintain sufficient catalytic activity, albeit at the expense of slower kinetics [255]. However, these trends are not absolute and strongly depend on the monomer structure and reaction medium. In several GMA-based systems, classical Cu/amine ligands such as PMDETA or bpy have also been successfully employed in less polar aromatic solvents such as anisole, which can provide an adequate balance between catalyst solubility and controlled radical generation [23]. This highlights that solvent selection in ATRP is not governed solely by polarity considerations, but rather by a complex interplay between ligand structure, monomer reactivity, and medium homogeneity.

An effective strategy to optimize these competing effects is the use of mixed solvents. For example, DMF/toluene mixtures have been successfully employed in supplemental activator and reducing agent-ATRP of GMA to achieve well-controlled polymerization at room temperature [240]. In such systems, DMF promotes dissolution of the copper complex and polar intermediates, while toluene moderates solvent polarity, thereby reducing radical concentration and limiting side reactions. Overall, careful tuning of solvent polarity in combination with ligand selection provides a powerful approach to control ATRP kinetics, regulate radical flux, and preserve the epoxide functionality during ATRP of GMA.

More recently, it has been demonstrated that solvent effects in ATRP extend beyond simple polarity considerations and also involve specific coordination interactions with the catalytic center. Racioppi et al. [262] showed that solvents such as acetonitrile and DMSO can coordinate directly to Cu(I)/Cu(II) complexes bearing ligands such as Me$_6$TREN, TPMA, and PMDETA, thereby significantly modifying the redox thermodynamics of the catalytic system. In particular, explicit inclusion of solvent molecules within the coordination sphere of the copper center is required to accurately reproduce experimental ATRP trends, whereas implicit solvation models fail to capture these effects. These findings indicate that solvent molecules can act as secondary ligands, dynamically interacting with the Cu–ligand complex and modulating the structure, stability, and reactivity of the catalytic species. As a result, ATRP kinetics is not only governed by bulk solvent polarity but also by specific coordination interactions, which operate synergistically with the primary ligand to fine-tune the activation–deactivation equilibrium.

Based on the above discussion, it becomes evident that polymerization control in ATRP of GMA is governed by a highly interconnected set of parameters rather than isolated factors. In this context, catalytic performance cannot be attributed solely to ligand activity or solvent polarity, but instead arises from a delicate interplay between the Cu(I)/Cu(II) redox equilibrium, radical concentration, and medium-specific interactions. These observations highlight the absence of universal design rules for ligand–solvent selection in ATRP of functional monomers such as GMA. Consequently, a deeper mechanistic understanding that integrates coordination chemistry, reaction kinetics, and medium effects is required to move beyond empirical optimization. Such an integrated approach is essential to achieve reliable control over polymer architecture while preserving the integrity of sensitive functional groups such as epoxide.

Despite these advances, however, several studies have demonstrated that undesired side reactions may still occur in GMA-based systems, even under well-optimized ATRP conditions. For example, Gomes et al. [23] demonstrated that pre-functionalization of GMA with oleic acid prior to polymerization leads to problematic reaction pathways, likely due to uncontrolled side reactions and branching, whereas significantly improved control and solubility are achieved when the epoxide groups are modified after polymerization. Importantly, these authors further showed that the occurrence of branching and crosslinking is not inherently related to the ATRP mechanism itself, as similar side reactions were also observed under RAFT conditions. This clearly indicates that these undesired processes are not associated with the presence of metal catalysts, but rather originate from the intrinsic reactivity of the glycidyl epoxide functionality and the chosen reaction pathway. As a result, even well-controlled ATRP conditions may lead to partial network formation, particularly at high monomer conversion or under reactive environments.

To mitigate undesired epoxide reactivity during polymerization, several alternative strategies have been developed. One approach consists in modifying or protecting the GMA monomer prior to polymerization in order to suppress epoxide reactivity during chain growth. Although this strategy can effectively reduce undesired side reactions, it generally requires additional synthetic steps such as epoxide protection/deprotection or monomer pre-functionalization, which increases synthetic complexity, may reduce overall efficiency, and can alter the intrinsic reactivity of the monomer, as highlighted by Gomes et al. [23] Consequently, this approach is less frequently adopted compared to post-polymerization functionalization strategies. A more robust and widely used strategy involves performing ATRP of unmodified GMA followed by post-polymerization functionalization of the epoxide groups under controlled conditions, which preserves polymer architecture while minimizing interchain coupling [259]. In addition, reducing GMA content in copolymer systems, limiting monomer conversion, and carefully controlling solvent polarity and impurity levels have been shown to significantly decrease the likelihood of branching and crosslinking. Overall, these approaches are essential to maintain well-defined block copolymer architectures and solubility in GMA-based ATRP systems.

To further mitigate these limitations while preserving the advantages of ATRP, two modern strategies have been developed as alternatives to conventional ATRP to enhance catalyst efficiency and reduce environmental impact: Activators Regenerated by Electron Transfer (ARGET-ATRP) and Initiators for Continuous Activator Regeneration (ICAR-ATRP). Both methods allow controlled radical polymerization with significantly reduced amounts of transition metal catalysts while maintaining precise control over molecular weight, dispersity and polymer architecture. These techniques are especially useful for the polymerization of functional monomers like GMA, where maintaining the integrity of sensitive functional groups, such as the epoxide ring, is essential.

Among these approaches, ARGET-ATRP has attracted particular attention due to its simplicity and efficiency in minimizing catalyst concentration while preserving a high level of control over the polymerization process. The key feature of this technique lies in the continuous regeneration of the active catalytic species during the reaction. In ARGET-ATRP, the concentration of the active Cu(I) species is continuously regenerated from Cu(II) through the use of a mild reducing agent, such as ascorbic acid, tin(II) 2-ethylhexanoate, or glucose, allowing the polymerization to proceed with extremely low catalyst loadings typically in the range of 10–100 ppm of copper. This modification significantly shifts the ATRP equilibrium toward the dormant state, resulting in a substantially reduced steady-state radical concentration compared to conventional ATRP [24,263]. From a mechanistic standpoint, ARGET-ATRP maintains the same reversible activation–deactivation equilibrium as classical ATRP, but the continuous in situ regeneration of Cu(I) ensures a more controlled and gradual radical

flux [264,265]. As a consequence, termination events, intermolecular coupling, and branching reactions are significantly suppressed. This aspect is particularly important for functional monomers such as GMA, where excessive radical concentrations can exacerbate side reactions involving the epoxide group. In addition, the lower concentration of copper species reduces the probability of Lewis acid–mediated activation of the epoxide ring, thereby indirectly limiting undesired RO reactions.

The advantages of ARGET-ATRP over conventional FRP have been clearly demonstrated in the synthesis of macroporous polymeric microspheres. Dai et al. [266] reported the preparation of GMA-based microspheres via ARGET-ATRP and compared them with those obtained by conventional suspension polymerization. The ARGET-ATRP route led to significantly improved structural features, including smaller and more uniform particle clusters (10–300 nm vs. 400–800 nm), larger average pore diameters (~275 nm vs. 234 nm), and higher specific surface areas (59.3 m^2/g vs. 37.5 m^2/g). These enhancements are attributed to the controlled radical nature of ARGET-ATRP, which suppresses premature termination and enables a more homogeneous formation of the polymer network, ultimately resulting in improved protein-binding capacity and enhanced performance in separation-related applications.

Beyond such porous microsphere systems, more advanced ARGET-ATRP strategies have also been developed in the context of polymerization-induced self-assembly (PISA). In these systems, controlled polymerization of GMA or GMA-containing mixtures in solution leads simultaneously to chain growth and in situ formation of well-defined nano-objects. For example, Wang et al. [267] reported the in situ crosslinking of nanoparticles during PISA via ARGET-ATRP of GMA in ethanol, using a $CuBr_2$/TPMA catalytic system and tin(II) 2-ethylhexanoate as reducing agent. In their system, poly(oligo(ethylene oxide) methyl ether methacrylate) was employed as a macroinitiator and stabilizer, enabling the formation of spherical nanoparticles with good colloidal stability. A key finding of this work is the strong decoupling between vinyl polymerization and epoxide RO: while the methacrylate double bond of GMA was fully consumed within ≈4 h, only ~33% of the epoxide groups reacted even after prolonged reaction times (~117 h). This kinetic mismatch allows the initial formation of well-defined nano-objects via PISA, followed by a gradual intra-particle crosslinking driven by epoxide RO, thereby stabilizing the particles without disrupting the self-assembly process.

Similarly, Liang et al. [268] reported the preparation of polyolefin-based nano-objects via an ARGET-ATRP mediated PISA process using hydrogenated polyisoprene-b-poly(GMA) (HPI-b-poly(GMA)/poly (GMA) blends. In this system, the incompatibility between non-polar polyolefin segments and polar poly(GMA) blocks is exploited to drive self-assembly during polymerization, leading to spherical nano-objects with tunable hydrodynamic diameters. The structure and size of the particles were precisely controlled by adjusting the wtpoly(GMA)/wtHPI ratio and the initiator-to-macroinitiator ratio. In addition, in situ RO of the epoxide groups allowed the incorporation of phosphoric acid into the poly(GMA) core, resulting in enhanced flame-retardant properties. This work further demonstrates the versatility of ARGET-ATRP PISA in simultaneously controlling morphology, composition, and post-functionalization of GMA-based nano-architectures.

Several studies have also explored grafting approaches using ARGET-ATRP despite the intrinsic complexity of these heterogeneous systems. As demonstrated by Jonsson et al. [264], surface-initiated ARGET-ATRP of GMA on thermally expandable microspheres can be successfully performed using very low catalyst loadings (~50 ppm Cu) in toluene at 30 °C, enabling controlled grafting while preserving both the epoxide functionality and the thermal expansion properties of the substrate. Under optimized conditions, this approach allows efficient surface functionalization without significantly compromising the integrity of the microspheres. However, when compared to solution polymerization of GMA under similar ARGET-ATRP conditions, which typically affords good control with dispersity values around Đ ≈ 1.4 and Mn ≈ 20,500 g/mol (using PMDETA as ligand), a clear decrease in polymerization control is generally observed in the presence of solid substrates. In the heterogeneous system, the dispersity increases to approximately Đ ≈ 1.9, even though comparable molar masses are reached (Mn ≈ 25,300 g/mol). This loss of control is mainly attributed to diffusion constraints, heterogeneous radical distribution, and local variations in catalyst accessibility at the surface of the microspheres.

Despite the reduction in control often observed when transitioning from homogeneous solution polymerization to heterogeneous systems, ARGET-ATRP has proven to be a robust and versatile technique for grafting-from reactions on a wide range of solid substrates clearly highlighting its advantages over conventional ATRP in terms of significantly reduced catalyst loadings, improved environmental compatibility, and better tolerance toward complex reaction media.

This robustness is further confirmed in more complex heterogeneous environments. In mesoporous confined systems such as SBA-15 [269], where surface-initiated ARGET-ATRP using a $CuCl_2$/PMDETA catalytic system leads to progressive and controllable growth of poly(GMA) chains within cylindrical pores. In this system, the use of PMDETA as ligand ensures a balanced Cu(I)/Cu(II) equilibrium, allowing controlled radical deactivation and relatively narrow dispersities at early stages (Đ ≈ 1.36–1.64). Size exclusion chromatography (SEC) analysis shows a gradual increase in molecular weight from 13,100 to 32,200 g/mol as polymerization time increases (6 h–24 h), confirming controlled chain growth. However, as pore filling progresses, diffusion of deactivating species becomes restricted, leading to a moderate increase in dispersity and reduced control at higher conversions. Despite this limitation, high grafting densities (up to 6.27 chains/nm^2) and successful block copolymer formation poly(GMA)-b-poly(MMA), Mn ≈ 74,500 g/mol) confirm preserved chain-end functionality and the efficiency of the ARGET-ATRP mechanism under confined conditions.

A similar behavior is observed in cellulose-based systems [270], where ARGET-ATRP grafting on fibrous supports further confirms the applicability of this methodology under complex heterogeneous conditions. A systematic comparison between different monomers clearly highlights the particular behavior of GMA. For non-functional monomers such as methyl methacrylate and styrene, well-controlled polymerizations are generally achieved, characterized by linear first-order kinetics, efficient regulation of the activation–deactivation equilibrium, and relatively narrow dispersities (Đ ≈ 1.09–1.35 for poly(methyl methacrylate) and Đ ≈ 1.14–1.60 for poly(styrene)). Moreover, the experimentally obtained molar masses closely follow theoretical values, confirming an effective balance between radical propagation and reversible deactivation under ARGET conditions. In contrast, GMA consistently exhibits lower control under similar experimental conditions, with broader molecular weight distributions (Đ up to ~1.7) and deviations from theoretical Mn even at moderate conversions. This behavior is mainly attributed to the high reactivity of the epoxide group, which can undergo side reactions such as RO and crosslinking, especially in the presence of copper species and protic impurities. These undesired processes are further enhanced in heterogeneous systems, where diffusion limitations and local variations in catalyst distribution reduce the efficiency of the control mechanism. Despite these limitations, cellulose-based ARGET-ATRP systems demonstrate that controlled grafting of poly(GMA) can still be achieved under carefully optimized conditions, typically involving low temperatures (~0 °C), reduced reducing agent concentrations, and very low copper loadings (<100 ppm). However, the system remains highly sensitive, with a narrow operational window and the onset of gelation observed at conversions above ~40%, confirming the intrinsic reactivity and structural sensitivity of GMA compared to styrene and methyl methacrylate. Overall, these findings highlight that although ARGET-ATRP is highly versatile across heterogeneous supports, GMA-based systems require stricter control of reaction parameters and inherently exhibit lower reproducibility.

In parallel with ARGET-ATRP, another strategy has been developed to further reduce catalyst loading while maintaining efficient control over the polymerization process, namely Initiators for Continuous Activator Regeneration (ICAR-ATRP). Although both approaches rely on the continuous regeneration of the active Cu(I) species, ICAR-ATRP differs fundamentally in the way this regeneration is achieved.

In ICAR-ATRP, a conventional radical initiator, typically a thermal initiator such as AIBN, is introduced into the reaction system to continuously generate a small flux of free radicals. These radicals act as reducing agents, converting the deactivator Cu(II) species into the active Cu(I) form in situ, thereby sustaining the activation–deactivation equilibrium throughout the polymerization. As a result, the concentration of copper catalyst can be significantly reduced, often to ppm levels, while still maintaining controlled radical polymerization behavior [271–273].

From a mechanistic perspective, ICAR-ATRP preserves the same fundamental equilibrium as classical ATRP; however, the regeneration of Cu(I) is indirectly driven by the decomposition of the added radical initiator rather than by a chemical reducing agent, as in ARGET-ATRP. This leads to a continuous but low radical flux, which contributes to limiting termination reactions and maintaining a relatively controlled growth of polymer chains [274]. Nevertheless, the presence of additional radicals originating from the initiator may slightly increase the probability of termination events compared to ARGET-ATRP, especially if the initiator concentration is not carefully optimized.

Several studies have demonstrated the successful application of ICAR-ATRP to GMA-based systems in homogeneous solution. For example, the synthesis of block copolymers incorporating GMA has been reported using a $CuBr_2$/ligand catalytic system in combination with AIBN as the radical source [273]. In this work, diblock copolymers of the type poly(oligo(ethylene oxide) methacrylate)-b-poly(oligo(ethylene oxide) methacrylate-*co*-GMA) were successfully prepared using a one-pot strategy, where both polymerization and subsequent chemical modification were carried out sequentially without intermediate purification. The polymerization proceeded with nearly complete monomer conversion (~100%), indicating high efficiency of the ICAR-ATRP process under ppm copper catalyst conditions (~100 ppm).

Furthermore, the epoxide groups of GMA were quantitatively transformed after polymerization through reaction with 3-mercaptopropionic acid, leading to nearly complete RO (~100%) and the introduction of carboxylic acid functionalities along the polymer chains. The resulting functional block copolymers were successfully employed as dispersants for pigment particles, highlighting the key role of the absolute number of functional groups per chain in determining dispersibility. This work demonstrates that ICAR-ATRP not only enables the synthesis of well-defined GMA-containing copolymers, but also allows efficient and quantitative post-polymerization functionalization within a single process.

ICAR-ATRP has also been extended to PISA involving GMA, providing access to functional nano-objects through a single-step process [275]. In a representative system, ethanol was used as the reaction medium, while poly(oligo(ethylene oxide) methyl ether methacrylate) served simultaneously as macroinitiator and stabilizing block. Copolymerization of 2-(perfluorohexyl)ethyl methacrylate (PFHEMA) with GMA enabled the in situ formation of amphiphilic block copolymers that self-assembled during chain growth. The resulting morphologies were strongly dependent on the monomer composition. At relatively low GMA content, the system predominantly formed spherical nanoparticles with comparatively large diameters. Increasing the proportion of PFHEMA favored the formation of smaller spheres or short worm-like structures due to enhanced amphiphilic contrast between the blocks. However, when the GMA content was further increased, the system lost colloidal stability and led to precipitation instead of well-defined nano-objects. This transition was attributed to side reactions involving the epoxide groups, which promote intermolecular crosslinking and disrupt the self-assembly process, particularly in ethanol, where the solvent environment enhances the accessibility and reactivity of the epoxide functions, thereby facilitating undesired coupling between growing chains. In addition to its influence on morphology, the preserved epoxide functionality of GMA allowed efficient post-polymerization modification. Reaction with thiol-containing compounds enabled the introduction of carboxylic acid groups within the nanostructures, which were subsequently used for coordination with iron ions, leading to the formation of organic–inorganic hybrid nanoparticles based on Fe_3O_4. Overall, this study illustrates both the versatility and the limitations of ICAR-ATRP in PISA systems involving GMA. While the method enables the formation of functional nanostructures and subsequent chemical modification, the high reactivity of the epoxide groups imposes constraints on monomer composition, as excessive GMA content can trigger crosslinking and compromise the self-assembly process.

In addition to homogeneous polymerization and self-assembly systems, ICAR-ATRP has also been extensively applied in heterogeneous environments for the surface functionalization of solid substrates, where precise control over interfacial architecture is crucial for tuning final material properties. In this context, surface-initiated ICAR-ATRP provides an efficient strategy to grow polymer chains directly from a wide variety of surfaces, including both organic and inorganic supports, while maintaining low catalyst loading and controlled chain growth.

In this framework, ICAR-ATRP has been successfully used to modify polyester substrates through the formation of poly(GMA) surface layers, which were subsequently functionalized with polyhedral oligomeric silsesquioxane [276]. This approach led to the construction of well-defined organic–inorganic hybrid coatings, resulting in significantly enhanced flame retardancy and smoke suppression performance. The formation of controlled poly(GMA) interlayers with relatively low dispersity ($Đ \approx 1.22$) highlights the ability of ICAR-ATRP to preserve polymerization control while maintaining the reactivity of epoxide groups for further chemical modification.

Similarly, surface-initiated ICAR-ATRP has been successfully applied to inorganic substrates. In one representative study, glass fibres were functionalized via a mussel-inspired polydopamine coating, which served as a versatile platform for immobilizing ATRP initiators [277]. Subsequent ICAR-ATRP of GMA enabled the growth of poly(GMA) chains directly from the fiber surface, with a progressive increase in graft thickness as the reaction proceeded, indicating controlled chain propagation. The presence of epoxide groups along the grafted layer allowed further chemical modification, leading to improved interfacial adhesion between the fibres and polymer matrices in composite materials. The resulting modified fibres exhibited well-defined interfacial polymer architectures, with a relatively low polydispersity index ($Đ \approx 1.2$), confirming the efficiency and robustness of the ICAR mechanism under strongly heterogeneous conditions.

Overall, both ARGET-ATRP and ICAR-ATRP represent significant advancements over conventional ATRP, particularly in terms of catalyst efficiency, reduced metal content, and environmental compatibility, while maintaining the fundamental features of controlled radical polymerization. These two strategies rely on different pathways for the continuous regeneration of the active Cu(I) species—chemical reduction in ARGET-ATRP and radical-mediated reduction via a thermal initiator in ICAR-ATRP—yet both enable efficient control over molecular architecture under low copper loading conditions.

Across the reported studies, both methods have demonstrated strong applicability for the polymerization of functional monomers such as GMA, as well as for their use in heterogeneous environments including surface-initiated grafting, confined systems, and self-assembly processes. In all cases, controlled growth, preservation of chain-end functionality, and the possibility of post-polymerization modification were key advantages, enabling the design of advanced functional materials and hybrid architectures. However, the performance of both techniques remains highly dependent on reaction conditions and system complexity. For GMA-based systems in particular, the high reactivity of the epoxide group introduces a persistent challenge, as side reactions such as RO and crosslinking can compromise control, especially at

higher conversions or under heterogeneous conditions where diffusion limitations become significant. In this context, ARGET-ATRP generally offers slightly better control due to the absence of an additional radical flux, whereas ICAR-ATRP provides operational simplicity and scalability but may require stricter optimization to minimize termination events.

Overall, ARGET-ATRP and ICAR-ATRP should be viewed as complementary strategies, each offering distinct advantages depending on the targeted application. Their combined development has significantly expanded the scope of controlled radical polymerization for GMA-based systems, enabling precise macromolecular design across solution, interface, and nanostructured materials.

3.3.2.1. Externally stimulated ATRP techniques. The class of ATRP techniques activated by external energy sources has emerged as a powerful alternative to conventional thermally driven ATRP, offering enhanced control over polymerization conditions and improved sustainability. In traditional ATRP systems, continuous regeneration of the active Cu(I) species typically relies on chemical reducing agents or elevated temperatures, which may lead to increased side reactions, catalyst accumulation, and limited compatibility with sensitive functional monomers. In contrast, externally stimulated ATRP methods-such as ultra-sonication, electrochemical activation, microwave irradiation, and plasma initiation-enable in situ and on-demand regeneration of the activator species under significantly milder and more environmentally benign conditions [278].

These approaches provide several key advantages, including reduced catalyst loading (often down to ppm levels), precise temporal control through external stimulus switching, and compatibility with aqueous or heterogeneous systems [279,280]. Such features are particularly beneficial for the polymerization of GMA, where the preservation of the highly reactive epoxy group is crucial. Indeed, harsh thermal conditions or uncontrolled radical fluxes can promote undesired side reactions such as RO, crosslinking, or branching, thereby compromising both the functionality and the structural integrity of the resulting polymers, as described in previous sections. The use of external energy sources allows fine regulation of radical generation and redox equilibria, thereby minimizing these side reactions while maintaining the controlled/living character of ATRP.

Among these externally stimulated ATRP strategies, photocatalyzed ATRP represents one of the most advanced and versatile approaches due to its ability to couple temporal control with efficient redox activation. In this context, visible-light irradiation enables the continuous regeneration of the active Cu(I) species from Cu(II) through a photocatalytic cycle, thereby eliminating the need for external chemical reducing agents and significantly lowering catalyst loading.

A representative example has been reported by Qizhi Yang and coworkers [281], who investigated the photocatalyzed ATRP copolymerization of methyl methacrylate and GMA using a bis(1,10-phenanthroline)copper(I)-based catalyst under blue LED irradiation. A distinctive feature of this system lies in the synergistic role of the epoxide functionality of GMA. Beyond its function as a reactive comonomer, the epoxy group acts as an internal reducing agent, promoting the in situ regeneration of the Cu(I) activator from Cu(II) and thereby accelerating the polymerization rate. This dual role leads to a noticeable increase in polymerization kinetics with increasing GMA content, while maintaining a well-controlled "living" character, as evidenced by narrow dispersity values (typically $Đ \approx 1.1–1.3$) and predictable molecular weight evolution. Importantly, despite its participation in the redox cycle, the epoxide functionality remains largely preserved due to its high excess relative to the catalyst concentration, ensuring that the resulting poly(GMA-*stat*-methyl methacrylate) retains reactive epoxy groups for subsequent post-polymerization modifications. Furthermore, the use of light as an external stimulus provides an additional level of control through "ON/OFF" switching experiments, enabling precise temporal regulation of radical generation and polymer growth. This feature is particularly advantageous for GMA-containing systems, as it minimizes uncontrolled propagation and suppresses side reactions such as epoxide ring opening or crosslinking, thereby preserving both the structural integrity and functional potential of the polymer chains.

More generally, other photo-ATRP systems reported in the literature further confirm the broad compatibility of functional methacrylates, including MMA [282–284] and poly(ethylene glycol) methyl ether methacrylate [284]. These studies collectively demonstrate that a wide range of functional methacrylate monomers can be efficiently incorporated under mild visible-light-mediated conditions without loss of their reactive functionalities, reinforcing the robustness and versatility of photocatalyzed ATRP as a platform for the synthesis of functional polymers. A representative example of visible-light-mediated Cu-based ATRP has been reported by Yang, Qizhi, et al. [283] where well-controlled polymerization of methacrylate monomers was achieved under blue LED irradiation at 0.9 W. In addition, metal-free photoinduced electron transfer ATRP enables controlled polymerization under visible light using an organic photocatalyst (e.g., fluorescein), eliminating the need for transition metals while maintaining "living" characteristics [282]. Notably, $Cu(dap)_2Cl$-based photo-ATRP (dap = 2,9-bis(p-anisyl)-1,10-phenanthroline) systems also provide well-controlled polymerizations of methacrylates, affording polymers with low dispersity (Đ as low as ~1.15). This further confirms the high efficiency and "living" character of visible-light-mediated ATRP, as well as its robustness for the synthesis of well-defined functional polymers.

In addition to light-mediated systems, ultrasound has also emerged as an efficient external stimulus for ATRP, enabling polymerization control through acoustic cavitation-induced redox activation of the catalytic species. For illustrative purposes, Wang et al. [285] demonstrated the use of ultrasonication to trigger sono-ATRP for the controlled polymerization of oligo(ethylene oxide) methyl ether methacrylate and 2-hydroxyethyl acrylate. In this system, acoustic cavitation generates hydroxyl radicals that continuously regenerate the active Cu(I) species from Cu(II), ensuring a dynamic redox equilibrium. This approach enables excellent control over molecular weight and dispersity while preserving chain-end fidelity. Ultrasound-assisted-ATRP further offers several advantages, including mild operating conditions (room temperature), the use of ppm-level copper concentrations, and precise temporal control via ultrasound on/off switching. Owing to its compatibility with aqueous media and hydrophilic monomers, this methodology is particularly suitable for the synthesis of block copolymers, nanostructured materials, and functional biohybrids.

Beyond to ultrasound, a number of other external energy-mediated ATRP techniques have been documented, including electrochemically mediated ATRP, microwave-assisted ATRP and plasma-induced ATRP. These approaches have been successfully applied to various methacrylates and acrylates, demonstrating excellent control over molecular weight and dispersity. For instance, electrochemically mediated ATRP has been used to polymerize hydrophilic monomers such as oligo(ethylene oxide) methyl ether methacrylate, offering precise control under mild aqueous conditions [286]. Similar strategies were also extended to more hydrophobic systems, including the polymerization of n-butyl acrylate in mini-emulsion using a dual catalytic system [287], as well as the efficient electrochemically mediated ATRP of n-butyl acrylate on non-platinum cathodes [288]. Additionally, microwave-assisted ATRP has been shown to accelerate the polymerization of styrene [289] and methyl methacrylate [290] while reducing reaction times. In another example, plasma-initiated ATRP was successfully employed for the surface grafting of poly(MMA) brushes onto solid substrates, confirming the feasibility of plasma as a clean, catalyst-regenerating energy source [291].

Despite these advances, externally energy-stimulated ATRP strategies remain comparatively less explored for GMA compared to chemically or thermally regulated systems, such as ARGET-ATRP and ICAR-ATRP described in the previous section. Nevertheless, several functional methacrylate monomers have already been successfully

polymerized using these externally driven ATRP techniques, demonstrating the general feasibility of this approach. These results highlight that GMA, owing to its well-established compatibility with controlled radical polymerization, is a promising candidate for further development within externally energy-mediated ATRP platforms.

3.3.3. Nitroxide mediated polymerization (NMP)

NMP is a RDRP technique based on the reversible homolytic cleavage of an alkoxyamine bond, generating a transient propagating radical and a persistent nitroxide radical that regulates chain growth through a dynamic activation–deactivation equilibrium. In this approach, the initiator is an alkoxyamine, which acts both as a thermal initiator and a control agent through reversible homolysis of the C–ON bond. While early systems relied on (2,2,6,6-tetramethylpiperidin-1-oxyl)-based alkoxyamines (TEMPO), their limited efficiency toward methacrylate monomers led to the development of more advanced nitroxides. A major breakthrough was achieved by Benoit et al. [292], who introduced "universal" alkoxyamines based on second-generation nitroxides; SG1-type nitroxides (2,2,5-trimethyl-4-phenyl-3-azahexane-3-oxy), significantly expanding the range of monomers amenable to controlled polymerization and enabling low dispersity values (Đ ≈ 1.05–1.15).

When applied to GMA, NMP proceeds via the methacrylate double bond while ideally preserving the epoxide functionality, making it particularly attractive for the synthesis of reactive polymers. The use of SG1-based alkoxyamine initiators, such as BlocBuilder derivatives, has enabled improved control over GMA-containing systems. For instance, Moayeri, Ali et al. [293] reported the controlled copolymerization of GMA and styrene using an SG1-based alkoxyamine bearing a succinimidyl ester group (NHS-BlocBuilder), achieving a linear increase of number-average molecular weight with conversion up to ~50% and relatively low dispersity (Đ ≈ 1.22–1.44), even at high GMA content (up to 0.92). Moreover, successful chain extension experiments demonstrated the preservation of chain-end fidelity and a substantial "living" character of the resulting copolymers. In addition, earlier work by Grubbs et al. [294] illustrated the importance of GMA incorporation in block copolymers, such as poly(methyl acrylate-*co*-GMA)-b-polyisoprene, which can form well-organized nanostructures and chemically bond to epoxy thermosetting matrices, highlighting the relevance of controlled polymerization strategies for functional material design.

Despite these advances, the polymerization of GMA by NMP remains challenging. Methacrylate monomers generally exhibit slower kinetics and reduced control compared to styrenic systems, often requiring relatively high temperatures to promote efficient alkoxyamine dissociation [295]. These conditions can compromise the stability of the epoxide ring, leading to side reactions such as RO in the presence of trace moisture or acidic impurities, potentially resulting in branching or crosslinking. Furthermore, conventional TEMPO-based initiators are poorly suited for GMA, reinforcing the need for more efficient nitroxide systems such as SG1 derivatives. However, even these SG1-based systems have been shown to be highly sensitive to reaction conditions; for instance, controlled copolymerization in aqueous acidic media is only maintained under strictly optimized conditions (low temperature and short reaction time), highlighting the narrow operational window of NMP systems [296,297]. This limitation is further exacerbated by the intrinsic sensitivity of GMA to basic media, where the epoxide group can undergo nucleophilic RO and hydrolysis, as explained in previous sections.

In order to better rationalize the reactivity of GMA in this polymerization approach, the mechanism of NMP is first described in detail, as it is crucial for understanding the limitations and challenges associated with this monomer.

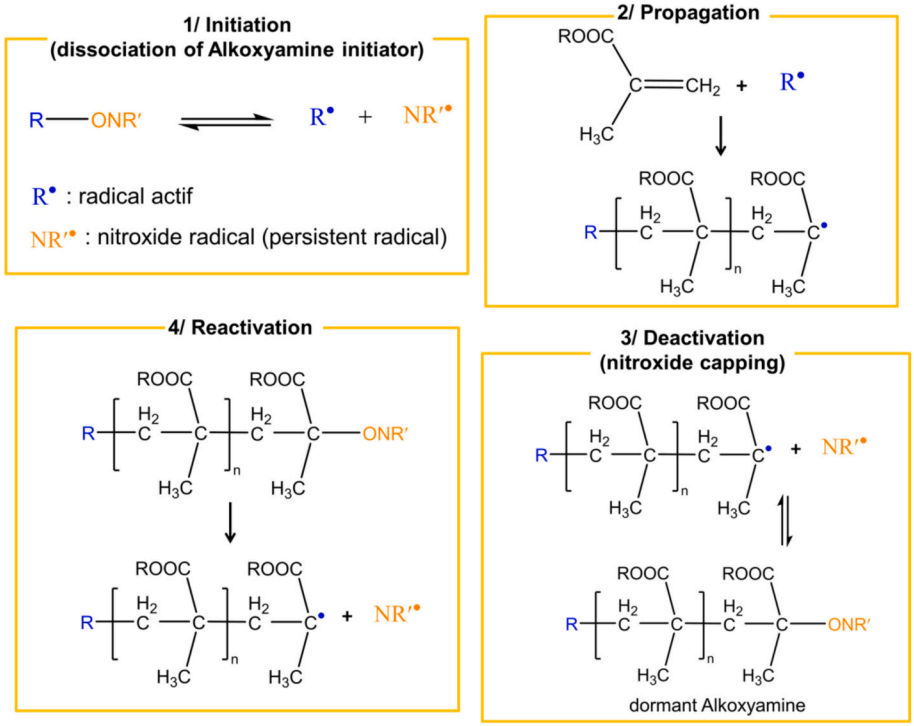
This process relies on a dynamic and reversible equilibrium between an active propagating radical and a dormant alkoxyamine species, which ensures controlled chain growth. Upon thermal activation, the alkoxyamine initiator undergoes homolytic cleavage of the C–ON bond to generate a carbon-centered radical and a persistent nitroxide radical as shown in Scheme 14 [298]. The former adds to the vinyl monomer (e.g., methacrylates) and propagates the chain, while the nitroxide rapidly recombines with the growing radical to reform the dormant alkoxyamine. This reversible activation–deactivation process maintains a low concentration of active radicals, thereby limiting termination reactions and enabling a certain degree of control over molecular weight and dispersity [299]. However, in the case of methacrylate-based monomers such as GMA, this ideal behavior is often compromised by the occurrence of side reactions, most notably backbiting followed by β-scission [22].

Backbiting corresponds to an intramolecular chain transfer reaction in which the propagating radical abstracts a hydrogen atom from its own polymer backbone, typically via a six-five membered transition state. This process leads to the formation of a mid-chain radical (MCR) as shown in Scheme 15, in which the radical center is no longer located at the chain end but within the polymer backbone. These MCR species are significantly less controlled by the nitroxide equilibrium and can undergo further undesired transformations. In particular, β-scission of the MCR results in the cleavage of the polymer chain, producing a macromonomer bearing a terminal double bond and a new radical species (Scheme 15) [22]. This mechanism has been extensively identified as one of the primary pathways responsible for the loss of "living" character in NMP systems, especially for (meth)acrylates [300]. This sequence not only reduces the number of living chains but also generates dead chains and unsaturated species, thereby increasing dispersity and compromising the ability to perform chain extension or synthesize well-defined block copolymers. At high conversion up to about 50%, the accumulation of such macromonomers and dead chains has been experimentally correlated with deviations from linear molecular weight growth and loss of chain-end fidelity in NMP of methacrylates [22].

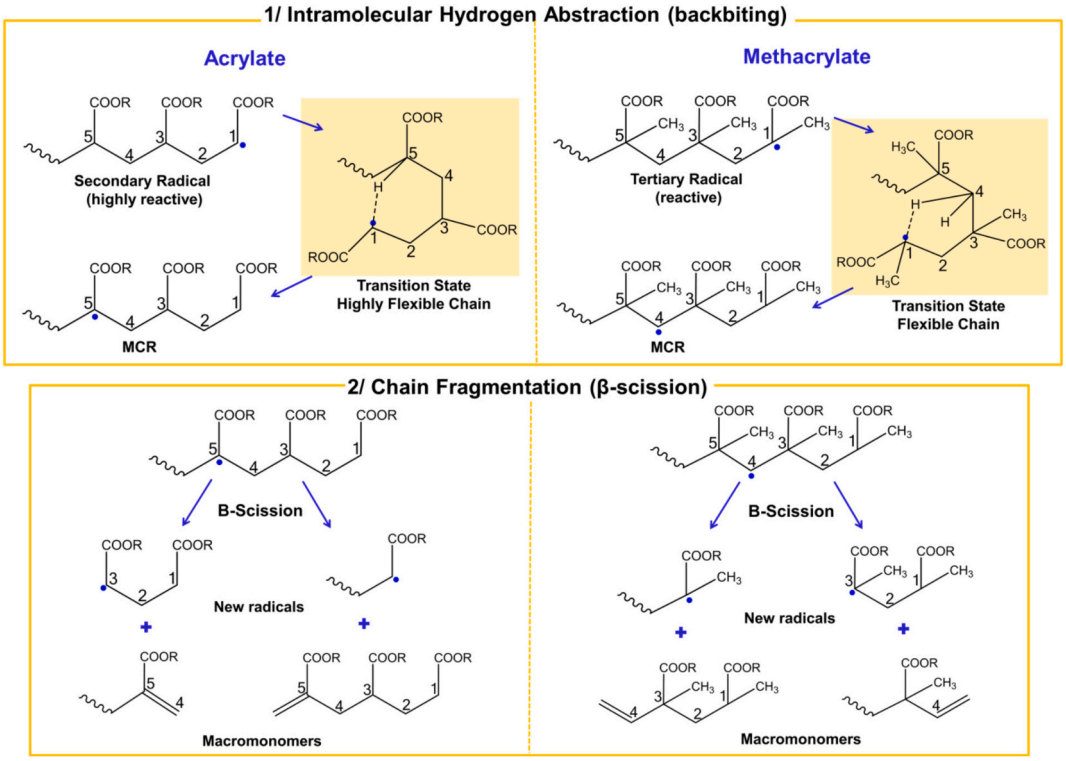
These side reactions are especially pronounced in acrylate and methacrylate systems due to the higher flexibility of the polymer backbone and the relatively high reactivity of the propagating radicals, which facilitate intramolecular hydrogen abstraction. Notably, acrylates are generally more prone to backbiting than methacrylates, as the secondary propagating radicals in acrylates more readily adopt the conformations required for six-membered transition states leading to MCR formation, as illustrated in Scheme 15. This higher susceptibility has been quantitatively supported by kinetic and computational studies showing lower activation barriers for backbiting in acrylates compared to methacrylates [300]. In contrast, the presence of the α-methyl group in methacrylates introduces steric hindrance, which reduces the rate of intramolecular hydrogen abstraction. In styrenic systems, the propagating radical is stabilized by resonance with the phenyl group and the polymer chain is more rigid, which significantly reduces the probability of backbiting and subsequent β-scission. As a result, NMP generally provides better control for styrenic monomers than for methacrylates and, even more so, than for acrylates. This explains why NMP is considered highly efficient for styrenics but intrinsically limited for methacrylate-based systems, particularly at elevated temperatures required for alkoxyamine hemolysis [301].

In the specific case of GMA, these intrinsic limitations are further exacerbated by the presence of the epoxide functionality, which is itself prone to side reactions such as RO under certain conditions. Consequently, the combined effects of backbiting, β-scission, and epoxide instability make the controlled polymerization of GMA by NMP particularly challenging, especially at high temperatures and high conversion. Experimental studies on GMA polymerization under NMP conditions have indeed reported a loss of control at moderate to high conversion, attributed to the combined contribution of backbiting reactions and epoxide-related side reactions [293]. In some cases, uncontrolled systems even lead to gelation due to intermolecular reactions involving epoxide groups, highlighting the sensitivity of GMA to reaction conditions [302].

The polymerization of GMA by NMP is strongly influenced by the nature of the nitroxide controlling agent. Conventional systems based on



Scheme 14. Schematic representation of the NMP mechanism for methacrylate monomers.



Scheme 15. Comparison of intramolecular hydrogen abstraction and β-scission mechanisms in acrylate and methacrylate radicals,

TEMPO typically exhibit poor control over methacrylate polymerization due to slow alkoxyamine homolysis and inefficient reversible deactivation, leading to a relatively high concentration of propagating radicals. As a result, side reactions such as intramolecular hydrogen abstraction (backbiting) via transition state and subsequent β-scission are significantly promoted. The high steady-state radical concentration in TEMPO-based systems has been directly correlated with increased rates of chain transfer and irreversible termination reactions in methacrylate NMP systems [301]. In contrast, alkoxyamines based on SG1 nitroxide provide improved control owing to a more favorable activation–deactivation equilibrium, which reduces the lifetime of propagating radicals and lowers their steady-state concentration

[298,303]. This reduction in radical lifetime limits, but does not eliminate, the occurrence of backbiting and β-scission reactions, as these processes remain kinetically accessible under typical NMP temperatures. For these reasons, it is important to note that these side reactions are not completely suppressed, as they are inherently related to the high reactivity of methacrylate radicals and the flexibility of the polymer backbone. Consequently, backbiting and β-scission may still occur, particularly at high temperatures and conversions, although to a significantly lesser extent than in TEMPO-based systems.

The influence of reaction conditions further amplifies these limitations. Temperature is a particularly critical parameter in NMP, as sufficiently high temperatures are required to ensure efficient alkoxyamine homolysis; however, increasing temperature simultaneously accelerates backbiting, β-scission, and epoxide RO reactions [300,301]. As a result, although higher temperatures improve polymerization rates, they significantly compromise structural control and favor the formation of branched or crosslinked structures [22]. Monomer concentration also plays a key role: bulk or highly concentrated conditions enhance intermolecular interactions between epoxide groups and increase the probability of radical–radical and radical–polymer reactions, thereby promoting gelation and network formation at high conversion [302]. In contrast, more diluted systems reduce these interactions and help limit crosslinking, albeit at the expense of slower kinetics. The nature of the solvent is equally important; polar solvents can facilitate nucleophilic or ionic RO of the epoxide group, further contributing to branching and structural defects [302], whereas less polar solvents better preserve epoxide integrity as explained previously. Therefore, careful selection of temperature, solvent, and monomer concentration is essential to mitigate side reactions, although these effects cannot be completely eliminated due to the intrinsic reactivity of methacrylate radicals under NMP conditions.

In summary, the NMP of GMA is inherently limited by the combined effects of backbiting, β-scission, and the high reactivity of the epoxide group. These side reactions, intensified by the elevated temperatures required for alkoxyamine homolysis, lead to loss of control, increased dispersity, and the formation of branched or crosslinked structures, particularly at high conversion. Although careful optimization of reaction conditions (temperature, solvent, and monomer concentration) and the use of more efficient nitroxides such as SG1 can improve control, these strategies only partially suppress the underlying limitations. Consequently, NMP remains less suitable for the synthesis of well-defined GMA-based polymers compared to other controlled radical polymerization techniques.

At the end of this section, a comparative analysis of the three main controlled / reversible deactivation radical polymerization techniques is provided, revealing distinct behaviors when applied to GMA-based systems, largely governed by the intrinsic reactivity of the epoxide functionality.

RAFT polymerization stands out as the most versatile and operationally robust approach for GMA. Its degenerative chain transfer mechanism ensures excellent tolerance toward functional groups, including epoxides, while maintaining low radical concentrations. As discussed above, fine control over molecular weight and dispersity can be achieved through careful selection of the *Z*-group and reaction conditions. However, a critical limitation of RAFT lies in the sensitivity of the equilibrium to the electronic nature of the RAFT agent. Over stabilization of the intermediate radical (e.g., strongly electron-withdrawing Z-groups) can significantly reduce propagation rates, whereas insufficient stabilization leads to poor control. In addition, the presence of thiocarbonylthio end groups may require post-polymerization removal for certain applications. Despite these drawbacks, RAFT remains the most adaptable method for GMA, particularly for complex architectures and high functional group fidelity.

ATRP provides comparable control over molecular weight and enables precise architectural design, especially for surface-initiated systems and advanced topologies. However, its application to GMA is inherently more complex due to the involvement of transition metal catalysts. The epoxide group can interact with the metal center or ligand environment, increasing the risk of side reactions such as RO under suboptimal conditions. Although modern variants such as ARGET and ICAR ATRP significantly reduce catalyst loading and improve system tolerance, ATRP remains highly sensitive to the interplay between ligand structure, solvent, and reaction medium. This creates a narrower optimization window compared to RAFT. From a critical standpoint, ATRP offers superior structural precision but at the cost of increased system complexity and stricter reaction control requirements.

In contrast, NMP is intrinsically less suited for GMA polymerization. While it offers the advantage of being metal-free, its mechanism requires elevated temperatures to ensure efficient alkoxyamine homolysis. These conditions directly conflict with the stability of the epoxide group, promoting side reactions such as RO, backbiting, and β-scission. Even with advanced nitroxides such as SG1, the control over methacrylate-based systems remains limited, particularly at high conversion. A key limitation is that these side reactions are not merely kinetic artifacts but are fundamentally linked to the nature of methacrylate radicals and the thermal requirements of NMP. Consequently, NMP suffers from reduced chain-end fidelity, broader dispersity, and a higher tendency toward branching or crosslinking in GMA systems.

From a critical perspective, the choice of RDRP technique for GMA is not solely dictated by control over molecular weight, but rather by the ability to balance radical reactivity with epoxide stability. RAFT achieves this balance most effectively due to its functional group tolerance and flexible equilibrium. ATRP can reach a similar level of control but requires precise tuning of catalytic and environmental parameters. In contrast, NMP is fundamentally constrained by its thermal requirements and radical side reactions, making it less competitive for GMA-based systems.

Overall, RAFT emerges as the most suitable technique for GMA polymerization, offering the best compromise between control, functional group compatibility, and operational simplicity. ATRP represents a powerful alternative when advanced architectures or surface functionalization are targeted, whereas NMP remains limited to specific cases where its inherent constraints can be carefully managed.

3.3.3.1. Externally stimulated NMP techniques. The combination of NMP with external energy inputs for the controlled polymerization of GMA remains a highly limited and emerging research area. Conventional NMP of GMA is typically performed under thermal activation, relying on the reversible homolytic cleavage of alkoxyamine species at elevated temperatures to generate propagating radicals. However, to reduce thermal stress and preserve the sensitive epoxy functionality of GMA, alternative activation pathways have been investigated.

In this context, Yoshida, Eri and co-workers reported the example of photo-induced NMP of GMA using a nitroxide system 4-methoxy-2,2,6,6-tetramethylpiperidine-1-oxyl (MTEMPO) in combination with a photosensitive onium salt, where light irradiation promoted radical generation through an electron-transfer mechanism instead of thermal bond dissociation. Importantly, this system enabled controlled polymerization at mild conditions while maintaining the epoxy group intact, with no evidence of epoxy RO side reactions observed throughout the process. The obtained polymers exhibited controlled characteristics with a relatively narrow molecular weight distribution, with Đ around 1.4–1.5, confirming a partial living/controlled nature of the polymerization. Despite this pioneering demonstration, no further development of electrochemical, ultrasonic, or plasma-assisted NMP of GMA has been reported. In contrast to ATRP, which has been extensively adapted to multiple external energy sources, NMP remains fundamentally limited by its thermally driven alkoxyamine activation mechanism. Therefore, external energy–assisted NMP of GMA is still at an early stage and represents a promising but underexplored direction in controlled radical polymerization [302].

Beyond GMA, the application of NMP under external energy inputs to methacrylate monomers remains highly limited and largely restricted to model systems such as MMA. In this regard, the pioneering studies of Eri Yoshida (2010) established the fundamental framework of photo-induced NMP based primarily on MMA, highlighting both the potential and limitations of this approach.

In the initial work [304], the photo-living radical polymerization of MMA was successfully demonstrated at room temperature using a three-component system consisting of racemic (2*RS*,2′*RS*)-azobis(4-methoxy-2,4-dimethylvaleronitrile) (r-AMDV) as the initiator, 4-methoxy-2,2,6,6-tetramethylpiperidine-1-oxyl (MTEMPO) as the nitroxide mediator, and (4-*tert*-butylphenyl)diphenylsulfonium triflate (tBuS) as a photo-acid generator. Under UV irradiation, the sulfonium salt acts as an electron-transfer agent, enabling reversible activation of the dormant species without thermal cleavage of the alkoxyamine bond. The polymerization exhibited typical living characteristics, including linear first-order kinetics, a linear increase of molecular weight with conversion, and relatively narrow dispersity values (Đ ≈ 1.4–1.5). Importantly, the system displayed a photo-latency effect, allowing temporal ON/OFF control of the polymerization upon light exposure.

Subsequent studies further elucidated the influence of reaction parameters on the control of the process [305]. In particular, the structure of azo-initiators and photo-acid generators was shown to play a critical role in regulating the molecular weight distribution, with optimized systems yielding dispersities as low as ~1.4, while less efficient combinations led to broader distributions (Đ up to ~1.6 or higher). The efficiency of the system was found to depend strongly on the UV absorption properties of the initiators and the redox characteristics of the sulfonium salts, confirming that electron transfer processes govern the activation/deactivation equilibrium.

In parallel, alternative photo-initiated NMP systems were explored using redox-active metal complexes, such as (η^6-benzene)(η^5-cyclopentadienyl)FeII hexafluorophosphate, which interact with MTEMPO through a redox cycle to mediate radical generation. Although these systems do not directly initiate polymerization, they contribute to the control of radical concentration via electron transfer, yielding polymers with comparable dispersity (Đ ≈ 1.4–1.5) and confirming the versatility of photo-induced activation pathways [306].

A critical aspect of the photo-initiated NMP process was further highlighted through chain-end fidelity studies based on block copolymerization experiments [307]. It was demonstrated that the stability of the dormant chain ends strongly depends on irradiation time, with prolonged exposure (6.5 h) leading to partial deactivation and bimodal molecular weight distributions, while shorter polymerization times (5 h) preserve chain-end activity and enable successful block copolymer formation. These findings reveal an inherent limitation of photo-NMP systems, namely the progressive loss of chain-end functionality under extended irradiation.

Beyond the pioneering photo-induced systems developed by Eri Yoshida, only a limited number of studies have explored NMP of methacrylates under alternative external energy inputs or non-conventional activation conditions. For instance, hybrid approaches combining NMP with photo-induced iniferter techniques have been reported for tert-butyl methacrylate, where UV irradiation promotes radical generation while the nitroxide species contributes to partial control of chain growth, leading to pseudo-living graft polymerization behavior [308].

In parallel, environmentally assisted NMP has been investigated through the dispersion polymerization of methyl methacrylate in supercritical CO_2, where the unique physicochemical properties of the medium enhance radical diffusion and polymerization control, although the process remains fundamentally thermally driven [309]. Additionally, efforts to reduce the energetic requirements of NMP have led to the development of low-temperature systems capable of polymerizing methacrylates at 40–50 °C via in situ generation of active nitroxide species, thereby limiting side reactions and improving control [310].

Despite these advances, such strategies remain indirect and do not constitute fully external energy–driven NMP processes comparable to photo- or electrochemically controlled systems. Overall, these studies highlight that, apart from a few hybrid or environmentally assisted approaches, the application of external energy inputs to NMP of methacrylates remains highly restricted and largely confined to model monomers.

4. Poly-GMA characterization

4.1. Spectroscopy FTIR

Fourier-transform infrared (FTIR) spectroscopy has been demonstrated to be a reliable technique for confirming the ROP of GMA via both cationic and anionic pathways. In both cases, the characteristic disappearance of the oxirane ring signal and the appearance of ether-related vibrations serve as confirmatory indicators of successful RO. However, this phenomenon is obscured by subtle spectral differences, which arise due to the distinct mechanisms of RO in each case.

In models of cationic and anionic ROP, the absorption band around 907-910 cm^{-1}, attributed to epoxy ring deformation vibration, disappears, indicating effective RO [311,312]. As the methacrylate moiety is sustained in both pathways, a notable band near 1730 cm^{-1} and 1630 cm^{-1} is evident, corresponding to the ester C═O and C═C stretching vibrations, respectively. A distinguishable variation is apparent in the C–O–C stretching region (1050–1250 cm^{-1}). In the course of cationic polymerization, the ring opens selectively at the more substituted carbon, resulting in the formation of tertiary ether linkages. Conversely, the anionic pathway involves nucleophilic attack at the less substituted position, resulting predominantly in primary or secondary ether structures. These structural variations can lead to shifts in the position and intensity of the C—O stretching region, although both types of polymers exhibit bands characteristic of ether functionalities. In certain instances, a broad band approximating 3400 cm^{-1} may be observed, correlating with hydroxyl groups that emerge during polymerization or side reactions as it is described previously in mechanism section [313,314,316]. It can be stated that, in general, FTIR analysis can be utilized to confirm RO nature of both polymerizations. Furthermore, it can provide insight into the underlying mechanism based on slight variations in the ether region of the spectrum.

In contrast to ROP mechanisms, radical polymerization of GMA primarily involves the vinyl (C=C) group, while leaving the epoxide ring largely unaffected. Consequently, the FTIR spectra of poly(GMA) synthesized via radical pathways generally exhibit the characteristic absorption band associated with the oxirane ring, situated at 907–910 cm^{-1}. This preservation indicates that no substantial RO has occurred during the polymerization process. Furthermore, the C═O stretching vibration at 1730 cm^{-1} and the C═C stretching vibration at ~1630 cm^{-1} may persist in unreacted monomer or partially polymerized samples. Concurrently, the gradual disappearance of the C═C band signifies vinyl polymerization. Overall, the persistent oxirane signal in radical-derived polymers provides a clear spectroscopic distinction from those obtained via cationic or anionic RO routes [315].

Transesterification in homopolymers or copolymers of GMA can be verified by distinct FT-IR spectral changes. The ester carbonyl absorption band, usually observed near 1720 cm^{-1}, often shows a slight shift or variation in intensity, indicating the formation of new ester linkages. At the same time, the C–O–C stretching vibrations in the 1240–1170 cm^{-1} and 1100–1050 cm^{-1} regions become more pronounced, consistent with the generation of additional ester groups. When alcohols or polyols participate in the reaction, the broad O—H stretching band around 3500 cm^{-1} decreases or disappears, confirming their consumption. Together, these spectral modifications provide strong evidence for transesterification in GMA-based polymers, independently of any epoxy RO reactions [123].

Overall, FTIR analysis not only confirms the occurrence of specific

reactions in GMA-based systems but also provides valuable insight into the underlying polymerization pathway and associated structural variations. In particular, the presence or disappearance of the oxirane band, combined with changes in the C–O–C and O—H regions, enables clear discrimination between radical polymerization, RO mechanisms, and transesterification processes. These spectral features are directly linked to the chemical structure of the resulting polymer, especially the nature of ether or ester linkages formed during reaction. Moreover, the epoxy RO mechanism can generate multiple isomeric structures (primary, secondary, or tertiary ethers), which may coexist within the same material and subtly influence the FTIR profile. Therefore, FTIR serves as a powerful tool not only for identifying reaction pathways but also for predicting the structural diversity of GMA-based polymers, which is critical for understanding and tailoring their final properties.

4.2. NMR spectroscopy (^1H and ^{13}C NMR)

Nuclear magnetic resonance (NMR) spectroscopy is a valuable technique for studying the structural changes of GMA during radical polymerization and epoxide RO processes. In particular, ^1H and ^{13}C NMR spectroscopy provides clear evidence of functional group transformations and enables reliable monitoring of reaction progression. For this purpose, this section will present the NMR signals characteristic of GMA-based structures.

In the monomeric state, GMA displays well-resolved signals that serve as reliable markers in its ^1H NMR spectrum (in $CDCl_3$). The vinyl protons exhibit distinct resonances at $\delta = 6.06$ ppm (1H, s, =CH_2, trans) and $\delta = 5.53$ ppm (1H, s, =CH_2, cis) [68]. These signals disappear upon successful radical polymerization, indicating complete consumption of the methacrylate double bond. Concurrently, the region between $\delta \sim 0.8$ and 2.2 ppm becomes more populated with broad, overlapping resonances corresponding to the aliphatic protons of the polymer backbone (–CH_2–C–) [302]. The methyl group (–CH_3) attached to the α-carbon of the methacrylate moiety appears in the monomer as a sharp singlet integrating to three protons at $\delta = 1.85$–1.95 ppm in $CDCl_3$ [316]. After polymerization, this methyl resonance remains present but appears broadened and less resolved due to the polymeric environment. In particular, the peaks at 3.85 and 4.31 ppm, which correspond to the methylene protons adjacent to the oxygen atom in the glycidyl group, as well as the signal at 3.26 ppm, which is attributed to the methyl proton of the oxirane ring, are clearly observed. In a parallel observation, the resonances at 2.7 and 2.9 ppm, which are attributed to the methylene protons of the epoxy ring, remained consistent. This observation indicates that the epoxy groups are not consumed during the polymerization process and remain available for subsequent chemical modifications [317]. These chemical shift values may vary depending on the solvent, due to differences in polarity and hydrogen-bonding interactions with the ester group[318]. In the ^13C NMR spectrum (in DMSO-d_6), the disappearance of the C═C vinyl carbon signals at $\delta =$ 136–138 ppm (C=C–) and $\delta = 125$–129 ppm (=CH_2), together with the emergence of new peaks at $\delta = 45$–55 ppm (–CH_2–CH–) from the polymer backbone, further confirm successful polymerization[319].

In the ^1H NMR spectrum of GMA (in $CDCl_3$), the oxirane ring is clearly identifiable. The methylene protons of the epoxide resonate as multiplets at $\delta = 2.60$–2.80 ppm (2H, –CH_2–), while the methine proton appears as a quintuplet at $\delta = 3.19$ ppm (1H, –CH– in the oxirane ring). The methylene protons adjacent to the epoxy oxygen are observed as double doublets at $\delta = 3.90$–4.45 ppm (2H, –CH_2–O–). Upon RO, typically induced by nucleophilic agents or under suitable catalytic conditions, these characteristic resonances decrease or shift, and new signals emerge in the $\delta = 3.50$–3.70 ppm range, corresponding to hydroxylated groups such as –CH_2–OH and –CH(OH)– [212].The hydroxyl proton itself generally appears as a broad, variable signal and is therefore less reliable for structural assignment. The chemical shift of the proton on the carbon bearing the hydroxyl group allows identification of the regioisomer formed. A resonance at $\delta \approx 3.55$ ppm indicates nucleophilic attack at the β-position, corresponding to the β-hydroxy-α-substituted isomer, whereas a resonance at $\delta \approx 3.70$ ppm is characteristic of nucleophilic attack at the α-position, leading to the α-hydroxy-β-substituted isomer. These results were also obtained in a recent unpublished study by the second author of this review, in which GMA was polymerized using Maghnite-Zn^{2+} and Maghnite-H^+ as catalysts in a comparative investigation (see Fig. 16). A more detailed discussion of this subject will be provided in the subsequent section of the Green Catalysts. In the ^13C NMR spectrum of the reaction product recorded in D_2O [8], the disappearance of the characteristic epoxide carbon resonances at $\delta \approx 44$ ppm (methylene, –CH_2–O–) and $\delta \approx 49$ ppm (methine, –CH–O–), together with the appearance of new signals in the $\delta \approx 60$–80 ppm region corresponding to carbons bonded to hydroxyl and ether oxygens, confirms successful epoxide RO. The ester carbonyl resonance at $\delta \approx 172$–178 ppm remains unchanged, indicating preservation of the methacrylate ester group [8].

According to the work of Ferreira. L et al. [126], the occurrence of transesterification during the polymerization of GMA can be clearly evidenced by comparative ^1H and ^13C NMR analyses in D_2O. In the ^1H NMR spectrum of the transesterified product, characteristic resonances of the methacrylate group are observed at $\delta \approx 1.90$ ppm (methyl protons) and at $\delta \approx 6.14$ and 5.72 ppm (vinyl protons), with an integral ratio of 3:2. At the same time, the oxirane proton signals typical of native GMA ($\delta \approx 2.6$–3.2 ppm) are no longer detected, indicating complete consumption of the glycidyl group. The ^13C NMR spectrum shows the expected carbonyl ($\delta \approx 171.0$ ppm), quaternary C═C ($\delta \approx 137.2$ ppm), vinylic CH_2 ($\delta \approx 128.8$ ppm), and methyl ($\delta \approx 19.0$ ppm) signals of the methacrylate moiety, while the characteristic carbons of the glyceryl/oxirane portion ($\delta \approx 76.0$, 70–72, and 66.0 ppm) are absent. In systems where polymerization occurs without transesterification, the epoxy-related resonances remain, reflecting only partial RO and the persistence of unreacted epoxide groups. Conversely, under transesterification conditions, extensive nucleophilic attack on the highly reactive epoxy groups leads to their complete consumption, resulting in the total disappearance of the epoxy signals. This set of observations confirms that the reaction proceeds predominantly through transesterification, directly attaching the methacryloyl group to hydroxyl functionalities rather than via nucleophilic epoxide RO.

Similar observations were reported by Crismi et al. [320], who investigated the transesterification of GMA with poly(vinyl alcohol) using ^1H and ^13C NMR analysis in DMSO-d_6. They observed the complete disappearance of epoxy-related ^1H resonances ($\delta \approx 2.6$–3.2 ppm) and corresponding ^13C signals ($\delta \approx 44$–55 ppm), confirming full consumption of the oxirane rings. In addition, they reported the appearance of new ^1H resonances at δ 1.6 ppm, 3.6 ppm, and 5.1 ppm, which were absent in both pure poly(vinyl alcohol) and pure GMA. These peaks were attributed respectively to the methyl group of the methacrylate moiety (δ 1.6 ppm), the –CH_2–O– protons of PVA adjacent to the newly formed ester linkage (δ 3.6 ppm), and the vinylic proton (=CH–) of the methacrylate group (δ 5.1 ppm). In addition, the formation of glycidol as a byproduct was confirmed by gas chromatography–mass spectrometry (GC–MS). These findings are consistent with a recent study on PVA-co-GMA [122] which also reported the same general set of new methacrylate-related signals, further supporting the successful grafting of methacrylate groups via transesterification by ^1H NMR in DMSO-d_6.

Overall, NMR spectroscopy not only confirms structural transformations but also enables estimation of the predominant reaction mechanism occurring during polymerization. This is particularly important for epoxide RO processes, where multiple pathways may coexist. By analyzing the position, intensity, and multiplicity of signals associated with hydroxylated carbons and adjacent protons, it becomes possible to distinguish between different regioisomeric products arising from nucleophilic attack at either the α- or β-position of the oxirane ring. Consequently, NMR provides valuable insight into the distribution of isomeric structures formed during polymerization and allows identification of the dominant ring-opening pathway. This capability is

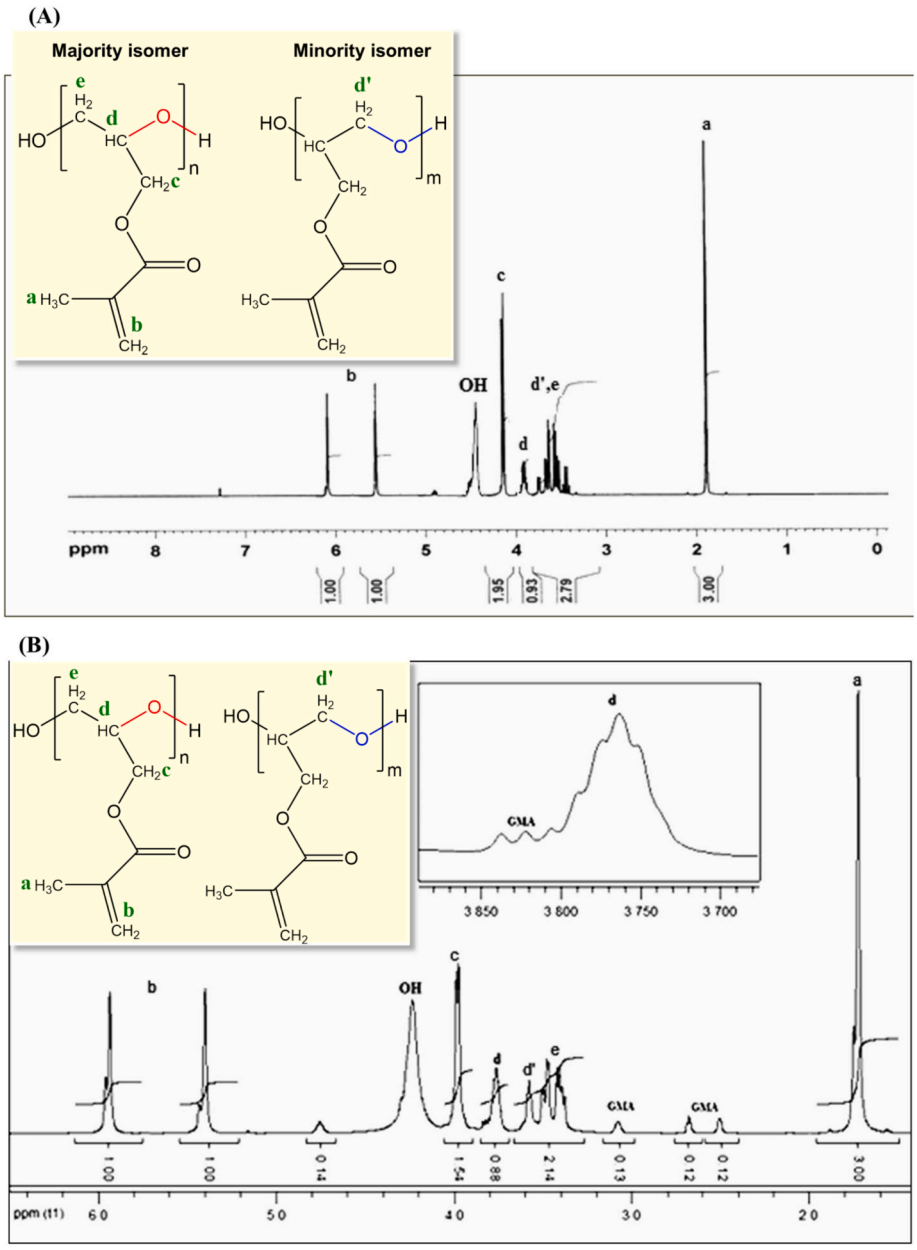


Fig. 16. ^{1}H NMR spectra of poly(GMA) synthesized using (A) Maghnite-H^{+} and (B) Maghnite-Zn^{2+}.

essential for understanding the complexity of GMA reactivity, where competing mechanisms can occur simultaneously and directly influence the resulting polymer structure and properties.

4.3. Thermal analysis (DSC and TGA)

To comprehensively evaluate the influence of GMA incorporation on the thermal behavior of polymer systems, it is essential to consider not only the comonomer type but also the polymerization pathway employed. In this context, TGA and DSC data were compared for materials synthesized via radical polymerization and those obtained through RO strategies. Such a comparative approach allows for a clearer understanding of how GMA's reactive epoxide functionality and methacrylate backbone contribute to differences in degradation temperatures, mass-loss profiles, and glass transition behavior, as well as how subsequent post-polymerization modifications can further tailor these thermal responses.

In radical copolymerization systems, the thermal behavior of GMA-based materials strongly depends on the nature and proportion of the comonomer. In the case of styrene-GMA copolymers [321], TGA analysis reveals a complex, composition-dependent degradation pattern reflecting the interplay between thermally stable aromatic styrene units and the more labile methacrylate segments bearing epoxy groups. Copolymers rich in styrene (70/30 St/GMA) typically exhibit a three-step degradation process: an initial mass loss below ~170 °C associated with moisture or residual impurities, followed by a second stage (≈170–340 °C) corresponding to ester group decomposition and CO_2 evolution, and a final stage above ~340 °C related to main-chain degradation. As the GMA content increases, the early-stage weight loss becomes more pronounced (up to ~21% below 180 °C for high GMA content), indicating enhanced sensitivity to thermal cleavage due to the higher concentration of epoxy and ester functionalities. Notably, the maximum degradation temperature (T_{max}) decreases significantly with increasing GMA fraction, dropping from ~400 °C in styrene-rich systems to ~180 °C in GMA-rich copolymers. This trend suggests that although moderate GMA incorporation may promote intermolecular

interactions or limited crosslinking, excessive GMA introduces a high density of thermally labile groups that act as initiation sites for degradation. The intermediate composition (50/50 St/GMA) exhibits a more uniform two-step degradation behavior with T_{max} around ~320 °C, indicative of a more homogeneous microstructure and distribution of functional groups.

A similar dependence on composition is observed in other copolymer systems, although the effect of GMA may differ depending on the comonomer. For instance, in GMA–acrylonitrile copolymers [142], the incorporation of GMA units generally reduces thermal stability compared to pure poly(acrylonitrile), as reflected by decreases in both T_{max} and T_{50} values with increasing GMA content. While pure poly(GMA) typically exhibits T_{max} around ~460 °C and T_{50} near ~276 °C, these values remain significantly lower than those of poly(acrylonitrile) (T_{max} ~810 °C, $T_{50} \approx 620$ °C). For example, a copolymer containing 75 mol% GMA may display intermediate values (T_{max} ~570 °C and T_{50} ~338 °C), confirming that the introduction of thermally labile ester groups reduces overall thermal resistance.

In block copolymer systems, thermal behavior is also governed by the nature of each segment. For example, a GMA–poly(ethylene glycol) block copolymer obtained via redox polymerization exhibits two distinct degradation steps [322]: the first at ~160 °C, attributed to the poly(ethylene glycol) segment, and the second near ~400 °C, associated with the poly(GMA) block. The glass transition temperature (T_g) of this material (~28 °C) is significantly lower than that of pure poly(GMA), highlighting the plasticizing effect of the flexible poly(ethylene glycol) chains. These observations further confirm that the thermal properties of GMA-based polymers are highly sensitive to the chemical identity and stability of the comonomer or block component.

Post-polymerization modification via RO of pendant epoxide groups provides a powerful strategy to tailor the thermal behavior of GMA-based polymers by directly eliminating one of their most thermally labile structural features. The oxirane ring, characterized by significant ring strain and high reactivity, acts as a preferential initiation site for thermal degradation, typically triggering an early decomposition step around 250–270 °C. Consequently, polymers containing intact epoxide functionalities commonly exhibit multi-step degradation profiles, with the first stage corresponding to the cleavage or rearrangement of oxirane-containing fragments. This effect is clearly illustrated in crosslinked GMA-based microspheres such as poly(GMA-*co*-1,4DMB) (crosslinked with benzene-1,4-diylbis(2-methylprop-2-enoate)) and poly(GMA-co-TRIM) (crosslinked with trimethylolpropane trimethacrylate) [323,324] in which the epoxide groups remain intact within the polymer structure, both display three-step degradation processes under inert conditions. For instance, poly(GMA-co-TRIM) exhibits a first degradation maximum at 251 °C associated with a substantial mass loss of 39.4%, directly attributed to the decomposition of oxirane groups. This is followed by a second degradation stage at 375 °C ($\Delta m \approx 49.3\%$) corresponding to network breakdown, and a final stage at 519 °C ($\Delta m \approx 9.6\%$) related to the degradation of more stable carbonaceous residues. A similar behavior is observed for poly(GMA-co-1,4DMB), although with slightly higher thermal resistance, as evidenced by a delayed first degradation step at 269 °C ($\Delta m \approx 33.8\%$) and an overall increase of ~15 °C in stability compared to the TRIM-based system. These results demonstrate that, regardless of the crosslinking agent, the presence of oxirane groups governs the onset of degradation and contributes significantly to early mass loss.

Importantly, DSC analysis provides additional insight into the complexity of this initial degradation stage. Although thermal decomposition is initiated by an endothermic depolymerization process, the presence of oxirane groups introduces competing exothermic contributions associated with their isomerization into carbonyl-containing species. As a result, the first thermal event may appear as an overall exothermic signal or as an exothermic–endothermic transition, as observed in poly(GMA-co-1,4DMB) in the 100–250 °C range. This behavior reflects the superposition of multiple processes, including depolymerization, oxirane rearrangement, and the evolution of volatile species such as GMA monomer and acrolein, thereby explaining the complex DSC profiles of unmodified systems.

Following RO modification with nucleophiles such as thioglycolic acid, the thermal degradation mechanism is profoundly altered. The epoxide groups are converted into more stable functionalities (hydroxyl and thiol groups), effectively suppressing the primary thermally weak sites. As a direct consequence, the degradation profile shifts from a three-step to a two-step process, indicating a more homogeneous and thermally stabilized network. Importantly, the characteristic degradation peak around ~250 °C completely disappears after functionalization, unequivocally confirming that this stage originates from oxirane decomposition. Consistently, DSC thermograms of the modified copolymers exhibit simplified thermal behavior, with predominantly endothermic transitions and the disappearance of the exothermic contributions previously associated with oxirane isomerization.

Quantitatively, this structural transformation leads to a significant enhancement in thermal resistance. The temperature at 5% mass loss (T_5) increases by up to 54 °C for poly(GMA-*co*-TRIM)-SH and by approximately 22 °C for poly(GMA-*co*-1,4DMB)-SH, highlighting the strong dependence of stabilization on the initial epoxide content. Indeed, the TRIM-based copolymer, which initially contains a higher epoxide concentration (1.23 $mmol \cdot g^{-1}$ vs 1.12 $mmol \cdot g^{-1}$), shows a more pronounced improvement upon modification, further confirming the critical role of oxirane groups as degradation-triggering sites. After modification, the main degradation step shifts to higher temperatures (≈344–355 °C) and is associated with the loss of the majority of the polymer mass (>80%), while the final degradation stage occurs between ~479 and 531 °C.

Supporting this, studies on GMA-co-ethylene glycol dimethacrylate [27] microspheres have shown that systems retaining intact oxirane groups exhibit multi-step degradation with a main T_{max} around 290 °C, along with endothermic and exothermic effects at ~250 °C and ~ 308 °C related to depolymerization and epoxide rearrangement. These findings highlight that the presence of intact oxirane groups in the structure can act as thermally labile sites, promoting degradation at relatively moderate temperatures. Therefore, epoxide RO through post-modification could mitigate this instability by reducing epoxy group reactivity and promoting the formation of more stable linkages, thereby enhancing the overall thermal resistance of the polymer.

To evaluate the influence of the polymerization method on the thermal properties of GMA-based polymers, it should be noted that no systematic studies directly comparing FRP, ATRP, and RAFT under identical conditions are available. Therefore, the discussion is based on a comparison of independent literature reports, which are summarized in Table 3. Overall, the collected data indicate that the thermal properties of poly(GMA), including both Tg and degradation behavior, are only weakly affected by the polymerization technique. Similarly, most poly(GMA) systems, regardless of synthesis method, exhibit a comparable two-step degradation mechanism, where the first step is associated with the decomposition of pendant epoxide groups and the second corresponds to main-chain scission. This consistency suggests that both Tg

Table 3
Thermal properties of poly(GMA) synthesized via different radical polymerization methods.

Polymerization method	Tg (°C)	Degradation behavior (TGA)	References
FRP	74–78	Two-step degradation ~300 °C (epoxide side groups)–450 °C (backbone)	[142]
ATRP	~65	Two-step degradation: ~340 °C ~ 400 °C	[244,325]
RAFT	~65–70	Two-step degradation: ~320 °C and ~ 420 °C	[25]

and degradation pathways are primarily governed by the intrinsic chemical structure of GMA rather than the polymerization process itself.

Overall, comparing the thermal properties of GMA-based polymers obtained through different synthetic routes provides a comprehensive understanding of how polymerization strategy influences material performance. In particular, evaluating systems prepared by radical polymerization and those modified via epoxide ring-opening highlights the key role of the epoxy functionality in governing thermal stability. The available data indicate that maintaining intact epoxide groups introduces thermally labile sites that can promote earlier degradation, whereas their transformation through ring-opening reactions generally enhances thermal resistance by eliminating these weak points. This comparative approach therefore enables a more reliable prediction of the thermal behavior of GMA-based materials depending on whether the epoxy functionality is preserved or chemically modified. Such understanding is essential for guiding the design of materials with targeted properties and for identifying suitable application domains requiring enhanced thermal stability or controlled degradation. These structure–property relationships are reflected in thermal analysis, where variations in degradation behavior and thermal stability can be directly correlated with the presence or transformation of epoxide groups.

4.4. Morphological characterization (SEM, TEM and AFM)

A variety of microscopic analytical techniques have been used to extensively investigate the morphology of poly(GMA). These techniques include SEM, TEM, and AFM. These methods provide complementary insights into the surface texture, internal porosity, and nanoscale topography of polymer network structures.

Several studies on GMA polymerization have emphasized that the choice of porogenic solvent is a key factor influencing the morphological outcome of the resulting polymer, as it governs the phase separation mechanism during polymerization and, consequently, affects pore size, distribution, and connectivity. In poly(GMA-*co*-ethylene glycol dimethacrylate) poly(GMA-co-EGDMA) networks, the dynamics of phase separation have been shown to directly determine the pore architecture. When apolar, non-solvent porogens such as cyclohexane are employed, early phase separation occurs prior to gelation, leading to opaque, highly porous microspheres with irregular, rough surfaces and interconnected macropores (Fig. 17-a). These materials typically exhibit high pore volumes (Vp ≈ 0.5 cm^3/g) and large specific surface areas (S_{BET} up to 260.4 m^2/g) [30], with submicron-scale open cavities and granular, highly porous surface textures observable by SEM (Fig. 17-a). Conversely, the use of polar or partially polar porogens, such as cyclohexanol and butan-2-ol, often results in delayed phase separation occurring after gelation, producing smooth, translucent microspheres with dense, low-porosity structures that frequently feature closed or collapsed pores. These polymers show lower pore volumes (Vp ≈ 0.1–0.4 cm^3/g) and reduced specific surface areas (S_{BET} ≈ 18.5–175.1 m^2/g) depending on the solvent, which can be attributed to the ability of polar solvents to interact with the polymer network through hydrogen bonding with the oxygen atoms of epoxy and ester groups. These interactions enhance solvation of the growing polymer chains and delay phase separation, ultimately promoting the formation of more compact and homogeneous polymer networks.

However, the general trend associating solvent polarity with porosity is not universal. Pączkowski and Gawdzik [29] reported that using a seed-swelling polymerization approach, in which polystyrene seeds are swollen with monomers and porogenic solvents prior to polymerization, can produce contrasting results. In this system, polar solvents such as cyclohexanol yielded microspheres with higher surface areas (S_{BET} ≈ 158.7 m^2/g) and pore volumes (Vp ≈ 0.56 cm^3/g) compared to toluene (S_{BET} ≈ 131.7 m^2/g, Vp ≈ 0.43 cm^3/g). These observations indicate that the polymerization methodology, presence of seed templates, and specific solute-solvent interactions significantly influence pore formation and can override the conventional correlation between solvent polarity and porosity. Therefore, while solvent polarity remains a critical factor, it must be considered in the broader context of polymerization method, phase separation dynamics, and thermodynamic compatibility.

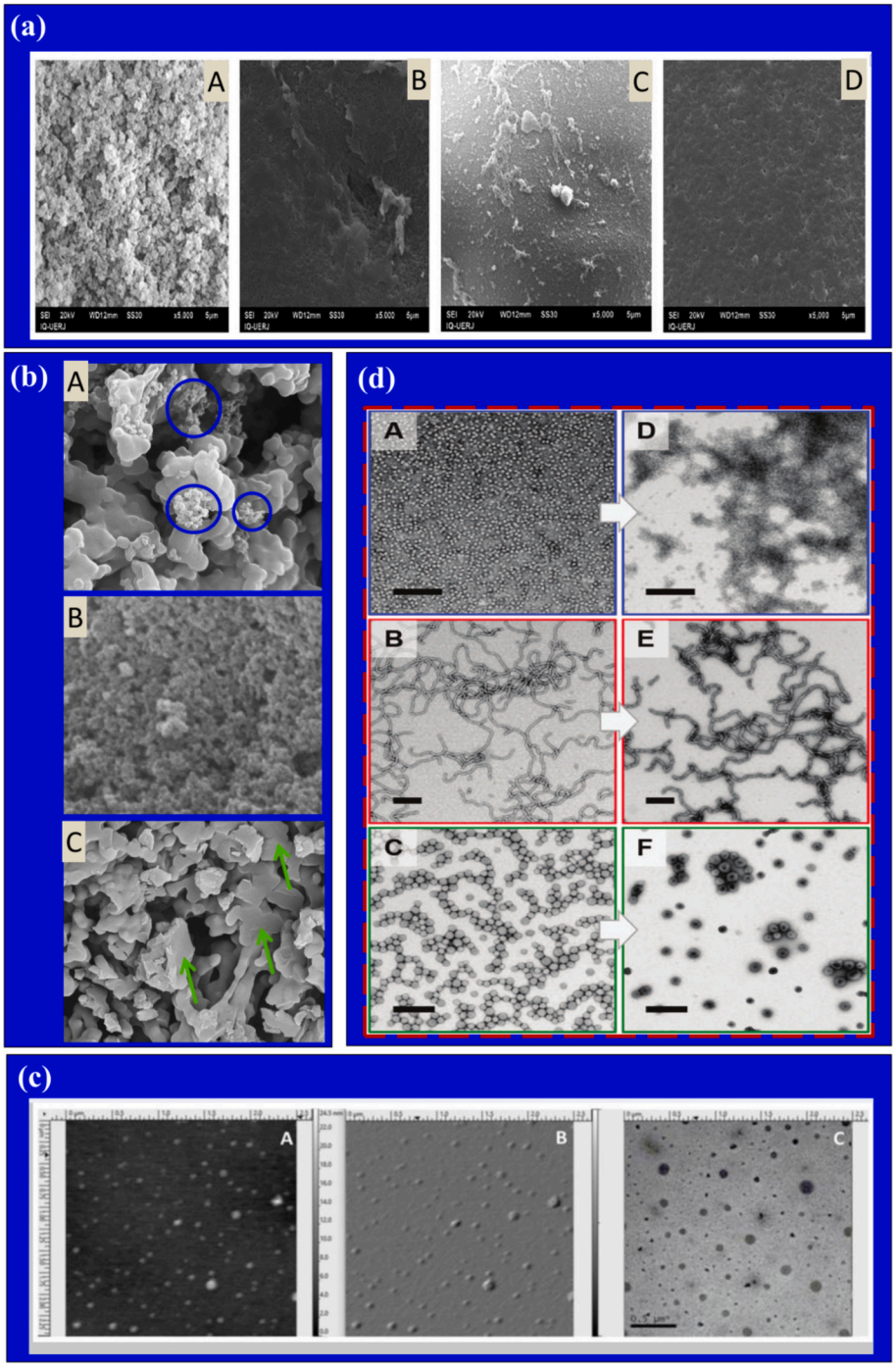
In agreement with previous studies on GMA-based polymers, our investigation of poly(GMA-*co*-maleic anhydride) (PGM) copolymers synthesized via FRP and subsequently precipitated in different solvents (hexane, cyclohexane, and toluene) revealed pronounced solvent-dependent morphological variations, as observed by SEM at identical magnification (8 μm × 20,000, Fig. 17-b). It should be noted that these results originate from an ongoing study that has not yet been published.

PGM—H, precipitated in hexane (a non-solvent; nonpolar aliphatic solvent) for the polymer-underwent rapid phase separation immediately upon chain growth. This fast precipitation did not allow polymer chains sufficient time to reorganize, resulting in broad particle size distribution, irregular aggregation, and a highly heterogeneous, rough surface. PGM—C, precipitated in cyclohexane (a non-solvent; nonpolar alicyclic solvent), exhibited intermediate behavior, with relatively uniform spherical particles and rough, porous surfaces due to slower precipitation. PGM-T, precipitated in toluene (a non-solvent; nonpolar aromatic solvent) in which the polymer is more soluble-underwent slow precipitation, allowing dense chain packing and formation of smooth, compact particles with reduced surface roughness. These differences in morphology directly reflect the kinetics of solvent-controlled precipitation: rapid precipitation in poor solvents produces heterogeneous, highly porous particles, whereas slow precipitation in better solvents yields denser, more uniform structures with limited porosity.

The adsorption efficiency of PGM copolymers correlates closely with their surface morphology. PGM—H, with its rough and heterogeneous surface and open pores, provided the highest number of accessible active sites, facilitating dye diffusion and resulting in superior adsorption capacity for cationic dyes such as methylene blue and crystal violet. PGM—C, with intermediate porosity and surface roughness, exhibited moderate adsorption performance, while PGM-T, with smooth and dense particles, showed the lowest adsorption efficiency due to limited accessibility of adsorption sites. These observations indicate that the controlled choice of precipitation solvent represents an effective strategy to tailor PGM copolymers for adsorption applications and environmental remediation.

A recent study reported results contradictory to our study, using hexane as a nonpolar, non-solvating medium. Cardil et al. [326] showed that covalently crosslinked poly(GMA) nano-assemblies synthesized in hexane adopt a distinct disk-like morphology. These nano-assemblies have smooth, homogeneous surfaces and lateral dimensions ranging from 10 to 40 nm, as confirmed by AFM and TEM imaging in Fig. 17-c. In this system, the polymer backbone is first prepared via the radical polymerization of GMA, which preserves the epoxide side groups. These epoxide moieties are then used to induce post-polymerization cross-linking. Combined with the poor solvation conditions in hexane, this process drives self-assembly into deformable, compact nano-discs rather than porous or aggregated structures. Despite the use of hexane, these assemblies lack internal porosity, which contrasts with previous studies. However, their regular and well-defined morphology suggests that the synthesis strategy plays a key role in determining the final architecture of poly(GMA)-based materials, as the synthesis strategy can be either radical polymerization followed by epoxide cross-linking or direct pore formation by phase separation.

Other research has demonstrated that poly(GMA) synthesized directly via radical polymerization exhibits a broader spectrum of morphological features largely dictated by polymerization conditions, crosslinker presence or absence, and surfactant or structured environment use. In bulk or surface-initiated polymerization systems, such as thin films or grafted layers, the morphology tends to be smooth and continuous, resembling a film, with little to no nanoscale texturing [327]. However, dispersed-phase radical polymerization, particularly emulsion, suspension, and microfluidic-assisted techniques, enables the production of both spherical and anisotropic particles. For example,



(caption on next page)

Fig. 17. a. Morphological variation of poly(GMA-*co*-EGDMA) microspheres as influenced by different porogenic solvents (adapted from Costa et al. [30]). SEM images of poly(GMA-*co*-EGDMA) microspheres prepared via radical polymerization in the presence of various porogenic solvents, showing distinct surface morphologies depending on solvent polarity and interaction with the growing polymer network: (A) Cyclohexane (apolar): rough, macroporous surface due to early phase separation; (B) Cyclohexanol (polar): smooth and dense morphology indicating delayed phase separation; (C) Butan-2-ol (moderately polar): intermediate porosity and texture; (D) Toluene (apolar): non-porous, compact beads with closed surface. These images highlight the crucial role of solvent selection in tuning pore structure and microsphere morphology. b. Morphological variation of PGM microspheres as influenced by different antisolvents of precipitation. SEM images of PGM copolymers prepared via FRP in the presence of different non-polar solvents, inducing precipitation of the copolymer, and showing distinct surface morphologies depending on solvent type and polymer-solvent interactions: (A) PGM-H (hexane; nonpolar aliphatic solvent): highly heterogeneous, rough morphology with spherical particles of varying sizes and irregular aggregation; (B) PGM-C (cyclohexane; nonpolar alicyclic solvent): porous, rough structure composed of relatively uniform spherical particles; (C) PGM-T (toluene; nonpolar aromatic solvent): smoother, more compact morphology with reduced apparent porosity. These SEM images highlight the critical role of antisolvent-induced precipitation in controlling the morphology and porosity of PGM copolymers. c. Morphological characterization of poly(GMA) nanoassemblies prepared via radical colymerization and epoxide RO Crosslinking (adapted from Cardil et al. [326]). AFM and TEM images illustrate the disk-like morphology of covalently crosslinked poly(GMA) nanoassemblies formed in hexane . A: An AFM height image showing isolated nanodiscs with smooth, flat surfaces and lateral dimensions ranging from 10 to 40 nm. B: An AFM 3D topography image emphasizing uniformity and surface regularity. C: A TEM micrograph confirms the well-defined, non-aggregated, discoidal shape and compact, nanoscale organization. This morphology results from self-assembly in a poor solvent (hexane), followed by controlled post-polymerization crosslinking via epoxide RO. This process offers a sharp contrast to the porous or amorphous morphologies obtained via conventional radical polymerization. d. TEM images of the poly(GMA)$_{328}$-poly(GlyMA) block copolymer nano-objects that were prepared via RAFT aqueous emulsion polymerization (adapted from Hatton et al., [205]). The representative TEM micrographs show the diverse morphologies that result from increasing the length of the hydrophobic poly(GlyMA) block: (A) spherical micelles (poly(GMA)$_{26}$–poly(GlyMA)$_{25}$) (B) worm-like micelles (poly(GMA)$_{28}$–poly(GlyMA)$_{40}$), (C) vesicles (poly(GMA)$_{28}$–poly(GlyMA)$_{80}$). (Scale bars = 200 nm).

batch emulsion polymerization of GMA with crosslinkers, such as ethylene glycol dimethacrylate, produces well-dispersed spherical microparticles [328], and microfluidic methods provide precise control over particle shape and access to non-spherical geometries, as demonstrated by Kilinceli et al. [328].

Furthermore, RAFT aqueous emulsion polymerization of GMA yields epoxy-functional nanoparticles that are typically uniform and spherical, as confirmed by Chan et al. [329]. In contrast, Hatton et al. [205] demonstrated that RAFT emulsion polymerization can form diverse nanostructures, such as spheres, worms, and vesicles, through adjusting the composition of the amphiphilic block copolymer as illustrate in Fig. 17-d. This morphological variation is primarily governed by the relative length of the hydrophobic block poly(GMA) to the hydrophilic block poly(glycerol monomethacrylate) poly(GlyMA) which influences the polymer's packing parameter during self-assembly in aqueous media. Short hydrophobic blocks tend to favor spherical morphologies due to their high interfacial curvature. As the hydrophobic block length increases, the packing constraints change, leading to the formation of worm-like micelles with lower curvature. When the hydrophobic segment becomes dominant enough, the system minimizes interfacial energy by organizing into bilayer vesicles with hollow interiors. These transitions reflect well-established thermodynamic principles of polymer self-assembly, in which the block ratio, concentration, and temperature are key determinants of morphology. The ability to fine-tune nanostructures in this manner highlights the versatility of RAFT polymerization in designing functional materials with tailored nanoscale architectures.

These diverse morphologies observed in GMA-based polymers highlight the remarkable versatility of this monomer. Depending on the polymerization method, solvent system, porogen nature, crosslinking density, monomer composition, and phase separation conditions, GMA can generate a wide range of structures, from dense and smooth films to highly porous networks and well-defined nano-objects. This morphological diversity significantly broadens the scope of potential applications. Importantly, the comparative analysis of these parameters provides valuable insight into the mechanisms governing structure formation and enables precise control over the final morphology. These structure–property relationships are directly evidenced by SEM, TEM, and AFM analyses, which reveal how variations in synthesis parameters translate into distinct morphological features such as surface roughness, porosity, and particle organization, ultimately impacting functional performance. Such understanding allows for the rational design of GMA-based materials with tailored architectures, making it possible to obtain the desired structure according to the targeted application, whether in adsorption, catalysis, drug delivery, or advanced functional materials.

At the end of this section, it can be stated that although the characterization techniques used in GMA-based polymer systems (FTIR, NMR, SEM, TEM, TGA, DSC, etc.) are generally standard in polymer science, their interpretation requires particular attention due to the specific structural features of GMA. In fact, the coexistence of a polymerizable vinyl group and highly reactive epoxide functionality introduces additional complexity compared to conventional monomers. This complexity is particularly pronounced in epoxide RO reactions, as discussed in previous sections, where multiple competing pathways and regioselective attacks can lead to the formation of different isomeric structures. As a result, radical polymerization, RO processes, and possible side reactions such as transesterification may occur simultaneously or sequentially, complicating structural analysis. Therefore, careful interpretation of FTIR and NMR data is required to distinguish between preserved and transformed epoxy groups and to correctly identify the resulting structures. In addition, the presence of such mechanistic complexity increases the structural heterogeneity of the final materials. Consequently, although the techniques themselves are conventional, their application to GMA-based systems requires a more critical, mechanism-oriented analysis to accurately establish structure–property relationships.

5. Green and sustainable polymerization mechanisms

The development of green and sustainable polymerization strategies has attracted considerable attention over the past two decades, driven by mounting environmental concerns and the demand for more sustainable materials. In this context, GMA represents a particularly attractive monomer due to its dual functionality: a polymerizable methacrylate double bond and a reactive epoxide moiety. This structural uniqueness fosters a vast array of polymerization pathways, encompassing radical polymerization of the vinyl group and ROP of the epoxide functionality. Conventional polymerization processes involving GMA frequently entail the use of organic solvents, metal-based catalysts, and elevated temperatures. These elements give rise to a number of environmental and safety concerns. In contrast, recent advancements have demonstrated the feasibility of implementing greener methodologies, with the objective of minimizing the ecological footprint of GMA-based materials. These strategies encompass the utilization of environmentally friendly solvents (e.g. water, bio-derived solvents), metal-free or bio-based catalysts, and mild reaction conditions, such as room-temperature processes, solvent-free systems, and light-mediated initiation. Moreover, organocatalysis, enzymatic catalysis, and hybrid catalytic systems have emerged as promising alternatives to traditional catalytic systems, offering enhanced selectivity, reduced toxicity, and compatibility with sensitive functional groups. These strategies not only enhance the environmental profile of GMA-based polymerizations but

also generate novel opportunities for tailoring polymer architectures and functionalities.

This section provides a comprehensive overview of recent developments in green and sustainable polymerization mechanisms involving GMA, regardless of the polymerization mode (i.e., radical polymerization or epoxide RO). The emphasis of this section is on solvent selection, catalytic systems, and process conditions aligned with the principles of green chemistry.

5.1. Green solvents

The replacement of hazardous organic solvents with environmentally benign alternatives represents one of the fundamental principles of green chemistry and has become an important strategy in the development of sustainable polymerization processes. In both radical polymerization and ROP systems, the choice of solvent plays a crucial role not only in reducing environmental impact and toxicity, but also in controlling reaction kinetics, catalyst activity, polymer molecular weight, dispersity, and monomer reactivity. Consequently, increasing attention has been devoted to the use of green solvents that combine low toxicity, biodegradability, renewability, and reduced volatile organic compound (VOC) emissions. Among these, solvents such as anisole, ethyl acetate, supercritical CO_2, ionic liquids, deep eutectic solvents, and other bio-based solvents have emerged as promising alternatives to conventional chlorinated and petroleum-derived solvents commonly employed in polymer synthesis [330–332].

In the polymerization of functional monomers such as GMA, environmentally friendly solvents can additionally improve process safety, facilitate purification, and enhance the sustainability of both FRP and ionic polymerization processes while maintaining high polymerization efficiency. For example, in cationic ROP of glycidol an analogue of GMA conversions above 99.9% were achieved in just 10 min using anisole, and in 24 h using ethyl acetate, while maintaining good control over molecular weight and low dispersity [333]. These solvents are much more environmentally friendly than traditional chlorinated or aromatic solvents.

In addition to their use in ROP, ethyl acetate has also been successfully employed in the radical polymerization of GMA. For instance, photopolymerization of GMA using benzophenone as the initiator in an ethyl acetate solution has been employed to produce low-molecular-weight poly(GMA) while retaining epoxy functionality [334]. Furthermore, USRP of GMA in an ethyl acetate/water biphasic system has been reported. In this system, ultrasound irradiation significantly enhances polymerization kinetics by improving mass transfer and radical generation [17]. These studies emphasize the versatility of ethyl acetate as a green solvent in homogeneous and biphasic systems, demonstrating its potential as an environmentally friendly medium for GMA radical polymerization strategies.

In more recent research, anisole has also been employed as a greener alternative in the controlled radical polymerization of GMA. For instance, a triphenylphosphine tris(4-methoxyphenyl)phosphine -mediated organocatalytic radical polymerization was conducted in anisole at 70 °C, thereby facilitating well-modulated polymer growth with narrow dispersity and preserved epoxy functionality. This was achieved through a chain-extension process from poly(MMA) to the *block* copolymer poly(MMA)-*block*-poly(GMA) in the same anisole medium [335]. This approach underscores the merits of anisole, which offers an environmentally benign solvent platform and facilitates efficient control over GMA polymerization. These properties serve to emphasize the value of anisole in sustainable polymer synthesis.

Other renewable and bio-derived solvents have also been explored. For example, 2-methyltetrahydrofuran (2-MeTHF), which is derived from biomass, has been successfully used in controlled radical polymerizations of methacrylate monomers. It offers a performance comparable to that of THF, but has a significantly lower environmental impact. More recently, Englezou et al. [336] demonstrated the use of 2-MeTHF as a versatile green solvent for tandem polymerizations (ROP, FRP and controlled RAFT), enabling the synthesis of well-defined amphiphilic copolymers under sustainable conditions. The potential extension of this to GMA polymerization offers promising prospects for designing fully sustainable solvent systems.

Supercritical carbon dioxide (CO_2) has demonstrated excellent potential as a green solvent and processing medium for GMA-based systems, offering a unique combination of properties including non-toxicity, abundance, low cost and ease of removal after the reaction. Unlike conventional organic solvents, supercritical CO_2 leaves no harmful residues, offers controllable solvation power through adjustments to pressure and temperature, and enables environmentally benign processing. For example, Kunita et al. [337] performed solid-state radical grafting of GMA onto poly(4-methyl-1-pentene) using supercritical CO_2 impregnation, achieving controlled surface functionalization at ≈130 bar, 70 °C and 7 h. Similarly, Daneshvar et al. [338] used supercritical CO_2 to graft GMA onto medium density polyethylene and facilitate the purification of the resulting polymer by effectively removing residual monomer and initiator. More recently, Versteeg et al. [339] developed a green reactive extrusion process in supercritical CO_2, enabling the grafting of GMA onto low-density polyethylene and recycled polyethylene. This study highlights the influence of processing parameters such as temperature and screw speed on grafting efficiency. Furthermore, the polymerization of GMA in supercritical CO_2 dispersion systems stabilized by block copolymers has been reported, further demonstrating the versatility of supercritical CO_2 as a sustainable solvent in polymerization processes [340].

In overview, the integration of green solvents into GMA polymerization represents an important step toward sustainable polymer chemistry. By reducing environmental impact while maintaining efficient and well-controlled polymerization, these solvents demonstrate that sustainable media can offer more than ecological benefits. Their versatility across different mechanisms from radical to ROP highlights their potential to enhance process efficiency, selectivity, and functional group preservation, thereby contributing to the development of advanced materials under environmentally responsible conditions.

5.2. Green catalysts

Earlier studies on the development of heterogeneous catalytic systems for the polymerization of epoxide-containing monomers have highlighted the feasibility of replacing conventional homogeneous and metal-based catalysts with more sustainable, efficient, and recyclable alternatives. This shift is strongly driven by the principles of green chemistry, which aim to minimize environmental impact, reduce toxicity, and improve process sustainability through the use of non-toxic, abundant, and reusable catalytic materials [341,342]. In this context, increasing attention has been devoted to solid acid catalysts, particularly naturally occurring clays and modified mineral-based systems, which combine catalytic efficiency with low cost and environmental compatibility. These green catalytic systems have demonstrated promising performance in ROP offering improved selectivity, mild reaction conditions, and easy recovery from the reaction medium.

In this context, several heterogeneous catalysts have been reported, including carbon whiskers [343] that have been functionalized with acylium perchlorate groups. It has been demonstrated that these catalysts can initiate the polymerization of lactones, including such compounds as ε-caprolactone, δ-valerolactone, and β-propiolactone. Another example is Maghnite-H^+, a proton-exchanged montmorillonite clay, which is employed as a versatile catalyst for the ROP of various cyclic monomers, including ε-caprolactone [27], oxetane [344], styrene oxide [345], tetrahydrofuran [346,347], ethylene oxide [348] and octamethylcyclotetrasiloxane [349] either individually or in copolymerization systems. The seminal works of the preceding scholars provided valuable insights into the efficiency of solid acid catalysts and their potential to advance environmentally friendly polymerization

processes.

Extending this line of research, an alternative solvent-free approach for the ROP of GMA has been investigated using a heterogeneous and environmentally sustainable catalytic platform based on activated Maghnite clay. Maghnite naturally occurring montmorillonite-type clay, can be activated with different agents to generate acidic or Lewis-acidic sites [350,351]. In a recent unpublished study conducted by the second author, the ROP of GMA was carried out under solvent-free conditions with a green heterogeneous catalyst. Two types of Maghnite-based catalysts were prepared and compared (see Fig. 18). Maghnite-H^+, obtained by activation with sulfuric acid, introduces protonic acid sites, whereas Maghnite-Zn^{2+}, prepared via zinc chloride treatment, generates Lewis acid centers. Both catalysts were evaluated under identical solvent-free conditions with a loading of 20 wt% relative to GMA (0.2 g). This approach is consistent with the principles of green chemistry, as it avoids the use of organic solvents and enables solid-state polymerization. A significant disparity was observed in the catalytic performance of the two systems, manifesting in distinct patterns of reaction kinetics and regioselectivity. While Maghnite-H^+ provided a 69% yield of polyether after 72 h, the reaction catalyzed by Maghnite-Zn^{2+} reached a significantly higher 88% yield within just 24 h, highlighting its superior kinetic profile. The ^{1}H NMR analysis of the resulting polymers which are presented in characterization section (see Fig. 16) revealed discrepancies in the regioisomeric distribution, contingent on the utilized catalyst. The [CH_2OH)/[CHOH] ratio, as determined from the integration of characteristic proton signals (a and d), was 7:93 for the polymer obtained with Maghnite-H^+, confirming that approximately 93% of the product corresponds to majority isomer, For the polymer synthesized using Maghnite-Zn^{2+}, the ratio was 14:86, indicating that 86% of the polymer was the major regioisomer. These values indicate that both catalysts promote regioselective RO, with Maghnite-H^+ offering slightly higher selectivity toward the major isomer, whereas Maghnite-Zn^{2+} enables faster conversion and greater overall efficiency.

The results obtained provide compelling evidence for the use of heterogeneous clay-based catalysts in sustainable polymerization processes. The additive catalytic behavior exhibited by these two systems suggests a high degree of tunability in achieving a balanced, selective and productive output, contingent upon the specific application under consideration. In addition, the solvent-free reaction conditions, in conjunction with the utilization of recyclable and abundantly available catalysts, at room temperature substantiate the viability of this method as a green alternative for the preparation of polyethers from epoxide-functional monomers, such as GMA.

This green catalytic strategy employing proton-exchanged clays such as Maghnite-H^+ has been explored not only for the homopolymerization of GMA via epoxide RO, but also for the synthesis of a wide range of functionalized copolymers through cationic polymerization of the epoxy group. These reactions, generally carried out in solution, have involved the use of various nucleophiles to generate value-added materials. In related work, Souli et al. investigated the reactivity of GMA with two different anhydrides using Maghnite-H^+: the reaction with pyromellitic dianhydride [352] was performed in acetone, while the reaction with maleic anhydride [70] was carried out under solvent-free conditions. In both cases, the catalyst effectively promoted the RO esterification of the epoxy moiety, leading to the formation of regioselective polyesters. These results demonstrated not only the chemical efficiency of Maghnite-H^+ but also its adaptability to different green processing conditions. Additionally, Cherifi et al. [68] investigated the cationic copolymerization of the epoxide group of GMA with methacrylic anhydride using proton-exchanged Bentonite-H^+, a protonated clay mineral, as a heterogeneous catalyst. This reaction led to the formation of macromonomers bearing pendant methacrylate double bonds. The reaction proceeded efficiently, highlighting the potential of protonated clay minerals as cost-effective, reusable, and environmentally friendly catalysts for the synthesis of GMA-based macromonomers.

In general, the studies discussed in this section demonstrate that green heterogeneous catalysts, particularly proton-exchanged clay minerals, represent efficient and sustainable alternatives for the polymerization and copolymerization of GMA and other cyclic monomers. Their catalytic activity under mild or solvent-free conditions, combined with their low cost, recyclability, and environmental compatibility, highlights their strong potential for the development of greener and more sustainable polymerization processes toward functional polymeric materials.

5.3. *Bio-based catalysts*

The use of bio-based and bio-inspired systems as sustainable catalysts for the polymerization of GMA has attracted increasing attention in recent years, driven by the growing demand for greener and more environmentally benign polymerization technologies. These systems are derived from renewable resources or designed to mimic natural catalytic

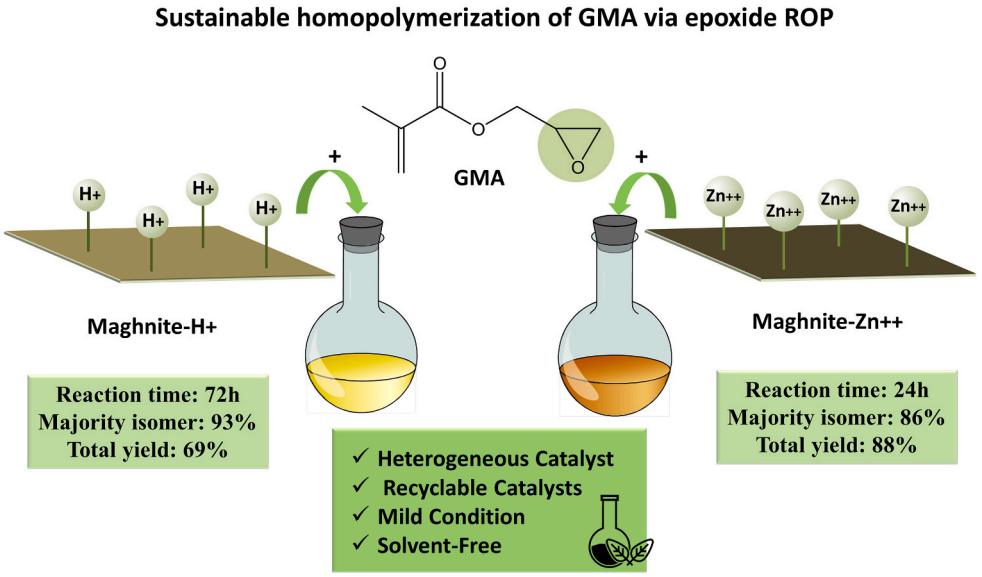


Fig. 18. Schematic representation of green GMA homopolymerization using Maghnite-H^+ and Maghnite-Zn^{2+}. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

processes, offering an attractive alternative to conventional metal-based or strongly acidic catalysts. In line with the principles of green chemistry, bio-based catalytic and catalytic-like systems aim to reduce toxicity, improve biodegradability, and enhance overall process sustainability while maintaining high efficiency and selectivity [353,354].

In this context, surface-initiated ARGET-ATRP of GMA from bacterial cellulose aerogels has been achieved using ascorbic acid (vitamin C) which serves as a bio-derived reducing agent to regenerate the active Cu (I)/ligand catalyst, thereby enabling the growth of well-defined poly (GMA) brushes with preserved epoxy groups [261].

As another strategy, enzyme-mediated RAFT polymerization in aqueous media has been reported, where an enzymatic redox system initiates the polymerization, leading to the formation of epoxy-functionalized poly(GMA)-containing triblock copolymer vesicles. This highlights the ability of enzymes to trigger radical processes under mild and environmentally benign conditions, while yielding functional nanostructures [355]. In a further bio-inspired approach, polydopamine chemistry has been combined with surface-initiated ARGET ATRP to graft dense poly(GMA) brushes from carbon nanotubes. Here, polydopamine acts as an adhesive and redox-active platform, enabling efficient grafting and generating epoxy-rich surfaces that can be further modified, providing enhanced composite performance such as increased uranium ion adsorption capacity [24,356].

In addition, riboflavin (vitamin B2) has emerged as a bio-based photo-initiator under visible light, offering an eco-friendly and biocompatible alternative to conventional photo-initiators for methacrylate curing [357]. Riboflavin functions by absorbing visible light and generating radicals under mild conditions, thereby avoiding cytotoxic by-products and making it particularly attractive for biomedical applications. Beyond its broad utility in polysaccharide [358] and protein-derived methacrylates [359], riboflavin has also been successfully employed in systems involving GMA. In one particular case, gelatin-GMA hydrogels combined with treated dentin matrix were photo-crosslinked with riboflavin and glycine under visible light, thereby producing stable and cytocompatible structures that are deemed suitable for dental tissue engineering [360]. In another example, riboflavin was used to initiate the crosslinking of hyaluronic acid–GMA composite hydrogels, which were designed for bone regeneration. The resulting biomaterials showed improved mechanical integrity and osteogenic performance [361]. The present studies illustrate the high potential of riboflavin-mediated photo-initiation to valorize GMA-based polymers in the field of regenerative medicine.

Collectively, these studies demonstrate the versatility of bio-based and bio-inspired systems whether acting as reducing agents (ascorbic acid), enzymatic redox catalysts (enzymes), adhesive and redox-active platforms (polydopamine), or photo-initiators (riboflavin) in enabling the controlled radical polymerization of GMA. Such advances underscore their crucial role in promoting sustainable and green polymer chemistry.

5.4. Sustainable routes toward GMA monomer production

While significant progress has been achieved in the development of sustainable polymerization processes, the environmental impact associated with monomer production remains comparatively underexplored. In this context, the synthesis of GMA deserves particular attention, as it is conventionally produced via industrial routes involving epichlorohydrin [362–365], which raise significant environmental and safety concerns due to the use of toxic reagents and the generation of hazardous chlorinated by-products. From a green chemistry perspective, these conventional routes suffer from poor atom economy, reliance on hazardous intermediates, and the generation of substantial waste streams, thereby highlighting the urgent need for safer and more sustainable alternatives.

In response to these limitations, more sustainable approaches for GMA synthesis have been developed, including enzymatic transesterification of methacrylate vinyl esters with glycidol in sponge-like ionic liquids as reaction media [366]. This process, typically conducted at around 60 °C for 6 h, enables high conversion and yields approaching 100%, while benefiting from mild reaction conditions, reduced solvent toxicity, and improved catalyst recyclability. Such strategies illustrate the effective implementation of green chemistry principles, including the use of benign solvents, energy-efficient conditions, and recyclable catalytic systems.

In parallel, increasing attention has been directed toward the sustainability of the monomer precursor, particularly glycidol, whose green production critically determines the overall environmental performance of GMA synthesis pathways. Significant advances have therefore been achieved in developing greener routes for glycidol production from glycerol, a renewable and abundant by-product of biodiesel production. As highlighted in recent studies, glycerol valorization into glycidol represents a key step in biorefinery strategies aimed at upgrading low-value feedstocks into high-value epoxide intermediates [367].

Among emerging technologies, continuous-flow processes have demonstrated clear advantages over conventional batch systems. In such configurations, reactants are continuously fed into the reactor while products are simultaneously removed, enabling precise control over residence time, temperature, and selectivity. This leads to enhanced heat and mass transfer, improved safety, and process intensification, which is particularly important for highly reactive epoxide intermediates such as glycidol and GMA. Sivo et al. [368] reported that continuous-flow glycerol upgrading to glycidol can achieve selectivities of ~80–90%, with glycerol conversions typically in the range of 70–85%, depending on reaction conditions and catalyst configuration.

In addition, gas-phase catalytic routes have emerged as highly attractive solvent-free alternatives. A one-step conversion of glycerol to glycidol has been demonstrated in packed-bed reactors over HZSM-5 zeolite catalysts modified with $CsNO_3$ [369]. In this system, glycerol is vaporized and converted through sequential dehydration and intramolecular cyclization steps. The introduction of cesium species plays a crucial role in moderating the strong Brønsted acidity of HZSM-5, thereby suppressing side reactions such as acrolein formation, coke deposition, and deep dehydration. Under optimized conditions, this system typically achieves glycidol selectivities of ~50–75%, with glycerol conversions often exceeding 60–80%.

A complementary gas-phase strategy employs $CsNO_3$-modified ZSM-5 zeolite catalysts in packed-bed reactors, where catalyst composition and alkali loading are finely tuned to optimize acid–base properties [370]. This balance is essential for selective glycerol activation toward epoxide formation while minimizing undesired cracking and polymerization pathways. Compared to unmodified zeolites, these systems exhibit improved stability, reduced deactivation by coking, and enhanced glycidol selectivity (>70%), along with sustained catalytic performance under continuous operation.

Overall, continuous-flow and gas-phase catalytic strategies represent efficient and more sustainable pathways for glycidol production. They align well with green chemistry principles through solvent elimination, improved energy efficiency, enhanced safety, and process intensification, making them promising candidates for industrial implementation. However, despite these advances, large-scale application remains limited. This gap between academic development and industrial deployment highlights the need for further efforts in process optimization, catalyst durability, and scale-up strategies. Bridging this gap is essential to accelerate the transition toward safer, more sustainable, and economically viable production processes for glycidol and, consequently, GMA.

6. Practical applications of poly(GMA)

Poly(GMA) and its functional derivatives have attracted significant attention due to the high reactivity of their pendant epoxy groups, which permit a wide range of chemical modifications. This tunable

functionality provides access to a wide spectrum of applications, including biomedical, environmental, industrial and electrochemical fields as illustrated in Fig. 19. This versatility arises not only from the ease with which the epoxy groups can be functionalized after polymerization, but also from the fact that GMA can participate in both radical and ring-opening polymerization pathways. Such dual reactivity allows fine adjustment of molecular structure and material properties according to the requirements of a given application.

The following subsections highlight recent progress in each application area, with particular attention to how controlled synthesis and structural adaptability enhance overall performance.

6.1. Biomedical applications

Poly(GMA) is a material that has found particular application in the biomedical field, due to its unique structure which combines a reactive epoxy group and a polymerizable methacrylate moiety. The dual functionality of the material under consideration provides accessible sites for post-polymerization modification, thereby enabling covalent immobilization of biomolecules, therapeutic agents, or targeting ligands, while maintaining the structural integrity of the polymer backbone. This tunability renders poly(GMA) a valuable platform for the design of multifunctional biomaterials. Poly(GMA) derivatives have been demonstrated to possess the capacity for self-assembly into nanostructures, including micelles, vesicles, and capsules. These morphologies have been exploited for drug and gene delivery as well as antimicrobial applications, due to their chemical reactivity and ability to encapsulate and release bioactive molecules in a controlled manner. The modular nature of these assemblies allows for the customisation of surface chemistry and degradation rates to suit specific biomedical requirements [224].

Recent advances have further demonstrated that the incorporation of fatty acids into GMA-based architectures enables the design of amphiphilic block copolymers with controlled self-assembly behavior [23]. In particular, oleic acid-containing block copolymers have been successfully synthesized using GMA chemistry, where the hydrophobic segment is derived from oleic acid and the hydrophilic block from 2-(dimethylamino)ethyl methacrylate. These materials spontaneously self-assemble in aqueous media into well-defined spherical nanoaggregates (up to ~100 nm), enabling efficient encapsulation and controlled release of bioactive molecules. Importantly, the post-polymerization modification strategy based on grafting oleic acid onto GMA-based polymers reduces undesired branching and crosslinking while providing scalable synthesis routes and enabling pH- and temperature-responsive behavior, making these systems promising candidates for smart drug delivery applications.

Beyond nanostructured delivery systems, GMA-based materials have also demonstrated significant relevance in bioanalytical and biochemical separation technologies, further highlighting their biomedical versatility. For example, zwitterionic GMA-divinylbenzene polymer-based stationary phases have been developed for hydrophilic interaction chromatography, enabling efficient separation of ionic and polar analytes, including simultaneous separation of anions and cations, which is highly relevant for metabolomic and clinical bioanalysis [371]. Similarly, weak anion-exchange monolithic columns derived from poly (GMA-*co*-ethylene dimethacrylate) have been employed for the separation of DNA fragments using high-performance liquid chromatography, offering high permeability, mechanical stability, and effective resolution of oligonucleotides, thus providing a valuable alternative to gel electrophoresis in genetic analysis [372]. Moreover, functionalized GMA-based resins have been explored for protein retention and separation, where post-modification with amine-containing reagents enables effective interaction with proteins, making these materials promising candidates for chromatographic purification in biotechnological applications [373].

In addition to separation-based bioanalytical systems, chemically functionalized poly(GMA) derivatives have been widely explored for targeted biomedical application. For instance, aminated poly(GMA), synthesized via RAFT polymerization, has been employed in cancer therapy as a nanocarrier for the co-delivery of doxorubicin and the

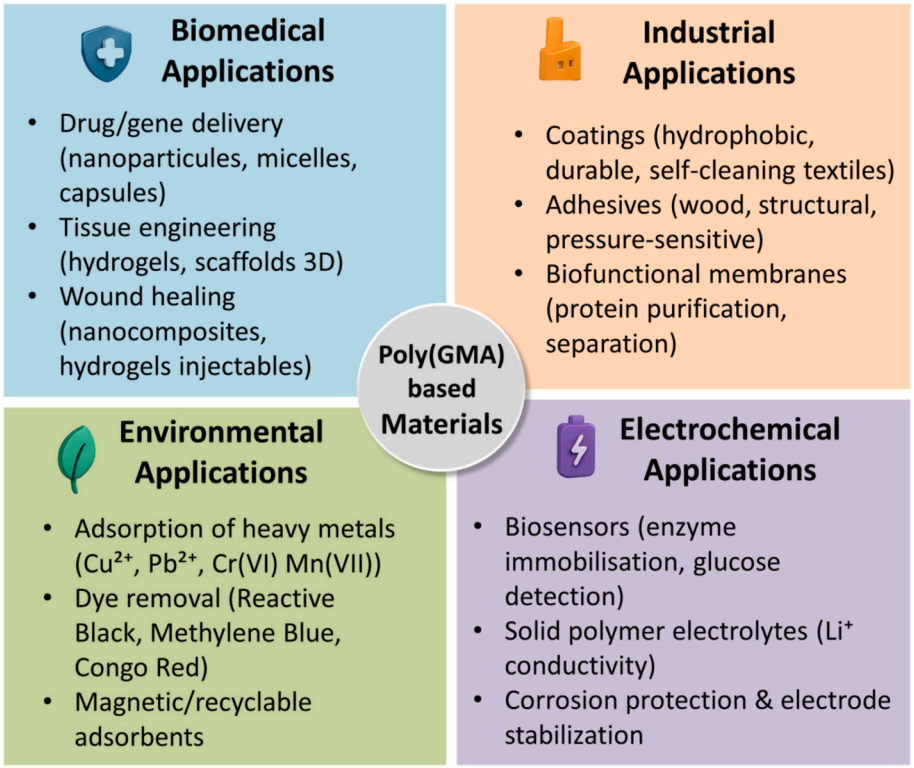


Fig. 19. Multifunctional applications of Poly(GMA) based materials.

tumour-suppressor gene p53, thereby significantly enhancing anti-tumour efficacy in vitro [374]. Similarly, multifunctional poly(GMA) microspheres functionalized with amine groups, Fe_3O_4 nanoparticles, and folic acid exhibit magnetic and fluorescent properties and a high affinity toward folate receptor–expressing cells (e.g., HeLa). The functionality of these features is such that they facilitate the targeted capture and magnetic separation of cancer cells under the influence of an external magnetic field [375].

Beyond oncology, poly(GMA)-based systems have shown great promise for regenerative medicine. Injectable hydrogels, based on GMA-modified poly(vinyl alcohol) or silk fibroin, have been developed for tissue engineering applications. These hydrogels exhibit the capacity for expeditious crosslinking when subjected to mild UV irradiation (365 nm, 6 mW/cm^2), resulting in the formation of networks characterized by gel fractions that exceed 80% [122]. The incorporation of cellulose nanofibers significantly enhances mechanical performance, increasing compressive strength from ~23 to ~127 kPa and stiffness from ~27 to ~130 kPa. This improvement is accompanied by enhanced proliferation of cartilage and stem/progenitor cells, as well as the formation of a microporous network with pore sizes ranging from 3 to 68 μm and porosity of 83–94%, providing a favorable environment for cell adhesion and nutrient transport [376]. Furthermore, gelatin modified with GMA has been widely used to fabricate photo-crosslinkable 3D hydrogel scaffolds. These scaffolds leverage the inherent bioactivity of gelatin and the chemical stability of methacrylate networks, exhibiting considerable promise in the field of tissue regeneration, including liver tissue engineering [162]. In parallel, dextran–GMA/gelatin nanoparticles have been engineered for the sustained delivery of bone morphogenetic protein. These offer a uniform size distribution, controlled release profile, and excellent cytocompatibility. These systems have been demonstrated to possess considerable potential for the purpose of promoting bone and periodontal tissue regeneration, a process that requires the sustained and localized availability of bone morphogenetic protein [377].

In the field of wound healing, poly(GMA)-based materials have demonstrated considerable potential in promoting cutaneous regeneration. For instance, the incorporation of gold nanoparticles into chitosan-graft-GMA nanocomposites has been demonstrated to expedite wound closure by integrating antibacterial activity, electrical conductivity, and regenerative capacity, thereby eliciting a synergistic effect on tissue repair [378]. Concurrently, an injectable hydrogel system based on oxidized hyaluronic acid–GMA/γ-polyglutamic acid–thiol has been developed, forming a stable in situ crosslinked network through Michael addition. The hydrogel under investigation has been demonstrated to promote fibroblast migration, neovascularization, and collagen deposition, thereby accelerating cutaneous wound healing in vivo [379]. Collectively, these studies highlight the versatility of GMA-modified biomaterials in designing multifunctional wound dressings with both structural and bioactive properties for effective skin tissue regeneration. Table 4 highlights some examples of poly(GMA) and derivatives applications in biomedical and healthcare applications.

The broad range of applications encompassing drug and gene delivery, tissue engineering, and wound healing emphasizes the versatility of poly(GMA) as a biomedical material. Its bifunctional structure provides a unique platform for post-polymerization modifications, enabling precise tailoring of chemical, mechanical, and biological properties. This adaptability allows the design of multifunctional systems that combine structural integrity with bioactivity, thereby positioning poly(GMA) and its derivatives as promising candidates for next-generation therapeutic and regenerative strategies.

Table 4
Applications of poly(GMA) and its derivatives in biomedical applications.

Sample	Considerations	Reference
Poly(N-isopropyl acrylamide-*co*-GMA)-coated magnetic nanoparticles	Good pH and temperature-dependent release behavior of doxorubicin. Impressive cytotoxicity against cancer HeLa cells.	[380]
N-isopropyl acrylamide -co-GMA-L-lysine hydrogel	Almost a complete curcumin release under combined temperature and pH (45 °C/pH 4.0) conditions. Compatibility proved by the MTT assay Potential use for green fluorescence-based bioimaging applications. Great potential for meniscus tissue engineering application.	[381]
Poly(vinyl alcohol) Grafted GMA/Cellulose Nanofiber Injectable Hydrogels	10%PVA-g-GMA/0.5%CNF and 10% PVA-g-GMA/0.7%CNF showed high cell viability of 81 and 78%, respectively, after culturing for 14 days.	[376]
Poly (vinyl alcohol)-g-GMA Hydrogels	Interest in biomedical applications, such as wound dressings and controlled release systems Minimal hemolytic activity for both Gel-GMA and Gel-GMA/GO hydrogels, Excellent hemocompatibility (0.54 and 0.50%, respectively).	[159]
Gelatin-GMA/graphene oxide composite hydrogels	No cytotoxicity over a wide range of concentrations. Promoted cell growth and proliferation. Potential of the hydrogels for applications such as vascular catheters.	[382]
Chitosan-g-GMA/AuNP	Greater skin wound healing capacity than a commercial film (TheraForm®) after 40 days. Presence of the collagen allows better cosmetic aspects in skin regeneration. Broad-spectrum antibacterial performance, with antibacterial rates over 90% against both *S. aureus* and *E. coli*.	[378]
Crosslinked cryogels (CS) based on GMA modified chitosan	Cytotoxicity and hemolysis assays confirmed the biocompatibility and hemocompatibility. The CS-based cryogels reduced blood loss in mice tail amputation models and accelerated tissue regeneration in full-thickness wound models.	[383]

6.2. Environmental applications

Poly(GMA) has received a great deal of attention in the field of environmental remediation as a result of its robust mechanical properties, such as high tensile strength and resistance to acidic and alkaline conditions, as well as its porous structure, which enables high adsorption capacities. The presence of reactive epoxy groups in GMA enables straightforward post-polymerization functionalization with amines, sulfonates, ureas, iminodiacetic acids, pyrazoles, and pyridine, for example, allowing for the creation of highly selective adsorbents for organic dyes, heavy or noble metals, and other pollutants. In light of this context, this section will provide a description of a select number of research studies conducted using poly(GMA) and its derivatives in the field of environmental remediation, intending to evaluate the value of incorporating GMA into these systems.

Poly(GMA) has been extensively utilized as a multifaceted precursor in the design of functional adsorbents, with the objective of targeting a range of water pollutants. In one study, the grafting of radiation-modified GMA onto bamboo fibres, followed by sulfonation, resulted in the production of a biosorbent that exhibited a notable affinity for copper ions [187]. This biosorbent achieved a maximum adsorption

capacity of 198 mg/g, as determined by the Langmuir model, and was able to remove up to 96% of Cu^{2+} under batch conditions. The adsorption process exhibited pseudo-second-order kinetics ($R^2 \approx 0.999$), indicating chemisorption as the predominant mechanism. Another study reported the use of amino-functionalized cellulose–poly(GMA) graft copolymers for the adsorption of anionic dyes [384]. The optimized system achieved maximum adsorption capacities of approximately 150 mg/g for Reactive Black 5 and 147 mg/g for Acid Black 194, thus confirming the Langmuir isotherm and obeying pseudo-second-order kinetics. This outcome is indicative of monolayer dye sorption on a homogeneously functionalized surface. More recently, crystal violet has been immobilized onto nano-sulfonated Poly (GMA) particles [385]. The nanocomposite demonstrated a high adsorption capacity for Cr(VI) (~110.78 mg/g, Langmuir model, $R^2 = 0.998$), exhibiting rapid kinetics and reusability. In a similar manner, methylene blue–immobilized sulfonated poly(GMA) composites have been created as multifunctional adsorbents [386]. These composites demonstrated adherence to the Freundlich isotherm, indicative of heterogeneous, multilayer adsorption. The kinetics followed pseudo-second order, exhibiting dual remediation capacity, effectively removing dyes and toxic anionic metal ions, such as Cr(VI) and Mn(VII), under mild conditions, supporting chemisorption as the dominant mechanism. In further research, magnetic poly (GMA)-g-poly(pyrrole) microspheres demonstrated an adsorption capacity of 502.5 mg/g for Congo Red, accompanied by exceptional recyclability (91.7% after five cycles) [387]. Moreover, GMA copolymers modified with chelating groups exhibited removal efficiencies of up to 85.6% and 77.8% for Cu(II) and Pb(II), respectively, when present in aqueous solutions [388]. In a recent study conducted by the first author, which is pending publication, a copolymer of GMA with maleic anhydride demonstrated a crystal violet adsorption efficiency of up to 100% within 60 min, corresponding to a removal capacity of approximately 200 mg/g under the tested conditions. This copolymer also exhibited strong affinity toward other cationic dyes, achieving 98.7% removal of methylene blue within 90 min. The PGM—H, which was precipitated in hexane as described in Section 4.4, was used in this study. The significance of this work lies in the finding that a simple GMA-based copolymer can provide highly efficient adsorption performance without the need for additional grafting or complex surface modifications to enhance its functionality. Collectively, these results demonstrate that GMA-based polymers, through tailored functionalization, can be engineered into high-performance adsorbents for the removal of dyes from aqueous solutions, underscoring their potential in advanced wastewater treatment applications. Collectively, these results demonstrate that GMA-based polymers, through tailored functionalization, can be engineered into high-performance adsorbents for the removal of dyes from aqueous solutions, underscoring their potential in advanced wastewater treatment applications.

Table 5 highlights some examples of poly(GMA) and derivatives applications in environmental applications..

6.3. Industrial applications

Poly(GMA) has been identified as a key component in the enhancement of performance across a range of industrial applications, attributable to the presence of reactive epoxide groups. To illustrate this point, consider the example of waterborne acrylic coatings. The incorporation of 15 wt% GMA into the resin led to a substantial enhancement in tensile strength, with a recorded increase of 78.6%. Concurrently, there was an improvement in hydrophobicity, as evidenced by an increase in the water contact angle from 87.7° to 99.5° [395]. In coating formulations where GMA is copolymerized with fluorinated methacrylates, the epoxy groups chemically bond to substrates, resulting in exceptional adhesion. The films exhibited high water repellency even after 10 peel tests, and their performance remained stable over a period of six months, demonstrating superb durability [396]. Extending these strategies to textile surface coatings, recent work reported the fabrication of hydrophobic cotton fabric via dispersion polymerization of poly(GMA) nanoparticles, which imparted self-cleaning properties without compromising mechanical flexibility. The treated fabrics exhibited durable water repellency, underscoring the broader potential of Poly (GMA)-based coatings and nanostructures in advanced and multifunctional surface treatments [397]. In particular, a novel approach has also been developed for biofunctional coatings, where poly(GMA) was grafted onto polybutylene terephthalate nonwovens using UV and heat. The resulting membranes formed uniform and conformable layers suitable for ion-exchange protein capture, exhibiting high binding capacities (up to 200 mg/g) and rapid adsorption kinetics, with equilibrium achieved within minutes. These findings highlight the promise of poly (GMA)-modified membranes as effective coating platforms for industrial bioseparations and protein purification [398].

The reaction between GMA and fatty acids has emerged as an efficient and sustainable strategy for the development of bio-based monomers and coating materials. In particular, oleic acid has been successfully functionalized with GMA through epoxide RO, leading to methacrylated derivatives that preserve the unsaturated fatty acid backbone. These monomers have been further processed via free-radical miniemulsion polymerization to obtain stable latex systems suitable for coating applications. The resulting films exhibited high conversion and improved mechanical properties, which were further enhanced through auto-oxidative curing of residual unsaturations within the polymer backbone [399]. In parallel, Can et al. [400] demonstrated the use of GMA–fatty acid derivatives and related oleic acid-based monomers) as reactive diluents in vinyl ester resin formulations. These bio-based components were introduced as partial replacements for styrene in bisphenol A and novolac-type vinyl ester systems, aiming to reduce volatile organic compound emissions while introducing additional functionalities such as flame retardancy. Although full substitution led to increased viscosity and reduced thermomechanical properties, optimized formulations containing both styrene and fatty acid–GMA derivatives exhibited balanced processability and mechanical performance suitable for liquid molding and coating applications. This approach has also been confirmed at the industrial level, where fatty acid–GMA adducts have been patented as reactive diluents for vinyl ester resins, enabling their use in gel coatings and composite systems with reduced volatile organic compound emissions and improved processing characteristics (including derivatives of oleic acid, stearic acid, lauric acid, linoleic acid, and fatty acid mixtures from vegetable oils such as safflower oil) [401].

In addition to coating and membrane-based applications, GMA-

Table 5
Applications of poly(GMA) and its derivatives in environment-related applications.

Sample	Considerations	Reference
Quaternary amine-functionalized polymeric nanosorbent resin	Optimum capacities of 47.26 mg/g and 66.83 mg/g for phenol and malathion, respectively.	[389]
Triethylamine hydrochloride functionalized cotton-poly (GMA)–graft copolymer	Q = 357.14 mg/g for C.I. Reactive Black 5	[384]
Imidazolium-based ionic liquid covalently grafted on poly (GMA) support	Perfluorooctanoic acid adsorption capacity from an aqueous solution = 769.23 mg/g	[390]
Poly(GMA) grafted sulfonamide based polystyrene resin with tertiary amine	Q = 1200 mg/g for Everzol red3BS	[391]
Thio-semi-carbazide functionalized Cellulose– poly (GMA)– graft copolymer	Q = 279.33 mg/g for Cd(II) Q = 505.50 mg/g for Hg(II) Q = 105.71 mg/g for acid fuchsin	[392]
Poly GMA/divinyl benzene	Q = 13.6 mg/g for malachite green	[393]
Diethylenetriamine-functionalised poly(GMA) hydrogels	Q = 196 mg/g for congo red	[394]

functionalized stationary phases have also been developed for chromatographic and bioseparation processes. For example, monodisperse poly(GMA-*co*-ethylene dimethacrylate) beads, after surface hydrolysis to introduce diol functionalities, have been employed as efficient stationary phases for high-performance liquid chromatography, exhibiting high stability and enhanced selectivity compared to silica-based materials, particularly under aqueous conditions. Such systems enable the separation of a wide range of analytes including isomers and polymers with fine structural differences, highlighting their robustness for industrial analytical applications [402].

In the area of adhesives, copolymers such as poly(GMA-co-butyl acrylate) have been demonstrated to exhibit enhanced cohesive strength and interfacial adhesion to metals, glass, and composites, a property attributable to the covalent interactions facilitated by epoxy groups. The aforementioned structural advantages are of particular pertinence within the automotive, aerospace and electronics industries, where the necessity for strong and durable bonding under mechanical and thermal stress is paramount [403]. Beyond these applications, several recent studies have highlighted the versatility of GMA-based adhesives in other industrial contexts. For instance, GMA grafted onto enzyme-treated soybean meal produced bio-based wood adhesives with markedly improved wet shear strength, providing a sustainable alternative for plywood manufacturing [404]. In a similar manner, emulsion copolymers of vinyl acetate, vinyl neodecanoate, and GMA were found to yield high-performance wood adhesives with enhanced resistance to moisture and heat. These adhesives were found to demonstrate superior performance in comparison to conventional polyvinyl acetate formulations [405]. In the field of structural adhesives, the incorporation of GMA into pressure-sensitive adhesives has enabled post-curing crosslinking reactions, significantly enhancing shear resistance while maintaining peel adhesion, a feature of particular relevance for packaging and electronics assembly [406].

The collective evidence from these studies, as referenced in this section, underscores the adaptability of GMA-based copolymers as pivotal constituents in the design of adhesives with adjustable properties, ranging from bio-based wood glues to high-performance structural and pressure-sensitive systems. Besides, a further approach concerns the application of GMA as a polymer compatibilizer. The incorporation of a compatibilizer tends to improve the interfacial adhesion as well as refine and stabilize the blend morphology [407]. GMA-containing copolymers are the most efficient species as a compatibilizer [408]. For example, the epoxy functionality of GMA can react with both hydroxyl and carbonyl end groups of poly(ethylene terephthalate) [409]. The application of ethylene-GMA copolymer is effective in increasing the interactions between recycled high-density polyethylene and recycled poly(ethylene terephthalate) [409]. Pracella et al. [410] studied the effect of chemical modification of fibres, as well as the addition of compatibilizers containing GMA reactive groups, on characteristics for polypropylene, polystyrene, poly(ethylene-vinyl acetate) composites containing hemp, cellulose, and oat. Yu et al. [411] focused on the assessment of the effect of GMA in (recycled linear low-density polyethylene /ethylene-octene copolymer)-g-(GMA-*co*- styrene) (RPGS) on the properties and microstructure of poly(lactic acid)/RPGS blends. The work highlights the reaction of the epoxy group of GMA with the terminal hydroxyl groups of PLA and the enhancement of mechanical and thermal properties with a notched impact strength and elongation at break reaching 10.182.4 J/m^2 and 231.7%, respectively, which represent 2.7 times and 30.2 times improvements over pure poly(lactic acid). Other works concern the fabrication of poly(lactide) melt blended with either polypropylene or a polypropylene based elastomer and ethylene–GMA–methyl acrylate terpolymer as compatibilizer through coupling to the end groups of poly (lactic acid) [412], poly(styrene-*co*-GMA)-co-(maleic anhydride) ternary copolymers containing both reactive epoxide groups and maleic anhydride groups incorporated into the immiscible poly(amide 11)/poly (l-lactic acid) blend [413], recycled polypropylene/poly(ethylene terephthalate) blends nanocomposites with polypropylene-*graft*-maleic anhydride [414] and poly(lactic acid)/polybutylene adipate co-terephthalate blend and its nanocomposites prepared using melt blending technique and GMA as a reactive compatibilizer [415].

6.4. Electrochemical applications

Poly(GMA) has recently attracted interest in the field of electrochemical applications, owing to the high reactivity of its pendant epoxy groups. These groups provide accessible sites for post-functionalization with redox-active molecules or biomolecules. Surface-attached poly (GMA) brushes have been exploited as versatile platforms for dual-functionalization, thereby enabling the immobilization of moieties such as ferrocene and nitrobenzene through epoxide RO reactions. The modified brushes exhibited efficient and stable electron-transfer kinetics ($k \approx 1.2\ s^{-1}$) with a surface density of 0.19 nmol cm^{-2}, independent of film thickness which demonstrates their reliability as electroactive coatings [416]. Furthermore, poly(GMA)-based films have been extensively utilized in the field of electrochemical biosensors, where their reactive epoxy groups facilitate covalent immobilization of enzymes and proteins while preserving their biological activity. For instance, silicon substrates grafted with poly(GMA) via surface-initiated ATRP were functionalized with glucose oxidase through an epoxide–amine coupling strategy, leading to high enzyme loadings of $\approx$ 0.2 mg·cm^{-2} and retention of ~55–65% of enzymatic activity relative to the free enzyme. The aforementioned characteristics were found to result in the production of stable and reproducible amperometric signals, thus demonstrating the suitability of poly(GMA)-modified surfaces for the sensitive detection of glucose [417]. Building on this strategy, block copolymer brushes integrating ferrocene-containing segments with poly(GMA) have been engineered to further enhance biosensor performance. Such hybrid architectures enabled the fabrication of glucose sensors with high sensitivities (up to 10.9 μA·(mM·cm^2)$^{-1}$), low detection limits (~0.4 mM), and rapid response times (< 20 s), highlighting the dual role of poly(GMA) in providing enzyme anchoring sites and supporting efficient electron transfer [418].

Beyond biosensing, the versatility of poly(GMA) extends to protective and energy-related materials. Poly(GMA)-based coatings and composites have shown promise for corrosion protection and electrode stabilization, where their crosslinking capability and tunable surface chemistry enable the formation of robust electrode/electrolyte interfaces with improved electrochemical stability [419]. In this context, GMA-based copolymers have also been explored as solid polymer electrolytes. For instance, poly(GMA-*co*-methyl methacrylate) doped with $LiClO_4$ exhibited a substantial enhancement in ionic conductivity, from approximately 10^{-12} S cm^{-1} (undoped) to approximately 8.7×10^{-6} S cm^{-1} at room temperature, thus substantiating their viability for utilization in solid-state electrochemical devices. This enhancement is attributable to the polar epoxy groups of GMA, which provide coordination sites for Li^+ ions and improve salt solubility. Concurrently, these groups permit crosslinking reactions that reinforce the mechanical stability of the polymer matrix [420]. Interestingly, while biosensing poly (GMA) primarily serves as a reactive platform for enzyme immobilization and electron-transfer mediation, in energy storage applications its epoxy-derived polar functionalities play a different role by enhancing salt dissociation and ion transport within the polymer matrix.

Overall, the multifunctionality of poly(GMA) renders it a highly adaptable material in the context of electrochemical technologies. The epoxy groups present in the molecule offer a versatile chemical handle that can be tailored for different applications. These include enzyme immobilization and electron-transfer mediation in biosensing platforms, as well as enhancing interfacial stability and ionic conductivity in energy-related systems (Table 6). This distinctive duality underscores poly(GMA)'s aptitude for integrating biological and energy-related applications, consequently establishing it as a prospective contender for the subsequent generation of multifunctional electroactive coatings, biosensors, and solid polymer electrolytes.

Table 6
Applications of poly(GMA) and its derivatives in electrochemical applications.

Sample	Considerations	Reference
p(GMA-co-vinylferrocene)/ reduced graphene oxide	Use for electrochemical amperometric biosensor electrode construction. Sensitivity (nA/mM) = 27.44 Detection limit = 0.020 nM	[421]
poly(vinylidene fluoride) /linear methyl methacrylate-co-GMA copolymer) PVDF/MG	Excellent dielectric and energy storage properties. Higher breakdown strength, which increased from 300 MV/m for PVDF to 540 MV/m for the PVDF/MG-40% composite. The discharge energy density increased from 3.75 J/cm^3 for PVDF to 9.2 J/cm^3 for the PVDF/MG-40% composite.	[422]
Tri-armed star poly(GMA) polymer-coated indium tin oxide-polyethylene terephthalate electrode	High efficiency of 84% even at a 540 MV/m electric field for PVDF/MG-40% Use for the determination of neuron-specific enolase (NSE) cancer biomarker. NSE concentration was determined in the range from 0.03 pg/mL to 6 pg/mL with an ultralow detection limit of 9.1 fg/mL. Application as an electrochemical formaldehyde biosensor.	[423]
poly(GMA-co-3-methylthienyl methacrylate)/formaldehyde dehydrogenase/ polypyrroleelectrode	Sensitivity = 15,000 per mmol L^{-1} (500 nA ppm^{-1}) in a linear range between 0.1 ppm and 3 ppm (3.3–100 μmol L^{-1}). A minimum detectable concentration of 4.5 ppb (0.15 μmol L^{-1}) (S/N = 3) with a relative standard deviation (RSD) of 0.73% (n = 5) was obtained. Use for the detection of cortisol in human sweat.	[280]
poly(GMA-co ethylene glycol dimethacrylate MIP-based capacitive sensor	Rapid in analyte response (3 min) The sensor has a limit of detection (LOD) of 2.0 ng/mL ± 0.4 ng/mL, a dynamic range of 10–66 ng/mL, and a sensor reproducibility of 2.6% relative standard deviation (RSD). Photoelectrochemical cell application.	[424]
Plasticized UV-curable GMA-based solid polymer electrolyte	Highest conductivity at room temperature and 373 K with the values of 3.4 × 10^{-4} S cm^{-1} and 1.2 × 10^{-3} S cm^{-1} at 80 wt% of ethyl carbonate.	[425]
Polymer electrolyte based on Poly(GMA-*co*-ethyl methacrylate)	Highest room temperature conductivity obtained at approximately 2.8 × 10^{-5} Scm^{-1} at 30 wt% $LiClO_4$. Poly(GMA-*co*-ethyl methacrylate)- 30 wt% $LiClO_4$ showed good electrochemical stability (±5v).	[426]

7. Conclusion

In conclusion, this review provides a comprehensive mechanistic overview of GMA polymerization, highlighting the fundamental differences between ionic, radical, controlled, and externally stimulated pathways. The bifunctional nature of GMA, combining a methacrylate double bond with a reactive epoxide group, governs its complex reactivity and strongly influences polymerization behavior, macromolecular architecture, and functional group preservation.

Ionic polymerization proceeds through cationic and anionic pathways, where monomer-activated anionic systems generally provide superior control over molecular weight and effectively suppress side reactions. In contrast, cationic activated monomer (AM) processes offer faster kinetics and simpler experimental operation, but with reduced structural precision. These mechanistic differences directly impact the degree of control achievable over the resulting polymer structures. ESRP approaches offer milder conditions, improved temporal control, and a greener alternative to conventional FRP, with plasma-initiated processes being particularly effective for GMA due to efficient surface activation while preserving epoxy groups.

Within RDRP techniques, RAFT emerges as the most suitable approach for GMA due to its excellent functional group tolerance, metal-free nature, and strong control over molecular weight and architecture. ESCRP further enhances RDRP strategies by combining external energy inputs with reversible deactivation mechanisms, thereby improving performance and overcoming some inherent limitations. However, although ATRP and NMP can be effective within ESCRP-type strategies, their application to GMA remains limited, and further dedicated studies are still required to fully optimize their use.

Importantly, green polymerization approaches involving the use of green solvents, green catalysts, and bio-based catalysts have demonstrated promising results in terms of sustainability, reduced environmental impact, and improved process safety, while still maintaining effective control over polymerization pathways. Finally, the application scope of GMA-based polymers has been significantly expanded, extending from industrial coatings and environmental materials to electrochemical systems and biomedical applications, highlighting their growing importance in advanced functional material design.

8. Future perspectives

Future research on poly(GMA) is expected to progress toward more mechanistically driven, sustainable, and application-oriented strategies. A first key direction concerns the development of fully green and metal-free controlled polymerization systems, where reversible deactivation radical polymerization could be further optimized using bio-derived redox mediators, organocatalysts, or catalyst-free externally stimulated processes. Particular attention should be given to improving control over GMA reactivity while minimizing side reactions associated with the epoxide functionality under environmentally benign conditions, including solvent-free or aqueous media.

A second important avenue lies in the integration of radical polymerization with epoxide RO chemistry in a single, synergistic process. A deeper mechanistic understanding of this coupling could enable one-pot synthesis of multifunctional polymers, where chain growth and post-functionalization occur simultaneously in a controlled manner, significantly improving synthetic efficiency and structural complexity.

Third, although ESCRP, including plasma-, photo-, electrochemical, and other triggered systems, has already been widely explored in the literature, its application to GMA remains not fully optimized. Further efforts are therefore required to achieve precise temporal and spatial control over radical generation specifically in GMA systems, particularly to better preserve epoxy functionality and control interfacial reactions. This could further expand opportunities for surface patterning, interface engineering, and high-resolution functional grafting of GMA-based materials.

In addition, although ATRP and NMP have demonstrated potential for controlled polymerization, their specific adaptation to GMA systems remains underdeveloped, particularly regarding catalyst-related effects and epoxy group stability. Systematic mechanistic investigations are therefore still required to fully exploit their potential and expand their applicability.

Finally, the integration of data-driven approaches, multiscale modelling, and advanced in situ characterization techniques is expected to play an increasing role in predicting structure–property relationships and guiding the rational design of next-generation GMA-based functional materials.

CRediT authorship contribution statement

Hadjer Chikh: Writing – original draft, Writing – review & editing, Conceptualization, Investigation, Visualization. **Zakaria Cherifi:** Project administration, Writing – review & editing, Conceptualization, Validation, Supervision. **Gianluca Viscusi:** Writing – review & editing, Investigation, Supervision. **Derdar Hodhaifa:** Writing – review & editing, Investigation. **Chebout Redouane:** Project administration, Validation. **Bachari Khaldoun:** Investigation, Validation. **Rachid Meghabar:** Investigation, Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] P. Kubisa, S. Penczek, Cationic activated monomer polymerization of heterocyclic monomers, Prog. Polym. Sci. 24 (10) (1999) 1409–1437, https://doi.org/10.1016/S0079-6700(99)00028-3.

[2] K. Ito, Polymeric design by macromonomer technique, Prog. Polym. Sci. 23 (4) (1998) 581–620, https://doi.org/10.1016/S0079-6700(97)00049-X.

[3] K. Matyjaszewski, Macromolecular engineering: from rational design through precise macromolecular synthesis and processing to targeted macroscopic material properties, Prog. Polym. Sci. 30 (8–9) (2005) 858–875, https://doi.org/10.1016/j.progpolymsci.2005.06.004.

[4] I. Cianga, Y. Yagci, New polyphenylene-based macromolecular architectures by using well defined macromonomers synthesized via controlled polymerization methods, Prog. Polym. Sci. 29 (5) (2004) 387–399, https://doi.org/10.1016/j.progpolymsci.2004.01.005.

[5] J. Kalal, F. Švec, V. Maroušek, Reactions of epoxide groups of glycidyl methacrylate copolymers, J. Polym. Sci. Polym. Symp. 47 (1) (1974) 155–166, https://doi.org/10.1002/polc.5070470120.

[6] G.F. Chen, F.N. Jones, Synthesis of acrylic macromonomers by free-radical-initiated polymerization. Conversion to comblike copolymers, Macromolecules 24 (9) (1991) 2151–2155, https://doi.org/10.1021/ma00009a002.

[7] A. Labbé, A.-L. Brocas, E. Ibarboure, T. Ishizone, A. Hirao, A. Deffieux, et al., Selective ring-opening polymerization of glycidyl methacrylate: toward the synthesis of cross-linked (co) polyethers with thermoresponsive properties, Macromolecules 44 (16) (2011) 6356–6364, https://doi.org/10.1021/ma201075n.

[8] A.V. Reis, A.R. Fajardo, I.T. Schuquel, M.R. Guilherme, G.J. Vidotti, A.F. Rubira, et al., Reaction of glycidyl methacrylate at the hydroxyl and carboxylic groups of poly (vinyl alcohol) and poly (acrylic acid): is this reaction mechanism still unclear? J. Organomet. Chem. 74 (10) (2009) 3750–3757, https://doi.org/10.1021/jo900033c.

[9] J. Herzberger, K. Niederer, H. Pohlit, J. Seiwert, M. Worm, F.R. Wurm, et al., Polymerization of ethylene oxide, propylene oxide, and other alkylene oxides: synthesis, novel polymer architectures, and bioconjugation, Chem. Rev. 116 (4) (2016) 2170–2243, https://doi.org/10.1021/acs.chemrev.5b00441.

[10] M.I. Childers, J.M. Longo, N.J. Van Zee, A.M. LaPointe, G.W. Coates, Stereoselective epoxide polymerization and copolymerization, Chem. Rev. 114 (16) (2014) 8129–8152, https://doi.org/10.1021/cr400725x.

[11] Y. Sarazin, J.-F. Carpentier, Discrete cationic complexes for ring-opening polymerization catalysis of cyclic esters and epoxides, Chem. Rev. 115 (9) (2015) 3564–3614, https://doi.org/10.1021/acs.chemrev.5b00033.

[12] I. Erol, H. Akbıyık, Kinetic parameters, thermal stability, biological activity, and dielectric properties of new methacrylate-based copolymers functionalized with methylparaben, J Polym Res 29 (3) (2022) 93, https://doi.org/10.1007/s10965-022-02950-6.

[13] I. Erol, D.N. Devrim, H. Ciftci, B. Ersoy, I.H. Cigerci, Novel functional copolymers based on glycidyl methacrylate: synthesis, characterization, and polymerization kinetics, J. Macromol. Sci. A 54 (7) (2017) 434–445, https://doi.org/10.1080/10601325.2017.1320747.

[14] I. Erol, A. Güldiken, Synthesis of novel functionalized methacrylate copolymers and their copolymerization kinetics, thermal stability, and biocidal properties, J. Appl. Polym. Sci. 138 (45) (2021) 51334, https://doi.org/10.1002/app.51334.

[15] İ. Erol, M. Özer, Copolymers of a novel amphiphilic methacrylate monomer based on the hydroxyl group: copolymerization kinetics, thermal properties, biological activity, and swelling behavior, J Polym Res 28 (10) (2021) 372, https://doi.org/10.1007/s10965-021-02712-w.

[16] Preparation of random and graft copolymers of GMA and MMA using reactive melt mixing and solution polymerization, in: A. Karimi, H. Garmabi, A. Javadi, M. Ahmadi (Eds.), 7th International Conference on Composites: Characterization, Fabrication and Application (CCF a-7), Amirkabir University, 2020.

[17] K. Sankar, V. Rajendran, Ultrasound assisted free radical polymerization of glycidyl methacrylate by a new disite phase-transfer catalyst system: a kinetic study, Ultrason. Sonochem. 19 (6) (2012) 1205–1212, https://doi.org/10.1016/j.ultsonch.2012.01.015.

[18] T.G. McKenzie, F. Karimi, M. Ashokkumar, G.G. Qiao, Ultrasound and sonochemistry for radical polymerization: sound synthesis, Chem. Eur. J. 25 (21) (2019) 5372–5388, https://doi.org/10.1002/chem.201803771.

[19] C. Hu, R. Ruan, W. Wang, A. Gao, L. Xu, Electrochemical grafting of poly (glycidyl methacrylate) on a carbon-fibre surface, RSC Adv. 10 (18) (2020) 10599–10605, https://doi.org/10.1039/D0RA00562B.

[20] N.V. Tsarevsky, K. Matyjaszewski, "Green" atom transfer radical polymerization: from process design to preparation of well-defined environmentally friendly polymeric materials, Chem. Rev. 107 (6) (2007) 2270–2299, https://doi.org/10.1021/cr050947p.

[21] D. Grishin, I. Kriulichev, M. Pavlovskaya, I. Grishin, Well-defined poly (stearyl methacrylate-co-glycidyl methacrylate) obtained by ATRP as a complex additive for hydrotreated diesel fuel, Chem. Pap. 78 (2) (2024) 897–907, https://doi.org/10.1007/s11696-023-03129-z.

[22] A. Simula, N. Ballard, M. Aguirre, J.R. Leiza, S. van Es, J.M. Asua, Nitroxide mediated copolymerization of acrylates, methacrylates and styrene: the importance of side reactions in the polymerization of acrylates, Eur. Polym. J. 110 (2019) 319–329, https://doi.org/10.1016/j.eurpolymj.2018.11.041.

[23] C.P. Gomes, M.R.P. Costa, R.C. Dias, Reaction routes with glycidyl methacrylate for the incorporation of oleic acid in amphiphilic block copolymers, React. Funct. Polym. (2025) 106492, https://doi.org/10.1016/j.reactfunctpolym.2025.106492.

[24] Y. Song, G. Ye, Y. Lu, J. Chen, J. Wang, K. Matyjaszewski, Surface-initiated ARGET ATRP of poly (glycidyl methacrylate) from carbon nanotubes via bioinspired catechol chemistry for efficient adsorption of uranium ions, ACS Macro Lett. 5 (3) (2016) 382–386, https://doi.org/10.1021/acsmacrolett.6b00099.

[25] M. Barsbay, Y. Kodama, O. Güven, Functionalization of cellulose with epoxy groups via γ-initiated RAFT-mediated grafting of glycidyl methacrylate, Cellulose 21 (6) (2014) 4067–4079, https://doi.org/10.1007/s10570-014-0416-y.

[26] P. Farquet, C. Padeste, H.H. Solak, S.A. Gürsel, GnG. Scherer, A. Wokaun, Extreme UV radiation grafting of glycidyl methacrylate nanostructures onto fluoropolymer foils by RAFT-mediated polymerization, Macromolecules 41 (17) (2008) 6309–6316, https://doi.org/10.1021/ma800202b.

[27] M. Grochowicz, P. Pączkowski, B. Gawdzik, Investigation of the thermal properties of glycidyl methacrylate–ethylene glycol dimethacrylate copolymeric microspheres modified by Diels–Alder reaction, J. Therm. Anal. Calorim. 133 (1) (2018) 499–508, https://doi.org/10.1007/s10973-017-6785-3.

[28] C. Soykan, H. Obuz, Spectroscopic characterization of glycidyl methacrylate with acrylonitrile copolymers and monomer reactivity ratios, Microsc. Res. Tech. 83 (1) (2020) 22–34, https://doi.org/10.1002/jemt.23384.

[29] P. Pączkowski, B. Gawdzik, Studies on preparation, characterization and application of porous functionalized glycidyl methacrylate-based microspheres, Materials 14 (6) (2021) 1438, https://doi.org/10.3390/ma14061438.

[30] L.C. Costa, R.C. Monteiro, H.M. Castro, T.S. Ribeiro, M.A. Oliveira, E.C. Torquato, et al., Glycidyl methacrylate-ethylene glycol dimethacrylate copolymers with varied pore structures prepared with different reaction parameters, Mater. Res. 23 (3) (2020) e20190550, https://doi.org/10.1590/1980-5373-MR-2019-0550.

[31] M.A. Meier, J.O. Metzger, U.S. Schubert, Plant oil renewable resources as green alternatives in polymer science, Chem. Soc. Rev. 36 (11) (2007) 1788–1802, https://doi.org/10.1039/B703294C.

[32] P. Anastas, N. Eghbali, Green chemistry: principles and practice, Chem. Soc. Rev. 39 (1) (2010) 301–312, https://doi.org/10.1039/B918763B.

[33] N. Bektenov, K. Sadykov, A. Baidullayeva, N. Chinibayeva, T. Chalov, G. Koszhanova, et al., Vermiculite modified with glycidyl methacrylate, acrylonitrile, and phosphoric acid for the adsorption of molybdenum and rhenium ions from aqueous solutions, Processes 13 (5) (2025) 1584, https://doi.org/10.3390/pr13051584.

[34] M. Karabıyık, G. Cihanoğlu, Ö. Ebil, CVD deposited epoxy copolymers as protective coatings for optical surfaces, Polymers 15 (3) (2023) 652, https://doi.org/10.3390/polym15030652.

[35] S. Mao, D. Zhang, X. He, Y. Yang, I. Protsak, Y. Li, et al., Mussel-inspired polymeric coatings to realize functions from single and dual to multiple

antimicrobial mechanisms, ACS Appl. Mater. Interfaces 13 (2) (2021) 3089–3097, https://doi.org/10.1021/acsami.0c16510.
[36] F. Khelifa, H.M. Ehmann, A. Lamaoui, M. Surin, J. Rubio-Magnieto, P. Dubois, et al., Controlled stratification and assembly of cellulose nanocrystals in polymeric films toward optically active coatings, Adv. Mater. Interfaces 12 (7) (2025) 2400608, https://doi.org/10.1002/admi.202400608.
[37] Y. Gao, D. Zhou, J. Lyu, S. A, Q. Xu, B. Newland, et al., Complex polymer architectures through free-radical polymerization of multivinyl monomers, Nat. Rev. Chem. 4 (4) (2020) 194–212, https://doi.org/10.1038/s41570-020-0170-7.
[38] S. Wang, M.W. Urban, Self-healing polymers, Nat. Rev. Mater. 5 (8) (2020) 562–583, https://doi.org/10.1038/s41578-020-0202-4.
[39] G.W. Coates, Y.D. Getzler, Chemical recycling to monomer for an ideal, circular polymer economy, Nat. Rev. Mater. 5 (7) (2020) 501–516, https://doi.org/10.1038/s41578-020-0190-4.
[40] E.M. Muzammil, A. Khan, M.C. Stuparu, Post-polymerization modification reactions of poly (glycidyl methacrylate) s, RSC Adv. 7 (88) (2017) 55874–55884, https://doi.org/10.1039/C7RA11093F.
[41] T. Hansen, A. Nin-Hill, J.D. Codée, T.A. Hamlin, C. Rovira, Rational tuning of the reactivity of three-membered heterocycle ring openings via SN2 reactions, Chem. Eur. J. 28 (60) (2022) e202201649, https://doi.org/10.1002/chem.202201649.
[42] R.-E. Parker, N. Isaacs, Mechanisms of epoxide reactions, Chem. Rev. 59 (4) (1959) 737–799, https://doi.org/10.1021/cr50028a006.
[43] T. Hansen, P. Vermeeren, A. Haim, M.J. van Dorp, J.D. Codée, F.M. Bickelhaupt, et al., Regioselectivity of epoxide ring-openings via SN2 reactions under basic and acidic conditions, Eur. J. Org. Chem. 2020 (25) (2020) 3822–3828, https://doi.org/10.1002/ejoc.202000590.
[44] M.H. Nasirtabrizi, S. Khodabandlou, L. Zargin, Jadid A. Parchehbaf, Modification of copolymers using nucleophilic reactions between glycidyl methacrylate and 9-anthracene carboxylic acid, Int. J. Ind. Chem. 5 (1) (2014) 6, https://doi.org/10.1007/s40090-014-0006-8.
[45] A. Duda, A. Kowalski, Thermodynamics and kinetics of ring-opening polymerization, in: Handbook of Ring-Opening Polymerization, 2009, pp. 1–51, https://doi.org/10.1002/9783527628407.
[46] T. Dudev, C. Lim, Ring strain energies from ab initio calculations, J. Am. Chem. Soc. 120 (18) (1998) 4450–4458, https://doi.org/10.1021/ja973895x.
[47] R. Klein, F.R. Wurm, Aliphatic polyethers: classical polymers for the 21st century, Macromol. Rapid Commun. 36 (12) (2015) 1147–1165, https://doi.org/10.1002/marc.201500013.
[48] R. Beaumont, B. Clegg, G. Gee, J. Herbert, D. Marks, R. Roberts, et al., Heat capacities of propylene oxide and of some polymers of ethylene and propylene oxides, Polymer 7 (8) (1966) 401–417, https://doi.org/10.1016/0032-3861(66)90055-3.
[49] I. Auzmendi-Murua, S. Charaya, J.W. Bozzelli, Thermochemical properties of methyl-substituted cyclic alkyl ethers and radicals for oxiranes, oxetanes, and oxolanes: C–H bond dissociation enthalpy trends with ring size and ether site, J. Phys. Chem. A 117 (2) (2013) 378–392, https://doi.org/10.1021/jp309775h.
[50] S. Aoki, Y. Harita, Y. Tanaka, H. Mandai, T. Otsu, Relative reactivities of cyclic ethers in cationic copolymerizations: effects of ring strain and basicity, J. Polym. Sci. 6 (9) (1968) 2585–2590, https://doi.org/10.1002/pol.1968.150060914.
[51] H. Shi, Z. Zhang, Y. Wang, Mechanism on epoxidation of alkenes by peracids: a protonation-promoted pathway and its quantum chemical elucidation, J. Mol. Catal. A Chem. 238 (1–2) (2005) 13–25, https://doi.org/10.1016/j.molcata.2005.04.046.
[52] W. Huang, Y. Zhou, D. Yan, Direct synthesis of amphiphilic block copolymers from glycidyl methacrylate and poly (ethylene glycol) by cationic ring-opening polymerization and supramolecular self-assembly thereof, J. Polym. Sci. 43 (10) (2005) 2038–2047, https://doi.org/10.1002/pola.20688.
[53] V. Byrikhin, A. Nesmelov, V. Murachev, E. Ezhova, A. Orlinkov, I. Akhrem, Cationic polymerization of hydrocarbon monomers under the action of acyl halide complexes with Lewis acids: 4. Kinetics of cationic polymerization of isobutylene induced by the AcBr· 2AlBr3 complex, Russ. Chem. Bull. 42 (5) (1993) 837–840, https://doi.org/10.1007/BF00698941.
[54] A.V. Orlinkov, I.S. Akhrem, M.E. Vol'pin, The structure of equimolar complexes of acyl halides with Lewis acids, Russ. Chem. Rev. 60 (5) (1991) 524, https://doi.org/10.1070/RC1991v060n05ABEH001090.
[55] B. Kaur, P. Singh, Epoxides: developability as active pharmaceutical ingredients and biochemical probes, Bioorg. Chem. 125 (2022) 105862, https://doi.org/10.1016/j.bioorg.2022.105862.
[56] R.A. Olofson, R. Kendall, Protection by acylation in the selective alkylation of heterocycles, J. Organomet. Chem. 35 (7) (1970) 2246–2248, https://doi.org/10.1021/jo00832a031.
[57] E.J. Moore, W.R. Pretzer, T.J. O'Connell, J. Harris, L. LaBounty, L. Chou, et al., Catalytic and regioselective acylation of aromatic heterocycles using carbon monoxide and olefins, J. Am. Chem. Soc. 114 (14) (1992) 5888–5890, https://doi.org/10.1021/ja00040a078.
[58] Y. Yamashita, S. Kozawa, M. Hirota, K. Chiba, H. Matsui, A. Hirao, et al., The mechanism of the initiation reaction of the cationic polymerization of tetrahydrofuran by stable cationic salts, Macromol. Chem. Phys. 142 (1) (1971) 171–181, https://doi.org/10.1002/macp.1971.021420115.
[59] H. Meerwein, G. Hinz, P. Hofmann, E. Kroning, E. Pfeil, Über tertiäre oxoniumsalze, I, J. Prakt. Chem. 147 (10−12) (1937) 257–285, https://doi.org/10.1002/prac.19371471001.
[60] G.A. Olah, J.A. Olah, J.J. Svoboda, Preparative carbocation chemistry VI: advantageous stable and soluble fluorophosphate salts: trialkyloxonium, trialkylcarboxonium, and dialkoxycarbenium hexafluorophosphates, in: Across Conventional Lines: Selected Papers of George a Olah Volume 2, World Scientific, 2003, pp. 880–882.
[61] V. Murachev, E. Ezhova, A. Nesmelov, V. Byrikhin, A. Orlinov, I. Akhrem, Cationic polymerization of hydrocarbon monomers induced by complexes of acyl halides with Lewis acids: the polymerization of isobutylene in the presence of superacids, Russ. Chem. Bull. 46 (5) (1997) 934–938, https://doi.org/10.1007/BF02494749.
[62] G. Odian, Principles of Polymerization, John Wiley & Sons, 2004.
[63] F.E. Du Prez, E.J. Goethals, R. Hoogenboom, Cationic polymerizations, Handb. Poly. Synth. Char. Process. (2013) 163–185, https://doi.org/10.1002/9781118480793.ch8.
[64] P.H. Plesch, The Chemistry of Cationic Polymerization, Elsevier, 2016.
[65] G. Ranoux, G. Tataru, X. Coqueret, Cationic curing of epoxy–aromatic matrices for advanced composites: the assets of radiation processing, Appl. Sci. 12 (5) (2022) 2355, https://doi.org/10.3390/app12052355.
[66] P. Kubisa, Hyperbranched polyethers by ring-opening polymerization: contribution of activated monomer mechanism, J. Polym. Sci. 41 (4) (2003) 457–468, https://doi.org/10.1002/pola.10605.
[67] S. Penczek, Cationic ring-opening polymerization (CROP) major mechanistic phenomena, J. Polym. Sci. 38 (11) (2000) 1919–1933, https://doi.org/10.1002/(SICI)1099-0518(20000601)38:11%3C1919::AID-POLA10%3E3.0.CO;2-W.
[68] Z. Cherifi, B. Boukoussa, A. Mokhtar, M. Hachemaoui, F.Z. Zeggai, A. Zaoui, et al., Preparation of new nanocomposite poly (GDMA)/mesoporous silica and its adsorption behavior towards cationic dye, React. Funct. Polym. 153 (2020) 104611, https://doi.org/10.1016/j.reactfunctpolym.2020.104611.
[69] Z. Cherifi, B. Boukoussa, A. Zaoui, M. Belbachir, R. Meghabar, Structural, morphological and thermal properties of nanocomposites poly (GMA)/clay prepared by ultrasound and in-situ polymerization, Ultrason. Sonochem. 48 (2018) 188–198, https://doi.org/10.1016/j.ultsonch.2018.05.027.
[70] L. Souli, R. Meghabar, M. Lahrech, Synthesis and characterization of dimethacrylate macromonomer using maghnite-H+ as an eco-friendly catalyst, J. Fundam. Appl. Sci. 12 (2) (2020) 814–824, https://doi.org/10.4314/jfas.v12i2.19.
[71] P. Kubisa, Activated monomer mechanism in the cationic polymerization of cyclic ethers, in: Makromol. Chem. Macromol. Symp. 13-14, Wiley Online Library, 1988, pp. 203–210.
[72] S. Penczek, J. Pretula, Activated monomer mechanism (AMM) in cationic ring-opening polymerization. The origin of the AMM and further development in polymerization of cyclic esters, ACS Macro Lett. 10 (11) (2021) 1377–1397, https://doi.org/10.1021/acsmacrolett.1c00509.
[73] M. Bednarek, P. Kubisa, S. Penczek, Polymerization of propylene oxide by activated monomer mechanism-kinetics, Macromol. Chem. Phys. 15 (S19891) (1989) 49–60, https://doi.org/10.1002/macp.1989.020151989106.
[74] T. Biedron, R. Szymanski, P. Kubisa, S. Penczek, Kinetics of polymerization by activated monomer mechanism, in: Makromol. Chem. Macromol. Symp. 32, Wiley Online Library, 1990, pp. 155–168.
[75] K. Brzezińska, R. Szymański, P. Kubisa, S. Penczek, Activated monomer mechanism in cationic polymerization, 1. Ethylene oxide, formulation of mechanism, Macromol. Rapid Commun. 7 (1) (1986) 1–4, https://doi.org/10.1002/marc.1986.030070101.
[76] M. Baśko, P. Kubisa, Cationic copolymerization of ε-caprolactone and L, L-lactide by an activated monomer mechanism, J. Polym. Sci. 44 (24) (2006) 7071–7081, https://doi.org/10.1002/pola.21712.
[77] T. Vidil, F. Tournilhac, S. Musso, A. Robisson, L. Leibler, Control of reactions and network structures of epoxy thermosets, Prog. Polym. Sci. 62 (2016) 126–179, https://doi.org/10.1016/j.progpolymsci.2016.06.003.
[78] H. Staudinger, H. Lohmann, 81. Mitteilung. Über eukolloides Polyäthylenoxyd, Justus Liebigs Ann. Chem. 505 (1) (1933) 41–51, https://doi.org/10.1002/jlac.19335050104.
[79] P.J. Flory, Molecular size distribution in ethylene oxide polymers, J. Am. Chem. Soc. 62 (6) (1940) 1561–1565, https://doi.org/10.1021/ja01863a066.
[80] J. Park, A. Kim, B.-S. Kim, Anionic ring-opening polymerization of functional epoxide monomers in the solid state, Nat. Commun. 14 (1) (2023) 5855, https://doi.org/10.1038/s41467-023-41576-0.
[81] S. Boileau, Anionic ring-opening polymerization: epoxides and episulfides, Compr. Poly. Sci. Synth. Char. React. Appl. Polymers 3 (1989) 467–487.
[82] A.-L. Brocas, C. Mantzaridis, D. Tunc, S. Carlotti, Polyether synthesis: from activated or metal-free anionic ring-opening polymerization of epoxides to functionalization, Prog. Polym. Sci. 38 (6) (2013) 845–873, https://doi.org/10.1016/j.progpolymsci.2012.09.007.
[83] A.J. Satti, P.I.R. Murano, C.J. Pérez, A. Freitas, C. Giacomelli, A.E. Ciolino, Anionic ring-opening polymerization following atom transfer radical polymerization: Poly (glycidyl methacrylate-g-dimethylsiloxane) obtained by "grafting from" strategy, Polym. Eng. Sci. 65 (3) (2025) 1424–1432, https://doi.org/10.1002/pen.27094.
[84] N.V. Tsarevsky, S.A. Bencherif, K. Matyjaszewski, Graft copolymers by a combination of ATRP and two different consecutive click reactions, Macromolecules 40 (13) (2007) 4439–4445, https://doi.org/10.1021/ma070705m.
[85] P. Dimitrov, E. Hasan, S. Rangelov, B. Trzebicka, A. Dworak, C.B. Tsvetanov, High molecular weight functionalized poly (ethylene oxide), Polymer 43 (25) (2002) 7171–7178, https://doi.org/10.1016/S0032-3861(02)00459-7.
[86] V. Rejsek, P. Desbois, A. Deffieux, S. Carlotti, Polymerization of ethylene oxide initiated by lithium derivatives via the monomer-activated approach: application to the direct synthesis of PS-b-PEO and PI-b-PEO diblock copolymers, Polymer 51 (24) (2010) 5674–5679, https://doi.org/10.1016/j.polymer.2010.09.061.

[87] C.G. Rodriguez, R.C. Ferrier Jr., A. Helenic, N.A. Lynd, Ring-opening polymerization of epoxides: facile pathway to functional polyethers via a versatile organoaluminum initiator, Macromolecules 50 (8) (2017) 3121–3130, https://doi.org/10.1021/acs.macromol.7b00196.
[88] A. de Lucas, L. Rodríguez, M. Pérez-Collado, P. Sanchez, J.F. Rodríguez, Synthesis of polyols by anionic polymerization: determination of kinetic parameters of propylene oxide polymerization using caesium and potassium alcoholates, Polym. Int. 51 (10) (2002) 1066–1071, https://doi.org/10.1002/pi.909.
[89] M. Hans, H. Keul, M. Moeller, Chain transfer reactions limit the molecular weight of polyglycidol prepared via alkali metal based initiating systems, Polymer 50 (5) (2009) 1103–1108, https://doi.org/10.1016/j.polymer.2009.01.012.
[90] R. Matthes, H. Frey, Polyethers based on short-chain alkyl glycidyl ethers: thermoresponsive and highly biocompatible materials, Biomacromolecules 23 (6) (2022) 2219–2235, https://doi.org/10.1021/acs.biomac.2c00223.
[91] O. Nuyken, S.D. Pask, Ring-opening polymerization—an introductory review, Polymers 5 (2) (2013) 361–403, https://doi.org/10.3390/polym5020361.
[92] K. Sagawa, M. Uchiyama, H. Watanabe, C. Homma, M. Kamigaito, Proton transfer anionic polymerization of methyl methacrylate with ligands for dual control of molecular weight and Tacticity, Precis. Chem. 2 (12) (2024) 628–633, https://doi.org/10.1021/prechem.4c00066.
[93] K. Ntetsikas, Y. Alzahrany, G. Polymeropoulos, P. Bilalis, Y. Gnanou, N. Hadjichristidis, Anionic polymerization of styrene and 1, 3-butadiene in the presence of phosphazene superbases, Polymers 9 (10) (2017) 538, https://doi.org/10.3390/polym9100538.
[94] S. Ménoret, M. Fontanille, A. Deffieux, P. Desbois, J. Demeter, Initiation of styrene retarded anionic polymerization using the combination of lithium alkoxides with organometallic compounds, Macromolecules 35 (12) (2002) 4584–4589, https://doi.org/10.1021/ma012057r.
[95] Z. Wu, X. Zhao, X. Wang, F. Ge, The (co) polymerization of (meth) acrylates via thiourea anion/Lewis acid ate complex, Polymer (2026) 129620, https://doi.org/10.1016/j.polymer.2026.129620.
[96] F. An, B.-Y. Han, Anionic polymerization and copolymerization of glycidyl methacrylate initiated by lithium organocuprate, J. Beijing Univ. Chem. Technol. 33 (5) (2006) 33–36.
[97] G. Hild, J.-P. Lamps, Diblock copolymers, triblock copolymers andmodel networks synthesized by sequential anionic polymerization of styrene and 2, 3-epoxypropyl methacrylate, Polymer 39 (12) (1998) 2637–2649, https://doi.org/10.1016/S0032-3861(97)00578-8.
[98] J. Zhang, H. Huang, D.-F. Wei, Y. Guan, F.-Z. Hu, A. Zheng, et al., Further studies on the anionic copolymerization of styrene and glycidyl methacrylate in toluene, J. Macromol. Sci. A Pure Appl. Chem. 47 (6) (2010) 626–632, https://doi.org/10.1080/10601321003659515.
[99] S. Antoun, J.-S. Wang, R. Jérôme, P. Teyssie, Anionic polymerization of various methacrylates initiated with LiCl-complexed sBuLi, Polymer 37 (25) (1996) 5755–5759, https://doi.org/10.1016/S0032-3861(96)00546-0.
[100] J. Tang, Y. Fu, J. Hua, J. Zhang, S. Peng, Z. Li, Unlocking regioselectivity: steric effects and conformational constraints of Lewis bases in alkyllithium-initiated butadiene polymerization, Chem. Sci. 15 (48) (2024) 20493–20502, https://doi.org/10.1039/D4SC05144K.
[101] N. Hadjichristidis, H. Iatrou, M. Pitsikalis, J. Mays, Macromolecular architectures by living and controlled/living polymerizations, Prog. Polym. Sci. 31 (12) (2006) 1068–1132, https://doi.org/10.1016/j.progpolymsci.2006.07.002.
[102] S. Ghosh, H. Lund, H. Jiao, E. Mejía, Rediscovering the isospecific ring-opening polymerization of racemic propylene oxide with dibutylmagnesium, Macromolecules 50 (3) (2017) 1245–1250, https://doi.org/10.1021/acs.macromol.6b01830.
[103] R.M. Thomas, P.C. Widger, S.M. Ahmed, R.C. Jeske, W. Hirahata, E.B. Lobkovsky, et al., Enantioselective epoxide polymerization using a bimetallic cobalt catalyst, J. Am. Chem. Soc. 132 (46) (2010) 16520–16525, https://doi.org/10.1021/ja1058422.
[104] N. Çolak, K. Alyürük, Polymerization of propylene oxide by the Pruitt-Baggett catalyst, Polymer 30 (9) (1989) 1709–1713, https://doi.org/10.1016/0032-3861(89)90335-2.
[105] K.L. Peretti, H. Ajiro, C.T. Cohen, E.B. Lobkovsky, G.W. Coates, A highly active, isospecific cobalt catalyst for propylene oxide polymerization, J. Am. Chem. Soc. 127 (33) (2005) 11566–11567, https://doi.org/10.1021/ja053451y.
[106] C.C. Price, M. Osgan, The polymerization of l-propylene Oxide1, J. Am. Chem. Soc. 78 (18) (1956) 4787–4792, https://doi.org/10.1021/ja01599a066.
[107] W. Kuran, Coordination polymerization of heterocyclic and heterounsaturated monomers, Prog. Polym. Sci. 23 (6) (1998) 919–992, https://doi.org/10.1016/S0079-6700(97)00031-2.
[108] T. Hagiwara, M. Ishimori, T. Tsuruta, 13C NMR study on the mechanism of stereoselective polymerization of methyloxirane catalyzed by a model compound for enantiomorphic catalysts, Macromol. Chem. Phys. 182 (2) (1981) 501–511, https://doi.org/10.1002/macp.1981.021820218.
[109] J. Furukawa, T. Tsuruta, R. Sakata, T. Saegusa, A. Kawasaki, Polymerization of propylene oxide by diethylzine in the presence of cocatalysts, Macromol. Chem. Phys. 32 (1) (1959) 90–94, https://doi.org/10.1002/macp.1959.020320109.
[110] Q. Song, J. Zhao, G. Zhang, D. Taton, F. Peruch, S. Carlotti, N-heterocyclic carbene/Lewis acid-mediated ring-opening polymerization of propylene oxide. Part 1: triisobutylaluminum as an efficient controlling agent, Eur. Polym. J. 134 (2020) 109819, https://doi.org/10.1016/j.eurpolymj.2020.109819.
[111] R. Colclough, G. Gee, A. Jagger, The polymerization of propylene oxide by trimethyl aluminum, J. Polym. Sci. 48 (150) (1960) 273–278, https://doi.org/10.1002/pol.1960.1204815025.
[112] E. Vandenberg, Organometallic catalysts for polymerizing monosubstituted epoxides, J. Polym. Sci. 47 (149) (1960) 486–489, https://doi.org/10.1002/pol.1960.1204714947.
[113] A. Labbé, S. Carlotti, C. Billouard, P. Desbois, A. Deffieux, Controlled high-speed anionic polymerization of propylene oxide initiated by onium salts in the presence of triisobutylaluminum, Macromolecules 40 (22) (2007) 7842–7847, https://doi.org/10.1021/ma070288d.
[114] S.S. Müller, C. Moers, H. Frey, A challenging comonomer pair: copolymerization of ethylene oxide and glycidyl methyl ether to thermoresponsive polyethers, Macromolecules 47 (16) (2014) 5492–5500, https://doi.org/10.1021/ma501280k.
[115] Meyer Jr.;Keul H.;Möller M. Poly (glycidyl amine) and copolymers with glycidol and glycidyl amine repeating units: synthesis and characterization. Macromolecules 2011; 44(11): 4082–4091. doi:https://doi.org/10.1021/ma200757v.
[116] A.L. Brocas, G. Cendejas, S. Caillol, A. Deffieux, S. Carlotti, Controlled synthesis of polyepichlorohydrin with pendant cyclic carbonate functions for isocyanate-free polyurethane networks, J. Polym. Sci. 49 (12) (2011) 2677–2684, https://doi.org/10.1002/pola.24699.
[117] M. Gervais, A.-L. Brocas, G. Cendejas, A. Deffieux, S. Carlotti, Synthesis of linear high molar mass glycidol-based polymers by monomer-activated anionic polymerization, Macromolecules 43 (4) (2010) 1778–1784, https://doi.org/10.1021/ma902286a.
[118] W. Kuran, Principles of Coordination Polymerisation, 2001.
[119] H. Sugimoto, C. Kawamura, M. Kuroki, T. Aida, S. Inoue, Lewis acid-assisted anionic ring-opening polymerization of epoxide by the aluminum complexes of porphyrin, phthalocyanine, tetraazaannulene, and schiff base as initiators, Macromolecules 27 (8) (1994) 2013–2018, https://doi.org/10.1021/ma00086a006.
[120] F.M. Rabagliati, F. López, Epoxide polymerization, 6. Structure of poly (propylene oxide) prepared with diphenylzinc/wáter as initiating system, Macromol. Rapid Commun. 6 (3) (1985) 141–144, https://doi.org/10.1002/marc.1985.030060305.
[121] W. van Dijk-Wolthuis, O. Franssen, H. Talsma, M. Van Steenbergen, J. Kettenes-Van Den Bosch, W. Hennink, Synthesis, characterization, and polymerization of glycidyl methacrylate derivatized dextran, Macromolecules 28 (18) (1995) 6317–6322, https://doi.org/10.1021/ma00122a044.
[122] R. Jeencham, J. Sinna, C. Ruksakulpiwat, T. Tawonsawatruk, P-o. Numpaisal, Y. Ruksakulpiwat, Development of biphasic injectable hydrogels for meniscus scaffold from photocrosslinked glycidyl methacrylate-modified poly (vinyl alcohol)/glycidyl methacrylate-modified silk fibroin, Polymers 16 (8) (2024) 1093, https://doi.org/10.3390/polym16081093.
[123] M.R. Guilherme, A.V. Reis, S.H. Takahashi, A.F. Rubira, J.P. Feitosa, E.C. Muniz, Synthesis of a novel superabsorbent hydrogel by copolymerization of acrylamide and cashew gum modified with glycidyl methacrylate, Carbohydr. Polym. 61 (4) (2005) 464–471, https://doi.org/10.1016/j.carbpol.2005.06.017.
[124] M.C.G. Pellá, A.R. Simao, M.R. Mauricio, R.A. Estrada, G.M. Pereira, R. da Silva, et al., A comparative study of starch-g-(glycidyl methacrylate)/synthetic polymer-based hydrogels, Carbohydr. Polym. 307 (2023) 120614, https://doi.org/10.1016/j.carbpol.2023.120614.
[125] M.C.G. Pellá, A.R. Simão, M.R. Maurício, D.C. Dragunski, Rd Silva, A.F. Rubira, Adjusting experimental parameters to modify corn starch with glycidyl methacrylate to tune specific properties, J. Braz. Chem. Soc. 34 (02) (2023) 263–275, https://doi.org/10.21577/0103-5053.20220105.
[126] L. Ferreira, M. Vidal, C. Geraldes, M. Gil, Preparation and characterisation of gels based on sucrose modified with glycidyl methacrylate, Carbohydr. Polym. 41 (1) (2000) 15–24, https://doi.org/10.1016/S0144-8617(99)00064-8.
[127] W. van Dijk-Wolthuis, J. Kettenes-Van Den Bosch, A. Van Der Kerk-Van Hoof, W. Hennink, Reaction of dextran with glycidyl methacrylate: an unexpected transesterification, Macromolecules 30 (11) (1997) 3411–3413, https://doi.org/10.1021/ma961764v.
[128] Q. Li, D-a. Wang, J.H. Elisseeff, Heterogeneous-phase reaction of glycidyl methacrylate and chondroitin sulfate: mechanism of ring-opening−transesterification competition, Macromolecules 36 (7) (2003) 2556–2562, https://doi.org/10.1021/ma021190w.
[129] T. Corner, Free Radical Polymerisation. The Synthesis of Graft Copolymers. Initiators—Poly-Reactions—Optical Activity, Springer, 2005, pp. 95–142.
[130] B.B. Noble, M.L. Coote, First principles modelling of free-radical polymerisation kinetics, Int. Rev. Phys. Chem. 32 (3) (2013) 467–513, https://doi.org/10.1080/0144235X.2013.797277.
[131] R.A. Hutchinson, S. Beuermann, D. Paquet, J. McMinn, Determination of free-radical propagation rate coefficients for alkyl methacrylates by pulsed-laser polymerization, Macromolecules 30 (12) (1997) 3490–3493, https://doi.org/10.1021/ma970176u.
[132] W. Wang, R.A. Hutchinson, PLP/SEC/NMR study of free radical copolymerization of styrene and glycidyl methacrylate, Macromolecules 41 (23) (2008) 9011–9018, https://doi.org/10.1021/ma801435t.
[133] R. Hutchinson, S. Beuermann, D. Paquet, J. McMinn, C. Jackson, Determination of free-radical propagation rate coefficients for cycloalkyl and functional methacrylates by pulsed-laser polymerization, Macromolecules 31 (5) (1998) 1542–1547, https://doi.org/10.1021/ma971307u.
[134] B. O'Shaughnessy, D. Vavylonis, Pulsed laser polymerization at low conversions: broadening and chain transfer effects, Macromol. Theory Simul. 12 (6) (2003) 401–412, https://doi.org/10.1002/mats.200350001.
[135] N. Prasoetsopha, P. Chumsamrong, N. Suppakarn, Effects of type and concentration of initiator on grafting of acrylic monomer onto depolymerized

natural rubber, Adv. Mater. Res. 264 (2011) 565–570, https://doi.org/10.4028/www.scientific.net/AMR.264-265.565.
[136] P.K. Dhal, M. Ramakrishna, G. Babu, Copolymerization of glycidyl methacrylate with alkyl acrylate monomers, J. Polym. Sci. 20 (6) (1982) 1581–1585, https://doi.org/10.1002/pol.1982.170200618.
[137] A. Darvishi, M.J. Zohuriaan Mehr, G.B. Marandi, K. Kabiri, H. Bouhendi, H. Bakhshi, Copolymers of glycidyl methacrylate and octadecyl acrylate: synthesis, characterization, swelling properties, and reactivity ratios, Des. Monomers Polym. 16 (1) (2013) 79–88, https://doi.org/10.1080/15685551.2012.705493.
[138] J.M. Jin, J.M. Lee, M.H. Ha, K. Lee, S. Choe, Highly crosslinked poly (glycidyl methacrylate-co-divinyl benzene) particles by precipitation polymerization, Polymer 48 (11) (2007) 3107–3115, https://doi.org/10.1016/j.polymer.2007.03.068.
[139] C.J. Selvamalar, T. Krithiga, A. Penlidis, S. Nanjundan, Copolymerization of 4-benzyloxycarbonylphenyl methacrylate with glycidyl methacrylate: synthesis, characterization, reactivity ratios and application as adhesives, React. Funct. Polym. 56 (2) (2003) 89–101, https://doi.org/10.1016/S1381-5148(03)00028-2.
[140] S. Nanjundan, C.S. Unnithan, C.J. Selvamalar, A. Penlidis, Homopolymer of 4-benzoylphenyl methacrylate and its copolymers with glycidyl methacrylate: synthesis, characterization, monomer reactivity ratios and application as adhesives, React. Funct. Polym. 62 (1) (2005) 11–24, https://doi.org/10.1016/j.reactfunctpolym.2004.08.006.
[141] O. Abdyev, Radical copolymerization of glycidyl methacrylate and carbonyl-containing phenyl methacrylates, Poly. Sci. Ser. B 53 (5) (2011) 278–282, https://doi.org/10.1134/S1560090411050010.
[142] M.S. Iqbal, Y. Jamil, T. Kausar, M. Akhtar, Thermal degradation study of glycidyl methacrylate acrylonitrile copolymers, J. Therm. Anal. Calorim. 96 (1) (2009) 225–233, https://doi.org/10.1007/s10973-008-9009-z.
[143] S. Srisuwan, N. Prasoetsopha, N. Suppakarn, P. Chumsamrong, Composite materials based on natural rubber modified epoxy resin and sisal fiber, Adv. Mater. Res. 410 (2012) 43–46, https://doi.org/10.4028/www.scientific.net/AMR.410.43.
[144] F. Shohraty, P.N. Moghadam, A.R. Fareghi, N. Movagharnezhad, J. Khalafy, Synthesis and characterization of new pH-sensitive hydrogels based on poly (glycidyl methacrylate-co-maleic anhydride), Adv. Polym. Technol. 37 (1) (2018) 120–125, https://doi.org/10.1002/adv.21648.
[145] R. Hasanzadeh, P.N. Moghadam, N. Bahri-Laleh, F. Ziaee, A reactive copolymer based on glycidylmethacrylate and maleic anhydride: 1-synthesis, characterization and monomer reactivity ratios, J Polym Res 23 (2016) 1–19, https://doi.org/10.1007/s10965-016-1048-8.
[146] T. Çaykara, F. Çakar, S. Demirci, A new type of poly (glycidyl methacrylate) microbeads with surface grafted iminodiacetic acid: synthesis and characterization, Polym. Bull. 61 (3) (2008) 311–318, https://doi.org/10.1007/s00289-008-0958-y.
[147] M.M. Nasef, I.A. Sugiarmawan, Radiation induced emulsion grafting of glycidyl methacrylate onto high density polyethylene: a kinetic study, Mal. J. Fundam. Appl. Sci. 6 (2) (2010), https://doi.org/10.11113/mjfas.v6n2.189.
[148] E. Pollert, K. Knížek, M. Maryško, K. Závěta, A. Lančok, J. Boháček, et al., Magnetic poly (glycidyl methacrylate) microspheres containing maghemite prepared by emulsion polymerization, J. Magn. Magn. Mater. 306 (2) (2006) 241–247, https://doi.org/10.1016/j.jmmm.2006.03.069.
[149] C. Lainé, V. Bounor-Legaré, C. Monnet, P. Cassagnau, Free radical polymerization of glycidyl methacrylate in plasticized Poly (vinyl chloride), Eur. Polym. J. 44 (10) (2008) 3177–3190, https://doi.org/10.1016/j.eurpolymj.2008.07.004.
[150] O. Gunaydin, F. Yilmaz, Copolymers of glycidyl methacrylate with 3-methyl-thienyl methacrylate: synthesis, characterization and reactivity ratios, Polym. J. 39 (6) (2007) 579–588, https://doi.org/10.1295/polymj.PJ2006180.
[151] S. Jiang, J. Deng, W. Yang, Block copolymers prepared by free radical polymerization using α-Methylstyrene-containing precopolymer as macroinitiator, Polym. J. 40 (6) (2008) 543–548, https://doi.org/10.1295/polymj.PJ2007192.
[152] I. Erol, B. Poyraz, M. Arif Koroğlu, C. Cifci, Copolymerization of 2-methyl-N-1, 3-thiazole-2-ylacrylamide with glycidyl methacrylate: synthesis, characterization, reactivity ratios and biological activity, J. Polym. Res. 16 (1) (2009) 19–28, https://doi.org/10.1007/s10965-008-9198-y.
[153] I. Erol, B. Gencer, Z. Gurler, Synthesis and characterization of novel methacrylate copolymers having pendant piperonyl group: monomer reactivity ratio, thermal degradation kinetics, and biological activity, Polym. Polym. Compos. 29 (9_suppl) (2021) S1432–S1445, https://doi.org/10.1177/09673911211053978.
[154] M.A. Tehfe, F. Louradour, J. Lalevée, J.-P. Fouassier, Photopolymerization reactions: on the way to a green and sustainable chemistry, Appl. Sci. 3 (2) (2013) 490–514, https://doi.org/10.3390/app3020490.
[155] G. Chatel, R.S. Varma, Ultrasound and microwave irradiation: contributions of alternative physicochemical activation methods to Green Chemistry, Green Chem. 21 (22) (2019) 6043–6050, https://doi.org/10.1039/B608891K.
[156] C. Aydogan, G. Yilmaz, A. Shegiwal, D.M. Haddleton, Y. Yagci, Photoinduced controlled/living polymerizations, Angew. Chem. Int. Ed. 134 (23) (2022) e202117377, https://doi.org/10.1002/ange.202117377.
[157] C. Wu, N. Corrigan, C.-H. Lim, W. Liu, G. Miyake, C. Boyer, Rational design of photocatalysts for controlled polymerization: effect of structures on photocatalytic activities, Chem. Rev. 122 (6) (2022) 5476–5518, https://doi.org/10.1021/acs.chemrev.1c00409.
[158] X. Pan, M.A. Tasdelen, J. Laun, T. Junkers, Y. Yagci, K. Matyjaszewski, Photomediated controlled radical polymerization, Prog. Polym. Sci. 62 (2016) 73–125, https://doi.org/10.1016/j.progpolymsci.2016.06.005.
[159] E.A. Kamoun, A.M. Omer, S.N. Khattab, H.M. Ahmed, A.A. Elbardan, In-situ UV-photopolymerized PVA-g-GMA hydrogels for biomedical applications: I. Synthesis, characterizations and grafting optimization, J. Appl. Pharm. Sci. 8 (1) (2018) 034–042, https://doi.org/10.7324/JAPS.2018.8106.
[160] W. Zhang, Y. Xu, Y. Gao, Y. Chen, Glycidyl methacrylate-modified Bis (2-hydroxyethyl) terephthalate acrylates for fast radical polymerization and robust films, ACS Appl. Polym. Mater. 7 (14) (2025) 9045–9054, https://doi.org/10.1021/acsapm.5c01138.
[161] N. Alemdar, Synthesis of chitosan-based hydrogel by using photopolymerization technique, Anadolu Univ. J. Sci. Technol. A Appl. Sci. Eng. 17 (2) (2016) 391–400, https://doi.org/10.18038/btda.96384.
[162] M.M. Sk, P. Das, A. Panwar, L.P. Tan, Synthesis and characterization of site selective photo-crosslinkable glycidyl methacrylate functionalized gelatin-based 3D hydrogel scaffold for liver tissue engineering, Mater. Sci. Eng. C 123 (2021) 111694, https://doi.org/10.1016/j.msec.2020.111694.
[163] X. Fei, Y. Shi, Y. Cao, Synthesis of photosensitive poly (methyl methacrylate-co-glycidyl methacrylate) for optical waveguide devices, Appl. Phys. A Mater. Sci. Process. 100 (2) (2010) 409–414, https://doi.org/10.1007/s00339-010-5851-5.
[164] B. Zhang, A. Serjouei, Y.-F. Zhang, J. Wu, H. Li, D. Wang, et al., Dual-stage thermosetting photopolymers for advanced manufacturing, Chem. Eng. J. 411 (2021) 128466, https://doi.org/10.1016/j.cej.2021.128466.
[165] R.F. Martínez, G. Cravotto, P. Cintas, Organic sonochemistry: a chemist's timely perspective on mechanisms and reactivity, J. Organomet. Chem. 86 (20) (2021) 13833–13856, https://doi.org/10.1021/acs.joc.1c00805.
[166] P. Riesz, T. Kondo, Free radical formation induced by ultrasound and its biological implications, Free Radic. Biol. Med. 13 (3) (1992) 247–270, https://doi.org/10.1016/0891-5849(92)90021-8.
[167] H. Feng, X. Shao, Z. Wang, Mechanochemical controlled radical polymerization: from harsh to mild, ChemPlusChem 89 (10) (2024) e202400287, https://doi.org/10.1002/cplu.202400287.
[168] E. Marimuthu, V. Murugesan, Influence of ultrasonic condition on phase transfer catalyzed radical polymerization of methyl methacrylate in two phase system–a kinetic study, Ultrason. Sonochem. 38 (2017) 560–569, https://doi.org/10.1016/j.ultsonch.2016.08.028.
[169] B.M. Teo, S.W. Prescott, M. Ashokkumar, F. Grieser, Ultrasound initiated miniemulsion polymerization of methacrylate monomers, Ultrason. Sonochem. 15 (1) (2008) 89–94, https://doi.org/10.1016/j.ultsonch.2007.01.009.
[170] K. Kempe, C.R. Becer, U.S. Schubert, Microwave-assisted polymerizations: recent status and future perspectives, Macromolecules 44 (15) (2011) 5825–5842, https://doi.org/10.1021/ma2004794.
[171] D. Lin, D.N. Futaba, K. Kobashi, M. Zhang, S. Muroga, G. Chen, et al., A microwave-assisted, solvent-free approach for the versatile functionalization of carbon nanotubes, ACS Nano 17 (4) (2023) 3976–3983, https://doi.org/10.1021/acsnano.2c12789.
[172] A. Singh, S.K. Sharma, S. Sharma, Microwave-and ultrasound-assisted synthesis of polymers, Green Chem. Synth. Microwaves Ultrasound (2024) 219–234, https://doi.org/10.1002/9783527844494.ch8.
[173] P. Lidström, J. Tierney, B. Wathey, J. Westman, Microwave assisted organic synthesis—a review, Tetrahedron 57 (45) (2001) 9225–9283, https://doi.org/10.1016/S0040-4020(01)00906-1.
[174] T.G. Oberti, M.J. Lavecchia, M.S. Cortizo, Polymerization of p-nitrobenzyl acrylate under microwave irradiation and their optical properties, J Polym Res 18 (6) (2011) 1293–1299, https://doi.org/10.1007/s10965-010-9532-z.
[175] W. Tanan, S. Saengsuwan, Microwave assisted synthesis of poly (acrylamide-co-2-hydroxyethyl methacrylate)/poly (vinyl alcohol) semi-IPN hydrogel, Energy Procedia 56 (2014) 386–393, https://doi.org/10.1016/j.egypro.2014.07.171.
[176] A. Chyzy, D. Pawelski, V. Vivcharenko, A. Przekora, M. Bratychak, O. Astakhova, et al., Microwave-assisted synthesis of modified glycidyl methacrylate–ethyl methacrylate oligomers, their physico-chemical and biological characteristics, Molecules 27 (2) (2022) 337, https://doi.org/10.3390/molecules27020337.
[177] S. Pramanik, R. Konwarh, R.C. Deka, L. Aidew, N. Barua, A.K. Buragohain, et al., Microwave-assisted poly (glycidyl methacrylate)-functionalized multiwall carbon nanotubes with a 'tendrillar'nanofibrous polyaniline wrapping and their interaction at bio-interface, Carbon 55 (2013) 34–43, https://doi.org/10.1016/j.carbon.2012.11.062.
[178] S.A. Baghban, M. Ebrahimi, S. Bagheri-Khoulenjani, M. Khorasani, A highly efficient microwave-assisted synthesis of an LED-curable methacrylated gelatin for bio applications, RSC Adv. 11 (25) (2021) 14996–15009, https://doi.org/10.1039/D1RA01269J.
[179] G. Gonzalez, M. Arévalo, A. Chiappone, E. Martínez Campos, C.F. Pirri, I. Roppolo, et al., A facile and green microwave-assisted strategy to induce surface properties on complex-shape polymeric 3D printed structures, Macromol. Mater. Eng. 308 (11) (2023) 2300118, https://doi.org/10.1002/mame.202300118.
[180] A. Ashfaq, M.-C. Clochard, X. Coqueret, C. Dispenza, M.S. Driscoll, P. Ulański, et al., Polymerization reactions and modifications of polymers by ionizing radiation, Polymers 12 (12) (2020) 2877, https://doi.org/10.3390/polym12122877.
[181] A.T. Naikwadi, B.K. Sharma, K.D. Bhatt, P.A. Mahanwar, Gamma radiation processed polymeric materials for high performance applications: a review, Front. Chem. 10 (2022) 837111, https://doi.org/10.3389/fchem.2022.837111.
[182] J.F. Madrid, G.M. Nuesca, L.V. Abad, Gamma radiation-induced grafting of glycidyl methacrylate (GMA) onto water hyacinth fibers, Radiat. Phys. Chem. 85 (2013) 182–188, https://doi.org/10.1016/j.radphyschem.2012.10.006.
[183] M. Kim, K. Saito, Radiation-induced graft polymerization and sulfonation of glycidyl methacrylate on to porous hollow-fiber membranes with different pore sizes, Radiat. Phys. Chem. 57 (2) (2000) 167–172, https://doi.org/10.1016/S0969-806X(99)00314-X.

[184] Z. Li, H. Wang, Radiation-induced grafting of glycidyl methacrylate onto high-density polyethylene (HDPE) and radiation lamination of HDPE, J. Appl. Polym. Sci. 96 (3) (2005) 772–779, https://doi.org/10.1002/app.21508.
[185] M. Tahir, A. Raza, A. Nasir, T. Yasin, Radiation induced graft polymerization of glycidyl methacrylate onto sepiolite, Radiat. Phys. Chem. 179 (2021) 109259, https://doi.org/10.1016/j.radphyschem.2020.109259.
[186] Gamma radiation induced simultaneous graft polymerisation of glycidyl methacrylate on poly-propylene non-woven fabric: a precursor for adsorbent, in: A. Kanjilal, K. Singh, V. Kumar (Eds.), Proceedings of the Sixteenth DAE-BRNS Biennial Trombay Symposium on Radiation and Photochemistry, 2022.
[187] R. Shaheen, T. Yasin, Z. Ali, A.S. Khan, B. Adalat, M. Tahir, et al., Synthesis, characterization, and adsorptive characteristics of radiation-grafted glycidyl methacrylate bamboo fiber composites, ACS Omega 8 (42) (2023) 38849–38859, https://doi.org/10.1021/acsomega.3c02466.
[188] B. Zhao, P. Wilson, Recent progress and applications enabled via electrochemically triggered and controlled chain-growth polymerizations, Polym. Chem. 14 (17) (2023) 2000–2021, https://doi.org/10.1039/D3PY00062A.
[189] S.K. Samal, B. Nayak, Electroinitiated polymerization of glycidyl methacrylate in dimethylformamide medium, Polymer 29 (11) (1988) 2105–2110, https://doi.org/10.1016/0032-3861(88)90189-9.
[190] E. Molle, S. Frech, T. Grüger, P. Theato, Electrochemically-initiated polymerization of reactive monomers via 4-fluorobenzenediazonium salts, Polym. Chem. 12 (41) (2021) 5970–5978, https://doi.org/10.1039/D1PY00536G.
[191] H. Zhou, B. Zhang, W. Liu, Y. Xue, C. Zhao, X. Zhang, et al., Grafting glycidyl methacrylate onto high-modulus carbon fiber by acid activation synergized with ionization radiation, Appl. Surf. Sci. 610 (2023) 155593, https://doi.org/10.1016/j.apsusc.2022.155593.
[192] H.K. Yasuda, Plasma Polymerization, Academic press, 2012.
[193] J. Friedrich, Mechanisms of plasma polymerization–reviewed from a chemical point of view, Plasma Process. Polym. 8 (9) (2011) 783–802, https://doi.org/10.1002/ppap.201100038.
[194] Y. Osada, M. Takase, Y. Iriyama, Effects and role of the solvents on the plasma-initiated solution polymerization of vinyl monomers, Polym. J. 15 (1) (1983) 81–86, https://doi.org/10.1295/polymj.15.81.
[195] H.-S. Ji, X. Liu, H.-S. Choi, J.-H. Kim, H.-O. Park, Emulsion grafting of glycidyl methacrylate onto plasma-treated polypropylene surface, Polymer 36 (1) (2012) 65–70.
[196] H. Chen, M. Guo, X. Yao, Z. Luo, K. Dong, Z. Lian, et al., Green and efficient synthesis of an adsorbent fiber by plasma-induced grafting of glycidyl methacrylate and its Cd (II) adsorption performance, Fibers Polym. 19 (4) (2018) 722–733, https://doi.org/10.1007/s12221-018-7785-z.
[197] X. Zou, E. Kang, K. Neoh, Y. Zhang, K. Tan, C. Cui, et al., Plasma polymerization and deposition of glycidyl methacrylate on Si (100) surface for adhesion improvement with polyimide, Polym. Adv. Technol. 12 (10) (2001) 583–595, https://doi.org/10.1002/pat.146.
[198] X. Zou, E. Kang, K. Neoh, C. Cui, T. Lim, Surface modification of poly (tetrafluoroethylene) films by plasma pre-activation and plasma polymerization of glycidyl methacrylate, Plasmas Polym. 5 (3) (2000) 219–234, https://doi.org/10.1023/A:1011354001424.
[199] S. Korpayev, C. Kavaklı, Ş. Çolak, P.A. Kavaklı, Preparation and characterization of ethylenediamine modified glycidyl methacrylate-grafted nonwoven cotton fabric adsorbent, Cellulose 25 (1) (2018) 813–828, https://doi.org/10.1007/s10570-017-1558-5.
[200] N. Corrigan, K. Jung, G. Moad, C.J. Hawker, K. Matyjaszewski, C. Boyer, Reversible-deactivation radical polymerization (controlled/living radical polymerization): from discovery to materials design and applications, Prog. Polym. Sci. 111 (2020) 101311, https://doi.org/10.1016/j.progpolymsci.2020.101311.
[201] K. Matyjaszewski, J. Xia, Atom transfer radical polymerization, Chem. Rev. 101 (9) (2001) 2921–2990, https://doi.org/10.1021/cr940534g.
[202] G. Moad, E. Rizzardo, S.H. Thang, Living radical polymerization by the RAFT process, Aust. J. Chem. 58 (6) (2005) 379–410, https://doi.org/10.1071/CH05072.
[203] B. Zhang, Y. Li, P. Ai, Z. Sa, Y. Zhao, M. Li, et al., Y-shaped diblock copolymer with epoxy-based block of poly (glycidyl methacrylate): synthesis, characterization, and its morphology study, J. Polym. Sci. 47 (20) (2009) 5509–5526, https://doi.org/10.1002/pola.23602.
[204] P.J. Docherty, M.J. Derry, S.P. Armes, RAFT dispersion polymerization of glycidyl methacrylate for the synthesis of epoxy-functional block copolymer nanoparticles in mineral oil, Polym. Chem. 10 (5) (2019) 603–611, https://doi.org/10.1039/C8PY01584H.
[205] F.L. Hatton, M.J. Derry, S.P. Armes, Rational synthesis of epoxy-functional spheres, worms and vesicles by RAFT aqueous emulsion polymerisation of glycidyl methacrylate, Polym. Chem. 11 (39) (2020) 6343–6355, https://doi.org/10.1039/D0PY01097A.
[206] H. Thankappan, M. Semsarilar, S. Li, Y. Chang, D. Bouyer, D. Quemener, Synthesis of block copolymer brush by RAFT and click chemistry and its self-assembly as a thin film, Molecules 25 (20) (2020) 4774, https://doi.org/10.3390/molecules25204774.
[207] G. Moad, E. Rizzardo, S.H. Thang, Living radical polymerization by the RAFT process–a second update, Aust. J. Chem. 62 (11) (2009) 1402–1472, https://doi.org/10.1071/CH09311.
[208] J. Chiefari, Y. Chong, F. Ercole, J. Krstina, J. Jeffery, T.P. Le, et al., Living free-radical polymerization by reversible addition-fragmentation chain transfer: the RAFT process, Macromolecules 31 (16) (1998) 5559.
[209] S. Perrier, 50th anniversary perspective: RAFT polymerization a user guide, Macromolecules 50 (19) (2017) 7433–7447, https://doi.org/10.1021/acs.macromol.7b00767.
[210] G. Moad, RAFT polymerization–then and now, in: Controlled Radical Polymerization: Mechanisms, ACS Publications, 2015, pp. 211–246.
[211] G. Moad, Trithiocarbonates in RAFT polymerization, RAFT Polymerization: Methods, Synthesis and Applications 1 (2021) 359–492, https://doi.org/10.1002/9783527821358.ch9.
[212] M. Benaglia, A. Alberti, L. Giorgini, F. Magnoni, S. Tozzi, Poly (glycidyl methacrylate): a highly versatile polymeric building block for post-polymerization modifications, Polym. Chem. 4 (1) (2013) 124–132, https://doi.org/10.1039/C2PY20646C.
[213] J. Zhu, D. Zhou, X. Zhu, G. Chen, Reversible addition–fragmentation chain transfer polymerization of glycidyl methacrylate with 2-cyanoprop-2-yl 1-dithionaphthalate as a chain-transfer agent, J. Polym. Sci. 42 (10) (2004) 2558–2565, https://doi.org/10.1002/pola.20119.
[214] C.S. Gudipati, M.B. Tan, H. Hussain, Y. Liu, C. He, T.P. Davis, Synthesis of Poly (glycidyl methacrylate)-block-Poly (pentafluorostyrene) by RAFT: precursor to novel amphiphilic Poly (glyceryl methacrylate)-block-Poly (pentafluorostyrene), Macromol. Rapid Commun. 29 (23) (2008) 1902–1907, https://doi.org/10.1002/marc.200800515.
[215] Y. Lee, C. Boyer, M.S. Kwon, Photocontrolled RAFT polymerization: past, present, and future, Chem. Soc. Rev. 52 (9) (2023) 3035–3097, https://doi.org/10.1039/D1CS00069A.
[216] A.M. Jazani, H. Murata, M. Cvek, A. Lewandowska-Andralojc, R. Bernat, K. Kapil, et al., Aqueous photo-RAFT polymerization under ambient conditions: synthesis of protein–polymer hybrids in open air, Chem. Sci. 15 (25) (2024) 9742–9755, https://doi.org/10.1039/D4SC01409J.
[217] G.K. Clothier, T.R. Guimarães, L.T. Strover, P.B. Zetterlund, G. Moad, Electrochemically-initiated RAFT synthesis of low dispersity multiblock copolymers by seeded emulsion polymerization, ACS Macro Lett. 12 (3) (2023) 331–337, https://doi.org/10.1021/acsmacrolett.3c00021.
[218] C. Bray, G. Li, A. Postma, L.T. Strover, J. Wang, G. Moad, Initiation of RAFT polymerization: electrochemically initiated RAFT polymerization in emulsion (emulsion eRAFT), and direct PhotoRAFT polymerization of liquid crystalline monomers, Aust. J. Chem. 74 (1) (2020) 56–64, https://doi.org/10.1071/CH20260.
[219] H. Yin, H. Zheng, L. Lu, P. Liu, Y. Cai, Highly efficient and well-controlled ambient temperature RAFT polymerization of glycidyl methacrylate under visible light radiation, J. Polym. Sci. 45 (22) (2007) 5091–5102, https://doi.org/10.1002/pola.22251.
[220] J. Zhou, Y. Lin, L. Ye, L. Wang, L. Zhou, H. Hu, et al., PVA hydrogel functionalization via PET-RAFT grafting with glycidyl methacrylate and immobilization with 2-hydroxypropyltrimethyl ammonium chloride chitosan via ring-open reaction, Macromol. Res. 27 (11) (2019) 1144–1154, https://doi.org/10.1007/s13233-019-7152-y.
[221] N.J. Warren, S.P. Armes, Polymerization-induced self-assembly of block copolymer nano-objects via RAFT aqueous dispersion polymerization, J. Am. Chem. Soc. 136 (29) (2014) 10174–10185, https://doi.org/10.1021/ja502843f.
[222] X. Dai, L. Yu, Y. Zhang, L. Zhang, J. Tan, Polymerization-induced self-assembly via RAFT-mediated emulsion polymerization of methacrylic monomers, Macromolecules 52 (19) (2019) 7468–7476, https://doi.org/10.1021/acs.macromol.9b01689.
[223] A. Takashima, Y. Maeda, S. Sugihara, Morphology control via RAFT emulsion polymerization-induced self-assembly: systematic investigation of Core-forming blocks, ACS Omega 7 (30) (2022) 26894–26904, https://doi.org/10.1021/acsomega.2c03440.
[224] Q.-L. Li, W.-X. Gu, H. Gao, Y.-W. Yang, Self-assembly and applications of poly (glycidyl methacrylate) s and their derivatives, Chem. Commun. 50 (87) (2014) 13201–13215, https://doi.org/10.1039/c4cc03036b.
[225] E.J. Cornel, J. Jiang, S. Chen, J. Du, Principles and characteristics of polymerization-induced self-assembly with various polymerization techniques, CCS Chem. 3 (4) (2021) 2104–2125, https://doi.org/10.31635/ccschem.020.202000470.
[226] J. He, Q. Xu, J. Tan, L. Zhang, Ketone-functionalized polymer Nano-objects prepared via photoinitiated polymerization-induced self-assembly (photo-PISA) using a poly (diacetone acrylamide)-based macro-RAFT agent, Macromol. Rapid Commun. 40 (2) (2019) 1800296, https://doi.org/10.1002/marc.201800296.
[227] V.L. Romero Castro, B. Nomeir, A.A. Arteni, M. Ouldali, J.-L. Six, K. Ferji, Dextran-coated latex nanoparticles via photo-RAFT mediated polymerization induced self-assembly, Polymers 13 (23) (2021) 4064, https://doi.org/10.3390/polym13234064.
[228] J. Tan, D. Liu, C. Huang, X. Li, J. He, Q. Xu, et al., Photoinitiated polymerization-induced self-assembly of glycidyl methacrylate for the synthesis of epoxy-functionalized block copolymer nano-objects, Macromol. Rapid Commun. 38 (15) (2017) 1700195, https://doi.org/10.1002/marc.201700195.
[229] J. Tan, X. Dai, Y. Zhang, L. Yu, H. Sun, L. Zhang, Photoinitiated polymerization-induced self-assembly via visible light-induced RAFT-mediated emulsion polymerization, ACS Macro Lett. 8 (2) (2019) 205–212, https://doi.org/10.1021/acsmacrolett.9b00007.
[230] K. Matyjaszewski, S.G. Gaynor, A. Kulfan, M. Podwika, Preparation of hyperbranched polyacrylates by atom transfer radical polymerization. 1. Acrylic AB* monomers in “living” radical polymerizations, Macromolecules 30 (17) (1997) 5192–5194, https://doi.org/10.1021/ma970359g.

[231] K. Matyjaszewski, Mechanistic and synthetic aspects of atom transfer radical polymerization, J. Macromol. Sci., Part A: Pure Appl. Chem. 34 (10) (1997) 1785–1801, https://doi.org/10.1080/10601329708010308.
[232] K. Matyjaszewski, P.J. Miller, N. Shukla, B. Immaraporn, A. Gelman, B.B. Luokala, et al., Polymers at interfaces: using atom transfer radical polymerization in the controlled growth of homopolymers and block copolymers from silicon surfaces in the absence of untethered sacrificial initiator, Macromolecules 32 (26) (1999) 8716–8724, https://doi.org/10.1021/ma991146p.
[233] J. Ran, L. Wu, Z. Zhang, T. Xu, Atom transfer radical polymerization (ATRP): a versatile and forceful tool for functional membranes, Prog. Polym. Sci. 39 (1) (2014) 124–144, https://doi.org/10.1016/j.progpolymsci.2013.09.001.
[234] J.-S. Wang, K. Matyjaszewski, Controlled/“ living” radical polymerization. Atom transfer radical polymerization in the presence of transition-metal complexes, J. Am. Chem. Soc. 117 (20) (1995) 5614–5615, https://doi.org/10.1021/ja00125a035.
[235] V. Coessens, T. Pintauer, K. Matyjaszewski, Functional polymers by atom transfer radical polymerization, Prog. Polym. Sci. 26 (3) (2001) 337–377, https://doi.org/10.1016/S0079-6700(01)00003-X.
[236] N.P. Truong, G.R. Jones, K.G. Bradford, D. Konkolewicz, A. Anastasaki, A comparison of RAFT and ATRP methods for controlled radical polymerization, Nat. Rev. Chem. 5 (12) (2021) 859–869, https://doi.org/10.1038/s41570-021-00328-8.
[237] Y.-N. Zhou, J.-J. Li, T.-T. Wang, Y.-Y. Wu, Z.-H. Luo, Precision polymer synthesis by controlled radical polymerization: fusing the progress from polymer chemistry and reaction engineering, Prog. Polym. Sci. 130 (2022) 101555, https://doi.org/10.1016/j.progpolymsci.2022.101555.
[238] K. Matyjaszewski, Advanced materials by atom transfer radical polymerization, Adv. Mater. 30 (23) (2018) 1706441, https://doi.org/10.1002/adma.201706441.
[239] S. Harrisson, R. Whitfield, A. Anastasaki, K. Matyjaszewski, Atom transfer radical polymerization, Nat. Rev. Methods Primers 5 (1) (2025) 2, https://doi.org/10.1038/s43586-024-00370-y.
[240] F. Catalão, J.R. Góis, A. Trino, A.C. Serra, J.F. Coelho, Facile synthesis of well-controlled poly (glycidyl methacrylate) and its block copolymers via SARA ATRP at room temperature, Polym. Chem. 6 (10) (2015) 1875–1882, https://doi.org/10.1039/C4PY01616E.
[241] Q. Lou, Block Copolymers: Synthesis and Applications in Nanotechnology, Clarkson University, 2011.
[242] M. Ma, F. Li, Fj. Chen, Sx. Cheng, Rx. Zhuo, Poly (ethylene glycol)-block-poly (glycidyl methacrylate) with oligoamine side chains as efficient gene vectors, Macromol. Biosci. 10 (2) (2010) 183–191, https://doi.org/10.1002/mabi.200900183.
[243] P.F. Cañamero, J.L. de la Fuente, E.L. Madruga, M. Fernández-García, Atom transfer radical polymerization of glycidyl methacrylate: a functional monomer. Macromol, Chem. Phys. 205 (16) (2004) 2221–2228, https://doi.org/10.1002/macp.200400186.
[244] D. Neugebauer, K. Bury, M. Wlazło, Atom transfer radical copolymerization of glycidyl methacrylate and methyl methacrylate, J. Appl. Polym. Sci. 124 (3) (2012) 2209–2215, https://doi.org/10.1002/app.35234.
[245] S.R. Woodruff, N.V. Tsarevsky, Synthesis of star polymers with epoxide-containing highly branched cores by low-catalyst concentration atom transfer radical polymerization and post-polymerization modifications, in: Controlled Radical Polymerization: Materials, ACS Publications, 2015, pp. 149–167.
[246] T. Erdogan, G. Hizal, U. Tunca, A new strategy for the preparation of multiarm star-shaped polystyrene via a combination of atom transfer radical polymerization and cationic ring-opening polymerization, Des. Monomers Polym. 9 (4) (2006) 393–401, https://doi.org/10.1163/156855506779521O1.
[247] M. Ejaz, S.C. Rai, K. Wang, K. Zhang, W. Zhou, S.M. Grayson, Surface-initiated atom transfer radical polymerization of glycidyl methacrylate and styrene from boron nitride nanotubes, J. Mater. Chem. C 2 (20) (2014) 4073–4079, https://doi.org/10.1039/C3TC32511C.
[248] G. Kocak, G. Solmaz, C. Tuncer, V. Bütün, Modification of glycidyl methacrylate based block copolymers and their aqueous solution behaviours, Eur. Polym. J. 110 (2019) 364–377, https://doi.org/10.1016/j.eurpolymj.2018.11.046.
[249] Q. Ren, Y. Xiang, C. Huang, J. Li, C. Wang, Epoxy-functionalized star-shaped polymers as novel toughens for epoxy resin, Polym. Bull. 72 (11) (2015) 2949–2965, https://doi.org/10.1007/s00289-015-1446-9.
[250] H. Ding, S. Park, M. Zhong, X. Pan, J. Pietrasik, C.J. Bettinger, et al., Facile arm-first synthesis of star block copolymers via ARGET ATRP with ppm amounts of catalyst, Macromolecules 49 (18) (2016) 6752–6760, https://doi.org/10.1021/acs.macromol.6b01597.
[251] R. París, B. Mosquera, J.L. de la Fuente, Atom transfer radical copolymerization of glycidyl methacrylate and allyl methacrylate, two functional monomers, Eur. Polym. J. 44 (9) (2008) 2920–2926, https://doi.org/10.1016/j.eurpolymj.2008.06.031.
[252] L. Jiang, X. Zhou, G. Wei, X. Lu, W. Wei, J. Qiu, Preparation and characterization of poly (glycidyl methacrylate)-grafted magnetic nanoparticles: effects of the precursor concentration on polyol synthesis of Fe_3O_4 and [PMDETA] 0/[$CuBr_2$] 0 ratios on SI-AGET ATRP, Appl. Surf. Sci. 357 (2015) 1619–1624, https://doi.org/10.1016/j.apsusc.2015.10.044.
[253] A. Brar, A.K. Goyal, Characterization and optimization of poly (glycidyl methacrylate-co-styrene) synthesized by atom transfer radical polymerization, Eur. Polym. J. 44 (12) (2008) 4082–4091, https://doi.org/10.1016/j.eurpolymj.2008.09.013.
[254] M. Flores-Guerrero, L.E. Elizalde, G.D.L. Santos-Villarreal, Synthesis of Poly (glycidyl methacrylate-r-photochromic monomer) by ATRP. Journal of macromolecular science®, part a, Pure Appl. Chem. 46 (3) (2009) 223–231, https://doi.org/10.1080/10601320802637045.
[255] D.J. Haloi, P. Mandal, Singha* NK., Atom transfer radical polymerization of glycidyl methacrylate (GMA) in emulsion, J. Macromol. Sci. A 50 (1) (2013) 121–127, https://doi.org/10.1080/10601325.2013.736270.
[256] W. Tang, K. Matyjaszewski, Effect of ligand structure on activation rate constants in ATRP, Macromolecules 39 (15) (2006) 4953–4959, https://doi.org/10.1021/ma0609634.
[257] K. Matyjaszewski, Atom transfer radical polymerization: from mechanisms to applications, Isr. J. Chem. 52 (3–4) (2012) 206–220, https://doi.org/10.1002/ijch.201100101.
[258] B. McCarthy, S. Sartor, J. Cole, N. Damrauer, G.M. Miyake, Solvent effects and side reactions in organocatalyzed atom transfer radical polymerization for enabling the controlled polymerization of acrylates catalyzed by diaryl dihydrophenazines, Macromolecules 53 (21) (2020) 9208–9219, https://doi.org/10.1021/acs.macromol.0c02245.
[259] I. Porello, F. Stucchi, R. Guarini, G. Sbaruffati, F. Cellesi, TCEP-enabled click modification of glycidyl-bearing polymers with biorelevant sulfhydryl molecules: toward chemoselective bioconjugation strategies, Biomacromolecules 26 (8) (2025) 5269–5286, https://doi.org/10.1021/acs.biomac.5c00766.
[260] P. Krys, Y. Wang, K. Matyjaszewski, S. Harrisson, Radical generation and termination in SARA ATRP of methyl acrylate: effect of solvent, ligand, and chain length, Macromolecules 49 (8) (2016) 2977–2984, https://doi.org/10.1021/acs.macromol.6b00345.
[261] X. Liu, Y. Li, Z. Chu, Y. Fang, H. Zheng, Surface modification of bacterial cellulose aerogels by ARGET ATRP, J. Appl. Biomater. Funct. Mater. 16 (1_suppl) (2018) 163–169, https://doi.org/10.1177/2280800018757337.
[262] S. Racioppi, L. Orian, C. Tubaro, A. Gennaro, A.A. Isse, Solvent coordination effect on copper-based molecular catalysts for controlled radical polymerization, Catalysts 12 (12) (2022) 1656, https://doi.org/10.3390/catal12121656.
[263] Z. Zhang, X. Wang, K.C. Tam, G. Sebe, A comparative study on grafting polymers from cellulose nanocrystals via surface-initiated atom transfer radical polymerization (ATRP) and activator re-generated by electron transfer ATRP, Carbohydr. Polym. 205 (2019) 322–329, https://doi.org/10.1016/j.carbpol.2018.10.050.
[264] M. Jonsson, D. Nyström, O. Nordin, E. Malmström, Surface modification of thermally expandable microspheres by grafting poly (glycidyl methacrylate) using ARGET ATRP, Eur. Polym. J. 45 (8) (2009) 2374–2382, https://doi.org/10.1016/j.eurpolymj.2009.05.002.
[265] S. Li, R. Colaco, A. Staubitz, ARGET ATRP of methyl acrylate and methyl methacrylate with diazocine-derived initiators, ACS Appl. Polym. Mater. 4 (10) (2022) 6825–6833, https://doi.org/10.1021/acsapm.2c00769.
[266] N. Dai, S. Wang, H. Li, L. Zhao, H. Jin, N. An, et al., Fabrication of poly (glycidyl methacrylate-co-ethylene glycol dimethacrylate) macroporous microspheres through activators regenerated by electron transfer atom transfer radical polymerization for rapid separation of proteins, J. Chromatogr. B 1128 (2019) 121794, https://doi.org/10.1016/j.jchromb.2019.121794.
[267] J. Wang, Z. Wu, G. Wang, K. Matyjaszewski, In situ crosslinking of nanoparticles in polymerization-induced self-assembly via ARGET ATRP of Glycidyl methacrylate, Macromol. Rapid Commun. 40 (2) (2019) 1800332, https://doi.org/10.1002/marc.201800332.
[268] X. Liang, J. Zhang, X. Wu, X. Fang, P. Qu, G. Wang, Polyolefin based Nano-objects via ARGET ATRP mediated polymerization-induced self-assembly process, Eur. Polym. J. 230 (2025) 113906, https://doi.org/10.1016/j.eurpolymj.2025.113906.
[269] M. Chen, L. Qin, Y. Liu, F. Zhang, Controllable preparation of polymer brushes from mesoporous silica SBA-15 via surface-initiated ARGET ATRP, Microporous Mesoporous Mater 263 (2018) 158–164, https://doi.org/10.1016/j.micromeso.2017.12.019.
[270] S. Hansson, E. Ostmark, A. Carlmark, E. Malmstrom, ARGET ATRP for versatile grafting of cellulose using various monomers, ACS Appl. Mater. Interfaces 1 (11) (2009) 2651–2659, https://doi.org/10.1021/am900547g.
[271] T. Guo, L. Zhang, X. Pan, X. Li, Z. Cheng, X. Zhu, A highly active homogeneous ICAR ATRP of methyl methacrylate using ppm levels of organocopper catalyst, Polym. Chem. 4 (13) (2013) 3725–3734, https://doi.org/10.1039/C3PY00309D.
[272] G.R. Jones, M.-N. Antonopoulou, N.P. Truong, A. Anastasaki, Initiators for continuous activator regeneration (ICAR) depolymerization, J. Am. Chem. Soc. 146 (51) (2024) 35023–35028, https://doi.org/10.1021/jacs.4c13785.
[273] M. Luan, D. Shen, P. Zhou, D. Li, P. Li, B. Shi, et al., One-pot synthesis of block copolymer dispersant by ICAR ATRP with ppm copper catalyst and the dispersibility on pigment, Prog. Org. Coat. 169 (2022) 106914, https://doi.org/10.1016/j.porgcoat.2022.106914.
[274] K. Kim, S.D. Sarkar, J. Strong, K. Takahashi, H.I. Coskun, R. Yin, et al., Comparison of various ATRP techniques for synthesis of polar block copolymers with polyolefins, Eur. Polym. J. (2026) 114723, https://doi.org/10.1016/j.eurpolymj.2026.114723.
[275] B. Shi, H. Zhang, Y. Liu, J. Wang, P. Zhou, M. Cao, et al., Development of ICAR ATRP–based polymerization-induced self-assembly and its application in the preparation of organic–inorganic nanoparticles, Macromol. Rapid Commun. 40 (24) (2019) 1900547, https://doi.org/10.1002/marc.201900547.
[276] Y. Zhou, Q. Hu, W. Wang, R. Wang, Controlled architecture of polyhedral oligomeric silsesquioxane-functionalized poly (glycidyl methacrylate)/polyester composites using surface-initiated ICAR ATRP technique for high flame Retardancy and smoke suppression, Polym. Degrad. Stab. 199 (2022) 109914, https://doi.org/10.1016/j.polymdegradstab.2022.109914.

[277] W. Wang, P. Julaiti, G. Ye, X. Huo, Y. Lu, J. Chen, Controlled architecture of glass fiber/poly (glycidyl methacrylate) composites via surface-initiated ICAR ATRP mediated by mussel-inspired polydopamine chemistry, Ind. Eng. Chem. Res. 56 (40) (2017) 11467–11476, https://doi.org/10.1021/acs.iecr.7b03065.
[278] X. Pan, M. Fantin, F. Yuan, K. Matyjaszewski, Externally controlled atom transfer radical polymerization, Chem. Soc. Rev. 47 (14) (2018) 5457–5490, https://doi.org/10.1039/C8CS00259B.
[279] A.J. Magenau, N.C. Strandwitz, A. Gennaro, K. Matyjaszewski, Electrochemically mediated atom transfer radical polymerization, Science (New York, N.Y.) 332 (6025) (2011) 81–84, https://doi.org/10.1126/science.1202357.
[280] D.A. Corbin, G.M. Miyake, Photoinduced organocatalyzed atom transfer radical polymerization (O-ATRP): precision polymer synthesis using organic photoredox catalysis, Chem. Rev. 122 (2) (2021) 1830–1874, https://doi.org/10.1021/acs.chemrev.1c00603.
[281] Q. Yang, S. Balverde, F. Dumur, J. Lalevée, J. Poly, Synergetic effect of the epoxide functional groups in the photocatalyzed atom transfer radical copolymerization of glycidyl methacrylate, Polym. Chem. 7 (39) (2016) 6084–6093, https://doi.org/10.1039/C6PY01443G.
[282] X. Liu, L. Zhang, Z. Cheng, X. Zhu, Metal-free photoinduced electron transfer–atom transfer radical polymerization (PET–ATRP) via a visible light organic photocatalyst, Polym. Chem. 7 (3) (2016) 689–700, https://doi.org/10.1039/C5PY01765C.
[283] Q. Yang, F. Dumur, F. Morlet-Savary, J. Poly, J. Lalevee, Photocatalyzed cu-based ATRP involving an oxidative quenching mechanism under visible light, Macromolecules 48 (7) (2015) 1972–1980, https://doi.org/10.1021/ma502384y.
[284] W. Ma, D. Chen, Y. Ma, L. Wang, C. Zhao, W. Yang, Visible-light induced controlled radical polymerization of methacrylates with cu (dap) 2 cl as a photoredox catalyst, Polym. Chem. 7 (25) (2016) 4226–4236, https://doi.org/10.1039/C6PY00687F.
[285] Z. Wang, Z. Wang, X. Pan, L. Fu, S. Lathwal, M. Olszewski, et al., Ultrasonication-induced aqueous atom transfer radical polymerization, ACS Macro Lett. 7 (3) (2018) 275–280, https://doi.org/10.1021/acsmacrolett.8b00027.
[286] M. Fantin, F. Lorandi, A.A. Isse, A. Gennaro, Sustainable electrochemically-mediated atom transfer radical polymerization with inexpensive non-platinum electrodes, Macromol. Rapid Commun. 37 (16) (2016) 1318–1322, https://doi.org/10.1002/marc.201600237.
[287] M. Fantin, S. Park, Y. Wang, K. Matyjaszewski, Electrochemical atom transfer radical polymerization in miniemulsion with a dual catalytic system, Macromolecules 49 (23) (2016) 8838–8847, https://doi.org/10.1021/acs.macromol.6b02037.
[288] F. Lorandi, M. Fantin, A.A. Isse, A. Gennaro, Electrochemically mediated atom transfer radical polymerization of n-butyl acrylate on non-platinum cathodes, Polym. Chem. 7 (34) (2016) 5357–5365, https://doi.org/10.1039/C6PY01032F.
[289] G. Wang, H. Wu, Microwave-assisted controlled copolymerization of styrene and acrylonitrile catalyzed by FeCl3/isophthalic acid, Polym. Bull. 67 (9) (2011) 1809–1821, https://doi.org/10.1007/s00289-011-0510-3.
[290] Z. Cheng, X. Zhu, J. Chen, J. Lu, Homogeneous atom transfer radical polymerization of methyl methacrylate under pulsed microwave irradiation, J. Macromol. Sci. A 40 (11) (2003) 1157–1171, https://doi.org/10.1081/MA-120024831.
[291] B.R. Coad, T. Bilgic, H.-A. Klok, Polymer brush gradients grafted from plasma-polymerized surfaces, Langmuir 30 (28) (2014) 8357–8365, https://doi.org/10.1021/la501380m.
[292] D. Benoit, V. Chaplinski, R. Braslau, C.J. Hawker, Development of a universal alkoxyamine for “living” free radical polymerizations, J. Am. Chem. Soc. 121 (16) (1999) 3904–3920, https://doi.org/10.1021/ja984013c.
[293] A. Moayeri, B. Lessard, M. Maric, Nitroxide mediated controlled synthesis of glycidyl methacrylate-rich copolymers enabled by SG1-based alkoxyamines bearing succinimidyl ester groups, Polym. Chem. 2 (9) (2011) 2084–2092, https://doi.org/10.1039/C1PY00190F.
[294] R.B. Grubbs, J.M. Dean, M.E. Broz, F.S. Bates, Reactive block copolymers for modification of thermosetting epoxy, Macromolecules 33 (26) (2000) 9522–9534, https://doi.org/10.1021/ma001414f.
[295] A.C. Schmidt, H. Turgut, D. Le, A. Beloqui, G. Delaittre, Making the best of it: nitroxide-mediated polymerization of methacrylates via the copolymerization approach with functional styrenics, Polym. Chem. 11 (2) (2020) 593–604, https://doi.org/10.1039/C9PY01458F.
[296] S. Brusseau, F. D'Agosto, S. Magnet, L. Couvreur, C. Chamignon, B. Charleux, Nitroxide-mediated copolymerization of methacrylic acid and sodium 4-styrene-sulfonate in water solution and one-pot synthesis of amphiphilic block copolymer nanoparticles, Macromolecules 44 (14) (2011) 5590–5598, https://doi.org/10.1021/ma2008282.
[297] B. Charleux, J. Nicolas, Water-soluble SG1-based alkoxyamines: a breakthrough in controlled/living free-radical polymerization in aqueous dispersed media, Polymer 48 (20) (2007) 5813–5833, https://doi.org/10.1016/j.polymer.2007.07.031.
[298] H.R. Lamontagne, B.H. Lessard, Nitroxide-mediated polymerization: a versatile tool for the engineering of next generation materials, ACS Appl. Polym. Mater. 2 (12) (2020) 5327–5344, https://doi.org/10.1021/acsapm.0c00888.
[299] G. Moad, E. Rizzardo, The history of nitroxide-mediated polymerization, 2015, https://doi.org/10.1039/9781782622635-00001.
[300] G. Gryn'ova, C.Y. Lin, M.L. Coote, Which side-reactions compromise nitroxide mediated polymerization? Polym. Chem. 4 (13) (2013) 3744–3754, https://doi.org/10.1039/C3PY00534H.
[301] R.B. Grubbs, Nitroxide-mediated radical polymerization: limitations and versatility, Polym. Rev. 51 (2) (2011) 104–137, https://doi.org/10.1080/15583724.2011.566405.
[302] E. Yoshida, Selective controlled/living photoradical polymerization of glycidyl methacrylate, using a nitroxide mediator in the presence of a photosensitive triarylsulfonium salt, Polymers 4 (3) (2012) 1580–1589, https://doi.org/10.3390/polym4031580.
[303] J. Nicolas, Y. Guillaneuf, C. Lefay, D. Bertin, D. Gigmes, B. Charleux, Nitroxide-mediated polymerization, Prog. Polym. Sci. 38 (1) (2013) 63–235, https://doi.org/10.1016/j.progpolymsci.2012.06.002.
[304] E. Yoshida, Nitroxide-mediated photo-living radical polymerization of methyl methacrylate using (4-tert-butylphenyl) diphenyl-sulfonium triflate as a photo-acid generator, Colloid Polym. Sci. 288 (2) (2010) 239–243, https://doi.org/10.1007/s00396-009-2161-6.
[305] E. Yoshida, Effects of initiators and photo-acid generators on nitroxide-mediated photo-living radical polymerization of methyl methacrylate, Colloid Polym. Sci. 288 (8) (2010) 901–905, https://doi.org/10.1007/s00396-010-2220-z.
[306] E. Yoshida, Nitroxide-mediated photo-living radical polymerization of methyl methacrylate in the presence of (η6-benzene)(η5-cyclopentadienyl) FeII hexafluorophosphate, Colloid Polym. Sci. 288 (18) (2010) 1745–1749, https://doi.org/10.1007/s00396-010-2292-9.
[307] E. Yoshida, Stability of growing polymer chain ends for nitroxide-mediated photo-living radical polymerization, Colloid Polym. Sci. 288 (9) (2010) 1027–1030, https://doi.org/10.1007/s00396-010-2230-x.
[308] D. Gromadzki, R. Makuška, M. Netopilík, P. Holler, J. Lokaj, M. Janata, et al., Comb copolymers of polystyrene-poly (tert-butyl (meth) acrylate) prepared by combination of nitroxide mediated polymerization and photoinduced iniferter technique, Eur. Polym. J. 44 (1) (2008) 59–71, https://doi.org/10.1016/j.eurpolymj.2007.10.024.
[309] B. Grignard, T. Phan, D. Bertin, D. Gigmes, C. Jerome, C. Detrembleur, Dispersion nitroxide mediated polymerization of methyl methacrylate in supercritical carbon dioxide using in situ formed stabilizers, Polym. Chem. 1 (6) (2010) 837–840, https://doi.org/10.1039/C0PY00066C.
[310] C. Detrembleur, C. Jérôme, J. De Winter, P. Gerbaux, J.-L. Clément, Y. Guillaneuf, et al., Nitroxide mediated polymerization of methacrylates at moderate temperature, Polym. Chem. 5 (2) (2014) 335–340, https://doi.org/10.1039/C3PY01133J.
[311] Z.-X. Zhou, Y.-W. Li, Y.-Q. Zheng, Z. Luo, C.-R. Gong, Y. Xu, et al., Synthesis and characterization of a dual-curing resin for three-dimensional printing, J. Mater. Sci. 54 (7) (2019) 5865–5876, https://doi.org/10.1007/s10853-018-3117-3.
[312] S.K. Dindar, T.M. Nasir, Z.S. Tofangdar, Synthesis and Characterization of Glycidyl Methacrylate Polymers Containing Tris (trimethylsilyl) Methyl Groups, Available online at: http://journal.ippi.ac.i, 2008.
[313] L. Cheng, J. Wu, H. Liang, Q. Yuan, Preparation of poly (glycidyl methacrylate) (PGMA) and amine modified PGMA adsorbents for purification of glucosinolates from cruciferous plants, Molecules 25 (14) (2020) 3286, https://doi.org/10.3390/molecules25143286.
[314] N. Erduran, Ö. Işılar, Novel magnetic poly (allyl methacrylate-co-glycidyl methacrylate) Fe3O4 nanoparticles: synthesis and evaluation as adsorbents for heavy metals, Polym. Bull. 81 (12) (2024) 10933–10952, https://doi.org/10.1007/s00289-024-05183-3.
[315] P. Bednarczyk, I. Irska, K. Gziut, P. Ossowicz-Rupniewska, Synthesis of epoxy methacrylate resin and coatings preparation by cationic and radical photocrosslinking, Molecules 26 (24) (2021) 7663, https://doi.org/10.3390/molecules26247663.
[316] T. Hunter, G. Price, Glycidyl methacrylate and N-vinylpyrrolidinone copolymers: synthesis and nuclear magnetic resonance characterization, Polymer 35 (16) (1994) 3530–3534, https://doi.org/10.1016/0032-3861(94)90919-9.
[317] I. Tzoumani, A. Soto Beobide, Z. Iatridi, G.A. Voyiatzis, G. Bokias, J.K. Kallitsis, Glycidyl methacrylate-based copolymers as healing agents of waterborne polyurethanes, Int. J. Mol. Sci. 23 (15) (2022) 8118, https://doi.org/10.3390/ijms23158118.
[318] N.R. Babij, E.O. McCusker, G.T. Whiteker, B. Canturk, N. Choy, L.C. Creemer, et al., NMR chemical shifts of trace impurities: industrially preferred solvents used in process and green chemistry, Org. Process. Res. Dev. 20 (3) (2016) 661–667, https://doi.org/10.1021/acs.oprd.5b00417.
[319] M. Espinosa, P. Del Toro, D. Silva, Microstructural analysis of poly (glycidyl methacrylate) by 1H and 13C NMR spectroscopy, Polymer 42 (8) (2001) 3393–3397, https://doi.org/10.1016/S0032-3861(00)00253-6.
[320] E.G. Crispim, J.F. Piai, I.T. Schüquel, A.F. Rubira, E.C. Muniz, Functionalization of poly (vinyl alcohol) by addition of methacryloyl groups: characterization by FTIR and NMR and optimization of reaction conditions by RSM, e-Polym 6 (1) (2006) 062, https://doi.org/10.1515/epoly.2006.6.1.793.
[321] A.D. Tulegenkyzy, P.S.M. Megat-Yusoff, K.M. Al Azzam, B.L. Kairatovna, A. Goyal, G. Eshmaiel, et al., Synthesis, characterization, and application of poly (styrene-co-glycidyl methacrylate) as reactive diluents to epoxy resin, Int. J. Technol. 15 (4) (2024), https://doi.org/10.14716/ijtech.v15i4.6920.
[322] M. Göktaş, C. Aykaç, Synthesis and characterization of the poly (GMA-b-EG) block copolymer containing epoxy-resin by redox polymerization, Kafkas Univ. Fen. Bilim Enst. Derg. 16 (1) (2023) 6–13, https://doi.org/10.58688/kujs.1239233.
[323] M. Maciejewska, E. Łastawiecka, M. Grochowicz, Thermal characterization of crosslinked polymeric microspheres bearing thiol groups studied by TG/FTIR/DSC under non-oxidative conditions, Materials 17 (6) (2024) 1372, https://doi.org/10.3390/ma17061372.

[324] M. Maciejewska, M. Grochowicz, Polymeric microspheres grafted with polyGMA chains containing thiol groups—synthesis and thermal characterization, J. Therm. Anal. Calorim. (2025) 1–15, https://doi.org/10.1007/s10973-025-14428-w.
[325] D.J. Haloi, S. Roy, N.K. Singha, Copper catalyzed atom transfer radical copolymerization of glycidyl methacrylate and 2-ethylhexyl acrylate, J. Polym. Sci. 47 (23) (2009) 6526–6533, https://doi.org/10.1002/pola.23695.
[326] A. Cardil, M. Palenzuela, J.F. Vega, M.E. Mosquera, Rheology of poly (glycidyl methacrylate) macromolecular nano assemblies, Polymers 14 (3) (2022) 455, https://doi.org/10.3390/polym14030455.
[327] F. He, B. Luo, S. Yuan, B. Liang, C. Choong, S.O. Pehkonen, PVDF film tethered with RGD-click-poly (glycidyl methacrylate) brushes by combination of direct surface-initiated ATRP and click chemistry for improved cytocompatibility, RSC Adv. 4 (1) (2014) 105–117, https://doi.org/10.1039/C3RA44789H.
[328] B. Kilincli, A.D. Cinar, B. Cetin, G. Kibar, Microfluidic vs. batch synthesis of fluorescent poly (GMA-co-EGDMA) micro/nanoparticles for biomedical applications, Emerg. Mater. (2024), https://doi.org/10.1007/s42247-024-00840-9.
[329] D.H. Chan, A.A. Cockram, R.R. Gibson, E.L. Kynaston, C. Lindsay, P. Taylor, et al., RAFT aqueous emulsion polymerization of methyl methacrylate: observation of unexpected constraints when employing a non-ionic steric stabilizer block, Polym. Chem. 12 (40) (2021) 5760–5769, https://doi.org/10.1039/D1PY01008E.
[330] U.A. Weerasinghe, T. Wu, P.L. Chee, P.Y.M. Yew, H.K. Lee, X.J. Loh, et al., Deep eutectic solvents towards green polymeric materials, Green Chem. 26 (15) (2024) 8497–8527, https://doi.org/10.1039/D4GC00532E.
[331] P. Verdía Barbará, H. Choudhary, P.S. Nakasu, A. Al-Ghatta, Y. Han, C. Hopson, et al., Recent advances in the use of ionic liquids and deep eutectic solvents for lignocellulosic biorefineries and biobased chemical and material production, Chem. Rev. 125 (12) (2025) 5461–5583, https://doi.org/10.1021/acs.chemrev.4c00754.
[332] A. Prabhune, R. Dey, Green and sustainable solvents of the future: deep eutectic solvents, J. Mol. Liq. 379 (2023) 121676, https://doi.org/10.1016/j.molliq.2023.121676.
[333] S.E. Kim, H.J. Yang, S. Choi, E. Hwang, M. Kim, H.-J. Paik, et al., A recyclable metal-free catalytic system for the cationic ring-opening polymerization of glycidol under ambient conditions, Green Chem. 24 (1) (2022) 251–258, https://doi.org/10.1039/D1GC03063A.
[334] Liang S-J, Deng J-P, W.-T. Yang, Preparation of low molecular weight poly (glycidyl methacrylate) by photopolymerization, Chin. J. Polym. Sci. 24 (05) (2006) 489–496.
[335] F. Zhong, L. Qiu, C.-Y. Hong, C.-Y. Pan, Allylthioketone mediated free radical polymerization of methacrylates, Polymers 9 (11) (2017) 608, https://doi.org/10.3390/polym9110608.
[336] G. Englezou, K. Kortsen, A.A. Pacheco, R. Cavanagh, J.C. Lentz, E. Krumins, et al., 2-Methyltetrahydrofuran (2-MeTHF) as a versatile green solvent for the synthesis of amphiphilic copolymers via ROP, FRP, and RAFT tandem polymerizations, J. Polym. Sci. 58 (11) (2020) 1571–1581, https://doi.org/10.1002/pol.20200183.
[337] M.H. Kunita, M.R. Guilherme, C. Lucio Filho, E.C. Muniz, E. Franceschi, C. Dariva, et al., Solid-state radical grafting reaction of glycidyl methacrylate and poly (4-methyl-1-pentene) in supercritical carbon dioxide: surface morphology and adhesion, J. Colloid Interface Sci. 361 (1) (2011) 331–337, https://doi.org/10.1016/j.jcis.2011.05.024.
[338] M. Daneshvar, H. Kamali, M. Masoomi, H. Ghaziaskar, Supercritical carbon dioxide grafting of glycidyl methacrylate onto medium density polyethylene and purification of residual monomer and initiator, J. Supercrit. Fluids 70 (2012) 119–125, https://doi.org/10.1016/j.supflu.2012.05.017.
[339] F.A. Versteeg, B.B. Benita, J.A. Jongstra, F. Picchioni, Reactive extrusion grafting of glycidyl methacrylate onto low-density and recycled polyethylene using supercritical carbon dioxide, Appl. Sci. 12 (6) (2022) 3022, https://doi.org/10.3390/app12063022.
[340] H. Shiho, J.M. DeSimone, Dispersion polymerization of glycidyl methacrylate in supercritical carbon dioxide, Macromolecules 34 (5) (2001) 1198–1203, https://doi.org/10.1021/ma0010074.
[341] C.M. Friend, B. Xu, Heterogeneous catalysis: a central science for a sustainable future, Acc. Chem. Res. 50 (3) (2017) 517–521, https://doi.org/10.1021/acs.accounts.6b00510.
[342] D.S. Su, J. Zhang, B. Frank, A. Thomas, X. Wang, J. Paraknowitsch, et al., Metal-free heterogeneous catalysis for sustainable chemistry, ChemSusChem 3 (2) (2010) 169–180, https://doi.org/10.1002/cssc.200900180.
[343] N. Tsubokawa, T. Yoshihara, Grafting of polyesters onto carbon whisker surface: cationic ring-opening polymerization of lactones initiated by acylium perchlorate groups introduced onto the surface, Polym. Bull. 30 (4) (1993) 421–428, https://doi.org/10.1007/BF00338475.
[344] A. Harrane, N. Naar, M. Belbachir, Ring opening polymerization of oxetane by the use of a montmorillonite clay as catalyst, Mater. Lett. 61 (17) (2007) 3555–3558, https://doi.org/10.1016/j.matlet.2006.11.118.
[345] A. Yahiaoui, M. Belbachir, Ring-opening polymerization of styrene oxide with Maghnite-H+ as ecocatalyst, J. Appl. Polym. Sci. 100 (2) (2006) 1681–1687, https://doi.org/10.1002/app.22946.
[346] A. Bouchama, M. Ferrahi, M. Belbachir, Maghnite, a green catalyst for synthesis of poly ([epsilon]-caprolactone-co-tetrahydrofuran), Chem. Sci. J. 3 (2012). http://astonjournals.com/csj.html.
[347] K. Benkenfoud, A. Harrane, M. Belbachir, Ring opening polymerization of tetrahydrofuran catalysed by Maghnite-H+, Chin. J. Polym. Sci. 30 (1) (2012) 56–62, https://doi.org/10.1007/s10118-012-1094-6.
[348] A. Yahiaoui, A. Hachemaoui, M. Belbachir, Cationic polymerization of ethylene oxide with Maghnite-H as a clay catalyst in the presence of ethylene glycol, J. Appl. Polym. Sci. 113 (1) (2009) 535–540, https://doi.org/10.1002/app.30106.
[349] D.E. Kherroub, M. Belbachir, S. Lamouri, K. Chikh, Cationic ring opening polymerization of octamethylcyclotetrasiloxane using a cost-effective solid acid catalyst (maghnite-H+), Iran J. Sci. Technol. Trans. A Sci. 43 (1) (2019) 75–83, https://doi.org/10.1007/s40995-017-0269-y.
[350] S. Haoue, H. Derdar, M. Belbachir, A. Harrane, Polymerization of ethylene glycol dimethacrylate (EGDM), using an Algerian clay as eco-catalyst (maghnite-H+ and Maghnite-Na+), Bull. Chem. React. Eng. Catal. 15 (1) (2020) 221–230, https://doi.org/10.9767/bcrec.15.1.6297.221-230.
[351] H. Derdar, M. Belbachir, A. Harrane, A green synthesis of polylimonene using Maghnite-H+, an exchanged montmorillonite clay, as eco-catalyst, Bull. Chem. React. Eng. Catal. 14 (1) (2019) 69–78, https://doi.org/10.9767/bcrec.14.1.2692.69-78.
[352] L. Souli, R. Meghabar, A. Djemoui, M.B. Lahrech, M. Belbachir, Water-soluble poly (HMBC-co-AM) prepared using tetrafunctional methacrylate macromonomer, J. Chem. Pharm. Res. 7 (10) (2015) 108–115.
[353] V. Ganesan, S.S. Vinothan, N. Lakshmaiya, M. Thangaraj, N. Pandiarajan, Nature-inspired bio-materials: an alternative resource, in: Green Manufacturing, CRC Press, 2025, pp. 121–157.
[354] S. Sharma, H. Shapouri, Bio-inspired alternatives in construction for a green built environment: a review, Innov. Infrastruct. Solut. 11 (3) (2026) 141, https://doi.org/10.1007/s41062-026-02538-9.
[355] Q. Xu, Y. Zhang, X. Li, J. He, J. Tan, L. Zhang, Enzyme catalysis-induced RAFT polymerization in water for the preparation of epoxy-functionalized triblock copolymer vesicles, Polym. Chem. 9 (39) (2018) 4908–4916, https://doi.org/10.1039/c8py01053f.
[356] W. Xie, L. Zhao, Y. Wei, J. Yuan, Advances in enzyme-catalysis-mediated RAFT polymerization, Cell Rep. Phys. Sci. 2 (7) (2021), https://doi.org/10.1016/j.xcrp.2021.100487.
[357] Y.B. Lee, S. Lim, Y. Lee, C.H. Park, H.J. Lee, Green chemistry for crosslinking biopolymers: recent advances in riboflavin-mediated photochemistry, Materials 16 (3) (2023) 1218, https://doi.org/10.3390/ma16031218.
[358] S.H. Kim, C.C. Chu, Fabrication of a biodegradable polysaccharide hydrogel with riboflavin, vitamin B2, as a photo-initiator and L-arginine as coinitiator upon UV irradiation, J Biomed Mater Res B Appl Biomater 91 (1) (2009) 390–400, https://doi.org/10.1002/jbm.b.31414.
[359] R. Goto, E. Nishida, S. Kobayashi, M. Aino, T. Ohno, Y. Iwamura, et al., Gelatin methacryloyl–riboflavin (Gelma–rf) hydrogels for bone regeneration, Int. J. Mol. Sci. 22 (4) (2021) 1635, https://doi.org/10.3390/ijms22041635.
[360] E.M. Sedek, E.A. Kamoun, N.M. El-Deeb, S. Abdelkader, A.E. Fahmy, S.R. Nouh, et al., Photocrosslinkable gelatin-treated dentin matrix hydrogel as a novel pulp capping agent for dentin regeneration: I. Synthesis, characterizations and grafting optimization, BMC Oral Health 23 (1) (2023) 536, https://doi.org/10.1186/s12903-023-03236-z.
[361] M.M. Abdul-Monem, E.A. Kamoun, D.M. Ahmed, E.M. El-Fakharany, F.H. Al-Abbassy, H.M. Aly, Light-cured hyaluronic acid composite hydrogels using riboflavin as a photoinitiator for bone regeneration applications, J. Taibah Univ. Med. Sci. 16 (4) (2021) 529–539, https://doi.org/10.1016/j.jtumed.2020.12.021.
[362] D.F. Othmer, Kirk-Othmer Encyclopedia of Chemical Technology, Wiley, 1978.
[363] E. Kester, C. Gaiser, M. Lazar, Glycidyl esters of aliphatic acids, J. Organomet. Chem. 8 (6) (1943) 550–556, https://doi.org/10.1021/jo01194a010.
[364] G. Maerker, E.J. Saggese, W.S. Port, Glycidyl esters. II. Synthesis of esters of commercial and pure fatty acids, J. Am. Oil Chem. Soc. 38 (4) (1961) 194–197, https://doi.org/10.1007/BF02633229.
[365] P. Lozano, E. García-Verdugo, From green to circular chemistry paved by biocatalysis, Green Chem. 25 (18) (2023) 7041–7057, https://doi.org/10.1039/D3GC01878D.
[366] R. Villa, R. Porcar, S. Nieto, A. Donaire, E. Garcia-Verdugo, S.V. Luis, et al., Sustainable chemo-enzymatic synthesis of glycerol carbonate (meth) acrylate from glycidol and carbon dioxide enabled by ionic liquid technologies, Green Chem. 23 (11) (2021) 4191–4200, https://doi.org/10.1039/D1GC01096D.
[367] P. Prete, D. Cespi, F. Passarini, C. Capacchione, A. Proto, R. Cucciniello, Glycidol syntheses and valorizations: boosting the glycerol biorefinery, Curr. Opin. Green Sustain. Chem. 35 (2022) 100624, https://doi.org/10.1016/j.cogsc.2022.100624.
[368] A. Sivo, I. Montanari, M.C. Ince, G. Vilé, Sustainable upgrading of glycerol into glycidol and its derivatives under continuous-flow conditions, Green Chem. 26 (13) (2024) 7911–7918, https://doi.org/10.1039/D4GC01565G.
[369] A. Kostyniuk, D. Bajec, P. Djinović, B. Likozar, One-step synthesis of glycidol from glycerol in a gas-phase packed-bed continuous flow reactor over HZSM-5 zeolite catalysts modified by CsNO3, Chem. Eng. J. 394 (2020) 124945, https://doi.org/10.1016/j.cej.2020.124945.
[370] A. Kostyniuk, A. Prašnikar, B. Likozar, Highly efficient and selective one-step production of glycidol from glycerol using Cs/zeolite catalysts in a packed-bed reactor, J. Environ. Health Sci. Eng. (2025) 120416, https://doi.org/10.1016/j.jece.2025.120416.
[371] S. Li, Z. Li, F. Zhang, H. Geng, B. Yang, A polymer-based zwitterionic stationary phase for hydrophilic interaction chromatography, Talanta 216 (2020) 120927, https://doi.org/10.1016/j.talanta.2020.120927.
[372] A.N. Tasfiyati, E.D. Iftitah, S.P. Sakti, A. Sabarudin, Evaluation of glycidyl methacrylate-based monolith functionalized with weak anion exchange moiety

inside 0.5 mm id column for liquid chromatographic separation of DNA, Anal. Chem. Res. 7 (2016) 9–16, https://doi.org/10.1016/j.ancr.2015.11.001.
[373] E. Suárez, B. Paredes, F. Rubiera, M. Rendueles, M. Villa-Garcıa, Díaz J., Functionalized glycidyl methacrylate based polymers as stationary phases for protein retention, Sep. Purif. Technol. 27 (1) (2002) 1–10, https://doi.org/10.1016/S1383-5866(01)00163-0.
[374] L. Li, H. Tian, J. He, M. Zhang, Z. Li, P. Ni, Fabrication of aminated poly (glycidyl methacrylate)-based polymers for co-delivery of anticancer drugs and the p53 gene, J. Mater. Chem. B 8 (41) (2020) 9555–9565, https://doi.org/10.1039/D0TB01811B.
[375] X. Dong, Y. Zheng, Y. Huang, X. Chen, X. Jing, Synthesis and characterization of multifunctional poly (glycidyl methacrylate) microspheres and their use in cell separation, Anal. Biochem. 405 (2) (2010) 207–212, https://doi.org/10.1016/j.ab.2010.06.022.
[376] J. Sinna, R. Jeencham, P. Mueangkhot, S. Sophon, P. Noralak, R. Raksapakdee, et al., Development of poly (vinyl alcohol) grafted glycidyl methacrylate/cellulose nanofiber injectable hydrogels for meniscus tissue engineering, Polymers 15 (21) (2023) 4230, https://doi.org/10.3390/polym15214230.
[377] F-m. Chen, Z-w. Ma, G-y. Dong, Z-f. Wu, Composite glycidyl methacrylated dextran (Dex-GMA)/gelatin nanoparticles for localized protein delivery, Acta Pharmacol. Sin. 30 (4) (2009) 485–493, https://doi.org/10.1038/aps.2009.15.
[378] H.A. López-Muñoz, M. Lopez-Romero, M.A. Franco-Molina, A. Manzano-Ramirez, C. Velasquillo, B.L. España-Sanchez, et al., Chitosan-G-glycidyl methacrylate/au nanocomposites promote accelerated skin wound healing, Pharmaceutics 14 (9) (2022) 1855, https://doi.org/10.3390/pharmaceutics14091855.
[379] R. Yang, J. Huang, W. Zhang, W. Xue, Y. Jiang, S. Li, et al., Mechanoadaptive injectable hydrogel based on poly (γ-glutamic acid) and hyaluronic acid regulates fibroblast migration for wound healing, Carbohydr. Polym. 273 (2021) 118607, https://doi.org/10.1016/j.carbpol.2021.118607.
[380] A. Pourjavadi, M. Kohestanian, C. Streb, pH and thermal dual-responsive poly (NIPAM-co-GMA)-coated magnetic nanoparticles via surface-initiated RAFT polymerization for controlled drug delivery, Mater. Sci. Eng. C 108 (2020) 110418, https://doi.org/10.1016/j.msec.2019.110418.
[381] M. Santhamoorthy, R. Vanaraj, K. Thirupathi, S. Ulagesan, T.-J. Nam, T.T. V. Phan, et al., L-lysine-modified pNIPAm-co-GMA copolymer hydrogel for pH- and temperature-responsive drug delivery and fluorescence imaging applications, Gels 9 (5) (2023) 363, https://doi.org/10.3390/gels9050363.
[382] A.K. Atef, T.B. Mostafa, H.M. El-Sherif, Hemocompatible gelatin-glycidyl methacrylate/graphene oxide composite hydrogels for vascular catheter applications, Sci. Rep. 15 (1) (2025) 10224, https://doi.org/10.1038/s41598-025-93040-2.
[383] J. Li, H. Zhong, D. Li, N. Peng, Double crosslinked cryogels based on glycidyl methacrylate modified chitosan for hemostasis and wound healing, J. Biomater. Appl. (2026) 08853282261427996, https://doi.org/10.1177/08853282261427996.
[384] A.I. Waly, M.A. Khedr, H.M. Ali, I.M. Ahmed, Application of amino-functionalized cellulose-poly (glycidyl methacrylate) graft copolymer (AM-cell-g-PGMA) adsorbent for dyes removal from wastewater, Cleaner Eng. Technol. 6 (2022) 100374, https://doi.org/10.1016/j.clet.2021.100374.
[385] T.M. Tamer, R.E. Khalifa, M.M. Abou-Krisha, A.M. Omer, A.G. Alhamzani, M. E. Youssef, et al., Dichromate contaminated water treatment using novel crystal violet azo dye-sulphonated poly (glycidyl methacrylate) nano-composite adsorbent, Water Air Soil Pollut. 236 (1) (2025) 35, https://doi.org/10.1007/s11270-024-07645-w.
[386] M.R. El-Aassar, T.M. Tamer, M.Y. El-Sayed, A.M. Omer, I.O. Althobaiti, M. E. Youssef, et al., Development of azo dye immobilized sulfonated poly (glycidyl methacrylate) polymer composite as novel adsorbents for water treatment applications: methylene blue immobilization isotherm, kinetic, thermodynamic, and simulations studies, Molecules 27 (23) (2022) 8418, https://doi.org/10.3390/molecules27238418.
[387] Y. Yu, J. Su, J. Liu, W. Li, Magnetic poly (glycidyl methacrylate) microspheres with grafted polypyrrole chains for the high-capacity adsorption of Congo red dye from aqueous solutions, Coatings 12 (2) (2022) 168, https://doi.org/10.3390/coatings12020168.
[388] N. Bektenov, A. Baidullayeva, T. Chalov, T. Jumadilov, S. Kanat, Modified adsorbents based on glycidyl methacrylate copolymers for the removal of copper and lead ions from wastewater, Eng. Sci. 31 (2024) 1237, https://doi.org/10.30919/es1237.
[389] S.A. Younis, M.M. Ghobashy, M. Samy, Development of aminated poly (glycidyl methacrylate) nanosorbent by green gamma radiation for phenol and malathion contaminated wastewater treatment, J. Environ. Health Sci. Eng. 5 (3) (2017) 2325–2336, https://doi.org/10.1016/j.jece.2017.04.024.
[390] M. Shahabi Nejad, H. Soltani Nejad, S. Arabnejad, H. Sheibani, Enhanced adsorption of perfluorooctanoic acid using functionalized imidazolium iodide ionic liquid-based poly (glycidyl methacrylate), J. Appl. Polym. Sci. 138 (38) (2021) 50962, https://doi.org/10.1002/app.50962.
[391] B.F. Senkal, F. Bildik, E. Yavuz, A. Sarac, Preparation of poly (glycidyl methacrylate) grafted sulfonamide based polystyrene resin with tertiary amine for the removal of dye from water, React. Funct. Polym. 67 (12) (2007) 1471–1477, https://doi.org/10.1016/j.reactfunctpolym.2007.07.042.
[392] Y. Zhou, X. Hu, M. Zhang, X. Zhuo, J. Niu, Preparation and characterization of modified cellulose for adsorption of cd (II), hg (II), and acid fuchsin from aqueous solutions, Ind. Eng. Chem. Res. 52 (2) (2013) 876–884, https://doi.org/10.1021/ie301742h.
[393] D. Husaain, M. Najam-ul-Haq, A. Saeed, F. Jabeen, M. Athar, M.N. Ashiq, Synthesis of poly GMA/DVB and its application for the removal of Malachite Green from aqueous medium by adsorption process, Desalin. Water Treat. 53 (9) (2015) 2518–2528, https://doi.org/10.1080/19443994.2013.867538.
[394] K. Gokkus, S.B. Sengel, Y. Yildirim, S. Hasanbeyoglu, V. Butun, Amine-functionalised poly (glycidyl methacrylate) hydrogels for Congo red adsorption, J. Environ. Eng. Sci. 18 (4) (2023) 204–214, https://doi.org/10.1680/jenes.23.00025.
[395] X. Guo, S. Ge, J. Wang, X. Zhang, T. Zhang, J. Lin, et al., Waterborne acrylic resin modified with glycidyl methacrylate (GMA): formula optimization and property analysis, Polymer 143 (2018) 155–163, https://doi.org/10.1016/j.polymer.2018.04.020.
[396] T.D. Caliskan, T. Caykara, G. Demirel, Toward water and oil repellent coating: synthesis of fluorinated methacrylate-glycidyl methacrylate copolymers, ACS Omega 9 (32) (2024) 34650–34660, https://doi.org/10.1021/acsomega.4c03275.
[397] A.P. Rananavare, J. Lee, Hydrophobic cotton fabric synthesized via dispersion polymerization from poly (glycidyl methacrylate) nanoparticles for self-cleaning applications, Prog. Org. Coat. 170 (2022) 107006, https://doi.org/10.1016/j.porgcoat.2022.107006.
[398] M. Heller, Q. Li, K. Esinhart, B. Pourdeyhimi, C. Boi, R.G. Carbonell, Heat induced grafting of poly (glycidyl methacrylate) on polybutylene terephthalate nonwovens for bioseparations, Ind. Eng. Chem. Res. 59 (12) (2020) 5371–5380, https://doi.org/10.1021/acs.iecr.9b04936.
[399] M. Moreno, M. Goikoetxea, M.J. Barandiaran, Biobased-waterborne homopolymers from oleic acid derivatives, J. Polym. Sci. 50 (22) (2012) 4628–4637, https://doi.org/10.1002/pola.26287.
[400] E. Can, J. La Scala, J. Sands, G. Palmese, The synthesis of 9–10 dibromo stearic acid glycidyl methacrylate and its use in vinyl ester resins, J. Appl. Polym. Sci. 106 (6) (2007) 3833–3842, https://doi.org/10.1002/app.26249.
[401] G.R. Palmese, J.J. LaScala, J.M. Sands, Fatty acid monomers to reduce emissions and toughen polymers, U.S. Patent No. 8, 372, 926, (2013).
[402] M. Petro, F. Svec, J.M. Fréchet, Monodisperse hydrolyzed poly (glycidyl methacrylate-co-ethylene dimethacrylate) beads as a stationary phase for normal-phase HPLC, Anal. Chem. 69 (16) (1997) 3131–3139, https://doi.org/10.1021/ac970365a.
[403] P. Cañamero, M. Fernández-García, J.L. de la Fuente, Adhesives based on Poly (glycidyl methacrylate-co-butyl acrylate) with controlled structure: curing behavior and adhesion properties on metal substrates, Macromol. Mater. Eng. 309 (2) (2024) 2300267, https://doi.org/10.1002/mame.202300267.
[404] Z. Qin, Q. Gao, S. Zhang, J. Li, Glycidyl methacrylate grafted onto enzyme-treated soybean meal adhesive with improved wet shear strength, BioResources 8 (4) (2013).
[405] R.V. Gadhave, Glycidyl methacrylate co-polymer emulsion-based heat and water-resistant wood adhesive, J. Adhes. Sci. Technol. 38 (23) (2024) 4236–4257, https://doi.org/10.1080/01694243.2024.2369270.
[406] S. Ghosh, N. Krishnamurti, Use of glycidyl methacrylate monomers for developing cross-linkable pressure sensitive adhesives, Eur. Polym. J. 36 (10) (2000) 2125–2131, https://doi.org/10.1016/S0014-3057(00)00003-3.
[407] Z. Starý, T. Pemsel, J. Baldrian, H. Münstedt, Influence of a compatibilizer on the morphology development in polymer blends under elongation, Polymer 53 (9) (2012) 1881–1889, https://doi.org/10.1016/j.polymer.2012.02.056.
[408] S. Mbarek, M. Jaziri, Y. Chalamet, C. Carrot, Effect of the viscosity ratio on the morphology and properties of PET/HDPE blends with and without compatibilization, J. Appl. Polym. Sci. 117 (3) (2010) 1683–1694, https://doi.org/10.1002/app.32050.
[409] S. Ahmad, Influence of blend composition and compatibilizer on mechanical and morphological properties of recycled HDPE/PET blends, Mater. Sci. Appl. (2023), https://doi.org/10.4236/msa.2014.513096.
[410] M. Pracella, M.M.-U. Haque, V. Alvarez, Functionalization, compatibilization and properties of polyolefin composites with natural fibers, Polymers 2 (4) (2010) 554–574, https://doi.org/10.3390/polym2040554.
[411] L. Yu, Y. Qiu, B. Yang, M. Gao, L. Song, Y. Jing, et al., Developing high-performance and sustainable polylactic acid/recycled polyolefin blends: tuning the degree of functional group reaction and performance optimization, Int. J. Biol. Macromol. 289 (2025) 138554, https://doi.org/10.1016/j.ijbiomac.2024.138554.
[412] Y. Xu, J. Loi, P. Delgado, V. Topolkaraev, R.J. McEneany, C.W. Macosko, et al., Reactive compatibilization of polylactide/polypropylene blends, Ind. Eng. Chem. Res. 54 (23) (2015) 6108–6114, https://doi.org/10.1021/acs.iecr.5b00882.
[413] X. Yang, H. Wang, J. Chen, Z. Fu, X. Zhao, Y. Li, Copolymers containing two types of reactive groups: new compatibilizer for immiscible PLLA/PA11 polymer blends, Polymer 177 (2019) 139–148, https://doi.org/10.1016/j.polymer.2019.05.074.
[414] D. Gwon, D. Kim, J. Koo, Compatibility study of recycled polypropylene (PP)/Poly (ethylene terephthalate)(PET) blends nanocomposites with PP-g-MAH: modeling of twin screw extrusion, Polym. Adv. Technol. 35 (9) (2024) e6557, https://doi.org/10.1002/pat.6557.
[415] M. Kumar, S. Mohanty, S. Nayak, M.R. Parvaiz, Effect of glycidyl methacrylate (GMA) on the thermal, mechanical and morphological property of biodegradable PLA/PBAT blend and its nanocomposites, Bioresour. Technol. 101 (21) (2010) 8406–8415, https://doi.org/10.1016/j.biortech.2010.05.075.
[416] M. Lillethorup, K. Shimizu, N. Plumeré, S.U. Pedersen, K. Daasbjerg, Surface-attached poly (glycidyl methacrylate) as a versatile platform for creating dual-functional polymer brushes, Macromolecules 47 (15) (2014) 5081–5088, https://doi.org/10.1021/ma500872b.
[417] F. Xu, Q. Cai, Y. Li, E. Kang, K. Neoh, Covalent immobilization of glucose oxidase on well-defined poly (glycidyl methacrylate)− Si (111) hybrids from surface-

initiated atom-transfer radical polymerization, Biomacromolecules 6 (2) (2005) 1012–1020, https://doi.org/10.1021/bm0493178.
[418] Z. Zhang, S. Yuan, X. Zhu, K. Neoh, E. Kang, Enzyme-mediated amperometric biosensors prepared via successive surface-initiated atom-transfer radical polymerization, Biosens. Bioelectron. 25 (5) (2010) 1102–1108, https://doi.org/10.1016/j.bios.2009.09.023.
[419] S. Mohammed Safiullah, K. Abdul Wasi, K. Anver Basha, A comparative study on corrosion-resistant performance of halogenated copolymers, J. Coat. Technol. Res. 11 (5) (2014) 685–696, https://doi.org/10.1007/s11998-014-9588-5.
[420] M. Imperiyka, A. Ahmad, S. Hanifah, M. Rahman, Preparation and characterization of polymer electrolyte of Glycidyl methacrylate-methyl methacrylate-LiClO4 plasticized with ethylene carbonate, Int. J. Polym. Sci. 2014 (1) (2014) 638279, https://doi.org/10.1155/2014/638279.
[421] M. Dervisevic, E. Çevik, Z. Durmuş, M. Şenel, Electrochemical sensing platforms based on the different carbon derivative incorporated interface, Mater. Sci. Eng. C 58 (2016) 790–798, https://doi.org/10.1016/j.msec.2015.09.052.
[422] X. Zhao, J. Hu, J. Xie, Y. Liu, S. Sun, Highly transparent PVDF films with enhanced dielectric and energy storage properties tailored by a PMMA-co-GMA copolymer, ACS Appl. Energy Mater. 5 (10) (2022) 12693–12706, https://doi.org/10.1021/acsaem.2c02322.
[423] M. Aydın, An ultrasensitive immunosensor based on tri-armed star poly (glycidyl methacrylate) polymer-coated ITO-PET electrode for detection of neuron-specific enolase in human serum, Int. J. Environ. Anal. Chem. 100 (4) (2020) 492–506, https://doi.org/10.1080/03067319.2019.1673744.
[424] S.M. Mugo, J. Alberkant, Flexible molecularly imprinted electrochemical sensor for cortisol monitoring in sweat, Anal. Bioanal. Chem. 412 (8) (2020) 1825–1833, https://doi.org/10.1007/s00216-020-02430-0.
[425] M. Imperiyka, A. Ahmad, S. Hanifah, N. Mohamed, M. Rahman, Investigation of plasticized UV-curable glycidyl methacrylate based solid polymer electrolyte for photoelectrochemical cell (PEC) application, Int. J. Hydrog. Energy 39 (6) (2014) 3018–3024, https://doi.org/10.1016/j.ijhydene.2013.03.059.
[426] M. Imperiyka, A. Ahmad, S. Hanifah, M. Rahman, Potential of UV-curable poly (glycidyl methacrylate-co-ethyl methacrylate)-based solid polymer electrolyte for lithium ion battery application, Int. J. Electrochem. Sci. 8 (9) (2013) 10932–10945, https://doi.org/10.1016/S1452-3981(23)13160-9.